PROGRAMME PROJECT REPORT

MBA TOURISM

DIRECTORATE OF DISTANCE EDUCATION
ALAGAPPA UNIVERSITY
KARAIKUDI – 630 003

PROGRAMME PROJECT REPORT

MBA TOURISM

a) PROGRAMME MISSION AND OBJECTIVES

Business Studies have fascinated humans for two reasons, namely generating interest and augmenting essentials of running a firm effectively. That is why their study is enchanting and glorifying. The primary objective of this programme is to provide ample exposure to subjects from the fields of business legacy and accountancy, equip the Students for entry level jobs in industry and to contribute to the economic development of the country.

b) RELEVANCE OF THE PROGRAMME WITH HEI'S MISSION AND GOALS:

The Alagappa University is functioning with following Vision and Mission:

Mission: Achieving Excellence in all spheres of Education, with particular emphasis on 'PEARL"- Pedagogy, Extension, Administration, Research and Learning

Vision: Affording High Quality Higher Education to the learners so that they are transformed into intellectually competent human resources that will help in the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence.

Therefore, the introduction of MBA TOURISMprogramme in the Directorate of Distance Education will contribute substantially in fulfilling the mission of Alagappa University. Such a higher education in subject with appropriate Practical Exposer will enrich the human resources for the uplift of the Nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

c) NATURE OF PROSPECTIVE TARGET GROUP OF LEARNERS

- Working Professionals
- Entrepreneurs
- Service Personnel
- Academic Faculty
- Government Officials
- Researchers
- Home makers
- Unemployed Graduates

d) APPROPRIATENESS OF PROGRAMME

To Attain Leadership in spearheading qualitative and responsible academic programs relevant to the society through cost effective off-campus distance mode of education. knowledge and understanding, skills, qualities and other attributes in the following areas:

- The fundamental concepts of Management
- The higher-level taxonomy and diversity of Business Studies.
- How principles of Business can be applied to problems
- Internship training in Industry
- Undertake Inter tasks and techniques.
- Inter-disciplinary knowledge like statistics, Mathematics, Computer and E-Banking.
- Using the SPSS package for the analysis of data
- It also improves the Intellectual skills of the students.
- In nutshell, these skills will improve the performance of the students parameters

E) INSTRUCTIONAL DESIGN
MBA TOURISM

Course Code	Title	CIA Max.	ESE Max.	TOT Max.	C
I Semester					
33811	Management – Principles and Practices	25	75	100	4
33812	Organizational Behaviour	25	75	100	4
33813	Managerial Economics	25	75	100	4
33814	Quantitative Techniques	25	75	100	4
33815	Financial and Management Accounting	25	75	100	4
	Total	125	375	500	20
II Semester					
33821	Research Methods	25	75	100	4
33822	Business Environment	25	75	100	4
33823	Business Laws	25	75	100	4
33824	Management Information System	25	75	100	4
33825	Human Resource Management	25	75	100	4
	Total	125	375	500	20
III Semester					
33831	Marketing Management	25	75	100	4
33832	Financial Management	25	75	100	4
33833	Tourism and Accommodation	25	75	100	4
33834	Tour Operators and Travel Agencies	25	75	100	4
33835	Quality in Tourism	25	75	100	4
	Total	125	375	500	20
IV Semester					
33841	Tourism Entrepreneurship	25	75	100	4
33842	Tourism Project Management	25	75	100	4
33843	Institutional Framework of Tourism	25	75	100	4
33844	Global Tourism	25	75	100	4
33845	Emerging Issues in Tourism	25	75	100	4
	Total	125	375	500	20
Grand Total				**2000**	**80**

Course Code Legend:

3	**3**	**8**	**Y**	**Z**

338- M.B.A Programme
Y -Semester Number
Z- Course Number in the Semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

No. of Credits per Course (Theory) - 5	Total No. of Credits per Semester- 20
	Total Credits for the Programme- 20 X 4 = 80

33811- MANAGEMENT PRINCIPLES AND PRACTICES

Objectives:

- To introduce the basic concepts of Management functions and principles
- To learn the scientific decision making and modern trend in the management process
- To understand the contemporary practices and issues in management

UNIT 1 Management:Definition – Nature, Scope and Functions – Evolution of Management – Management thought in modern trend – Patterns of the management analysis – Management Vs. Administration - Management and Society: The external Environment, Social Responsibility and Ethics.

UNIT 2 Management Science and Theories : Contributions of FW Taylor, Henri Fayol, Elton Mayo, Roethilisberger, H.A.Simon and P.F Drucker - Universality of Management - Relevance of management to different types of organization.

UNIT 3 Planning: Nature and Purpose – Principles and planning premises – Components of planning as Vision, Mission, Objectives, Managing By Objective (MBO) Strategies, Types and Policies -Planning and Decision Making: Planning process.

UNIT 4 Decision making: Meanings and Types – Decision-making Process under Conditions of Certainty and Uncertainty – Rational Decision Making Strategies, Procedures, Methods, Rules, Projects and Budgets.

UNIT 5 Organizing: Nature, Importance, Principles, purpose and Scope - Organizing functions of management – Classifications of organization – Principles and theories of organization – Effective Organizing – Organizational Culture and Global Organizing.

UNIT 6 Organizational Structure – Departmentalization – Span of control – Line and staff functions – Formal and Informal Groups in Organizations - Authority and responsibility - Centralization and decentralization – Delegation of authority – Committees – Informal organization.

UNIT 7 Staffing: General Principles of Staffing- Importance, techniques, Staff authority and Empowerment in the organization – Selection and Recruitment - Orientation - Career Development - Career stages – Training – Performance Appraisal.

UNIT 8 Creativity and Innovation – Motivation - Meaning – Importance – Human factors of Motivation – Motivation Theories: Maslow, Herzberg, Mc Gregor (X&Y), Ouchi (Z) ,Vroom, Porter-Lawler, McClelland and Adam – Physiological and psychological aspects of motivation .

UNIT 9 Directing : Meaning, Purpose, and Scope in the organization – Leadership: Meaning, Leadership styles, Leadership theories: Trait, Contingency, Situation, Path-Goal, Tactical, Transactional, Transformational and Grid.

Leaders: Type, Nature, Significance and Functions, Barriers, Politics and Ethics. Leader Vs. Manager.

UNIT 10 Communications: Meaning – Types – Process – Communication in the decision making – Global Leading - Effective communication in the levels of management. – Uses of Communication to Planning, Organizing, coordinating and controlling.

UNIT 11 Co-ordination: Concept; Meaning, Characteristics, Importance in the organization, Co-ordination process and principles - Techniques of Effective co-ordination in the organization - Understanding and managing the group process.

UNIT 12 Business ethics: Relevance of values in Management; Holistic approach for managers indecision-making; Ethical Management: Role of organizational culture in ethics – Ethics Committee in the organization.

UNIT 13 Controlling: Objectives and Process of control Devices of control – Integrated control – Special control techniques- Contemporary - Perspectives in Device of Controls

UNIT 14 New Perspectives in Management - Strategic alliances – Core competence – Business process reengineering – Total quality management – Six Sigma- Benchmarking- Balanced Score-card.

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1. Stoner, et-al, Management, Prentice Hall, 1989.
2. Koontz and O'Donnell, Management: A Systems Approach, McGraw Hill, 1990
3. **Weihrich and Koontz,**Management: A Global Perspective, McGraw Hill, 1988
4. Peter F. Drucker, Management, 2008.
5. Gene Burton and Manab Thakur, Management Today: Principles and Practice, Tata McGraw Hill.
6. Ricky W. Griffin, Management, South-WesternCollege Publications, 2010
7. Stephen P. Robbins and Mary Coulter, Management, 9th Edition, 2006.
8. Kaplan and Norton, The Strategy-Focused Organization: How Balanced Scorecard Companies Thrive in the New Business Environment, HBP, 2000.

33812 - ORGANIZATIONAL BEHAVIOUR

Objectives:

- To understand the personality trades and influence on the organization.
- To imbibe the necessary conceptual understanding of behaviour related people
- To learn the modern trends, theories and changes in organizational Behaviour.

UNIT 1 Organizational Behaviour: History – Meaning Elements – Evolution, Challenges and opportunities – Trends – disciplines – Approaches – Models – Management functions relevance to organizational Behaviour – Global Emergence of OB as a discipline.

UNIT 2 Personality – Determinants, Structure, Behaviour, Assessment, Individual Behaviour: Personality & Attitudes- Development of personality – Nature and dimensions of attitude – Trait Theory – Organizational fit – Organizational Commitment

UNIT 3 Emotions – Emotional Intelligence – Implications of Emotional Intelligence on Managers – EI as Managerial tool – EI performance in the organization – Attitudes: Definitions – Meaning – Attitude relationship with behaviour – Types – Consistency

UNIT 4 Individual Behaviour and process of the organization: Learning, Emotions, Attitudes, Perception, Motivation, Ability, Job satisfaction, Personality, Stress and its Management – Problem solving and Decision making – Interpersonal Communication - Relevance to organizational behaviour.

UNIT 5 Group Behaviour: Group Dynamics - Theories of Group Formation - Formal and Informal Groups in organization and their interaction - Group norms – Group cohesiveness – Team: Importance and Objectives - Formation of teams – Team Work- Group dynamics – Issues - Their relevance to organizational behaviour.

UNIT 6 Organizational Power: Organizational Power: Definition, Nature, Characteristics - Types of powers - Sources of Power - Effective use of power – Limitations of Power – Power centre in Organization.

UNIT 7 Organizational Politics: Definition – Political behaviour in organization - Factors creating political behaviour – Personality and Political Behaviour - Techniques of managing politics in organization – Impact of organizational politics.

UNIT 8 Organizational Conflict Management: Stress Management: Meaning – Types – Sources and strategies resolve conflict – Consequences – Organizational conflict: Constructive and Destructive conflicts - Conflict Process - Strategies for encouraging constructive conflict - Strategies for resolving destructive conflict.

UNIT 9 Organizational Dynamics: Organizational Dynamics – Organizational Efficiency, Effectiveness and Excellence: Meaning and Approaches – Factors affecting the organizational Climate.

UNIT 10 Organizational Culture: Meaning, significance – Theories – Organizational Climate – Creation, Maintenance and Change of Organizational Culture – Impact of organizational culture on strategies – Issues in Organizational Culture.

UNIT 11 Inter personal Communication: Essentials, Networks, Communication technologies – Non-Verbal communications Barriers – Strategies to overcome the barriers. Behavioral Communication in organization - Uses to Business

UNIT 12 Organizational Change: Meaning, Nature and Causes of organizational change Organizational Change –Importance – Stability Vs Change – Proactive Vs. Reaction change – the change process – Resistance to change – Managing change.

UNIT 13 Organizational Behaviour responses to Global and Cultural diversity, challenges at international level, Homogeneity and heterogeneity of National cultures, Differences between countries.

UNIT 14 Organizational Development: Meaning, Nature and scope – Features of OD – OD Interventions- Role of OD – Problems and Process of OD – process OD and Process of Intervention - Challenges to OD- Learning Organizations - Organizational effectiveness Developing Gender sensitive workplace

REFERENCES

1. Fred Luthans, Organizational Behaviour, McGraw-Hill/Irwin, 2006.
2. Stephen P. Robbins, Organizational Behaviour, Prentice Hall; 2010
3. Keith Davis, Organizational Behavior: Human Behavior at Work, McGraw Hill, 2010
4. Griffin and Moorhead, Organizational Behavior: Managing People and Organizations, 2006.
5. Judith R. Gordon, Organizational Behavior: A Diagnostic, Prentice Hall, 2001.
6. K. Aswathappa, Organizational Behaviour, Himalaya Publishing, Mumbai, 2010
7. Judith R. Gordon, A Diagnostic Approach to Organizational Behaviour, Allyn & Bacon, 1993.

33813 - MANAGERIAL ECONOMICS

Objectives:

- To understand the economic principles and its applications in business
- To develop economics based analytic skills for business
- To make the learners to strong in economical approach

UNIT 1 Economics: Introduction – Meaning, nature and scope of Managerial Economics – General Foundations of managerial Economics – Economic Approach – Working of Economic system - Circular flow activities - Economics & Business Decisions - Relationship between Economic theory and Managerial Economics.

UNIT 2 Business Decisions: Role of managerial Economics in Decision making – Decision making under Risk and Uncertainty - Concepts of Opportunity cost, - Production possibility curve – Incremental Concepts - Cardinal and Ordinal approaches to consumer Behaviour Time Value of Money –

UNIT 3 Consumer Behaviour: Marginalism – Equilibrium and Equi-marginalism and their role in business decision making. – Equi-Marginal principles – Utility analysis – Total and Marginal Utility – Law of diminishing marginal utility – Marshallian approach and Indifference curve analysis.

UNIT 4 Demand analysis: Meaning, Functions - Determinants of demand-Law of Demand – Demand Estimation and Forecasting - Applications of demand in analysis - Elasticity of Demand: Types, Measures and Role in Business Decisions.

UNIT 5 Supply Analysis: Determinants of supply- Elasticity of Supply- Measures and Significance - Derivations of market demand – Demand Estimation and Fore casting- Demand and Supply equilibrium – Giffen Paradox

UNIT 6 Production Functions: Managerial uses of production function - Cobb-Douglas and other production functions - Isoquants – Short run and long run production function – Theory of production – Empirical estimations of production functions.

UNIT 7 Forms of Markets: Meaning and Characteristics - Market Equilibrium: Practical Importance, Market Equilibrium and Changes in Market Equilibrium. Pricing Functions: Market Structures - Pricing and output decisions under different competitive conditions: Monopoly Monopolistic completion and Oligopoly

UNIT 8 Strategic Behaviour of the firms and Game Theory - Nash Equilibrium: Implications – Prisoner's Dilemma: Types of strategy – Price and Non price competition – Relation to the firm behaviour.

UNIT 9 Cost and Return: Cost function and cost output relationship – Economics and Diseconomies of scale - Cost control and cost reduction- Cost Behaviour and Business Decision- Relevant costs for decision-making- Traditional and Modern theory of Cost.

UNIT 10 New Product Penetrative Decision and Skimming the cream Pricing- Government control over pricing - Concept of Profit- Types and Theories of Profit by Knight (Uncertainty), Schumpeter (Innovation), Clark (Dynamic) and Hawley (Risk) - Profit maximization – Cost volume profit analysis – Risk and Return Relationship.

UNIT 11 Profit and Investment Analysis: Meaning – Measurement of profit – Theories of Pricing- Profit planning and forecasting- Profit and Wealth maximization – Cost volume profit analysis – Investment analysis and Evaluation: IRR, NPV and APV techniques.

UNIT 12 Macro-economic Factors: Nature, Importance ; Economic Growth and Development - Business cycle – Phases and Business Decision- Inflation - Factors causing Inflation and Deflation - Control measures – Balance of payment Trend and its implications in managerial decision.

UNIT 13 National Income: Introduction Meaning – Theories – Methods of Measurement - Sectoral and Population distributions – Per capita Income: Definition – Calculations – Uses – Limitations – GDP – GNP - Recent developments in Indian Economy.

UNIT 14 Economic Regulations of Business: Introduction – Antitrust theory and Regulations – The structure – Conduct – Performance paradigm – Concentration: Overview – Measuring concentration – Regulation of Externalities.

REFERENCES

1. Dominick Salvatore, Managerial Economics in a Global Economy,OxfordUniversity Press, 2011.
2. Ivan Png and Dale Lehman, Managerial Economics, Wiley-Blackwell, 2007.
3. Truett Lila J., Truett, Dale B. and Truett J. Lila (2006), Managerial Economics: Analysis Problems, Cases, 8th Editon, John Wiley & Sons.
4. Atmanand (2008), Managerial Economics, 2nd Edition, Excel Books.
5. Christopher R Thomas & S Charles Maurice (2008), Managerial Economics, 9th edition, McGraw Hill Co.
6. Petersen, H. C., Cris, L W and Jain, S.K. (2008), Managerial Economics, 1st edition Pearson
7. Gupta G S, Managerial Economics, Tata McGraw-Hill.
8. Varshney and Maheswari, Managerial Economics, Sultan Chand and Sons.
9. Mehta P L, Managerial Economics, Sultan Chand and Sons.
10. Joel Dean, Managerial Economics, Prentice-Hall.

33814 - QUANTITATIVE TECHNIQUES

Objectives:

- To help develop analytical skills based on problem solving approach
- To learn quadrature problems solving of business issues.
- To acquire the knowledge in statistics and their use inbusiness decision making.

UNIT 1 Basic Quantitative Concepts: Place of quantitative analysis in the practice of management – Problem definition: Models and their development. Variables notion of Mathematical models – concept of trade off – Notion of constants – concept of Interest.

UNIT 2 Basic Concept of differentiation – integration – Optimization concepts – use of differentiation for optimization of business problem Optimization Statistics: Meaning and Applications of Statistics in business decision making and research - Collection, Tabulation and presentation of data - Measures of central tendency: Mean, Median and Mode. Measures of dispersion

UNIT 3 Variables and function: Linear and Non-linear –Graphical representation of functions and their applications in cost and revenue behavior. Slope and its relevance –Use of functional relationships to understand elasticity of demands, Relationship between costs and level of activity, Decisions on Minimizing Costs and Maximizing output/profits.

UNIT 4 Linear Programming: Introduction to the linear programming – Concepts of optimization- Formulation of different types of linear programming –Standard from of LP problems - Importance and practical implementation in Industry

UNIT 5 Simple regression and Correlation analysis: Introduction, Correlation, Correlation analysis, linear regression analysis and Co-efficient. Duality and sensitivity analysis for decision-making- Solving LP using graphical and simplex method (only simple problems) – Interpreting the solution for decision-making

UNIT 6 Special Algorithms of LPP: Transportation Algorithm - Balanced and Unbalanced Problem Formulation and solving methods: North West Corner, Vogel's Approximation-MODI method- Assignment and Travelling Executive Algorithms

UNIT 7 Theory of Probability: Introduction to the Concept – Development of probability – Areas and Utilisation of probability theories in the Business – Sample space – terminology – Types of probability.

UNIT 8 Theoretical Probability Distributions: Introduction - Concept of events – Probability of events – Joint, conditional and marginal probabilities Probability distributions: Binomial, Poisson and Normal – Features and Applications – Use of Normal Tables.

UNIT 9 Operational research for Decision Making: Historical background and Developments – Definition – Phases in the use of Operations research – Models – Characteristics of quantitative methods - Benefits and Limitations of Quantitative methods.

UNIT 10 Sequencing /Scheduling Methods : Concepts – terminology – Notations – Assumption for scheduling models – Job sequencing priorities – Processing the job and Mass production system.

UNIT 11 Simulation Techniques: Introduction to simulation as an aid to decision-making- Advantages and Disadvantages of Simulation – Applications of simulations models – Types: Inventory, Cash, and Project – Random Numbers.

UNIT 12 Queuing Theory: Introduction – Definition – Queue priorities Product launching problems using Monte Carlo simulation- Queuing Theory: M/M/1 queuing model and applications.

UNIT 13 Decision Analysis: Concepts – Definition – Decision Tables Pay-off and Loss tables – Expected value of pay-off – Expected value of Perfect Formation – decision making process

UNIT 14 Decision Tree Analysis: Decision making environments – Concept of Posterior probabilities Decision Tree approach to choose optimal course of action Criteria for decision – Mini-max, Maxi-max, Minimizing Maximal Regret and their applications.

REFERENCES

1. David R. Anderson, et al, An Introduction to Management Science: Quantitative Approaches to Decision Making, Cengage Learning, 2008.
2. Lucey, Quantitative Techniques Cengage Learning Business Press, 2002
3. Sharma, Operations Research: Theory and Applications.
4. Richard I Levin, & C. Atkinson Kirkpatrick, Quantitative Approaches to Management, McGraw-Hill.
5. K. Gupta and D.S. Hira, Operations Research.
6. Srivastava, Shenoy and Sharma, Quantitative Techniques for Managerial Decision-making, New Age International, 2006.
7. N.D. Vohra, Quantitative Techniques in Management, Tata McGraw-Hill Education.
8. V.K. Kapoor, Operations Research.
9. Dharani Venkatakrishnan, Operations Research: Principles and Problems.
10. Hamdy A. Taha, Operations Research: An Introduction, Prentice Hall, 2002.

33815 - FINANCIAL AND MANAGEMENT ACCOUNTING

Objectives:

- To enable the students to learn basic accounting principles, concepts.
- To practice Financial and Management accounting applications
- To make the learners familiarize in managerial decision making.

UNIT 1 Accounting: Definition – Accounting for historical function and managerial function - Types of Accounting- Management, Management and Cost accounting – Scope for Accounting-Managerial Uses of Management accounting and Financial Accounting.

UNIT 2 Accounting Concepts and Conventions – Accounting standards - Financial Accounting Definitions – Principles – Accounting standards - Double entry system of accounting: Accounting books – Preparation of journal and ledger, subsidiary books.

UNIT 3 Preparation of Trial Balance – Errors and rectification – Classifications of capital and Revenue – Fixed Assets and Depreciation accounting – Preparation of Manufacturing accounting- Preparation of Final Accounts - Accounting from incomplete records – Statements of affairs methods

UNIT 4 Conversion methods – Preparation of Trading, Profit & Loss Account and Balance Sheet from incomplete records – Depreciation methods - Straight line method, Written down value method, Sinking fund method.

UNIT 5 Financial Statement Analysis - Objectives - Reorganizing the Financial Statement information -Techniques of Financial Statement Analysis: Comparative Statements, Common – Size statement, Trend Percentage -

UNIT 6 Management Statement Analysis: Management statements – Nature of management statements – Limitations of management statements – Analysis of interpretation -Types of analysis- Tools of analysis: Trend analysis, Common size statements and Comparative statements;

UNIT 7 Accounting Ratios: Construction of balance sheet using ratios (problems) – Financial ratios – Types: Profitability ratios – Turnover ratios – Liquidity ratios – Proprietary ratios – Market earnings ratios- Uses and limitations of ratios - Dupont analysis.

UNIT 8 Fund Flow Analysis: Need and meaning – Preparation of schedule of changes in working capital and the fund flow statement – Workings for Computation of various sources and uses - Preparation of Fund Flow Statement

UNIT 9 Cash flow Analysis: Meaning and importance Managerial uses of cash flow statement – Differences between fund flow and cash flow analysis - Uses and limitation of fund flow statement- Preparation of cash flow statement

UNIT 10 Cost Accounting: Cost Accounting - Meaning - Distinction between Financial Accounting and Cost Accounting - Cost Terminology: Cost, Cost Centre, Cost Unit - Elements of Cost - Cost Sheet – Problems - Overhead Cost Allocations: Over and under Absorption. Job and Contract Costing,

UNIT 11 Operating Costing: Material Cost Accounting, Perpetual Inventory Control, Inventory Valuation, EOQ, ABC Analysis, Setting of Reorder Level, Maximum Level, Minimum Level, Labour Cost Accounting, Remuneration and Incentive Schemes- Reconciliation of Financial and Cost Accounting

UNIT 12 Marginal Costing: Definition – Difference between marginal costing and absorption costing – Break- even point Analysis - Contribution, p/v Ratio, margin of safety - Decision making under marginal costing system-key factor analysis, make or buy decisions, export decision, sales mix decision-Problems

UNIT 13 Budgeting and Budgetary Control: Concept and Need for Budgeting-Classification of budgets – Preparation of Sales, Production, Material, Purchase and Cash Budgets –Budgetary control system – Mechanism – Master budget.

UNIT 14 Capital Budgeting System: Importance – Methods of capital expenditure appraisal – Payback period method – ARR method – DCF methods – NPV and IRR methods – Their rationale – Capital rationing.

REFERENCES

1. Arulanandam& K.S. Raman,Advanced Accounting, Himalaya Publishing House.
2. Gupta &Radhasamy,Advanced Accounting, Sultan Chand & Sons.
3. Shukla & T.S. Grewal,Advanced Accounting, S.Chand&Company.
4. Jain &Narang,Advanced Cost Accounting, Kalyani. Publications.
5. Ravi M. Kishore, Cost Management, Taxman Publications
6. S.N. Maheswari, Management Accounting &Management Accounting, Vikas Publishers.
7. Manmohan & Goyal, Principles of Management Accounting, Shakithabhavan Publication.
8. N. K. Prasad,Advanced Cost Accounting, Book Syndicate Pvt. Ltd., Calcutta.
9. Andrew A Haried, Advanced Accounting, Atlantic Publishers.
10. Hoyle,Advanced Accounting, McGraw Hill.

33821 - RESEARCH METHODS

Objectives:

- To Understand the basic principles of research and design
- To practice the research process, tools and techniques
- To facilitate managerial decision making

UNIT 1 Research Bases: Definition and applications of business research; Types of research –descriptive, exploratory, correlational, explanatory, quantitative, qualitative; Steps in the research process; establishing operational definitions

UNIT 2 Research scope - Recent advancements in research. Distinction between Pure & Applied, Historical & Futuristic, Analytical & Synthetic, Descriptive & Prescriptive, Survey & Experimental and Case & Generic Researches

UNIT 3 Planning of Research: Research problem – Identification, selection and formulation of research problem – Review of literature in the field of business - Identifying objectives of the research.

UNIT 4 Economic management: Use in identifying Research Gaps and Techniques – Hypothesis – Meaning – Sources and Types of Hypothesis – Hypothesis Formulation for testing – Research design – Factors affecting research design – Evaluation of research design

UNIT 5 Variables construction for Hypothesis: Identifying variables - Constructing hypotheses – functions, characteristics, types of hypotheses - Significance of research in social sciences – Induction and deduction.

UNIT 6 Sampling Design: Census method and sampling method for investigation – Principle of sampling – Essentials of a good sampling – sampling frame; Methods of sampling: Probability, non-probability, mixed sampling designs;

UNIT 7 Construction of sampling for Finite and Infinite populations – Sample size determination– Calculations - Factors affecting the size of the sample – Biased sample – Sampling and non-sampling errors.

UNIT 8 Sources and Collection of Data: Sources of data – Primary and secondary data – Modes of data collection – Observation: Types and Techniques –Interview: Types and conduct – Preparation for an interview – Effective interview techniques – Limitations of interview

UNIT 9 Schedule: Meaning and kinds – Essentials of a good schedule – Procedure for the formulation of a schedule – Questionnaire: Meaning and types – Format of a good questionnaire– Schedules Vs. Questionnaires

UNIT 10 Scaling techniques: Meaning, Importance, Types of measurement scales – Nominal, Ordinal, Interval, Ratio; Methods of their construction of Questionnaires or Schedules – Pre-testing of Data Collection Tools- Validity and Reliability – Methods.

UNIT 11 Processing and Analysis of Data: Meaning – Importance – Process of data analysis – Editing – Coding – Tabulation – Diagrams – Univariate, Bivariate and Multi-variant analysis

UNIT 12 Test of Significance: Fundamentals on Test Procedure- Testing for significance of Mean/Proportion and difference between Means/Proportions- F Test for Means and Chi-square test Contingency Table - Parametric Test: T test, F Test and Z test

UNIT 13 Non-parametric Test: Concept and Types: Mann Whitney Test- Test, Kruskal Wallis, sign test. Multivariate analysis-factor, cluster, MDS, Discriminant analysis - The process of interpretation of Test Results– Guidelines for making valid interpretation

UNIT 14 Report Writing : Role and types of reports – Contents of research report – Steps involved in drafting reports – Principles of good report writing – Grammatical Quality – Language flow- Data Support- Diagrammatic Elucidation- References and Annotations – Clarity and Brevity of expressions- Features of a good Report- Criteria for evaluating research reports/ research findings.

REFERENCES

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2. Anderson et-al, Thesis and Assignment Writing, Wiley, New Delhi, 1989.
3. William Josiah Goode and Paul K. Hatt, Methods of Social Research, McGraw Hill, 1981.
4. Wilkinson and Bhandarkar, Methods and Techniques of Social Research, 2003, HPH.
5. Earl R. BabbieRobert, ThePractice of Social Research, Cengage Learning, 2010.
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7. Krishnaswami and Ranganatham, Research Methodology in social Sciences, HPH, Mumbai
8. Bryman & Bell: Business Research Methods, OUP.
9. Pauline V Young, Scientific Social Surveys and Research, Prentice-Hall, (Digitalized) 2007.
10. C.R.Kothari, Research Methodology: Methods and Techniques, 2009

33822 - BUSINESS ENVIRONMENT

Objectives:

- To understand the concepts and constituents of Business environment
- To know the environmental issues in the business context
- To analyze the changes in the global environmental relating to business

UNIT 1 Business Environment: Introduction: Concepts – Significance - Dynamic factors of environment – Importance of scanning the environment – Macro and Micro Environment – Micro and Macro Economics to the business – Constituents of Business environment

UNIT 2 Fundamental issues captured in PESTLE– Political, Economic, Socio-cultural, Technological, Legal and Ecological environment- Opportunities and Threats as environmental issues to address by Businesses.

UNIT 3 Political Environment: Government and Business – Political Systems, Political Stability and Political Maturity as conditions of business growth - Role of Government in Business: Entrepreneurial, Catalytic, Competitive, Supportive, Regulative and Control functions

UNIT 4 Government and Economic planning: Industrial policies and promotion schemes – Government policy and SSI – Interface between Government and public sector - Guidelines to the Industries – Industrial Development strategies; salient features, Role of public and private sectors, Comparative cost dynamics.

UNIT 5 Economic Environment: Phase of Economic Development and its impact- GDP Trend and distribution and Business Opportunities – capacity utilisation – Regional disparities and evaluation - Global Trade and investment environment.

UNIT 6 Financial System and Business capital: Monetary and Fiscal policies - Financial Market structure – Money and Capital markets – Stock Exchanges and Its regulations – Industrial Finance - Types, Risk - Cost-Role of Banks; Industrial Financial Institutions - Role of Management Institutions

UNIT 7 Role of Central Bank- Fiscal System: Government Budget and Taxation Measures- Fiscal Deficits and Inflation- FDI and collaboration –Foreign Capital tapping by businesses- Export-Import policy – Foreign Exchange and Business Development.

UNIT 8 Labour Environment: Labour Legislation – Labour and social securities – Industrial Relations – Trade Unions – Workers participation in management – Exit Policy – Quality Circles.

UNIT 9 Social and Technological Environment: Societal Structure and Features- Entrepreneurial Society and its implications for business – Social and cultural factors and their implications for business- Technology Development Phase in the Economy as conditioner of Business Opportunities

UNIT 10 Technology Environment: Technology Policy- Technology Trade and transfer- Technology Trends in India- Role of Information Technology – Clean Technology. – Time lag in technology – Appropriate technology and Technology adoption- Impact of technology on globalization.

UNIT 11 Legal and Ecological Environment: Legal Environment as the all-enveloping factor from inception, location, incorporation, conduct, expansion and closure of businesses – IDRA and Industrial licensing – Public, Private, Joint and Cooperative Sectors.

UNIT 12 Legal Aspects of Entering Primary and Secondary Capital Markets- Law on Patents- Law on Consumer Protection- Law on Environmental Protection- Need for Clean energy and Reduction of Carbon footprint.

UNIT 13 New Economic Policy Environment in India: Liberalization, Privatization and Globalization (LPG): Efficiency Drive through Competition- Facets of Liberalization and impact on business growth

UNIT 14 Aspects of Privatization and impact on business development– Globalization and Enhanced Opportunities and Threats – Extended competition in Input and Output Markets Role of WTO, IMF and World Bank in global economic development.

REFERENCES

1. Brooks, Weatherston, Wilkinson, International Business Environment, Pearson, 2010.
2. Steiner & Steiner, Business, Government and Society: A Managerial Perspective, McGraw-Hill, 2008.
3. Mohinder Kumar Sharma, Business Environment in India, South Asia Books.
4. Adhikary M, Economic Environment of Business, Sultan Chand & Sons.
5. Amarchand D, Government and Business, TMH.
6. Francis Cherunilam, Business Environment and Development, Himalaya Publishing House, 2008.
7. Maheswari & Gupta, Government, Business and Society.

33823 - BUSINESS LAWS

Objectives:

- To understand the legal structure and provision for running a business
- To learn various acts, enactments and amendments of mercantile law
- To know the various aspects of Business law for legal process.

UNIT 1 Indian Contract Act 1872: Contract – Meaning – Essential elements – Nature and formation of contract: Nature, elements, Classifications of Contracts on the basis of Validity, Formation and Performance– offer and acceptance

UNIT 2 Offer and Acceptance: Introduction – Proposal – acceptance – Communications of offer, Acceptance and Revocations – Offer and acceptance by Post.

UNIT 3 Consideration: Definitions, Types of consideration – essentials of Consideration – Privity of Contracts: Exceptions – Capacity: Consent – Legality of object – Quasi contract Discharge of contract - Remedies for breach of contract – Quasi contracts.

UNIT 4 Special Contracts: Contract of Indemnity and Guarantee – Bailment and Pledge – Law of Agency-Definition – Rights of Surety -Discharge of Surety – Bailment and Pledge: Introduction, Classifications, Duties and Rights of Bailer and Bailee – termination of Bailment -

UNIT 5 Formation of contract under Sale of Goods Act, 1930: Contract of sale - Conditions and Warranties - Transfer of property - Performance of the contract: Essentials of valid tender performance, Performance reciprocal promise- Rights of an unpaid seller.

UNIT 6 Laws on Carriage of Goods: Duties, Rights and Liabilities of Common Carriers under: (i) The Carriers Act, 1865. (ii) The Railways Act, 1989, (iii) The Carriage of Goods by Sea Act, 1925, (iv) The Carriage by Air Act, 1972 and (v) The Carriage By Road Act, 2007

UNIT 7 Negotiable Instruments Act, 1881: Negotiable Instruments: Features – Types- Parties – Material alteration – Parties to negotiable instruments – Presentations of negotiable instrument.

UNIT 8 Insurance: Definition and sources of Law – Judicial set up in India — Insurance as a contract -History of Insurance Legislation in India - Legal principles - Fundamental Principles of Life Insurance Fire Insurance and Marine Insurance.

UNIT 9 Indian Partnership Act, 1932: Meaning and test of partnership – registration of firms Life Insurance Corporation Act 1956 – General Insurance Business Nationalization Act 1973.

UNIT 10 Partners Relations: Introduction – Eligibility to be a partner – Registration of change in partner – Limited Liabilities of partnership - Dissolution of firms - Characteristics – Kinds – Incorporation of Companies – Memorandum of Association – Articles of Association

UNIT 11 Companies Act 1956: Nature and kinds of companies – Prospectus – Disclosure Needs - Management and Administration – Director – Appointment, Powers and Duties

UNIT 12 Formation of a Company : Introduction – process - Minutes and Resolutions – E-Filling of documents under Ministry of Corporate Affairs (MCA) 21- Management of companies –Meetings- Types- Requirements -AGM and EGM – Board Meeting

UNIT 13 Law of Information Technology: Introduction – Rationale behind IT act 2000 – Information technology Act 2000: Scheme of the IT Act 2000: Digital signature: attribution; Acknowledgement and dispatch of Electronics Record – Regulation certifying authorities.

UNIT 14 Protection of minority interest: Introduction - Methods of Winding-up - The Right to Information Act, 2005 Right to know, Salient features of the Act, obligation of public Authority, Designation of Public Information officer, Request for obtaining information,

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1. M.S.Pandit and ShobhaPandit, Business Law, Himalaya Publishing House, Mumbai, 2010.
2. Pathak, Legal Aspects of Business, TMH, 2009.
3. N.D. Kapoor, Mercantile Law, Sultan Chand & Sons, New Delhi.
4. M.C. Shukla, Mercantile Law, S. Chand & Co., New Delhi.
5. Relevant Bare Acts.
6. Balachandran and Thothadri, business Law, TMH, 2010

33824 - MANAGEMENT INFORMATION SYSTEM

Objectives:

- To learn the principles of Management Information System for organizations
- To understand the uses , function of application MIS in organization
- To analyze the scope of MIS for business organizations

UNIT 1 Foundations of Information System: Information system: Meaning, Role – System concepts – Organization as a system – Components of Information system – Various activities of IS and Types of IS

UNIT 2 Information System: Concepts of Information System and Management information systems design and development-Implementation testing and conversion- Evolution and element of MIS

UNIT 3 MIS : Definition – Characteristics and basic requirements of MIS – Structure of MIS- Approaches to MIS development- Computerized MIS- Pre-requisites of an effective MIS- Limitations of MIS.

UNIT 4 MIS and Decision support System (DSS): MIS Vs. data processing – MIS and decision support system – MIS and information resource management – DSS and AI – Overview of AI - DSS models and software.

UNIT 5 MIS and Operations Research- Executive information and Decision support systems – Artificial intelligence and expert system – Merits and De Merits – Pitfalls in MIS.

UNIT 6 MIS in Indian organizations – Recent developments in information technology - Installation of Management Information & Control System in Indian organization

UNIT 7 Computers and Communication: Information technology and Global integration –On-line information services – Electronic bulletin board systems – The internet, electronic mail, interactive video

UNIT 8 Communication Channels: Advantages disadvantages – Communication networks – Local area networks – Wide area networks – Video conferencing- Relevance to MIS- Usage in Business process.

UNIT 9 Functional Information systems: MIS for Research Production - MIS for Marketing - MIS for Personnel - MIS for Finance - MIS for Inventory- MIS for Logistics- MIS for Product Development- MIS for Market Development.

UNIT 10 Client/ Server Computing: Communication servers – Digital networks – Electronic data interchange and its applications - Enterprise resource planning systems (ERP Systems) – Inter-organizational information systems – Value added networks – Networking.

UNIT 11 Electronic Commerce and Internet: E-Commerce bases – E-Commerce and Internet – M-Commerce- Electronic Data Inter-change (EDI) - Applications of internet and website management - Types of Social Media - uses of social media in business organization

UNIT 12 Computer System and Resources: Computers systems: Types and Types of computer system processing - Secondary storage media and devices – Input and output devices – Hardware standards – Other acquisition issues.

UNIT 13 Managing Information Technology: Managing Information Resources and technologies – IS architecture and management - Centralized, Decentralized and Distributed - EDI, Supply chain management & Global Information technology Management.

UNIT 14 Security and Ethical Challenges: IS controls - facility control and procedural control - Risks to online operations - Denial of service, spoofing - Ethics for IS professional - Societal challenges of Information technology

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2. Kenneth Laudon & Jane Laudon, Essentials of MIS, Prentice Hall, 2010.
3. Lisa Miller, MIS Cases: Decision Making with Application Software, Prentice Hall, 2008.
4. David M. Kroenke, Experiencing MIS, Prentice Hall, 2011.
5. Kenneth C. Laudon, MIS: Managing the Digital Firm, Prentice Hall, 2005.
6. Sadogopan S, Management Information Systems, 2001PHI.
7. Murdie and Ross, Management Information Systems, Prentice Hall.
8. Henri C. Lucas, Information Systems Concepts for Management, McGraw Hill, 1994.
9. Stephen Haag, Management Information Systems, 2008.

33825 - HUMAN RESOURCE MANAGEMENT

Objective:

- To understand the concepts and methods and techniques of Human Resource Management
- To know the Human resource management theories and real time practices
- To identify the contemporary issues in human resource management

UNIT 1: Introduction to Human Resource Management: Concept, Definition, Objectives, Nature and Scope of HRM - Functions of HRM – Evolution of human resource management - Role and structure of Human Resource Function in organizations- Challenges in Human Resource Management

UNIT 2 Human Resource Management Approaches: Phases of human resource Management- The importance of the human factor – Competitive challenges of HRM – HRM Models – Roles and responsibilities of HR department.

UNIT 3 Human Resource Planning: Personnel Policy - Characteristics - Role of human resource manager – Human resource policies – Need, Scope and Process – Job analysis – Job description – Job specification- Succession Planning.

UNIT 4 Recruitment and Selection Process: Employment planning and fore casting Sources of recruitment- internal Vs. External; Domestic Vs. Global sources- Selection process Building employee commitment : Promotion from within - Sources, Developing and Using application forms – IT and recruiting on the internet.

UNIT 5 Employee Testing & selection : Selection process, basic testing concepts, types of test, work samples & simulation, selection techniques, interview, common interviewing mistakes, Designing & conducting the effective interview, small business applications, computer aided interview.

UNIT 6 Training and Development: Orientation & Training: Orienting the employees, the training process, need analysis, Training techniques, special purpose training, Training via the internet. - Need Assessment - Training methods for Operatives and Supervisors

UNIT 7 Executive Development: Need and Programs - Computer applications in human resource management – Human resource accounting and audit. On-the - job and off-the-job Development techniques using HR to build a responsive organization

UNIT 8 Employee Compensation : Wages and Salary Administration – Bonus – Incentives – Fringe Benefits –Flexi systems - and Employee Benefits, Health and Social Security Measures,

UNIT 9 Employee Retention: Need and Problems of Employees – various retention methods– Implication of job change. The control process – Importance – Methods – Employment retention strategies for production and services industry

UNIT 10 Appraising and Improving Performance: Performance Appraisal Programs, Processes and Methods, Job Evaluation, Managing Compensation, Incentives Performance appraisal: Methods - Problem and solutions - MBO approach - The appraisal interviews - Performance appraisal in practice.

UNIT 11 Managing careers: Career planning and development - Managing promotions and transfers - Sweat Equity- Job evaluation systems – Promotion – Demotions – Transfers- Labour Attrition: Causes and Consequences

UNIT 12 Employee Welfare, Separation: Welfare and safety – Accident prevention – Employee Grievances and their Redressal – Industrial Relations - Statutory benefits - non-statutory (voluntary) benefits – Insurance benefits - retirement benefits and other welfare measures to build employee commitment

UNIT 13 Industrial relations and collective bargaining: Trade unions – Collective bargaining - future of trade unionism - Discipline administration - grievances handling - managing dismissals and workers Participation in Management- Separation: Need and Methods.

UNIT 14 Human Resource Information System- Personnel Records/ Reports- e-Record on Employees – Personnel research and personnel audit – Objectives – Scope and importance.

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1. Mathis and Jackson, Human Resource Management,South-WesternCollege, 2004.
2. Nkomo, Fottler and McAfee, Human Resource Management, South-WesternCollege, 2007.
3. R. Wayne Mondy, Human Resource Management,Prentice Hall, 2011.
4. Venkataraman& Srivastava, Personnel Management & Human Resources
5. ArunMonappa, Industrial Relations
6. Yodder&Standohar, Personnel Management & Industrial Relations
7. Edwin B. Flippo, Personnel Management , McGraw-Hill, 1984
8. Pigors and Myers, Personnel Administration
9. R.S. Dwivedi, Manpower Management
10. Lynton &Pareek, Training and Development, Vistaar Publications, 1990.

33831 - MARKETING MANAGEMENT

Objectives:

- To help the learners understand markets, consumers and marketing principles.
- To understand the buyer behaviour and influencing factors
- To learn marketing plan, pricing, promotion and distribution in global context

UNIT 1 Introduction to Marketing: Meaning and Scope of Marketing; Marketing Philosophies; Marketing Management Process-an overview; Modern Marketing Concept: Social marketing concept – Approaches to the study of marketing.

UNIT 2 Marketing segmentation: Meaning – Bases for segmentation, benefits – Systems approach - Four Ps of Product and Seven Ps Service marketing mix and Extensions- Targeting and Positioning - meaning and importance.

UNIT 3 Marketing Environment: Internal and External and Demographic factors – Adopting marketing to new liberalized and globalized economy – Digitalization – Customization and E business settings.

UNIT 4 Consumer Behaviour : Meaning and importance – Consumer buying process – Determinants and Theories of consumer behaviour – Psychological, sociological determinants – Theories and their relevance to marketing-

UNIT 5 Marketing Research: Procedure. Meaning – Objectives – Process- Demand Forecasting- Marketing Information System – Strategic marketing plan and organization – Changing marketing practices.

UNIT 6 Product Mix Management: Product planning and development – Meaning and process – Test marketing – Product failures – Product line management: Practices – Implications and Strategies for current market condition.

UNIT 7 Product life cycles: Meaning and Stages – Strategies – Managing PLC-Product-Market Integration: Strategies – Product positioning – Diversification – Product line simplification – Planned obsolescence – Branding Policies and Strategies – Packing.

UNIT 8 Price Mix Management: Pricing and pricing policies – Objectives – Procedures – Bases for and Methods of price fixing. Cases for Free Pricing, Administered and Regulated pricing – Pricing and product life cycle

UNIT 9 Physical Distribution Mix: Types of physical Distribution - Importance of Physical Distribution- Distribution channel policy – Logistics Decisions – Methods – Strategic alliance for Logistic cost reduction.

UNIT 10 Marketing Channel system: Marketing channel decisions: Choice considerations– Managing Conflict and Cooperation in channels – Middlemen functions- Modern Trends in Retailing- Malls and Online.

UNIT 11 Promotional Mix: Personal selling Vs. impersonal selling – Personal selling – Process – Steps in selling – Management of sales force – Recruitment and selection – Training – Compensation plans – Evaluation of performance

UNIT 12 Integrated marketing communication Process: Advertising and sales promotion – Online Sales promotional activities – Public relationships – Direct marketing: Meaning, Nature, Growth and Channels.

UNIT 13 Advertising: Importance – Objectives – Media planning and selection – Factors influencing selection – Advertisement copy – Layout – Evaluation of advertising – Advertising budget – Sales promotion – Methods and practices.

UNIT 14 Competitor analyses: Identifying and analyzing the competitors – Types of Competitors – Competitive strategies framing for leaders, challengers, followers and nichers. Customer relationship marketing: Customer data base, Data ware housing and data mining

REFERENCES

1. Etzel, Walker and Stanton, Fundamentals of Marketing, McGraw Hill, 2004
2. Philip Kotler & Gary Armstrong, Principles of Marketing, Prentice Hall, 2010.
3. Jerome Mccarthy, Basic Marketing, Richard D. Irwin.
4. Cundiff, Still &Govani, Fundamentals of Modern Marketing, Prentice Hall.
5. Memoria & Joshi, Fundamental of Marketing.
6. Paul Peter and James Donnelly Jr, Marketing Management, McGraw-Hill, 2010.
7. William O. Bearden, Marketing: Principles & Perspectives, McGraw-Hill, 2006.
8. William Arens, et al, Contemporary Advertising, McGraw-Hill, 2008.
9. Perreault and McGarthy - Basic Marketing - Tata McGraw Hill, 2002\
10. Michael J Etzel, Bruce J Walker, William J Stanton and Ajay Pandit, Marketing concepts and cases - TMH 13th Edition, New Delhi, 2007.

33832 - FINANCIAL MANAGEMENT

Objectives:

- To help the students to know the basic concepts of financial management
- To understand capital structure, dividend policy and working capital management.
- To learn the various concepts of financial management along with applications

UNIT 1 Introduction: Financial management: objectives - Concept, nature, evaluation and significance – Finance Functions: Managerial and operative – Role of Financial management in the organization – Indian Financial system.

UNIT 2 Financial System: Legal and Regulatory frame work – Financial Functions: Meaning and scope – Finance and Tax Management Nexus- Tax Avoidance and Tax evasion- Tax incentive and business decisions.

UNIT 3 Investment Function: Meaning and scope - Time value of Money concepts and applications –Risk return relationship - Dividend function – Risk return trade off – Management planning- Global management environment

UNIT 4 Long-term Capital Resources: Equity and debt sources – Equity share, preference shares – types of preference share - debentures – types - sources of long-term capital.

UNIT 5 Capital Issues: Meaning, Nature, Purpose – Roles and Guidelines of SEBI in capital issues- Bridge finance, loan syndication, Book building – Borrowings from the term lending institutions and International capital market- Tax considerations in financing decision areas.

UNIT 6 Cost of Capital : Concept of cost of capital- Cost of debt, equity, preference share capital, retaining earning - Weighted average cost: EBIT –EPS Analysis- Tax, Capital structure and Value nexus - Computation of overall cost of capital – Tax and cost of capital.

UNIT 7 Capital structure: Determinates - Concept and Types- Optimum capital structure – Theories of capital structure – Net income and net operative income approach – M.M. Approach – Traditional theory – Their assumptions – Significance and limitations – Management leverage operating leverage – Combined leverage.

UNIT 8 Capital budgeting: Meaning, Nature and Types of Capital Investment- Methods of appraisal under certainty conditions: PBP, ARR, IRR and NPV techniques - Basic and International capital budgeting.

UNIT 9 Uncertainty and Risk models: Simulation Analysis- Sensitivity analysis- Decision tree analysis- Certainty equivalent and risk-adjusted return measures- Tax considerations in Investment Decisions Cost of capital and Investment Decisions.

UNIT 10 Working Capital Management: Definitions and Objectives - Concept and types – Determinants – Financing approaches – Conservative approaches - Sources of working capital finance Factors affecting working capital requirements- Working capital financing by commercial banks – Types of assistance

UNIT 11 Inventories and receivables Management under conditions of certainty and uncertainty – Operating cycle – Planning of funds through the management of assets – Various techniques used.

UNIT 12 Cash and liquidity management: Credit Management and evaluation alternative credit variables Methods and Functions- Tax considerations in Remittances and Purchases.

UNIT 13 Dividend Theories: Valuation under Gordon and Walter theories – Dividend irrelevance under M.M. Theory – Assumptions – Limitations - Implications and contributions of theories in financial decision making process.

UNIT 14 Dividend Policy: Types – Share valuation practices – Factors affecting dividend decision – Tax considerations in dividend decision when tax is levied at the hands of companies and recipients.

REFERENCES

1. Brigham and Ehrhardt, Financial Management: Theory & Practice, Thomson ONE, 2010
2. Brigham and Houston, Fundamentals of Financial Management, Thomson ONE, 2009.
3. Van Horne: Fundamentals of Financial Management, Prentice Hall, 2008
4. Jeff Madura, International Financial Management,South-WesternCollege Pub., 2010
5. Prasanna Chandra, Financial Management, McGraw Hill, 2008.
6. Khan and Jain, Financial Management ,Tata McGrawHill,2009
7. Pandey I M, Financial Management, Vikas Publishers,2009
8. Sheeba Kapil(2010), Financial Management, Pearson Education.
9. B J Camsey, Engene F.Brigham, “Introduction to Financial Management”, The Gryden Press

33833 - TOURISM AND ACCOMMODATION

Objective:

- To understand the accommodation facilities in tourism
- To exam the restaurant service , cuision and hygiene

UNIT 1 Tourism Industry and its structure: attractions, accommodation, transportation. Present trends in domestic and global tourism. Concepts, definitions, origin and development of Tourism - History -Pleasure travel Types of tourism, Forms of tourism: domestic, international, regional, inbound, outbound, Tourism net work and components of tourism, Interdisciplinary approaches to tourism. Tourism system (Leiper's Model)

UNIT 2 Accommodation facilities in tourism: Importance of accommodation in tourism- Types of accommodation- Basic and additional facilities provided – Nature of demand for and supply of accommodation facilities.

UNIT 3 Elements of Tourism: (attraction, accessibility, accommodation), tourism product, characteristics of tourism products, types of products and tourism. Hotel Industry, Hotel Chains, Departments of Hotel. Tourist Guide and Escort

UNIT 4 Star Hotels and Tourism: StarClassification of hotels- Criteria for classification – Features of different star hotels- Profile of guests- Guest relationship – MICE tourism in hotels- Tariff plans- Rate structure and concessions – Marketing by hotels

UNIT 5 Occupancy and penetration levels and determinants- Scarcity of Rooms. Meaning- Importance – Implication – Need-–Types- Scope, Features

UNIT 6 Supplemental accommodation facilities and tourism: Concept, types and significance of supplemental accommodation facilities-Profile of tourists- Factors influencing the choice.

UNIT 7 Housekeeping and Tourism: Cleaning of guest room/bathrooms- Bed making- cleaning public of areas- Food service area - Elevators and Corridors– Maintenance of equipments - Cleaning agents: safety in use and storage.

UNIT 8 Budgetary control-Define budget &budgetary control-Objectives-Types of budgets- Inventory control: Importance-Objectives-Methods-Pricing of commodities

UNIT 9 Food menus & Beverage lists-Introduction-Basic menu criteria-Types of food menus- The content of food menus-beverage menu/list-Menu planning-factors influencing menu planning-Menu merchandising-Pricing of menu-constrains of menu planning

UNIT 10 Restaurant Service and tourism: Tourist preferences, including that of Fastidious tourists for multi cuisine services - Indian cuisines - Cuisine: Basics, Made ups and Aesthetics- Food Personnel: kitchen hierarchy - Catering to different categories: Individuals including etiquettes) & groups

UNIT 11 Food and Beverage control-introduction-Objectives-Problems-The essentials of control system- Food and Beverage production controlling-calculation of food cost methods of food control-calculation of Beverage cost-methods of Beverage control

UNIT 12 Food and Beverage management in Hotel industry, quality Restaurants-fast foods functions- Caterings-Industrial catering-Institutional catering-Hospital catering

UNIT 13 Hygiene: Food Items and Personnel - Cooking Tasks, Hazard and Precautions- Approval of Restaurants - Restaurant industry ways to keeping customers.

UNIT 14 Human Resources in hotels and Tourism: HRs in Hotels and tourism development-Job positions- Recruitment- Compensation- Satisfaction level.

REFERENCES

1. Philip Kotler, et.al., Marketing for Hospitality and Tourism, Ed.3.
2. Anand, Tourism and Hotel Industry, India
3. Douglas Robert Brown , The Restaurant Managers Handbook : How to Set Up, Operate, and Manage a Financially Successful Food Service Operation
4. Denney G. Rutherford, Hotel Management and Operations, Ed.3.

33834 - TOUR OPERATORS AND TRAVEL AGENCIES

Objective:

- To underrated the role and service of tour operators
- To gain knowledge on specific area travels association, travel agenciesassociation

UNIT 1 Introduction to travel and tourism – meaning – nature – definitions – basic components – elements – domestic tourism – international tourism – future of tourism – technology and tourism. History of travel and tourism – travel steps in the evolution modern tourist travel for trade and commerce – religious purpose – seeking knowledge – paid holidays – mass tourism. The advent of steam age, impact of industrial revolution, rail transport, sea transport, development of sea side resorts, the auto mobiles, air travel, causes of rapid growth – post second world war period.

UNIT 2 Travel Agency and Tour Operator, Travel related documents, Passport, Visa,currency regulations, custom, health regulations, baggage regulations etc.,

UNIT 3 Travel industry – Air, Road, Rail and Water, segments of travel industry, water transportation – Cruise categories – Volume, premium, luxury, researching cruises – cruise facilities – inland water ways.

UNIT 4 Concepts of package tourism and Free Independent Tourists (FIT) Significance- Customised and Regularized packages.Economics of Tour Operation: Choice of travel circuits – Up-linking – Procurement of services

UNIT 5 Motivation to travel – definition of the term motivation – travel motivators, factors influencing the growth of tourism, social tourism, role of state in promoting social tourism – evolution of demand – barriers to travel

UNIT 6 TravelAgency-concept, -Travel agencies-role-functions-types ownership linkages- arrangements, negotiations with Airlines, Transparent Agencies-CRS functions-various computerized reservation systems

UNIT 7 Tourism in India – sergeant committee – formation of ministry of tourism, department of tourism – tourism information offices in India, its functions, tourism offices overseas – its functions, criteria for opening overseas offices, role and functions of ITDC. State level Tourism Development Corporation

UNIT 8 Organizations in tourism – need for the organization - factors influencing typeorganizations, the national tourist organization – role and function. History of international tourist organization – International Union of Official Travel Organization (IUOTO). World Tourism Organization (WTO), International Air Transport Association (IATA), International Civil Aviation Organization (ICAO), Pacific Area Travel Association (PATA), India Convention Promotion Bureau (ICPB), Federation of Hotel and Restaurant Association of India (FHRAI), Travel Agent Association of India (TAAI), Universal Federation of Travel Agents Association (UFTAA).

UNIT 9 Travel Agents Association Of India (TAAI) – Structure, Composition and functions of these organisations.

UNIT 10 Setting up of Travel Agency-setting up-organization structure-procedures or the approval from IATA,IATO and different organizations, services offered by Travel Agency-Air, water, surface, outbound, insurance, travel documents-travel agency managing-definition-objective-unique features.

UNIT 11 Tour Operation-Meaning-definition-functions-Tour operator product-component features-types of tour operation-overseas, domestic, specialist-incoming tour operator's main types of tour packages-nested, escorted, independent-Two centre or multi centre holidays self drive package-campsite holidays- adventure and overland holidays-crossing-honeymoon package, luxury arias package-pilgrim package

UNIT 12 Problems of Tour operator and Travel Agency Services: General Problems arising out of the nature of tourism industry like seasonality

UNIT 13 Tour Packaging Management - development-definition-types of - tour design process- destination research-market research- preparation-handling agency-negotiation costing and pricing of tour-Tour Brochure-development of research-marketing of tour-actual tour operation

UNIT 14 Tour cost and pricing-definition-components of tour cost-travel cost accommodation cost-transfer cost-administrative cost-marketing cost-category of tour cost-direct and indirect, fixed and variable cost-costing of tour package-marketing of tour.

REFERENCES

1. Charles R. Goeldner, Brent Ritchie, J.R., Tourism : Principles, Practices, Philosophies
2. Philip Kotler, et.al., Marketing for Hospitality and Tourism, Ed.3.
3. Roy A. Cook, et.al., Tourism : The Business of Travel, Ed.2.

33835 - QUALITY IN TOURISM

Objective:

- To understand the important of quality in tourism
- To acquire knowledge on safety and security in tourism

UNIT 1 Quality in Tourism : Quality Management : Concept, need & importance Quality Management in Urban Tourism

UNIT 2 QualityDeterminants in Tourism: Safety, Security, Hygiene, Accessibility, Transparency, Authenticity, Harmony.Quality Management applied to tourist destinations.Attraction & land use management

UNIT 3 Safety in Tourism : Food Safety – Quality in food stuff – Prescribing standards.Security in Tourism : Protection to tourists – Detection and prevention of offences against tourists

UNIT 4 Project Management Managing recourses, time, Quality Project Management techniques & skills.&Providing secured transport services to the tourists.

UNIT 5 The role of manager and management style.Application of service quality in managing tourist destination.

UNIT 6 Tourist destination life cycle and quality management. Total Quality Management & TourismISO.

UNIT 7 Importance of quality Management in developing and tourist destination - Concept of service Definitions & Meaning -Types of service Classifications of services.

UNIT 8 Growing importance of services in Future Societies & impact of service in daily life - Role of customers in Service Process.

UNIT 9 Service Quality -Deterioration of quality - Productivity & quality ` improvement - Management challenges in service sector

UNIT 10	Event Operations: Registration. Seating Arrangements, Documentation, Press Relations, Audio-Visual Arrangements, Use of Films, Videos, CDs, etc., Videoconferencing. Service quality management.
UNIT 11	Health and Tourism : Sanitary and health requirements of the Tourists – Prescribing standards-Protection against health & hazards – Providing emergency medical assistance.
UNIT 12	Information and Tourism : Providing adequate and authentic information to Tourists -Travel assistance
UNIT 13	Tourists rights for consumer protection-International Co-operation in Tourism : – Areas of Co-operation
UNIT 14	Key elements of Managerial skill in Tourism & hospitality Industry Tourism and crises Management

REFERENCE

1. Charles R. Goeldner, Brent Ritchie, J.R., Tourism : Principles, Practices, Philosophies
2. Selvam, M., Tourism Industry in India, Himalaya Publication, India.

33841 - TOURISM ENTREPRENEURSHIP

Objective:

- To understand the concept and scope of tourism entrepreneurship
- To gain knowledge onentrepreneurial opportunités in tourism

UNIT 1 Entrepreneur & Entrepreneurship: Definition and Theories;. Entrepreneurship environment – Socio-economic, Cultural, Political & Natural, Characteristics of Entrepreneur & Entrepreneurial Behaviour

UNIT 2 Ownership structure and organizational framework of small scale enterprises in Tourism and Travel Business- Venture Creation and Management

UNIT 3 Preparation of business plan and managerial process in small scale enterprise. Entrepreneurial performance assessment. Managing family enterprises in Tourism industry. Promotional agencies for SMEs in India Opportunity Identification – Business Plan - Feasibility Report – Funding options

UNIT 4 Entrepreneurial Opportunities in Tourism –I (Accommodation): Entrepreneurial opportunities in Tourism: An overview- Entrepreneurial opportunities in Accommodation sector- Nature, Scope, Risk and Return aspects of the opportunity- Sources of finance

UNIT 5 Entrepreneurial Opportunities in Tourism –II(Transportation): Entrepreneurial opportunities in Transportation sector: Determinants of success of the venture- Case studies of selected Hotel Projects

UNIT 6 Risk and Return aspects of the opportunity- Sources of finance – Determinants of success of the venture- Case studies of selected Tourist cab services.

UNIT 7 Entrepreneurial Opportunities in Tourism –III (Shopping and Restaurant): Entrepreneurial opportunities in Shopping and Restaurant services sector- Extent of tourist spending on these aspects- Sources of finance – Determinants of success of the venture.

UNIT 8 Entrepreneurial Opportunities in Tourism –IV (Tourism Attraction Development): Entrepreneurial opportunities in tourism attraction development:

UNIT 9 Sources of finance – Determinants of success of the venture- Case studies of selected Theme parks, Resorts Hotels, Tour operators, etc.

UNIT 10 Entrepreneurial Development in Tourism : Programmes for developing entrepreneurship – Entrepreneurial culture

UNIT 11Tourism industry and business ideas; business strategy-understanding customers and analyzing competition

UNIT 12 Feasibility; Writing a business plan- marketing, financial, operations, people, etc. Financial requirements and sources of finance;

UNIT 13 Form of organisation and legal considerations; networkingand collaboration; good business practices;Setting up a tourism enterprise-steps, procedures, licenses, registration etc

UNIT 14 Intrapreneurship – Special Programmes of assistance.

REFERENCES

1. Peter F Drucker, Innovation and Entrepreneurship

2. Charles R. Goeldner, Brent Ritchie, J.R., Tourism : Principles, Practices, Philosophies.

3. Philip Kotler , et.al., Marketing for Hospitality and Tourism, Ed.3

4. Peter Mason, Tourism Impacts, Planning and Management

5. Roy A. Cook, et.al., Tourism : The Business of Travel, Ed.2

6. Douglas Robert Brown, The Restaurant Managers Handbook : How to setup, Operate and Manage a Financially Successful Food Service Operation

7. Denney G. Rutherford , Hotel Management and Operations, Ed.3

33842 - TOURISM PROJECT MANAGEMENT

Objective:

- To understand the pre - investment stage
- To gain knowledge onproject evaluation and appraisal

UNIT 1 Concept of project: Basic concepts, classification, characteristics of project, Project life cycleProject management, Tools & Techniques of project management, project organization

UNIT 2 Project identification: Identification, generation of ideas, SWOT analysis,Preliminaryscreening, project rating index. Market& Demand Analysis: Collection of data, market.survey, market planning, market environment, project risk analysis, demand forecasting techniques.

UNIT 3 Technical Analysis: selection of technology, material input and utilities, plant capacity, location & site, machinery and equipment, structures and civil work, environmental aspects, project charts and layouts. Financial Estimation: Project cost, source of finance, cost of production.

UNIT 4 Financial Analysis: Characteristics of financial statement, Working Capital, Project income statement, projected cash flow statement, projected balance sheet, projected profitability.

UNIT 5 Investment Evaluation: Investment decision rule, techniques of evaluation, payback period, accounting rate of return, profitability index method, Net profitability index, Internal rate of return, discounted payback period.

UNIT 6 Social Cost Benefit Analysis: Concept of social cost benefit, significance of SCBA, Approach to SCBA, UNIDO approach to SCBA, Shadow pricing of resource, the little miracle approach.

UNIT 7 Project Implementation: Schedule of project implementation, Project Planning, Project Control, Human Aspects of project management, team building, high performance team.

UNIT 8 Pre – Investment Stage – I : Industrial background in India-

Government policies, guidelines and investment proceduresIdentification of project opportunities – Capital budgeting.

UNIT 9 Pre – Investment Stage – II : Preliminary feasibility study – Preparation of feasibility report-Project Evaluation and Appraisal : Facets of appraisal – Appraisal Criteria

UNIT 10 Analysis of Project Risk – Preparation of model projects Project Implementation : Network techniques – Multiple projects and scheduling

UNIT 11 Resource scheduling – Project management software- Project Control : PERT – CPM -Resource monitoring and control

UNIT 12 Integrated resource management.-Project Evaluation : Evaluation under uncertainty

UNIT 13 Monitoring and MIS – Project audit

UNIT 14 Project review and interface with industrial sickness.

REFERENCE:

1. Gopalakrishnan and Ramamoorthy, V.E., Text book of Project Management, Macmillan India Ltd.,
2. Prasanna Chandra, Project Appraisal Management, Tata McGraw Hill
3. Ahuja, Project Management : Techniques in Planning and Controlling Construction Projects, Wiley
4. Nair, Project Management, Scheduling & Monitoring , Vani.

33843 - INSTITUTIONAL FRAMEWORK OF TOURISM

Objective:

- To understand the functions of world tourism
- To gain knowledge onrole and fonction of tour operators

UNIT 1 Travel & Tourism Management Syllabus (Old) The Measurement Of Tourism - Need For Measuring Tourism Phenomenon -Importance Of Tourist Statistics - Definition Of The Term 'Tourist' -Definition By The League Of Nation & The U.N

UNIT 2 European Travel Commission - United Nations Rome Conference - Domestic Tourism - General Problems Of Measurements - Types Of Tourist Statistics - Methods of Measurement

UNIT 3 Global Institutional Framework I : World Tourism Organization : Organization and functions- Policies and Achievements- Role of World Trade Organization in streamlining global tourism trade.

UNIT 4 Global Institutional Framework II : Global level associations of travel agents, transport operators and hoteliers- Pacific Area Travel Association PATA .

UNIT 5 IATA,- Travel agents - International Hotel Association- Role and functions.

UNIT 6 Manila Declaration on World Tourism - International Organisations & Tourism - Early History Of Co-Operative Endeavour -International Union of official Travel Organisation (IUOTO) International Civil Aviational Organisation

UNIT 7 National Level Institutions - I: Ministry of Tourism: Role and functions Tourism Development Board/Corporation.

UNIT 8 National Level Institutions II: Associations of hoteliers,- Tour operators, Travel agents at the national level: Role and Functions.

UNIT 9	State Level Institutions : Tourism and Development Corporation Ltd.- Role and functions – State level Hoteliers Association -
UNIT 10	Local Institutions: Local bodies -Role and functions in augmenting and regulating tourist facilities and creating tourist friendly local environment.
UNIT 11	Tourism Planning & Development -Planning For Tourism - Co-Ordination In Planning - Assessment Of Tourist Demand & Supply - Establishing Objectives -Territorial Planning - Basic Infrastructure = Financial Planning
UNIT 12	Human Resource Planning - Administrative Structure , Tourism Marketing & Promotion - Monitoring Progress - Time Factor - Environment Planning - Regional Planning Considerations
UNIT 13	Economic & Social Significance Of Tourism - Economic Benefits - The Multiplier Effect -Development Of Infrastructure - Regional Development - Effects On Employment -
UNIT 14	Tourism & Economic Value Of Cultural Resources - Cultural Tourism In India - Tourism & International Understanding

Reference Books:

1. Charles R. Goeldner, and Brent Ritchie, J. R., Tourism : Principles, Practices, Philosophies
2. Publications of relevant Institutions.

33844 - GLOBAL TOURISM

Objective:

- To understand the Global tourism
- To gain knowledge onoutbound and inbound organisation

UNIT 1 Concept of Sustainable Development**:** Evolution, Meaning, Principles, Key Dimensions ofSustainability, Stockholm Conference 1972 (Human & Environment), World Conservation Union 1980 – World Commission on Environment & Development (WCED) 1987 and Brundtland Commission - Rio Declaration 1992 – Kyoto Protocol 1997 – World Summit on Sustainable Development (WSSD) 2002 - Global Warming & Climate Change

UNIT 2 Sustainable Tourism Development: Meaning- Principles - 10 Rs-Agenda 21 for Travel andTourism Industry - World Conference on Sustainable Tourism 1995 - Globe 90 Conference - Berlin Declarations - Bali Declarations 2005 - Cape Town Declarations 2002 and Kerala Declarations, Ecotourism- Quebec Declaration 2002 - Kyoto Protocol 1997 - Oslo Declaration 2007.

UNIT 3 Planning for Sustainable Tourism: - Topographical Analysis - Analysis of Local Resources - LandUse Pattern – Environmental Impact Assessment (EIA), Environmental Information System (EIS), Environmental Management System (EMS) & Community Participation and Types of Community Participation and Socio- Economic and Cultural Conditions - Evaluation of Impact of Tourism Site - Zoning System - Carrying capacity & its Type

UNIT 4 Approaches of Sustainable Tourism- Standardization and Certification – Alternative Tourism -Responsible Tourism - Collaboration and Partnership - Waste Management – Eco-friendly Practices - Basic Laws & ideas in Ecology- Function and Management of Ecosystem- Biodiversity and its Conservation -Pollution-Ecological Foot Prints - Relationship between Tourism & Ecology, Sustainable Tourism and Poverty Alleviation - Pro-poor Tourism and Community Participation.

UNIT 5 Eco- tourism - Evolution, Principles, Trends. Functions of Ecotourism - Mass Tourism VsEcotourism -Typology of Eco-tourists - Ecotourism Activities & Impacts -Western Views of Ecotourism. Eco- tourism travel essentials. Eco- tourism and protected areas: visitor management for sustainability.Major Eco tourism destinations of India.

UNIT 6 Global tourism markets (Western Europe and North American Countries):Western Europe and North American market segment Outbound and inbound tourism – Share in global tourism- Renowned Tourism attractions in the market segment

UNIT 7 Global tourism markets (Far East and Middle East Countries): Far East and Middle East Countries market segment Outbound and inbound tourism – Share in global tourism- Renowned Tourism attractions in the market segment .

UNIT 8 Global tourism markets (North and South and South East Asian Countries): North and South and South East Asian Countries market segment Outbound and inbound tourism – Share in global tourism- Renowned Tourism attractions in the market segment.

UNIT 9 Global Tourism: World tourism trend – Growth and prospects of tourism- Share of major regions and trend thereof Trend in inbound and outbound tourism – Mechanism – Legal challenges – Business scope and Contemporary developments.

UNIT 10 Recent Trends in International and Domestic tourism across the globe – Business strategy and techniques used to attract tourist- advertisement benefits.

UNIT 11 Global Tourism Resources: Global level accommodation and transportation facilities- Distribution over major regions

UNIT 12 Global tourism earnings and distribution thereof amongst major countries and regions- Barriers to Global Tourism : Problems of Epidemics

UNIT 13 Financial and legal barriers – Business competitiveness -Natural calamities like Tsunami, Cyclones, Earthquakes

UNIT 14 Manmade calamities and internal disturbance.

REFERENCE

1. Charles R. Goeldner and Brent Ritchie, J.R., Tourism : Principles, Practices, Philosophies
2. Publications of World Tourism Organization

33845 - EMERGING ISSUES IN TOURISM

Objective:

- To understand the international, peace and role of tourism
- To gain knowledge ontourism and politics

UNIT 1 Contemporary issues in tourism:New trends and emerging patterns; current initiatives ofMinistry of Tourism, Department of Tourism; Climate change and environmental issues: Community Based Tourism; Concept of Home-stays, Responsible tourism; Tourism and poverty alleviation; Socialmedia for tourism promotion; Social responsibility and ethical concerns; Special Tourism Zones

UNIT 2 Sustainable Tourism DevelopmentBackground, concept and definition; Dimensions of sustainability- social, economic, environmental; Global warming and sustainable development; Sustainability and climate change issues in tourism ; Benefits and issues of Sustainable tourism development; World Conference on Sustainable Tourism; Sustainable tourism planning-principles, basic concept inSustainable design

UNIT 3 Standardization and Certification for tourism sustainability ISO 14000; World Conferences on Sustainable Development; Need and importance of tourism sustainability; Role of tourism organizations in Sustainable tourism development

UNIT 4 Approaches to Sustainable TourismEco-tourism; Global initiative for sustainable tourism development; Responsible tourism; Eco- friendly practices and energy waste management; Benefits of sustainable tourism development,

UNIT 5 Tourism and International understanding: Global peace and role of tourism – International tourists as ambassadors Cultural exchanges across the world – Cross border tourism to reduce tensions across borders.

UNIT 6 Tourism and International Spread of Economic well-being: Tourist flows tantamount to global resources flowSpecial importance to tourism for less developed countries economies - Tourism as a means of localized development

UNIT 7 Tourism as means of employment generation-Tourism and Ecological and Environmental Balance: Fragile nature of ecology and environment

UNIT 8 Harmful effects of unfettered tourism on ecology and environment-Methods of mitigating ill effects- Sustainable tourism UNIT 8Pollution free tourism- Rebuilding tourism affected by natural catastrophes.

UNIT 9 Tourism and Politics : Politics – Political Policies and their impact on tourism

UNIT 10 Civil unrest – Terrorism and their impact on tourism.

UNIT 11 Tourism and Economic Openness: Economic policy – Economic reforms – Globalisation and its impact on tourism.

UNIT 12 Emerging Tourism Opportunities: Medical tourism: Health spas, Specialist medical facility

UNIT 13 Global Trade and Investment leading to spurt in Global tourism

UNIT 14 Overseas Education and tourism growth.

Reference Books:

1. Charles R. Goeldner and Brent Ritchie, J.R., Tourism : Principles, Practices, Philosophies
2. Publications of World Tourism Organization.

3. DURATION OF THE PROGRAMME

The course shall consist of two academic years divided into four semesters

4. Faculty and Support Staff Requirements

This programme requires the following faculty and supporting staffs

Staff Category	Required
Core Faculty *	3
Faculty for Specialization*	2
Clerical Assistant	1

*At least Assistant Professor Level (Either permanent or part time)

5. Instructional Delivery Mechanism

Each semester there will be one contact programme of 80 hours duration in theory. The SLM (Self Learning Material) will be supplied to the students in print form as well as in CD form. The face to face contact sessions of the programme for theory courses will be held at the head quarter / learning centers. The conduct of end semester examinations, evaluation and issuance of certificates will be done by office of the Controller of examinations, Alagappa University, Karaikudi.

F) PROCEDURE FOR ADMISSION, CURRICULUM TRANSACTION, AND EVALUATION

Procedure of Admission

A candidate who has passed any Bachelor Degree from a recognized University in the Pattern of 10+2+3 shall be permitted to appear and qualify for the programme.

Curriculum Transactions:

The class room teaching would be through conventional lecture, use of OHP, power point presentation and novel innovative teaching ideas like television and computer aided instruction. Student seminars would be arranged to improve their awareness and communicative skill.

Face to face contact session will be conducted as given in below table.

Course Type	Face to Face Contact Session/semester (in Hours)
5 Theory courses with 4 credits	80
Total	80

Evaluation

The examinations shall be conducted for theory to assess the knowledge acquired during the study. There shall be two systems of examinations viz., internal and external examinations. In the case of theory courses, the internal evaluation shall be conducted as Continuous Internal Assessment via. Student assignments preparation. The internal assessment shall comprise of maximum 25 marks for each course. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

f. 3.2. Distribution of Marks in Continuous Internal Assessments:

The following procedure shall be followed for awarding internal marks for **theory** courses

Component	Marks
Assignments(2) (12.5+12.5)	25
Total	**25**

Question paper pattern (Theory)

- The question paper carries a maximum of 75 marks.
- The question paper consists of three sections namely Part-A, Part-B and Part-C.
- Part-A consists of 10 questions of 2 marks each (10 x 2 = 20 marks) with no choice. The candidate should answer all questions.

- Part-B consists of 5 either or choice questions. Each question carries 5 marks (5 x 5=25 marks).
- Part-C consists of 5 questions. Each question carries 10 marks. The candidate should Answer any three questions (10 x 3 = 30 marks).

Passing minimum

- There shall be no Passing Minimum for Internal.
- For External Examination, Passing Minimum shall be of 50% (Fifty Percentage) of the maximum marks (75) prescribed for the paper.
- In the aggregate (External + Internal) the passing minimum shall be of 50 Mark for each Paper
- Grading shall be based on overall marks obtained (internal + external).

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

Marks and Grades:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester

G_i = Grade Point obtained for course i in any semester.
n refers to the semester in which such courses were credited

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

Grade Point Average = Sum of the multiplication of grade points by the credits of the courses

Sum of the credits of the courses in a semester

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = Sum of the multiplication of grade points by the credits of the entire programme

Sum of the credits of the courses for the entire programme

CGPA	**Grad**	**Classification of Final Result**
9.5-10.0 9.0 and above but below 9.5	O+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D++ D+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A++ A+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

Maximum duration for completion of the course

The maximum duration for the programme shall not exceed five years after the completion of the minimum duration of the programme.

Commencement of this regulation

These regulations shall come into effect from the academic year 2018-19 for students who are admitted to the first year of the course during the academic year 2018-19.

Fee structure

Sl. No.	Fees Detail	Amount in Rs.	
		First Year	**Second Year**
1	Admission Processing Fees	300.00	-
2	Course Fees	13200.00	13200.00
3	ICT Fees	150.00	150.00
	TOTAL	**13650.00**	**13350.00**

G) REQUIREMENT OF THE LIBRARY RESOURCES:

LIBRARY RESOURCES

The Central Library is one of the important central facilities of Alagappa University. It has text book, reference books, conference proceedings, back volumes, standards, and non-book material such as CD-ROMs and audios. The central Library procured several e-books in different areas. The library also subscribes to about 250 current periodicals. The Directorate of Distance

Education of Alagappa University has adequate number of copies of books related to Management Programme.

COST ESTIMATE OF THE PROGRAMME AND THE PROVISIONS:

Sl. No.	Nature of Expenditure	Amount in Rs. (Approx.)
1	Programme Development	10,00,000/-
2	Programme Delivery	20,00,000/-
3	Programme Maintenance	3,00,000/-

i) QUALITY ASSURANCE MECHANISM AND EXPECTED PROGRAMME OUTCOMES

- The feedback from students on teaching will be collected every semester using standard formats.
- Feedback on the curriculum will also be collected from the experiences of the students which help teachers in fine tuning of deliverables in the classroom.
- It helps in improving the standard of teaching as expected by the students.
- Exit survey feedback on various parameters to improve and quality of the programme and support services like course material, library and infrastructure.
- It helps to Strengthen the contents of the program to meet the requirements of the employment market and keep the curriculum as a treasure of knowledge.
- This programme provides opportunities for students to develop and demonstrate knowledge and understanding, skills, qualities and other attributes.

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ALAGAPPA UNIVERSITY

[Accredited with 'A+' Grade by NAAC (CGPA:3.64) in the Third Cycle]

KARAIKUDI

Minutes of the Meeting of the Board of Studies in Management (Distance Education) held at the Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 06.09.2017 at 11.00 am.

Members Present

1. Dr. S. Kaliyamoorthy - Chairman
2. Dr.G. Jayabal - Member
3. Dr.R. Perumal - Member
4. Dr.S. Rajmohan - Special Invitee
5. Mr.S. Prabhu - Special Invitee

At the outset, the Chairman has extended a warm welcome to all the Members of the Board and briefed the need and purpose of the meeting.

The board has considered and scrutinized the Regulations and Syllabi for the following UG, PG and PG Diploma Programmes in the field of Management to be offered through ODL mode.

S.No.	U.G. Programmes	P.G. Programmes	PG Diploma Programmes
1.	BBA	M.A.(Public Administration)	Hospital Administration
2.	B.A. (Public Administration)	M.A.(Personnel Management & Industrial Relations)	Human Resource Management
3.	BBA(Banking)	M.B.A (General)	Business Management
4.	-	MBA(International Business)	Personnel Management & Industrial Relations
5.	-	MBA(Corporate Secretaryship)	-
6.		MBA(Banking & Finance)	
7.	-	MBA(Project Management)	-
8.	-	MBA(Hospital Management)	-
9.	-	MBA (HumanResourceManagement)	-
10.	-	MBA(Education Management)	-
11.	-	MBA(Retail Management)	-
12.	-	MBA(Technology Management)	-
13.	-	MBA(Logistics Management)	-
14.	-	MBA(Corporate Management)	-
15.	-	MBA(Financial Management)	-
16.	-	MBA(Marketing Management)	-
17.	-	MBA(System Management)	-
18.		MBA(Production and OperationManagement)	-
19.	-	MBA (Tourism)	-
20.	-	MBA (Cooperative Management)	-
21.	-	MBA Five Years Integrated	-

The board has unanimously resolved to approve the Regulations and Syllabi of the various above mentioned UG, PG and PG Diploma Programmes proposed to be offered through ODL mode. The approved Regulations and Syllabi of the above mentioned programmes are provided in the Annexure-I

Finally the meeting came to end with a formal vote of thanks.

6/9/2017

(R. PERUMAL)

6/9/17.

(G. JAYABAL)

(S. PRABHU)

(R.S. RAJMOHAN)

6/9/17

(S. KALIYAMOORTHY)

ALAGAPPA UNIVERSITY

(A State University Accredited with 'A+ Grade by NAAC (CGPA: 3.64) in the third cycle)

Karaikudi – 630 003, Tamilnadu, India

DIRECTORATE OF DISTANCE EDUCATION

PROGRAMME PROJECT REPORT

M.Com

Programme Code: 310

REGULATIONS AND SYLLABUS

[From the academic year 2018-2019 onwards]

Credit Based System

Programme's Mission & Objectives

Mission: To bestow an exhaustive acquaintance by blending the subjects of Accounting, Banking, Insurance, Taxation, Marketing, Services Marketing and Mercantile Law in a pragmatic manner to the students so as to emerge as efficient Professionals, Entrepreneurs, Managers, Finance Experts, etc….

Objectives: i). To impart a comprehensive knowledge in Commerce to the students in a pragmatic manner.

ii). To be a strapping pedestal to key-in and fetch an assortment of job opportunities in the Public and Private Sectors.

iii). To craft Accounting, Finance, Tax and Management experts through captivating professional and cerebral associations.

Relevance of the Programme with HEI's Mission and Goals:

Affording quality higher Education to the learners who are interested in pursuing higher education through distance mode, so that they are transformed into intellectually competent human resources that will help in the uplift of the nation in terms of Educational, Social, Technological, Environmental and Economic magnificence (ESTEEM). This programme is very much effective in imparting quality education through flexi-timings.

In accordance with the mission of Alagappa University as a research-intensive institution, the teaching programme of the master's programme in Commerce is based on state of the art of scientific research and maintains a strong emphasis on the acquisition of academic and research skills.

Nature of prospective target group of learners:

The curriculum has been designed to fulfill the needs of diverse class of learners including a class of learners. In order to fulfill the needs of the learners, professionals who are in teaching, business professionals, chartered accountant practitioners, banking, Insurance and management professionals.

M.Com., Employment Areas:

- Banking Sector
- Insurance
- Budget Planning
- Business Consultancies
- Educational Institutes
- Foreign Trade
- Industrial Houses
- Inventory Control
- Investment Banking
- Marketing
- Merchant Banking
- Public Accounting Firms
- Working Capital Management
- Policy Planning
- Public Accounting Firms
- Treasury and FOREX Department
- Auditing
- Banking and finance
- Electronic marketing
- Funds management
- Management consulting
- Stock Broking
- Teaching
- Accounting

M.Com., Job Types:

- Auditors
- Book Keepers
- Budget Analysts
- Business Consultants

- Chartered Management Accountant
- Chief Financial Officer
- Finance Managers
- Financial Analyst
- Human Resource Managers
- Investment Analysts
- Marketing Managers
- Stock Brokers
- Teacher / Professors

Appropriateness of the programme to be conducted in Open and Distance Learning mode to acquire specific skills and competence in Master of Commerce programme focuses both in-depth study of theory and acquisition of professional and research skills.

- This programme makes learners to develop skill oriented entrepreneurship knowledge, business communication at critical thinking and proficiency in the field of business sectors.

- This programme helps learners to acquire necessary skills to perform research, and start up entrepreneurship in the field of commerce and industry.

It is a good base Master degree course for the purpose of higher research studies like M.Phil., and Ph.D. They can avail wide employment opportunities and employability skills in the field of Commerce and Industry.

e. Instructional Design:

Sl. No	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C.
FIRST YEAR						
I Semester						
1	**31011**	Management Concepts	25	75	100	4
2	**31012**	Managerial Economics	25	75	100	4
3	**31013**	Advanced Accounting	25	75	100	4
4	**31014**	Business Environment	25	75	100	4
		Total	**100**	**300**	**400**	**16**
II Semester						
5	**31021**	Marketing Management	25	75	100	4
6	**31022**	Research Methodology	25	75	100	4
7	**31023**	Advanced Cost Accounting	25	75	100	4
8	**31024**	Financial Services	25	75	100	4
		Total	**100**	**300**	**400**	**16**
SECOND YEAR						
III Semester						
9	**31031**	Investment Analysis and Portfolio Management	25	75	100	4
10	**31032**	Financial Management	25	75	100	4
11	**31033**	Principles of Personnel Management	25	75	100	4
12	**31034**	Corporate Accounting	25	75	100	4
		Total	**100**	**300**	**400**	**16**
IV Semester						
13	**31041**	Management Accounting	25	75	100	4
14	**31042**	Principles and Practice of Services Marketing	25	75	100	4
15	**31043**	Organizational Behaviour	25	75	100	4
16	**31044**	Retail and Distribution Management	25	75	100	4
		Total	**100**	**300**	**400**	**16**
Grand Total			**400**	**1200**	**1600**	**64**

Detailed Syllabi:

FIRST SEMESTER

Course Code	Title of the Course
31011	**Management Concepts**

Learning objectives:

1. To make the students to understand the elements of effective management

2. To familiarize the students with the basics principles of management

UNIT – I: Management: Definition – Nature – Scope and functions – Evolution of management thought – Relevance of management to different types of organization like, Insurance, Hospitals, Universities, Hotels, Social Service organizations etc.

UNIT – II: Planning: Nature, importance and strategic considerations in planning – Planning premises – Components of planning as objectives - Planning process – Methods of planning – Limitations of planning – Planning premises.

Unit – III: Policies, strategies, procedures, methods, rules, projects and budgets – Decision making – Meaning – Importance – Types – Process – Elements – Principles of decision making.

UNIT – IV: Organizing: Nature, purpose and kinds of organization – Structure – Principles and theories of organization – Formal organization – Informal organization.

Unit – V: Departmentation – Process of departmentation – Departmentattion by functions - Span of control – Line and staff functions – Authority and responsibility – Centralisation and decentralization.

Unit – VI: Delegation - Importance of delegation – Types of delegation – Problems of delegation – Effective delegation.

UNIT - VII: Staffing – Elements of staffing – Functions of staffing - General principles and importance – Motivation: Meaning – Importance – Types – Theories.

UNIT – IX: Communication: Meaning – Types – Process – Barriers – Overcoming barriers – Leadership: Meaning – Styles – Essential qualities.

UNIT – X: Controlling: Objectives and process of control – Devices of control – Integrated control – Special control techniques – Coordination – Need and techniques.

UNIT – XI: Directing – Principles of direction – Techniques of direction – Importance of direction – Characteristics of direction – Supervision – Functions of supervisor – Qualities of supervisor – Principles of supervisor.

UNIT – XII: Management Audit – Definition – Objectives – Scope of management audit – Need for management audit – Advantages of management audit.

UNIT – XIII: Business Ethics – Meaning – Need for business ethics – Principles of business ethics – Factors affecting business ethics – Management information system – Objectives of MIS – Importance of MIS – Process of MIS – Causes for poor MIS.

UNIT – XIV: Recent trends and new perspectives in management - Strategic alliances – Core competence – Business process reengineering – Total quality management – Bench marking.

REFERENCE BOOKS:

1. Ramaswamy.T, 2012, Principles of Management [Eight Edition] Himalaya Publishing Home Pvt Ltd, Mumbai.

2. Dinkar Pagare, 2011, Business Management [Fifth Edition] Sultan Chand & Sons, New Delhi.

3. Govindarajan. M, 2008, Principles of Management [First Edition] PHI, New Delhi.

4. Prasad L.M, 2015, Principles and Practice of Management [Eight Edition], Sultan Chand & Sons, New Delhi.

Course Code	Title of the Course
31012	**Managerial Economics**

Learning Objectives:

1. To discuss the relationship between economics and business.
2. To analyze the application of economic theories in modern business.

UNIT – I: Nature and Scope of Managerial Economics – Concept of Managerial Economics - Role and Responsibilities of Managerial Economist – Goals of Corporate Enterprises: Maximization of profit - Value of enterprise.

UNIT – II: Demand analysis - Elasticity of demand- Factors influencing elasticity of demand - Demand determinants – Demand distinctions – Types, methods – Applications – Demand forecasting for industrial goods – Consumer goods – Consumer durables.

UNIT – III: Cost and production analysis - Cost concepts – Cost and output relationship - cost control – Short run and Long run - cost functions - production functions – Break-even analysis – Economies of scale of production.

UNIT – IV: Pricing and output decisions in different market situations – Monopoly and Duopoly competition - Perfect and Imperfect competition - Pricing policies – Types - Factors affecting pricing policies.

UNIT – V: Business cycles – National income, monetary and fiscal policy – Public finance - Government Institutions, State Governments, Local Self-Government, Central Government – Industrial Sickness – Causes – Remedies.

UNIT – VI: Theory and nature of profit – Theories of profit – Depreciation and Profit – Valuation of stock and profit – Profit policies – Planning and forecasting – Calculation of break even point.

UNIT – VII: Managerial decision making – Risk and uncertainty – Decision theory – Classification of managerial decision problems – Analysis of decisions – Probability theorems.

UNIT – VIII: Economic planning and MRTP Act – Objectives of economic planning – Control of big business – Evil effects of economic concentration – Industrial policy – Features.

UNIT – IX: Consumer – Consumer's sovereignty – Consumer's surplus – Consumer's surplus in a market – Assumption of consumer's surplus.

UNIT – X: Business cycle and business policy – Characteristics of business cycle – Phases of business cycle – Depression – Inflation – Characteristics of inflation – Types of inflation – Demand pull inflation – Cost push inflation – Control of deflation – Effects on business cycle.

UNIT – XI: Product line – Reasons required changing existing product mix – Factors determining the scope of product line – Product line pricing – Special order pricing – Pricing Complementary products.

UNIT – XII: Economic Concentration – Meaning of economic power – Domination of markets – Product-wise concentration – Country-wise concentration – Growth of Large industrial houses and concentration of economic power.

UNIT – XIII: Foreign investment – Types of foreign investment – FDI in developing countries – Guidelines for foreign investments – FII investments – Euro issues – FDI inflow in India.

UNIT – XIV: Government and business – Performance of public enterprises in India – Price policy in public utilities, government measures to control monopoly in India.

REFERENCE BOOKS:

1. Sankaran.S.2004, Managerial Economics. Margham Publications.

2. Gupta.G.S .2009, Managerial Economics. Tata McGraw-Hill Education Pvt.Ltd.

3. Cauvery.R., Dr. Sudha Nayak. U.K., Girija.M.and Dr.Meenakshi.R. 2010, Managerial Economics. S.Chand, New Delhi.

4. Maheswari.K.L and Varshney.R.L. 2014, Managerial Economics. [22nd Edition, Chand Sons, New Delhi.

Course Code	Title of the Course
31013	**Advanced Accounting**

Learning Objectives:

1. To make the students to understand the basic concepts of accounting applied in the competitive corporate world.
2. To understand accounting software easily.

UNIT – I: Accounting Concepts and Conventions - Accounting Systems: Cash and mercantile systems – Advantages and Limitations of Accounting - Principles – Accounting standards.

UNIT – II: Fundamentals of Book Keeping – Journal – Ledger – Subsidiary books – Cash book - Trial balance.

UNIT – III: Single Entry System: Calculation of profit – Statement of Affairs method – Conversion of single entry into double entry system.

UNIT – IV: Preparation of profit and loss account and balance sheet under single entry system.

UNIT – V: Final Accounts of a Sole Trader – Trading Account, Profit and Loss Account and Balance sheet with adjustments – Differences between Trial Balance and Balance sheet – Errors and their rectification – Types of errors.

UNIT – VI: Bills of Exchange : Promissory notes and bills of exchange – Recording of transactions relating to bills – Books of Drawer and Acceptor – Honour and dishonor of bills –Renewal of bills – Retiring of bills under rebate.

UNIT – VII: Account Current and Average Due Date – Meaning – Advantages – Calculation of Account Current and Average Due Date.

UNIT – VIII: Bank Reconciliation Statement –Need –Reasons for difference between cash book and pass book balances – Preparation of Bank Reconciliation Statement.

UNIT – IX: Non-Trading Accounts - Receipts and Payments and income and expenditure account and Balance sheet– Difference between Receipts and Payments and income and expenditure account.

UNIT – X: Depreciation – Meaning – Causes – objects of providing for depreciation – Factors affecting depreciation – Accounting Treatment – Methods of providing depreciation – Straight line method – Diminishing Balance Method – Change in method of Depreciation – Sinking fund method.

UNIT – XI: Accounting for consignments- Consigner's book – Consignee's book – Accounting for joint ventures - Joint ventures account – co-venture's account – Joint bank account.

UNIT – XII: Investment accounts – Ex-interest – Cum-interest Types of securities – Fixed interest securities – Variable income securities – Royalty excluding Sublease - Methods of recoupment - Fixed recoupment of short workings – Flexible recoupment.

UNIT – XIII: Departmental accounts – Apportionment of common expenses – Ascertaining cost of departmental purchases Inter departmental transfers at cost or selling price – Branch – Dependent branches – Independent branches - excluding foreign branches.

UNIT – XIV: Hire purchase - Calculation of interest - installment systems - Default and Repossession including Hire Purchasing Trading account - Goods on sale or Return.

REFERENCE BOOKS:

1. Reddy, T. S. and Murthy, A. 2014, Financial Accounting [Sixth Edition], Margham Publications, Chennai.

2. Nagarajan, K. L., Vinayaka, N., and Mani P.L. 2013, Principles of Accountancy [First Edition], Sultan Chand & Company Ltd, New Delhi.

3. Jain, S. P., and Narang, K. 2005, Financial Accounting. [Fifth Edition], Kalyani Publishers, Ludhiana.

4. Grewal T.S. 2007, Introduction to Accountancy [Fifth Edition], Sultan Chand & Company Ltd, New Delhi.

Course Code	Title of the Course
31014	**Business Environment**

Learning Objectives:

1. To enhance the knowledge of business opportunities in current situation.
2. To enhance the capacity to making business polices.

UNIT – I: Concept of Business Environment- Significance - Types of Environment- External and Internal – Inter-Relationship between economic and non-economic environment.

UNIT – II: Impact of environment on business and Strategic Decisions - Culture and business - Social Responsibilities of Business.

UNIT – III: Economic Systems – Meaning – Characteristics -Types of economic systems-Capitalism-Socialism-Mixed economy.

UNIT – IV: Economic planning - Nature, Scope and Significance of Economic Planning in India - Achievements and Failures of Economic Planning.

UNIT – V: Industrial Policies and Regulations - Industrial Policy upto 1991, 2014 - Public, Private, Joint and Co-operative Sectors.

UNIT – VI: Privatization and Disinvestment - Ways of Privatization - Benefits and Arguments against Privatization - Privatization in India.

UNIT – VII: Technological environment-Factors Governing Technological Environment-Management of technology - Patents and Trademarks.

UNIT – VIII: Indian Contract Act 1872 – Indian Companies Act 1956 – Consumer Protection Act 1986 – Consumer Rights – World Trade Organization (WTO).

UNIT – IX: Industrial Finance – Short term finance – medium term finance - long term finance – Corporate securities.

UNIT – X: Reserve Bank of India - Financial Institution in India – IFCI – ICICI – IDBI – IIBI – SIDBI – LIC – SIDCO – Commercial Banks – DFHI - SFCs.

UNIT – XI: Globalization - Meaning and Dimensions - Features of Current Globalization - Essential Conditions for Globalization - Globalization of Indian business.

UNIT – XII: Foreign Direct Investment - Concept, Advantages and Disadvantages and Determinants- India's policy towards FDI - Multinational Corporation – Meaning - Merits and Demerits - Control over MNCs - MNC in India.

UNIT – XIII: International Environment – Meaning and concept – World Bank – International Monetary Fund (IMF) – The General Agreement on Tariffs and Trade (GATT) – The World Trade Organization.

UNIT – XIV: Current Issues in Business Environment – Urbanization – Population – Public distribution system – Natural environment – Various aspects of the natural environment – Pollution – Environmental management.

REFERENCE BOOKS:

1. Francis Cherunilam. 2000, Business Environment, Himalaya Publishers.

2. Gupta. C. B., 2014, Business environment, (8th Edition), McGraw Hill Education India Private Limited.

3. Avadhani.V. A., 2004, Essentials of Business Environment, (2nd Edition), Himalaya Publication, Mumbai.

4. Shaikh Saleem, 2009, Business Environment, (2nd Edition), Dorling Kindersley (India) Private Limited.

SECOND SEMESTER

Course Code	Title of the Course
31021	**Marketing Management**

Learning Objectives:

1. To give the knowledge of updated marketing scenario.
2. To enhance the knowledge of marketing behavior of consumer among the students.

UNIT – I: Marketing: Meaning – Scope – Importance – Approaches to the study of Marketing – Marketing Concept –

UNIT – II: Market Segmentation: Meaning – Bases for segmentation – Uses. Marketing Mix: Four P's in marketing – Marketing Planning – Importance – Types of planning.

UNIT – III: Marketing Environment – External factors – Internal factors – Consumer Behaviour – Meaning and importance – Consumer buying process – Determinants of consumer behaviour — Theories and their relevance to marketing.

UNIT – IV: Product Mix Management: Product planning and development – New Product development – Product Life Cycle – Meaning – Stages –Managing PLC – Product positioning – Branding – Packaging.

UNIT – V: Price Mix Management: Factors affecting pricing – Pricing and pricing policies – Objectives – Procedures – Methods of price fixation – Administered and regulated prices.

UNIT – VI: Physical Distribution Mix: Distribution channel policy – Types – Factors determining choice of channel – Channel management – Middlemen functions.

UNIT – VII: Promotional Mix: Personal selling Vs Impersonal selling – Personal selling process – Steps in selling – Compensation plans – Evaluation of salesmen performance.

UNIT – VIII: Advertising: Importance – Objectives – Media planning and selection – Factors influencing selection – Advertising copy – Layout – Evaluation of advertising – Advertising budget – Sales Promotion methods – Publicity – Sales promotion tools.

UNIT - IX: Marketing Research & Marketing Information System: Meaning – Scope – Need – Elements – Research Process – Steps involved.

UNIT – X: Consumerism: Meaning – Consumer rights – Consumer movement in India – Salient provisions of Consumer Protection Act.

UNIT – XI: Co-operative marketing – Meaning – Features of Co-operative marketing – Advantages – Structure and organization – National Agricultural Co-operative Marketing Federation (NAFED).

UNIT – XII: E-Marketing – Meaning – Business models – Websites associated with e-marketing – Benefits of e-marketing – Limitations of e-marketing – Scope of e-marketing.

UNIT – XIII: Concept of customer relationship management – Evolution – Scope – Differences between marketing and CRM – Types of CRM.

UNIT – XIV: Technology for customer relationship – Contact centre technology – CRM technology – Customer data management – Managing customer relationship – CRM measurement.

REFERENCE BOOKS:

1. Philip Kotler, 2014 Principles of Marketing (15th Edition 2014), Pearson Education Private Limited, New Delhi.

2. Pillai. R. S. N and Baghavathy .N, Modern Marketing (Edition 1987, Reprint 2012), Sultan Chand and sons Publishers, New Delhi.

4. Gupta C. B and Rajan Nair .N, Marketing Management, (Edition 1996 Reprint 2012), Sultan Chand and Sons Publishers, New Delhi.

5. Ramasamy. R. and Namakumari V. S, Marketing Management, (3rd Edition), McMillan India, Limited, New Delhi.

Course Code	Title of the Course
31022	**Research Methodology**

Learning Objectives:

1. To provide students with a firm foundation and understanding of business research methods and the research process.

2. To understand the relevance of and be able to apply a range of both quantitative and qualitative research methods.

UNIT – I: Research: Meaning of research; Types of research- Exploratory research, Conclusive research; The process of research; Research applications in social and business sciences; Features of a Good research study.

UNIT – II: Research Problem and Formulation of Research Hypotheses: Defining the Research problem; Management Decision Problem Vs Management Research Problem; Problem identification process; Components of the research problem; Formulating the research hypothesis- Types of Research hypothesis; Writing a research proposal- Contents of a research proposal and types of research proposals.

UNIT – III: Research Design: Meaning of Research Designs; Nature and Classification of Research Designs; Exploratory Research Designs: Secondary Resource analysis, Case study Method, Expert opinion survey, Focus group discussions; Descriptive Research Designs: Cross-sectional studies and Longitudinal studies; Experimental Designs, Errors affecting Research Design.

UNIT – IV: Primary and Secondary Data: Classification of Data; Secondary Data: Uses, Advantages, Disadvantages, Types and sources; Primary Data Collection: Observation method, Focus Group Discussion, Personal Interview method.

UNIT – V: Attitude Measurement and Scaling: Types of Measurement Scales; Attitude; Classification of Scales: Single item Vs Multiple Item scale, Comparative Vs Non-Comparative scales, Measurement Error, Criteria for Good Measurement.

UNIT – VI: Questionnaire Design: Questionnaire method; Types of Questionnaires; Process of Questionnaire Designing; Advantages and Disadvantages of Questionnaire Method.

UNIT – VII: Sampling: Sampling concepts- Sample Vs Census, Sampling Vs Non Sampling error; Sampling Design- Probability and Non Probability Sampling design;

Determination of Sample size- Sample size for estimating population mean, Determination of sample size for estimating the population proportion.

UNIT – VIII: Data Processing: Data Editing - Field Editing, Centralized in house editing; Coding - Coding Closed ended structured Questions, Coding open ended structured Questions; Classification and Tabulation of Data.

UNIT – IX: Univariate and Bivariate Analysis of Data: Descriptive Vs Inferential Analysis, Descriptive Analysis of Univariate data- Analysis of Nominal scale data with only one possible response, Analysis of Nominal scale data with multiple category responses, Analysis of Ordinal Scaled Questions, Measures of Central Tendency, Measures of Dispersion; Descriptive Analysis of Bivariate data.

UNIT – X: Testing of Hypotheses: Concepts in Testing of Hypothesis – Steps in testing of hypothesis, Test Statistic for testing hypothesis about population mean; Tests concerning Means- the case of single population; Tests for Difference between two population means; Tests concerning population proportion- the case of single population; Tests for difference between two population proportions.

UNIT – XI: Chi-square Analysis: Chi square test for the Goodness of Fit; Chi-square test for the independence of variables; Chi-square test for the equality of more than two population proportions.

UNIT – XII: Analysis of Variance: Completely randomized design in a one-way ANOVA; Randomized block design in two-way ANOVA; Factorial design.

UNIT – XIII: Research Report Writing: Types of research reports – Brief reports and Detailed reports; Report writing: Structure of the research report- Preliminary section, Main report, Interpretations of Results and Suggested Recommendations; Report writing: Formulation rules for writing the report: Guidelines for presenting tabular data, Guidelines for visual Representations.

UNIT – XIV: Ethics in Research: Meaning of Research Ethics; Clients Ethical code; Researchers Ethical code; Ethical Codes related to respondents; Responsibility of ethics in research – Uses of library and internet in research.

REFERENCE BOOKS:

1. Gupta S. L and Hitesh Gupta (2015), “Research Methodology: Text and Cases with SPSS Application”, International Book House Private Limited, New Delhi.

2. Peer Mohamed and Shazuli Ibrahim (2013), “Research Methodology”, Pass Publications, Madurai.

Course Code	Title of the Course
31023	**Advanced Cost Accounting**

Learning Objectives:

1. To enable the students to understand the Costing Terms in business.
2. To provide adequate knowledge on Cost Accounting Practice

UNIT – I: Cost Accounting Principles: Meaning of cost and cost accounting – Objectives of cost accounting – Installation of a costing system.

UNIT – II: Elements of cost – Cost concepts – Cost classifications – Methods, systems and techniques of costing – Cost sheet.

UNIT – III: Cost Accounting for material cost control – Need for material cost control – Purchase control – Stores control – Stock levels – EOQ analysis.

UNIT – IV: Pricing of stores issues – Perpetual inventory control – ABC analysis – VED analysis – Treatment of waste, scrap, defectives and spoilage.

UNIT – V: Labour Cost Control – Time keeping and time booking – Treatment of idle time and overtime cost – Wage rates for costing – Systems of wage payment – Time wage and piece rate – Incentive schemes of wage payment – Labour turnover.

UNIT – VI: Overhead Cost Control – Classification of overheads – Allocation and appointment – Absorption of overheads – Different methods – Treatment of under absorption and over absorption of overheads.

UNIT – VII: Methods of Costing – Job costing – Contract costing – Profit on incomplete contracts – Cost plus contracts – Target costing – Escalation clause.

UNIT – VIII: Unit costing – Meaning – Cost accumulation – Procedure in unit costing – Preparation of cost sheet.

UNIT – IX: Process Costing – Features – Job costing Vs Process costing – Process cost accounts – Inter-process profits – Accounting for joint products and by products.

UNIT – X: Contract costing – Types – Procedure for contract costing – Contract plus costing – Profit from incomplete contract.

UNIT – XI: Operating Costing – Meaning – Features – Objectives – Cost Unit – Transport costing – Operating cost sheet.

UNTI – XII: Batch costing – Definition – Economic batch quantity – Applicability of batch costing.

UNIT - XIII: Standard Costing – Definition –Advantages and limitations of standard costing – Variance analysis.

UNIT – XIV: Reconciliation of cost and final accounts – Cost control and cost reduction – Meaning – Tools and techniques – Essentials for success of cost control and cost reduction – Distinction between cost control and cost reduction – Areas of cost reduction and control – Advantages.

REFERENCE BOOKS:

1. Jain, S. P., and Narang, K. L. 2001, Cost and Management Accounting [Fifth Edition], Kalyani Publishers, New Delhi.
2. Pillai, R. S. N., and Bagavathi, 2009, Cost Accounting [First Edition], Sultan Chand Company Ltd., NewDelhi.
3. Sharma, and Shashi K. Gupta, 2012, Management Accounting [Twelfth Edition], Kalyani Publishers, New Delhi.
4. Maheswari, S. N. 2003, Cost and Management Accounting [First Edition], Sultan Chand Company Ltd., New Delhi.

Course Code	Title of the Course
31024	**Financial Services**

Learning Objectives:

1. On successful completion of this course, the student should know about the methods of financing by the agencies and the key role Played by them in Corporate Financing.
2. Understand the tradeoff between risk and reward in investing.

UNIT – I: Nature and various facets of financial service industry – Analysis of financial services – Need for financial innovation.

UNIT – II: Money Market in India – Indian Capital Markets – Difference between Money Market and Capital Market – Classification and objectives of Indian Money Markets and Structure of Capital Markets.

UNIT – III: Financial services and market environment – Development of financial markets – Global integration of financial market – Finance Companies: Functions, strengths and weaknesses.

UNIT – IV: Secondary Markets – Stock Exchange – Role of Secondary Market – Trading in Stock Exchange – Various Speculative Transactions – Role of SEBI – Regulation of Stock Exchange.

UNIT – V: Banks as Financial Intermediaries – Commercial Banks Role in Financing – IDBI – IFCI – LIC – GIC – UTI – Investments Companies.

UNIT – VI: Markets for Corporate Securities – New Issue Markets – Functions Issue Mechanism – Under writing.

UNIT – VII: Commercial Banking and their fund based and non-fund based financial services – Leasing – Steps in leasing transactions – Lease finance – Accounting and reporting for lease.

UNIT – VIII: Hire purchase financing: Salient features, guidelines, functions – Components of hire purchase contract – Hire purchase agreement – Cost of hire purchase.

UNIT – IX: Mutual Funds: Types of mutual funds – Floatation – Asset Management Company of mutual funds – Regulations.

UNIT – X: Factoring – Forfeiting – Securitization – Venture capital – Consumer finance and credit cards: Salient features, guidelines, functions – Strategies involved in financing.

UNIT - XI: Merchant Banking including public issue management – Underwriting – Portfolio management – Stock and security broking – Merger and Takeover: Salient features – Guidelines – Functions.

UNIT - XII: Foreign Exchange Broking – Bills discounting – Financial consultancy – Corporate advisory services – Credit rating services – Salient features – Guidelines – Functions.

UNIT – XIII: Housing Finance – Advantages – Method of housing finance – Role of NHB – Borrowing powers of national housing bank – Export finance – Need for export finance – Different types of export finance.

UNIT - XIV: Non-banking Financial Companies: Regulations of RBI – Role of NBFCs – Chit funds – Functions of chit fund companies and Finance companies.

REFERENCE BOOKS:

1. Gordon E, and Natarajan K, 2009, "Financial Markets and Services", Himalaya Publishing House Pvt. Ltd., India.

2. Guruswamy S, 2009, "Financial Services", Tata McGraw-hill Education, New Delhi.

3. Prasanna Chandra, 2011 "Financial Management Theory and Practice", Tata Mc Graw-Hill Education, New Delhi.

4. Khan M Y and Jain P K, 2008, "Financial Management Text, Problems and Cases", Tata McGraw-Hill Education, New Delhi.

5. Banerjee G and Banerjee S. Borrowing from Financial Institutions, UDH publishing house Delhi.

6. Bhole .L. M, "Financial Institutions' and Markets: Structure Growth and Innovations", Tata McGraw-Hill Publishing Co Ltd.

THIRD SEMESTER

Course Code	Title of the Course
31031	**Investment Analysis and Portfolio Management**

Learning Objectives:

1. To understand the characteristics of securities markets and the instruments traded therein.
2. To be able to analyze the risk, return of securities and to manage portfolios of investments.

UNIT – I: Investment: Concepts and goals – Types of investment: Financial, Real, Business, Personal and Institutional – Comparison of investments, speculation, gambling and hedging – Concept of portfolio management: Goals – Risk and return trade-off.

UNIT – II: Financial Investment Avenues: Fixed income and Varying income securities - Factors influencing Investment – Investment media – Features of investment Programme – Investment Process – Development of Financial system in India.

UNIT - III: Investment Analysis - Aspects of analysis – Return Analysis: Concepts, measures and computation of return of individual security and portfolio.

UNIT – IV: Capital Market – New issue Market and Stock Exchange in India – B.S.E - N.S.E – OTCEI – Kinds of Trading activity – Listing of Securities – SEBI and its Role and guidelines.

UNIT – V: Risk Analysis: Concepts, types, measure, computation of risk of individual security and portfolio – Valuation Analysis: Share and bond valuation – Price Earnings Analysis.

UNIT – VI: Investment Alternatives – Investment in Equity Shares, Preference shares, Bonds, Government Securities.

UNIT – VII: Mutual Funds – Real Estate – Gold – Silver – Provident fund – Unit Trust – National Savings Scheme – LIC.

UNIT – VIII: Approaches to Investment Analysis - Fundamental Analysis - Concept and components – Tools of economy, industry and company analysis.

UNIT – IX: Technical Analysis - Concept and tools – Assumption – Theories - Dow theory – Contrary opinion – Confidence index, Breadth of market and Relative strength analysis – Moving average analysis – Chart patterns.

UNIT – X: Options and Futures – Types of options – Call option – Advantages of options – Limitations – Valuation of options – Characteristics of options – Future – Forwards and futures – Differences between futures and options.

UNIT – XI: Portfolio Construction and Choice - Markowtz diversification – Efficient frontier – Risk-return indifferent curves – Portfolio choice – Single and two factorial models – Lagrange multiplier method.

UNIT – XII: Portfolio Performance Measures: Sharpe, Treynor and Jensen – Portfolio Audit and Portfolio Revision: Need and methods – Formula plans.

UNIT – XIII: Capital Asset Pricing Model – Assumptions and application – Capital market line and security market line.

UNIT – XIV: Efficient market hypotheses – The weakly efficient, semi strongly efficient and strongly efficient market forms – Random-walk theory.

REFERENCE BOOKS:

1. Preeti Singh, 2008, Investment Management [Sixteenth Edition], Himalaya Publishing House Pvt. Ltd., Mumbai.

2. Bhalla, V.K. 2008, Investment Management, Security Analysis and Portfolio Management [Fourteenth Edition]. Sultan Chand & Company Ltd., New Delhi.

3. Avadhani, V.A. 2008, Investment Management [Seventh Edition], Himalaya Publishing House, Mumbai.

4. Gangadhar, V., and Ramesh Babu, G. 2003, Investment Management [First Edition], Anmol Publication Pvt. Ltd., New Delhi.

Course Code	Title of the Course
31032	**Financial Management**

Learning Objectives:

1. To reveal the knowledge on fund utilization and management.
2. To analyze the various concepts and techniques for better financial decision.

UNIT – I: Financial Management: Concept, nature, evaluation and significance – Finance functions – Managerial and operative – Investment – Function, meaning and scope – Financing function – Meaning and scope – Dividend function.

UNIT – II: Goals of Financial Management – Types – Maximization of profit, profitability / wealth / liquidity / solvency – Minimization of risk, cost of capital, dilution of management control etc. – Risk – Return trade off – Maximization and minimization Vs optimization.

UNIT – III: Sources of finance – Long term sources – Short term sources – Sources of working capital – Equity shares – merits and demerits – Preference shares – merits and demerits – Debentures – Merits and demerits – Elements of financial system – Structure of Indian capital market.

UNIT – IV: Capitalization – Meaning – Need – Theories of capitalization – Over capitalization – Merits and demerits - Under capitalization – Merits and demerits.

UNIT – V: Financial decisions – Relationship between Risk and Return – Sources of finance – Short-term and Long-term finance.

UNIT – VI: Long Term Capital Resources – Equity and debt sources – Equity share, preference shares and debentures as sources of long term capital – Relative merits, demerits and uses.

UNIT – VII: Significance of convertible issues and right issues – Borrowings from term lending institutions – The institutional framework – Types of assistance – Public deposits.

UNIT – VIII: Working Capital: Concept and types – Determinants – Financing approaches – Conservative – Aggressive and hedging approaches – Their risk – Return

features and significance – Sources of working capital finance – Working capital financing by commercial banks.

UNIT – IX: Capital Planning – Determinants of capital structure – Optimum capital structure – Capital structure theories – Net income and net operative income theories – M.M. Theory – Traditional theory – Their assumptions – Significance and limitations.

UNIT – X: Cost of Capital Concept – Cost of debt, equity, preference share capital, retraining earning – Weighted average cost – Book weight, market weight – Marginal cost of capital use and computations.

UNIT – XI: Capital Budgeting: Concept – Significance – Methods of appraisal: Pay-back periods, ARR, IRR, NPV, Simulation and Certainty equivalent methods.

UNIT - XII: Leasing: Concept – Types – Significance – General considerations – Economics of leasing – Evaluation – Present value and IRR methods – Leverage – Concept – Types – Degree of operative leverage – Financial leverage and total leverage – Implications of high and low degrees of leverages.

UNIT – XIII: Dividend Theories: Valuation under Gordon and Walter theories – Dividend irrelevance under M.M. Theory – Assumptions – Limitations – Dividend policy – Different policies and practices – Factors affecting dividend decision.

UNIT – XIV: International Financial Management – Meaning – FOREX Market operations – Export credit needs – India's export finance – Methods of international payments – Letter of credit – Kinds of letter of credit – Foreign currency finance Export documents.

REFERENCE BOOKS:

1. Sharma R. K and Shasi, K. Gupta, 2014, Financial Management, (4th Edition), Kalyani Publishers, New Delhi.

2. Pandey I. M. 2009, Financial Management, (10th Edition), Vikas Publishing House.

3. Prasanna Chandra, 2012, Financial Management, (4th Edition), Tata McGraw-Hill Publishers.

4. Maheswari S.N. 2014, Financial Management, (26th Reprint) Vikas Publishing House Pvt. Ltd, New Delhi.

Course Code	Title of the Course
31033	**Principles of Personnel Management**

Learning Objectives:

1. To evaluate and apply theories of social science disciplines to workplace issues.
2. To enhance their effectiveness for optimizing the human resource potential of their organization in order to achieve business and strategic objectives.
3. To examine current issues, trends, practices, and processes in personnel management.

UNIT – I: Personnel Management – Definition – Objectives and functions – Role and structure of personal function in organizations – Personnel principles and policies – Managerial functions – Operative functions.

UNIT – II: Job Evaluation – Job Analysis – Job Design – Job Description – Job Specification – Methods of job evaluation – Ranking method – Advantages and disadvantages of job evaluation.

UNIT – III: Human Resource Planning: Characteristics – Need for planning – Human Resource Planning Process – Factors influencing human resource plan – Limitations of human resource planning – Forecasting the demand for HR.

UNIT – IV: Organizing the human resource – Definitions – Process of organization – Importance of organization – Organization Structure – Principles of organization – Theories of organization.

UNIT – V: Recruitment - Selection Process - Placement and induction – Training and development – Need for training – Importance of training – Essentials of good training programme - Promotion – Demotions – Transfers – Separation.

UNIT – VI: Performance appraisal – Meaning – Features of performance appraisal – Methods of performance appraisal - Human relations - approaches to good human relations – Punishment.

UNIT – VII: Wage and Salary Administration: Factors – Principles – Compensation plan – Individual –Group – Incentives – Bonus – Fringe benefits – Job evaluation systems – Wage and salary administration in relation to personal taxation.

UNIT – VIII: Motivation – Definition – Characteristics of motivation – Importance of motivation – Process of motivation – Types of motivation – Theories of motivation.

UNIT – IX: Leadership – Meaning – Need and characteristics of leadership – Importance of leadership – Qualities of a leader – Different kinds of leadership styles – Theories of leadership.

UNIT – X: Morale and job satisfaction – Meaning – Characteristics of morale – Morale and productivity – Measurement of morale – Determinants of job satisfaction – Personal factors – Organizational factors.

UNIT – XI: Absenteeism – Meaning – Causes of absenteeism – Measures to control absenteeism – Discipline and grievance – Objectives of discipline – Characteristics of grievance – Cause of grievance.

UNIT – XII: Employee maintenance and integration – Welfare and society – Accident prevention – Administration of discipline – Employee motivation – Need and measures.

UNIT –XIII: Industrial relations - Trade unionism - Collective bargaining and worker's participation in management – Quality of work life.

UNIT – XIV: Personnel Records / Reports: Personnel Research and Personnel Audit: Objectives – Scope and importance.

REFERENCE BOOKS:

1. Tripathi P.C., 2013, Personnel Management and Industrial Relations, Sultan Chand and Sons, New Delhi.
2. Aswathappa K., 2013, Human Resource Management: Text and Cases, McGraw Hill Education, New Delhi.
3. Memoria C.B. & Rao V.S.P., 2014, Personnel Management - Text & Cases, Himalaya Publishing house, New Delhi.
4. Khanka S.S., 2007, Human Resource Management - Text & Cases, S. Chand & Company Ltd., New Delhi.

Course Code	Title of the Course
31034	**Corporate Accounting**

Learning Objectives:

1. To give a clear understanding and knowledge to the students in the area of corporate accounting and other related matters.
2. To provide the knowledge based on corporate need to have a global perspective and grow accordingly.

UNIT – I: Issue of shares: Par, Premium and Discount - Forfeiture - Reissue - Surrender of Shares - Right Issue – Underwriting.

UNIT – II: Redemption of Preference Shares – Provisions of the companies Act - Debentures - Issue – Treatment of different items relating to debenture in final accounts – Redemption – Methods of redemption of Debenture - Sinking Fund Method – Insurance Policy method.

UNIT – III: Final Accounts of Companies – Trading Account – Profit and Loss Account – Profit and Loss Appropriation Accounts – Balance sheet - Managerial Remuneration – Remuneration payable to different categories of managerial personnel – Calculation of Managerial remuneration.

UNIT – IV: Valuation of Goodwill - Need - Methods of valuation of Goodwill - Average Profit method – super profit method – capitalization method.

UNIT – V: Shares - Methods of valuation of Shares – Net asset method – Yield Method – Fair value Method.

UNIT – VI: Liquidation of Companies –Modes of Winding up - Statement of Affairs - Deficiency account or Surplus Account.

UNIT – VII: Company Final accounts – Schedule VI Part I and Part II – Profit prior to incorporation - Managerial remuneration – Preparation of profit and loss account and Balance Sheet.

UNIT – VIII: Amalgamation – Meaning – Advantages – Purchase consideration – Types of amalgamation – Net present value method - Absorption (Excluding inter – company holdings).

UNIT – IX: External reconstruction and Internal reconstruction – Meaning – Accounting treatment – Alteration of share capital – Capital reduction account.

UNIT – X: Holding company accounts excluding inter-company holdings – Mutual Owings - Contingent Liability - Unrealized Profit - Revaluation of Assets.

UNIT – XI: Liquidation of companies – Meaning – Reasons for winding up – Liquidator – Preferential creditors – Calculation of liquidator's remunerations – Liquidator's final statement of accounts.

UNIT – XII: Accounts for banking companies - Preparation of profit and loss account and balance sheet - Accounts for Insurance Companies - Preparation of profit and loss account and balance sheet.

UNIT – XIII: International Financial Reporting Standards (IFRS) – Meaning – Advantages - Disadvantages.

UNIT – XIV: Human resource accounting - Characteristic, Applications methods - Principles of Government accounting – Principles of Responsibility accounting.

REFERENCE BOOKS:

1. Dr. M. A. Arulanandam, and Dr. K. S. Raman, 2003, "Advanced Accountancy, Part-I" - Himalaya Publications, New Delhi.
2. Jain S. P. & Narang K. L., 2004 - "Advanced Accounting" - Kalyani Publishers, New Delhi.
3. Gupta R. L. & Radhaswamy M., 2006, "Corporate Accounts – Theory, Method and Application", Sultan Chand & Co., New Delhi.
4. Reddy & Murthy, 2004, "Financial Accounting" - Margham Publicatuions, Chennai.
5. Jain S. P and Narang K. L. 2004, Corporate Accounting, (First Edition) Kalyani Publications, Chennai.
6. Joseph. T. 2009, Corporate Accounting, Vol. 1, (1st Edition), Tata McGraw-Hill Education Pvt. Limited, New Delhi.

FOURTH SEMESTER

Course Code	Title of the Course
31041	**Management Accounting**

Learning Objectives:

1. To enable the students to understand the conceptual framework of Management Accounting.
2. To acquaint the students with the Management Accounting Techniques that facilitates managerial decision making.

UNIT – I: Management Accounting: Definition – Scope – Objectives – Functions – Role – Importance – Limitations – Management accounts Vs Financial accounting – Management and Cost accounting.

UNIT – II: Installation of management accounting system – Tools of management accounting – Reporting.

UNIT - III: Financial Statement Analysis: Financial Statements – Nature and limitations of financial statements – Analysis and Interpretation.

UNIT – IV: Comparative statements – Common size statements – Criticism of published accounts and cost accounting.

UNIT - V: Ratio Analysis – Significance and uses – Important managerial uses of ratio analysis.

UNIT – VI: Types of ratios – Profitability ratios – Turnover ratios – Liquidity ratios – Proprietary ratios – Market earnings ratios – Factors affecting efficiency of ratios.

UNIT – VII: Uses of ratio analysis – Uses and limitations – Construction of profit and loss account and balance sheet with ratios and relevant figures.

UNIT - VIII: Fund Flow Analysis - Need and meaning – Preparation of schedule of change in working capital and the fund flow statement – Projected fund flow statement – Managerial uses and limitations of fund flow analysis

UNIT – IX: Cash Flow Analysis - Need – Meaning – Preparation of cash flow statement – Managerial uses of cash flow statement – Limitations – Differences between fund flow and cash flow analysis.

UNIT - X: Budgeting: Meaning of Budget and Budgetary Control – Importance – Limitations – Classification of budgets and budgets preparation – Cash budget – Sales budget – Production budget.

UNIT – XI: Materials purchase budget - Fixed and flexible budgeting – Performance budgeting – Master budget - Zero-base budgeting.

UNIT - XII: Marginal Costing and Break Even Analysis: Definition – Marginal costing Vs Absorption costing – Justification for marginal costing – Marginal cost sheet – Segregation of semi-variable costs – Contribution – Key factor.

UNIT – XIII: Managerial uses of marginal costing – Pricing decisions – Level of activity planning – Mix of sale – Profit planning techniques – Make or buy decisions.

UNIT – XIV: Break-even Analysis – Break-even meanings, chart and graph – P/V ratio – Margin of Safety – Assumptions of break – Even analysis – Limitations of break-even analysis – Advantages and limitations of marginal costing – Differential costing.

REFERENCE BOOKS:

1. Sharma., and Gupta, S. K. (2006), Management Accounting, Kalyani Publishers, New Delhi.
2. Reddy, T. S., and Hari Prased Reddy, Y. (2010), Management Accounting, Margham publications, Chennai.
3. Maheswari, S. N. (2004), Management Accounting, Sultan Chand & Sons, New Delhi.
4. Jain, S. P., and Narang. K. L. (2001), Cost and Management Accounting, Kalyani Publishers, New Delhi.

Course Code	Title of the Course
31042	**Principles and Practice of Services Marketing**

Learning Objectives:

1. The objective of this course is to facilitate understanding of the conceptual framework of marketing
2. Its applications in decision making under various environmental constraints.

UNIT – I: Services: Meaning and definition of services – Importance of services in Indian Environment – Classification of services – Characteristic features of services – Growth of the service sector – Economic policy on services – Differences between goods and services.

UNIT – II: Service Marketing: Concept – Significance – Customer's expectation in Service Marketing – Managing demand and supply in service business.

UNIT – III: Service design – Factors to be considered in designing service process – Guiding principles in service design – Service marketing system – Types of services layouts – Service mapping.

UNIT – IV: Marketing Mix for Services – Characteristics of service marketing mix – Sales promotion – Physical evidence – Types of physical evidence.

UNIT – V: Service product – Meaning – Customer benefit concept – Core service level – The service offer – Product planning and development - Quality management.

UNIT – VI: Pricing in services – Meaning of price – Objectives of pricing – Characteristics of services and prices – Service cost – Approaches to pricing services – Methods of demand based pricing.

UNIT – VII: Service promotion – Market communication – Communication process – Determining the communication objectives – Selecting the communication channels – Service communication – Steps in advertising process.

UNIT – VIII: Location of services and channels of distribution – Location – Factors to be considered in choosing a service location – Classification of services by location – Direct distribution – Franchising – Agents – Customer segmentation.

UNIT – IX: People in services marketing mix – Service personnel – Contact personnel – Role of frontline employees – Hiring the right people – Service culture.

UNIT – X: Physical evidence – Classification of physical evidence – Role of the service cape – Peripheral evidence – Guidelines for physical evidence strategy.

UNIT – XI: Internal marketing – Objectives – Role of internal marketing – Components of internal marketing – External marketing – Relationship marketing – Goals of relationship marketing – Customer relationship management – Objectives of CRM.

UNIT – XII: Marketing mix of selected services - Personal care marketing – Entertainment Marketing – Education Marketing – Communication Marketing – Electricity Marketing.

UNIT – XIII: Key Services Marketing: Banking services – Insurance services – Transport services – Tourism services – Hotel services- Consultancy services – Hospital services - Market segmentation.

UNIT – XIV: Service Quality: Measurement of Service Quality – Scope of Service Quality – Tools for achieving Service Quality – Causes of Service Quality – Problems – Principles of service quality.

REFERENCE BOOKS:

1. Reddy P. N. Appannaiah, H. R. Anil Kumar, S and Nirmala, 2000, Services Marketing, Himalaya Publishing House.

2. Jha S. M., 2014, Services Marketing, (7th Edition) Himalaya Publishing House.

3. Dr. Shajahan. S., 2001, Services Marketing, (Revised Edition 2003) Himalaya Publishing House.

4. Ramneek Kappor, Justin Paul, and Biplab Halder, 2011, Service Marketing, Tata McGraw Hill Education Limited.

Course Code	Title of the Course
31043	**Organizational Behaviour**

Learning Objectives:

1. To grasp the organizational theories that would enlighten the understanding of human behaviour at work.
2. To understand team and group process and to be able to address issues arising from individual and collective organizational behaviour behavior.

UNIT – I: Organizational behaviour – Characteristics of human behaviour - Meaning – Definition – Nature of organizational behaviour – Basic objectives of organizational behaviour – Key elements of organizational behaviour – Importance of organizational behaviour.

UNIT – II: Foundations of Individual behaviour – Positive individual behaviour – Negative individual behaviour – Factors influencing individual behaviour – Personal factors – Environmental factors – Behavioural models.

UNIT – III: Personality – Definitions – Determinants of personality – Influence of personality on behaviour – Personality traits – Influencing behaviour – Personality development – Personality theories.

UNIT – IV: Perception – Definitions – Sensation and perception – Process of perception – Determinants of perception – Qualities of perceiver - Learning – Meaning and definition – Nature and characteristics of learning – Theories of learning.

UNIT – V: Attitude and Values – Definitions – Nature and characteristics of attitude and values – Measurement of attitude – Functions of attitude – Attitude change – Values and attitudes – Types of values – Formation of values.

UNIT – VI: Groups in organization – Meaning – Characteristics – Reasons for formation of groups – Types of groups – Different stages of groups – Group norms – Group cohesiveness – Decision making and the group – Individual and group decisions.

UNIT – VII: Work Stress – Stress and councelling – Causes of stress – Personal factors – Organizational factors – Stress-performance relationship – Psychological problems – Behavioural changes – Escaping stress – Coping with stress – Couselling.

UNIT – VIII: Organizational change – Meaning – Factors influencing change – internal factors – External factors – Resistance to change – Possible benefits of resistance – Organizational development – Objectives of OD – Evaluation and follow up – Organizational development – Merits and Demerits.

UNIT – IX: Organizational Culture and Climate – Types – Determinants – Changing organizational culture – Organizational climate – Determinants of organizational culture – Impact of organizational climate – Measures.

UNIT – X: Organizational Conflicts – Definitions – Causes of conflicts – Different stages of conflict – Conflict and performance – Measures to stimulate conflicts – Conflict outcomes.

UNIT – XI: Career planning – Meaning and characteristics – Need for career planning – Process of career planning – Preparing and implementing action – plans – Evaluation of career planning Limitations.

UNIT – XII: Emotional Intelligence – Emotions – Types of emotions – Managing emotions – Emotional intelligence – Dimensions of emotional intelligence – Advantages and limitations of emotional intelligence.

UNIT – XIII: Power, Politics and Impression management – Power, authority and influence – Sources – Organizational politics – Nature of organizational politics – Negative impact of organizational conflicts.

UNTI – XIV: Communication and knowledge management – Meaning of communication – Functions – Communication process – Directions of communication – Types of communication – Knowledge management – Dimensions of knowledge management – Knowledge management processes.

REFERENCES:

1. Khanka S. S, “Organisational Behaviour” Sultan Chand & Sons Publications, New Delhi (2012).

2. Aswathappa K, “Organisational Behaviour”, Himalaya Publications, New Delhi (2011).

3. Varma, “Organisational Behaviour”, Forward Book Depot, New Delhi (2013).

4. Sharma, “Organisational Behaviour”, Tata McGrew-Hill Publications, New Delhi (2012).

Course Code	Title of the Course
31044	**Retail and Distribution Management**

Learning Objectives:

1. To enhance the knowledge in basic strategies of retail management
2. To analyze the multiple channels of distribution system in present marketing scenario.

UNIT – I: Retailing – Definition – Retail industry and economy – Retail industry in India – Characteristics of retailing – Role of services in retailing – Functions of retailing – Categorizing retailers – Trends in retail formats – Retail strategy.

UNIT – II: Retail organizations – Changing structure of retailing – Theories of structural change in retailing – Classification of retail units – Method customer interaction.

UNIT – III: Retail in India – Emergence of organized retailing – Traditional retail formats – Modern retail formats in India – Retailing to rural India – Product categories – Challenges in retail business in India.

UNIT – IV: Retail customer – Consumer behaviour – Factors affecting consumer decision making – Stages of the consumer decision process – Types of consumer decision making – Shopping behaviour.

UNIT – V: Retail market segmentation – Benefits – Criteria for effective market – Kinds of markets – Dimensions for segmentation – Customer profile – Market segmentation in India.

UNIT – VI: Retail location strategy – Importance of location decision – Determining factors – Types of retail location – Site selection analysis – Selection of particular shopping centre – Retail location theories.

UNIT – VII: Product management – Brand management and retailing – Merchandise management – Model stock plan – Types of suppliers – Category management – Various retail segments.

UNIT – VIII: Retail pricing – External influences on retail pricing strategy – Developments in retail prices – Retail pricing objectives – Role of price elasticity.

UNIT – IX: Retail promotion strategy – Selection of promotion mix – Advertising – Media selection- Sales promotion – Personal selling – Publicity.

UNIT – X: Relationship marketing in Retailing – Evolution of relationship marketing – Relationship marketing strategies in retailing – Organized and unorganized retail store.

UNIT – XI: Marketing Channels - Structure, Functions and Significance - Basic Channel, Role in the dynamic market place - Designing the Market Channel system.

UNIT – XII: Managing Marketing Channels - Channel Policies - Choice of the channel - Organizational Pattern in the Channel - Assessing Channel Performance - Causes for Channel Conflict - Techniques to overcome Channel Conflict- Channels for Consumer Goods, Industrial Goods and Services- Multi-level Marketing - Concepts, Role and Significance.

UNIT – XIII: Primary participants of the Channel - Manufacturer, Wholesaler and Retailer - Logistics and Operational dimensions - Material Handling, Transportation, Warehousing, Inventories, Logistics interface-Marketing Information System (MIS).

UNIT – XIV: Franchisee - Significance and importance of Franchisee in Channel Decision - Advantages of Franchisee - Process of appointment of Franchisee – Relationship between Franchiser and Franchisee.

REFERENCE BOOKS:

1. Kulkarni M. V., 2011, Physical Distribution Management, (3rd Edition), Vikas Publishing.
2. Gibson G. Vedamani, 2003, Retail Management, (4th Edition), JBA Publishers.
3. Ronald W. 1996, Retail Marketing Hasty, McGraw-Hill Publication.
4. Eliton S .2009, Sales and Distribution Management, (3rd Edition), Himalaya Publishing House Private Limited.

Duration of the Programme: The programme for the degree of Master of Commerce consists of two academic / calendar years divided into four semesters. Each semester is having four core papers carries four credits each. The duration of course covers 64 credits which are divided equally sixteen credits into four semesters.

Faculty and Support Staff Requirements: The programme for the degree of **Master of Commerce** requires the following faculty and supporting staff:

Staff Category	Required
Core Faculty	3
Faculty for Specialization	2
Clerical Assistant	1

Instructional Delivery Mechanisms:

- Self Learning Materials
- Web based Teaching and Learning
- Electronic study materials

6. Student Support Services: The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like counselling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issue of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact programme for theory courses will be held at the university campus. The conduct of end-semester examinations, evaluation and issue of certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

Procedure for Admission, curriculum transaction and evaluation:

f. 1. Procedure for Admission:

A candidate who has passed any UG Degree is eligible B.Com / B.Com (CA) / BCS / B.Com (CS) / BBM / BBA / B.Com. (C.S and C.A) shall be given preference, as per the norms set by the Government of Tamil Nadu or an Examination accepted as equivalent thereto by the Academic Council, subject to such conditions as may be prescribed thereto are permitted to appear and qualify for the Master of Commerce Degree Examination of this university after a course of study of two academic / calendar years.

f. 2. Curriculum Transactions:

The classroom teaching would be lecture method by using electronic devices. The face to face contact classes cover group discussion, seminars and paper presentation.

The face to face contact classes will be conducted in following durations;

Course Type	**Face to Face Contact Classes Per Semester (in Hours)**
Theory Courses (4 courses with 4 credits each)	**64**

3. Evaluation: The evaluation method which comprises both internal assessment and end semester examination. The internal assessment shall contain maximum 25 marks for each paper. The end semester examination shall contain 3 hours duration to each course at the end of each semester. The end semester examination shall comprise maximum 75 marks for each paper. The candidate who failed in any papers are permitted to appear in subsequent examinations to clear the failed papers.

Question Paper Pattern:

Answer all questions (one question from each unit with internal choices

Time: 3 Hours; Maximum Marks: 75

Part A - 10 x 2 Marks = 20 Marks

Part B - 5 x 5 Marks = 25 Marks

Part C - 3 x 10 Marks = 30 Marks

Distribution of Marks in Continuous Internal Assessments

The following procedure shall be followed for awarding internal marks for the courses

Component	Marks
Assignments (per course)	25

Passing Minimum:

The students will be evaluated by examinations. There shall be no passing minimum for internal. The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50% marks (50% out of 75 marks) in the University end semester examination. Then the marks secured by the candidate in the end semester examination will be taken and added with his/her internal marks (Maximum marks 25). In aggregate (External + Internal) the passing minimum shall be 50% for each course. Grading shall be based on overall marks obtained.

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

Marks and Grade:

The following table gives the marks, grade points, letter grades and classification to indicate the performance of the candidate.

Conversion of marks to grade points & Letter grade (Performance in a paper / Course)

Range of Marks	Grade Points	Letter Grade	Description
90 – 100	9.0 – 10.0	O	Outstanding
80 – 89	8.0 – 8.9	D+	Excellent
75 – 79	7.5 – 7.9	D	Distinction
70 – 74	7.0 – 7.4	A+	Very Good
60 – 69	6.0 – 6.9	A	Good
50 – 59	5.0 – 5.9	B	Average
00 – 49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

Ci = Credits earned for course i in any semester

Gi = Grade point obtained for course i in any semester

n = refers to the semester in which such courses were credited.

Grade Point Average (GPA): Grade point average (GPA) is calculated for each part taking into account all the courses studied under each part.

$$GPA = \frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$$

$$GPA = \frac{\sum_i (C_i \times G_i)}{\sum_i C_i}$$

Where Ci = Credits earned for course i in any semester.

G_i= Grade points obtained for course i in any semester

Cumulative Grade Point Average (CGPA): For the entire program CGPA is calculated in the following manner.

$$\textbf{CGPA} = \frac{\sum_n \sum_I C_{ni} \times G_{ni}}{\sum_n \sum_i C_{ni}}$$

$$\text{CGPA} = \frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses of the entire programme}}$$

Classification of CGPA: Based on the Cumulative Grade Points Average the performance of the students is classified as follows.

CGPA	Grade	Classification of Final Result
9.5 - 10.0	O+	First Class – Exemplary
9.0 and above but below 9.5	O	
8.5 and above but below 9.0	D++	First Class with Distinction
8.0 and above but below 8.5	D+	
7.5 and above but below 8.0	D	
7.0 and above but below 7.5	A++	First Class
6.5 and above but below 7.0	A+	
6.0 and above but below 6.5	A	
5.5 and above but below 6.0	B+	Second Class
5.0 and above but below 5.5	B	
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

Maximum duration for the completion of the course: The maximum duration for completion of M.Com Degree programme shall not exceed ten semesters from their fourth semester.

Commencement of this Regulation: These regulations shall come into force from the academic year 2018 - 2019 (June session) i.e., for students who are to be admitted to the

first year of the course during the academic year 2018 - 2019 (June session) and thereafter.

Fee Structure: The programme has the following Fee Structure:

Sl. No.	Fees Details	Amount in Rs.	
		First Year	Second Year
1.	Admission Processing Fees	300.00	-
2.	Course Fees	4400.00	4400.00
3.	ICT Fees	150.00	150.00
	Total	**4850.00**	**4550.00**

The above mentioned fee structure is exclusive of Exam fees.

Requirement of the laboratory support and Library Resources: Alagappa University, Karaikudi having well established Library facility with adequate number of copies of books in relevant titles for Commerce programme.

Cost estimate of the programme and the provisions:

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with M.Com., Programme are as follows:

Sl. No.	Expenditure Heads	Approx. Amount in Rs.
1.	Programme Development (Single Time investment)	10,00,000/-
2.	Programme Delivery (Per Year)	20,00,000/-
3.	Programme Maintenance	3,00,000/-

Quality Assurance Mechanism and Expected Programme Outcomes: Separate Internal Quality Assurance Cell (IQAC) has been established with certain parameters in the Directorate of Distance Education in order to provide quality teaching to the stakeholders.

University Motto: ‘Excellence in Action’

University Vision Statement: Achieving Excellence in all spheres of Education, with particular emphasis on **“PEARL”-** Pedagogy, Extension, Administration, Research and Learning.

University Objectives: Providing for Instructions and Training in such Branches of Learning as the University may determine. Fostering Research for the Advancement and Dissemination of Knowledge

University Quality Policy: Attaining Benchmark Quality in every domain of **‘PEARL’** to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

University Quality Quote: Quality Unleashes Opportunities towards Excellence **(QUOTE).**

Minutes of the Meeting of the Board of Studies in Commerce (For M.Com., Programme to be offered through ODL Mode) held at the Directorate of Distance Education, Alagappa University, Karaikudi 630 004, on 04.09.2017 (Monday) at 4:25 p.m.

Members Present:

1. Dr. T. R. Gurumoorthy	-	Chairman
2. Dr. S. Ganapathy	-	Member
3. Dr. A. Muthusamy	-	Member
4. Dr. R. Ganapathi	-	Member

After the deliberation and discussion the board resolved the following:

1. The board considered the curriculum design and detailed syllabi of M.Com., programme, prepared as per the norms by the Chairman and the Board Members, scrutinized and suitably modified the same.

2. The board resolved to approve curriculum design, detailed syllabi and other regulations for the M.Com., programme to be offered by the Directorate of Distance Education of Alagappa University are given in Annexure - I.

Dr. T. R. Gurumoorthy

Dr. S. Ganapathy

Dr. A. Muthusamy

Dr. R. Ganapathi

ALAGAPPA UNIVERSITY

(A State University Accredited with 'A+ Grade by NAAC (CGPA: 3.64) in the third cycle)

Karaikudi – 630 003, Tamilnadu, India

DIRECTORATE OF DISTANCE EDUCATION

PROGRAMME PROJECT REPORT

M.Com (Finance and Control)

Programme Code: 335

REGULATIONS AND SYLLABUS

[From the academic year 2018-2019 onwards]

Credit Based System

Programme's Mission & Objectives

Mission: To bestow an exhaustive acquaintance by blending the subjects of Accounting, Management, Operations Research, Business Law Banking, Insurance, Taxation, Marketing, Services Marketing and Investment Analysis in a pragmatic manner to the students so as to emerge as efficient Professionals, Entrepreneurs, Managers, Finance Experts, etc….

Objectives:

i). To impart a comprehensive knowledge in Commerce with finance and control to the students in a pragmatic manner.

ii). To be a strapping pedestal to key-in and fetch an assortment of job opportunities in the Public and Private Sectors.

iii). To craft Accounting, Finance, Tax and Management experts through captivating professional and cerebral associations.

Relevance of the Programme with HEI's Mission and Goals:

Affording quality higher Education to the learners who are interested in pursuing higher education through distance mode, so that they are transformed into intellectually competent human resources that will help in the uplift of the nation in terms of Educational, Social, Technological, Environmental and Economic magnificence (ESTEEM). This programme is very much effective in imparting quality education through flexi-timings.

In accordance with the mission of Alagappa University as a research-intensive institution, the teaching programme of the master's programme in Commerce (Finance and Control) is based on state of the art of scientific research and maintains a strong emphasis on the acquisition of academic and research skills.

Nature of prospective target group of learners:

The curriculum has been designed to fulfill the needs of diverse class of learners including a class of learners. In order to fulfill the needs of the learners, professionals

who are in teaching, business professionals, chartered accountant practitioners, banking, Insurance, chief financial officer and management professionals.

M.Com., (F & C) Employment Areas:

- Securities Analyst
- Auditor
- Budget Analyst
- Corporate Analyst
- Marketing Manager
- Risk Analyst
- Operations Manager
- Accountant
- Money Manager
- Revenue Agent
- Tax Advisor

M.Com., (F & C) Job Types:

- Banking and Finance
- Equity Research
- Insurance Management
- Leasing
- Merchant Banking
- Insurance Industry
- Mutual Funds
- Portfolio Management
- Project Appraisal
- Risk Management

Appropriateness of the programme to be conducted in Open and Distance Learning mode to acquire specific skills and competence in Master of Commerce (Finance and Control) programme focuses both in-depth study of theory and acquisition of professional and research skills.

- This programme makes learners to develop skill oriented entrepreneurship knowledge, business communication at critical thinking and proficiency in the field of business sectors.
- This programme helps learners to acquire necessary skills to perform research, and start up entrepreneurship in the field of commerce and industry.

It is a good base Master degree course for the purpose of higher research studies like M.Phil., and Ph.D. They can avail wide employment opportunities and employability skills in the field of Commerce and Industry.

e. Instructional Design:

Sl. No	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
FIRST YEAR						
I Semester						
1	**33511**	Management Concepts	25	75	100	4
2	**33512**	Business Law	25	75	100	4
3	**33513**	Advanced Accounting	25	75	100	4
4	**33514**	Business Environment	25	75	100	4
		Total	**100**	**300**	**400**	**16**
II Semester						
5	**33521**	Operations Research	25	75	100	4
6	**33522**	Organizational Behaviour	25	75	100	4
7	**33523**	Advanced Cost Accounting	25	75	100	4
8	**33524**	Financial Services	25	75	100	4
		Total	**100**	**300**	**400**	**16**
SECOND YEAR						
III Semester						
9	**33531**	Investment Analysis and Portfolio Management	25	75	100	4
10	**33532**	Financial Management	25	75	100	4
11	**33533**	Principles of Personnel Management	25	75	100	4
12	**33534**	Corporate Accounting	25	75	100	4
		Total	**100**	**300**	**400**	**16**
IV Semester						
13	**33541**	Management Accounting	25	75	100	4
14	**33542**	E-Commerce	25	75	100	4
15	**33543**	Research Methodology	25	75	100	4
16	**33544**	Banking and Insurance	25	75	100	4
		Total	**100**	**300**	**400**	**16**
Grand Total			**400**	**1200**	**1600**	**64**

Detailed Syllabi:

FIRST SEMESTER

Course Code	Title of the Course
33511	**Management Concepts**

Learning objectives:

1. To make the students to understand the elements of effective management

2. To familiarize the students with the basics principles of management

UNIT – I: Management: Definition – Nature – Scope and functions – Evolution of management thought – Relevance of management to different types of organisation like, Insurance, Hospitals, Universities, Hotels, Social Service organisations etc.

UNIT – II: Planning: Nature, importance and strategic considerations in planning – Planning premises – Components of planning as objectives - Planning process – Methods of planning – Limitations of planning – Planning premises.

Unit – III: Policies, strategies, procedures, methods, rules, projects and budgets – Decision making – Meaning – Importance – Types – Process – Elements – Principles of decision making.

UNIT – IV: Organising: Nature, purpose and kinds of organisation – Structure – Principles and theories of organization – Formal organization – Informal organization.

Unit – V: Departmentation – Process of departmentation – Departmentattion by functions - Span of control – Line and staff functions – Authority and responsibility – Centralisation and decentralization.

Unit – VI: Delegation - Importance of delegation – Types of delegation – Problems of delegation – Effective delegation.

UNIT - VII: Staffing – Elements of staffing – Functions of staffing - General principles and importance – Motivation: Meaning – Importance – Types – Theories.

UNIT – IX: Communication: Meaning – Types – Process – Barriers – Overcoming barriers – Leadership: Meaning – Styles – Essential qualities.

UNIT – X: Controlling: Objectives and process of control – Devices of control – Integrated control – Special control techniques – Coordination – Need and techniques.

UNIT – XI: Directing – Principles of direction – Techniques of direction – Importance of direction – Characteristics of direction – Supervision – Functions of supervisor – Qualities of supervisor – Principles of supervisor.

UNIT – XII: Management Audit – Definition – Objectives – Scope of management audit – Need for management audit – Advantages of management audit.

UNIT – XIII: Business Ethics – Meaning – Need for business ethics – Principles of business ethics – Factors affecting business ethics – Management information system – Objectives of MIS – Importance of MIS – Process of MIS – Causes for poor MIS.

UNIT – XIV: Recent trends and new perspectives in management - Strategic alliances – Core competence – Business process reengineering – Total quality management – Bench marking.

REFERENCE BOOKS:

1. Ramaswamy. T, 2012, Principles of Management [Eighth Edition] Himalaya Publishing Home Pvt Ltd, Mumbai.
2. Dinkar Pagare, 2011, Business Management [Fifth Edition] Sultan Chand & Sons, New Delhi.
3. Govindarajan. M, 2008, Principles of Management [First Edition] PHI, New Delhi.
4. Prasad L.M, 2015, Principles and Practice of Management [Eight Edition], Sultan Chand & Sons, New Delhi.

Course Code	Title of the Course
33512	**Business Law**

Learning objectives:

1. To enable students to read, interpret and understand laws relevant to business made by Parliament, and decisions by the courts on the application of those laws.

2. To equip students with the tools to recognize the legal implications of business activities.

3. To emphasize the important areas of contract law, company and partnership law.

UNIT – I: Contract Act – Evolution – Need for Law – Mercantile Law – Principles of equity – Law merchant – Precedents – Indian Statute Law – Kinds contracts – Elements of valid contract – Classification of contracts.

UNIT - II: Offer and acceptance – Essentials for a valid offer – Acceptance – Essentials for a valid acceptance – Communication of offer and acceptance – Revocation of offer and acceptance.

UNIT – III: Consideration – Essentials for a valid consideration – Stranger to consideration and stranger to contract – Contractual capacity – Persons not competent to contract – Minor – Law relating to minor – Persons of unsound mind.

UNIT – IV: Free consent – Coercion – Effect of coercion – Duress – Duress Vs coercion – Undue influence – Fraud – Silence as a fraud – Consequences of fraud – Mistake – Classification – Mistake of law – Mistake of foreign law.

UNIT – V: Void agreement – Void contract – Uncertain agreement – Reciprocal promises – Contingent contract – Performance of contract – Actual performance – Attempted performance – Performance o joint promises.

UNIT – VI: Quasi contract – Basis – Types – Claims – Payment by an interested person – Discharge of contracts – Modes of discharges of a contract – Remedies of breach of contract – Suit for damages – Rules for damages.

UNIT – VII: Contract of bailment and pledge – Kinds of bailment – Consideration – Duties of bailer – Rights of bailer – Particular lien – General lien – Termination of lien.

UNIT – VIII: Contract of sale of goods Act – Scope – Goods – Classification – Contract of sale – Sale and agreement to sell – Conditions and warranties – Implied condition – Implied warranty – Caveat emptor.

UNIT – IX: Nature of partnership – Essential elements of partnership – Duties of partnership – Partnership Vs Joint Hindu family – Partnership Vs Company – Formation of firm – Types of partnership.

UNIT – X: Dissolution of a firm – Mode of dissolution – Consequences of dissolution – Rights of partners – Liabilities of partners – Mode of settlement of accounts – Sale of goodwill.

UNIT – XI: Nature of negotiable instruments – Promissory note – Bill of exchange – Cheque – Crossing of cheque – Types of crossing – Parties of negotiable instruments – Capacity of parties – Minor – Unsound persons – Insolvent – Corporation – Agent – Partners – Hindu joint family.

UNIT – XII: Carriage of goods Act – Carriage by land – Common carrier – Duties – Rights and liabilities – Private carrier – Goods – Railway as carrier – Duties and liabilities.

UNTI – XIII: Law of insolvency – Object – Procedure – Act of insolvency – Insolvency Court – Secured creditors – Procedure on admission of petition – Dismissal of petition.

UNIT – XIV: Property and debts of insolvent – Official assignee – Official receiver – Powers and duties – Committee of inspection – Debts of inspection – Rights of secured creditors.

REFERENCES:

1. Kapoor N. D., 2007, Business Law, - Sultan Chand Co & Ltd, New Delhi.

2. Pillai S. N. & Bhagavathy, 2009, Business Law - Sultan Chand Co & Ltd, New Delhi.

3. Saravanavel P. & Syed Bandre, 2007, Business Law, Himalayan Publishing House, Chennai.

4. Kapoor N. D., 2008, Elements of Mercantile Law, Sultan Chand Co & Ltd, New Delhi.

Course Code	Title of the Course
33513	**Advanced Accounting**

Learning Objectives:

1. To make the students to understand the basic concepts of accounting applied in the competitive corporate world.
2. To understand accounting software easily.

UNIT – I: Accounting – Definition – Rules – Advantages and Limitations of Accounting – Accounting Concepts and Conventions - Accounting Systems: Cash and mercantile systems – Accounting concepts – Principles – Accounting standards.

UNIT – II: Fundamentals of Book Keeping – Journal – Ledger – Subsidiary books – Cash book - Trial balance.

UNIT – III: Single Entry System: Calculation of profit – Statement of Affairs method – Conversion of single entry into double entry system.

UNIT – IV: Preparation of profit and loss account and balance sheet under single entry system.

UNIT – V: Final Accounts of a Sole Trader – Trading Account, Profit and Loss Account and Balance sheet with adjustments – Differences between Trial Balance and Balance sheet– Errors and their rectification–Types of errors.

UNIT – VI: Bills of Exchange: Promissory notes and bills of exchange – Recording of transactions relating to bills – Books of Drawer and Acceptor – Honour and dishonor of bills –Renewal of bills – Retiring of bills under rebate.

UNIT – VII: Account Current and Average Due Date – Meaning – Advantages – Calculation of Account Current and Average Due Date.

UNIT – VIII: Bank Reconciliation Statement –Need –Reasons for difference between cash book and pass book balances – Preparation of Bank Reconciliation Statement.

UNIT – IX: Non Trading Accounts - Receipts and Payments and income and expenditure account and Balance sheet– Difference between Receipts and Payments and income and expenditure account.

UNIT – X: Depreciation – Meaning – Causes – objects of providing for depreciation – Factors affecting depreciation – Accounting Treatment – Methods of providing depreciation – Straight line method – Diminishing Balance Method – Change in method of Depreciation – Sinking fund method.

UNIT – XI: Accounting for consignments - Consigner's book – Consignee's book – Accounting for joint ventures - Joint ventures account – co venture's account – Joint bank account.

UNIT – XII: Investment accounts – Ex-interest – Cum-interest Types of securities – Fixed interest securities – Variable income securities – Royalty excluding Sublease - Methods of recoupment - Fixed recoupment of short workings – Flexible recoupment.

UNIT – XIII: Departmental accounts – Apportionment of common expenses – Ascertaining cost of departmental purchases Inter departmental transfers at cost or selling price – Branch – Dependent branches – Independent branches - excluding foreign branches.

UNIT – XIV: Hire purchase - Calculation of interest - installment systems - Default and Repossession including Hire Purchasing Trading account - Goods on sale or Return.

REFERENCE BOOKS:

1. Reddy, T. S. and Murthy, A. 2014, Financial Accounting [Sixth Edition], Margham Publications, Chennai.

2. Nagarajan, K. L., Vinayaka, N., and Mani P.L. 2013, Principles of Accountancy [First Edition], Sultan Chand & Company Ltd, New Delhi.

3. Jain, S. P., and Narang, K. 2005, Financial Accounting. [Fifth Edition], Kalyani Publishers, Ludhiana.

4. Grewal T.S. 2007, Introduction to Accountancy [Fifth Edition], Sultan Chand & Company Ltd, New Delhi.

Course Code	Title of the Course
33514	**Business Environment**

Learning Objectives:

1. To enhance the knowledge of business opportunities in current situation.
2. To enhance the capacity to making business polices.

UNIT – I: Concept of Business Environment- Significance - Types of Environment- External and Internal – Inter-Relationship between economic and non-economic environment.

UNIT – II: Impact of environment on business and Strategic Decisions - Culture and business - Social Responsibilities of Business.

UNIT – III: Economic Systems – Meaning – Characteristics -Types of economic systems-Capitalism-Socialism-Mixed economy.

UNIT – IV: Economic planning - Nature, Scope and Significance of Economic Planning in India - Achievements and Failures of Economic Planning.

UNIT – V: Industrial Policies and Regulations - Industrial Policy upto 1991, 2014 - Public, Private, Joint and Co-operative Sectors.

UNIT – VI: Privatization and Disinvestment - Ways of Privatization - Benefits and Arguments against Privatization - Privatization in India.

UNIT – VII: Technological environment-Factors Governing Technological Environment- Management of technology - Patents and Trademarks.

UNIT – VIII: Indian Contract Act 1872 – Indian Companies Act 1956 – Consumer Protection Act 1986 – Consumer Rights – World Trade Organization (WTO).

UNIT – IX: Industrial Finance – Short term finance – medium term finance - long term finance – Corporate securities.

UNIT – X: Reserve Bank of India - Financial Institution in India – IFCI – ICICI – IDBI – IIBI – SIDBI – LIC – SIDCO – Commercial Banks – DFHI - SFCs.

UNIT – XI: Globalization - Meaning and Dimensions - Features of Current Globalization - Essential Conditions for Globalization - Globalization of Indian business.

UNIT – XII: Foreign Direct Investment - Concept, Advantages and Disadvantages and Determinants - India’s policy towards FDI - Multinational Corporation – Meaning - Merits and Demerits - Control over MNCs - MNCs in India.

UNIT – XIII: International Environment – Meaning and concept – World Bank – International Monetary Fund (IMF) – The General Agreement on Tariffs and Trade (GATT) – The World Trade Organization.

UNIT – XIV: Current Issues in Business Environment – Urbanization – Population – Public distribution system – Natural environment – Various aspects of the natural environment – Pollution – Environmental management.

REFERENCE BOOKS:

1. Francis Cherunilam, 2000, Business Environment, Himalaya Publishers.
2. Gupta. C. B., 2014, Business environment, (8th Edition), McGraw Hill Education India Private Limited.
3. Avadhani.V. A., 2004, Essentials of Business Environment, (2nd Edition), Himalaya Publication, Mumbai.
4. Shaikh Saleem, 2009, Business Environment, (2nd Edition), Dorling Kindersley (India) Private Limited.

SECOND SEMESTER

Course Code	Title of the Course
33521	**Operations Research**

Learning Objectives:

1. The objective of the course is to acquaint the student with the application of operations research to business and industry and help them to grasp the significance of analytical techniques in decision making.

UNIT – I: Operations Research: Historical Background, Scope of Operations Research, Features of Operations Research, Phases of Operations Research, Types of Operations Research Models, Operations Research Methodology, Operations Research Techniques and Tools, Structure of the Mathematical Model, Limitations of Operations Research.

UNIT – II: Linear Programming: Linear Programming Problem, Requirements of LPP, Mathematical Formulation of LPP, Case Studies of LPP, Graphical Methods to Solve Linear Programming Problems, Applications, Advantages, Limitations.

UNIT – III: Graphical Analysis of Linear Programming Problems: Graphical Analysis, Some Basic Definitions, Graphical Methods to Solve LPP, Some Exceptional Cases, Important Geometric Properties of LPP.

UNIT – IV: Simplex Method: Standard Form of LPP, Fundamental theorem of LPP, Solution of LPP – Simplex Method, The Simplex Algorithm, Penalty Cost Method or Big M-method, Two Phase Method, Solved Problems on Minimisation.

UNIT – V: Duality in Linear Programming Problem: Importance of Duality Concepts, Formulation of Dual Problem, Economic Interpretation of Duality, Sensitivity Analysis.

UNIT – VI: Transportation Problem: Formulation of Transportation Problem (TP), Transportation Algorithm (MODI Method), the Initial Basic Feasible Solution, Moving Towards Optimality.

UNIT – VII: Assignment Problem: Mathematical Formulation of the Problem, Hungarian Method Algorithm, Routing Problem, Travelling Salesman Problem.

UNIT – VIII: Integer Programming Problem: Types of Integer Programming Problems, Gomory's All-IPP Method, All IPP Algorithm, Branch and Bound Technique.

UNIT – IX: Infinite Queuing Models: Queuing Theory, Operating Characteristics of a Queuing System, Constituents of a Queuing System, Service Facility, Queue Discipline.

UNIT – X: Mathematical Analysis of Queuing Theory: Mathematical Analysis of Queuing Process, Properties of Queuing System, Notations, Service System, Single Channel Models, Multiple Service Channels, Erlang Family of Distribution of Service Times, Applications of Queuing Theory, Limitations of Queuing Theory.

UNIT – XI: Finite Queuing Models: Finite Queuing Models - Simulation: Methodology of Simulation, Basic Concepts, Simulation Procedure, Application of Simulation.

UNIT – XII: Simulation Monte-Carlo Method: Monte-Carlo Simulation, Applications of Simulation, Advantages of Simulation, Limitations of Simulation.

UNIT – XIII: Project Scheduling and PERT - CPM: Basic Difference between PERT and CPM, PERT / CPM Network Components and Precedence Relationship, Project Management – PERT

UNIT – XIV: Game Theory: Competitive Situations, Characteristics of Competitive Games, Maximin – Minimax Principle, Dominance.

REFERENCE BOOKS:

1. Havinal Veerabhadrappa (2012), “An Introduction to Operations Research”, New Age International Private Limited, New Delhi.
2. Gurusamy S, (2015), “Operations Research”, Vijay Nicole Imprints Private Limited, Chennai.
3. Sharma J. K, (2016), “Operations Research – Theory and Applications”, Laxmi Publications, New Delhi.

Course Code	Title of the Course
33522	**Organizational Behaviour**

Learning Objectives:

1. To grasp the organizational theories that would enlighten the understanding of human behaviour at work.
2. To understand team and group process and to be able to address issues arising from individual and collective organizational behaviour behavior.

UNIT – I: Organizational behaviour – Characteristics of human behaviour - Meaning – Definition – Nature of organizational behaviour – Basic objectives of organizational behaviour – Key elements of organizational behaviour – Importance of organizational behaviour.

UNIT – II: Foundations of Individual behaviour – Positive individual behaviour – Negative individual behaviour – Factors influencing individual behaviour – Personal factors – Environmental factors – Behavioural models.

UNIT – III: Personality – Definitions – Determinants of personality – Influence of personality on behaviour – Personality traits – Influencing behaviour – Personality development – Personality theories.

UNIT – IV: Perception – Definitions – Sensation and perception – Process of perception – Determinants of perception – Qualities of perceiver - Learning – Meaning and definition – Nature and characteristics of learning – Theories of learning.

UNIT – V: Attitude and Values – Definitions – Nature and characteristics of attitude and values – Measurement of attitude – Functions of attitude – Attitude change – Values and attitudes – Types of values – Formation of values.

UNIT – VI: Groups in organization – Meaning – Characteristics – Reasons for formation of groups – Types of groups – Different stages of groups – Group norms – Group cohesiveness – Decision making and the group – Individual and group decisions.

UNIT – VII: Work Stress – Stress and counselling – Causes of stress – Personal factors – Organizational factors – Stress-performance relationship – Psychological problems – Behavioural changes – Escaping stress – Coping with stress – Couselling.

UNIT – VIII: Organizational change – Meaning – Factors influencing change – internal factors – External factors – Resistance to change – Possible benefits of resistance – Organizational development – Objectives of OD – Evaluation and follow up – Organizational development – Merits and Demerits.

UNIT – IX: Organizational Culture and Climate – Types – Determinants – Changing organizational culture – Organizational climate – Determinants of organizational culture – Impact of organizational climate – Measures.

UNIT – X: Organizational Conflicts – Definitions – Causes of conflicts – Different stages of conflict – Conflict and performance – Measures to stimulate conflicts – Conflict outcomes.

UNIT – XI: Career planning – Meaning and characteristics – Need for career planning – Process of career planning – Preparing and implementing action – plans – Evaluation of career planning Limitations.

UNIT – XII: Emotional Intelligence – Emotions – Types of emotions – Managing emotions – Emotional intelligence – Dimensions of emotional intelligence – Advantages and limitations of emotional intelligence.

UNIT – XIII: Power, Politics and Impression management – Power, authority and influence – Sources – Organizational politics – Nature of organizational politics – How to overcome negative impact of organizational conflicts.

UNTI – XIV: Communication and knowledge management – Meaning of communication – Functions – Communication process – Directions of communication – Types of communication – Knowledge management – Dimensions of knowledge management – Knowledge management processes.

REFERENCES:

1. Khanka S. S, “Organisational Behaviour” Sultan Chand & Sons Publications, New Delhi (2012).

2. Aswathappa K, “Organisational Behaviour”, Himalaya Publications, New Delhi (2011).

3. Varma, “Organisational Behaviour”, Forward Book Depot, New Delhi (2013).

4. Sharma, “Organisational Behaviour”, Tata McGrew-Hill Publications, New Delhi (2012).

Course Code	Title of the Course
33523	**Advanced Cost Accounting**

Learning Objectives:

1. To enable the students to understand the Costing Terms in business.
2. To provide adequate knowledge on Cost Accounting Practice

UNIT – I: Cost Accounting Principles: Meaning of cost and cost accounting – Objectives of cost accounting – Installation of a costing system.

UNIT – II: Elements of cost – Cost concepts – Cost classifications – Methods, systems and techniques of costing – Cost sheet.

UNIT – III: Cost Accounting for material cost control – Need for material cost control – Purchase control – Stores control – Stock levels – EOQ analysis.

UNIT – IV: Pricing of stores issues – Perpetual inventory control – ABC analysis – VED analysis – Treatment of waste, scrap, defectives and spoilage.

UNIT – V: Labour Cost Control – Time keeping and time booking – Treatment of idle time and overtime cost – Wage rates for costing – Systems of wage payment – Time wage and piece rate – Incentive schemes of wage payment – Labour turnover.

UNIT – VI: Overhead Cost Control – Classification of overheads – Allocation and appointment – Absorption of overheads – Different methods – Treatment of under absorption and over absorption of overheads.

UNIT – VII: Methods of Costing – Job costing – Contract costing – Profit on incomplete contracts – Cost plus contracts – Target costing – Escalation clause.

UNIT – VIII: Unit costing – Meaning – Cost accumulation – Procedure in unit costing – Preparation of cost sheet.

UNIT – IX: Process Costing – Features – Job costing Vs Process costing – Process cost accounts – Inter-process profits – Accounting for joint products and by products.

UNIT – X: Contract costing – Types – Procedure for contract costing – Contract plus costing – Profit from incomplete contract.

UNIT – XI: Operating Costing – Meaning – Features – Objectives – Cost Unit – Transport costing – Operating cost sheet.

UNTI – XII: Batch costing – Definition – Economic batch quantity – Applicability of batch costing.

UNIT - XIII: Standard Costing – Definition –Advantages and limitations of standard costing – Variance analysis.

UNIT – XIV: Reconciliation of cost and final accounts – Cost control and cost reduction – Meaning – Tools and techniques – Essentials for success of cost control and cost reduction – Distinction between cost control and cost reduction – Areas of cost reduction and control – Advantages.

REFERENCE BOOKS:

1. Jain, S. P., and Narang, K. L. 2001, Cost and Management Accounting [Fifth Edition], Kalyani Publishers, New Delhi.
2. Pillai, R. S. N., and Bagavathi, 2009, Cost Accounting [First Edition], Sultan Chand Company Ltd., NewDelhi.
3. Sharma, and Shashi K. Gupta, 2012, Management Accounting [Twelfth Edition], Kalyani Publishers, New Delhi.
4. Jain, S. P., and Narang, K. L. 2010, Cost Accounting [Twenty First Edition], Kalyani Publishers, New Delhi.
5. Maheswari, S. N. 2003, Cost and Management Accounting [First Edition], Sultan Chand Company Ltd., New Delhi.

Course Code	Title of the Course
33524	**Financial Services**

Learning Objectives:

1. On successful completion of this course, the student should know about the methods of financing by the agencies and the key role Played by them in Corporate Financing.
2. Understand the tradeoff between risk and reward in investing.

UNIT – I: Nature and various facets of financial service industry – Analysis of financial services – Ned for financial innovation.

UNIT – II: Money Market in India – Indian Capital Markets – Difference between Money Market and Capital Market – Classification and object of Indian Money Markets and Structure of Capital Markets.

UNIT – III: Financial services and market environment – Development of financial markets – Global integration of financial market – Finance Companies: Functions, strengths and weaknesses.

UNIT – IV: Secondary Markets – Stock Exchange – Role of Secondary Market – Trading in Stock Exchange – Various Speculative Transactions – Role of SEBI – Regulation of Stock Exchange.

UNIT – V: Banks as Financial Intermediaries – Commercial Bank s Role in Financing – IDBI – IFCI– LIC – GIC – UTI – Investments Companies.

UNIT – VI: Markets for Corporate Securities – New Issue Markets – Functions Issue Mechanism – Under writing.

UNIT – VII: Commercial Banking and their fund based and non-fund based financial services – Leasing – Steps in leasing transactions – Lease finance – Accounting and reporting for lease.

UNIT – VIII: Hire purchase financing: Salient features, guidelines, functions – Components of hire purchase contract – Hire purchase agreement – Cost of hire purchase.

UNIT – IX: Mutual Funds: Types of mutual funds – Floatation – Asset management company of mutual funds – Regulations.

UNIT – X: Factoring – Forfeiting – Securitization – Venture capital – Consumer finance and credit cards: Salient features, guidelines, functions – Strategies involved in financing.

UNIT - XI: Merchant Banking including public issue management – Underwriting – Portfolio management – Stock and security broking – Merger and Takeover: Salient features – Guidelines – Functions.

UNIT - XII: Foreign Exchange Broking – Bills discounting – Financial consultancy – Corporate advisory services – Credit rating services – Salient features – Guidelines – Functions.

UNIT – XIII: Housing Finance – Advantages – Method of housing finance – Role of NHB – Borrowing powers of national housing bank – Export finance – Need for export finance – Different types of export finance.

UNIT - XIV: Non-banking Financial Companies: Regulations of RBI – Role of NBFCs – Chit funds – Functions of chit fund companies – Finance companies.

REFERENCE BOOKS:

1. Gordon E, and Natarajan K, 2009, "Financial Markets and Services", Himalaya Publishing House Pvt. Ltd., India.
2. Guruswamy S, 2009, "Financial Services", Tata McGraw-hill Education, New Delhi.
3. Prasanna Chandra, 2011 "Financial Management Theory and Practice", Tata Mc Graw-Hill Education, New Delhi.
4. Khan M Y and Jain P K, 2008, "Financial Management Text, Problems and Cases", Tata McGraw-Hill Education, New Delhi.
5. Banerjee G and Banerjee S. Borrowing from Financial Institutions, UDH Publishing House, New Delhi.
6. Bhole .L. M, "Financial Institutions' and Markets: Structure Growth and Innovations", Tata McGraw-Hill Publishing Co Ltd.

THIRD SEMESTER

Course Code	Title of the Course
33531	**Investment Analysis and Portfolio Management**

Learning Objectives:

1. To understand the characteristics of securities markets and the instruments traded therein.

2. To be able to analyze the risk, return of securities and to manage portfolios of investments.

UNIT – I: Investment: Concepts and goals – Types of investment: Financial, Real, Business, Personal and Institutional – Comparison of investments, speculation, gambling and hedging – Concept of portfolio management: Goals – Risk and return trade-off.

UNIT – II: Financial Investment Avenues: Fixed income and Varying income securities - Factors influencing Investment – Investment media – Features of investment Programme – Investment Process – Development of Financial system in India.

UNIT - III: Investment Analysis - Aspects of analysis – Return Analysis: Concepts, measures and computation of return of individual security and portfolio.

UNIT – IV: Capital Market – New issue Market and Stock Exchange in India – BSE - NSE – OTCEI – Kinds of Trading activity – Listing of Securities – SEBI and its Role and guidelines.

UNIT – V: Risk Analysis: Concepts, types, measure, computation of risk of individual security and portfolio – Valuation Analysis: Share and bond valuation – Price Earnings Analysis.

UNIT – VI: Investment Alternatives – Investment in Equity Shares, Preference shares, Bonds, Government Securities.

UNIT – VII: Mutual Funds – Real Estate – Gold – Silver – Provident fund – Unit Trust – National Savings Scheme – LIC.

UNIT – VIII: Approaches to Investment Analysis - Fundamental Analysis - Concept and components – Tools of economy, industry and company analysis.

UNIT – IX: Technical Analysis - Concept and tools – Assumption – Theories - Dow theory – Contrary opinion – Confidence index, Breadth of market and Relative strength analysis – Moving average analysis – Chart patterns.

UNIT – X: Options and Futures – Types of options – Call option – Advantages of options – Limitations – Valuation of options – Characteristics of options – Future – Forwards and futures – Differences between futures and options.

UNIT – XI: Portfolio Construction and Choice - Markowtz diversification – Efficient frontier – Risk-return indifferent curves – Portfolio choice – Single and two factorial models – Lagrange multiplier method.

UNIT – XII: Portfolio Performance Measures: Sharpe, Treynor and Jensen – Portfolio Audit and Portfolio Revision: Need and methods – Formula plans.

UNIT – XIII: Capital Asset Pricing Model – Assumptions and application – Capital market line and security market line.

UNIT – XIV: Efficient market hypotheses – The weakly efficient, semi strongly efficient and strongly efficient market forms – Random-walk theory.

REFERENCE BOOKS:

1. Preeti Singh, 2008, Investment Management [Sixteenth Edition], Himalaya Publishing House Pvt. Ltd., Mumbai.
2. Bhalla, V.K. 2008, Investment Management, Security Analysis and Portfolio Management [Fourteenth Edition]. Sultan Chand & Company Ltd., New Delhi.
3. Avadhani, V.A. 2008, Investment Management [Seventh Edition], Himalaya Publishing House, Mumbai.
4. Gangadhar, V., and Ramesh Babu, G. 2003, Investment Management [First Edition], Anmol Publication Pvt. Ltd., New Delhi.

Course Code	Title of the Course
33532	**Financial Management**

Learning Objectives:

1. To reveal the knowledge on fund utilization and management.
2. To analyze the various concepts and techniques for better financial decision.

UNIT – I: Financial Management: Concept, nature, evaluation and significance – Finance functions – Managerial and operative – Investment – Function, meaning and scope – Financing function – Meaning and scope – Dividend function.

UNIT – II: Goals of Financial Management – Types – Maximization of profit, profitability / wealth / liquidity / solvency – Minimization of risk, cost of capital, dilution of management control etc. – Risk – Return trade off – Maximization and minimization Vs optimization.

UNIT – III: Sources of finance – Long term sources – Short term sources – Sources of working capital – Equity shares – merits and demerits – Preference shares – merits and demerits – Debentures – Merits and demerits – Elements of financial system – Structure of Indian capital market.

UNIT – IV: Capitalization – Meaning – Need – Theories of capitalization – Over capitalization – Merits and demerits - Under capitalization – Merits and demerits.

UNIT – V: Financial decisions – Relationship between Risk and Return – Sources of finance – Short-term and Long-term finance.

UNIT – VI: Long Term Capital Resources – Equity and debt sources – Equity share, preference shares and debentures as sources of long term capital – Relative merits, demerits and uses.

UNIT – VII: Significance of convertible issues and right issues – Borrowings from term lending institutions – The institutional framework – Types of assistance – Public deposits.

UNIT – VIII: Working Capital: Concept and types – Determinants – Financing approaches – Conservative – Aggressive and hedging approaches – Their risk – Return features and significance – Sources of working capital finance – Working capital financing by commercial banks.

UNIT – IX: Capital Planning – Determinants of capital structure – Optimum capital structure – Capital structure theories – Net income and net operative income theories – M.M. Theory – Traditional theory – Their assumptions – Significance and limitations.

UNIT – X: Cost of Capital Concept – Cost of debt, equity, preference share capital, retraining earning – Weighted average cost – Book weight, market weight – Marginal cost of capital use and computations.

UNIT – XI: Capital Budgeting: Concept – Significance – Methods of appraisal: Payback periods, ARR, IRR, NPV, Simulation and Certainty equivalent methods.

UNIT - XII: Leasing: Concept – Types – Significance – General considerations – Economics of leasing – Evaluation – Present value and IRR methods – Leverage – Concept – Types – Degree of operative leverage – Financial leverage and total leverage – Implications of high and low degrees of leverages.

UNIT – XIII: Dividend Theories: Valuation under Gorden and Walter theories – Dividend irrelevance under M.M. Theory – Assumptions – Limitations – Dividend policy – Different policies and practices – Factors affecting dividend decision.

UNIT – XIV: International Financial Management – Meaning – FOREX Market operations – Export credit needs – India's export finance – Methods of international payments – Letter of credit – Kinds of letter of credit – Foreign currency finance Export documents.

REFERENCE BOOKS:

1. Sharma R.K and Shasi, K. Gupta. 2014, Financial Management, (4th Edition), Kalyani Publishers, New Delhi.
2. Pandey I.M. 2009, Financial Management, (10th Edition), VikasPublishing House
3. Prasanna Chandra, 2012, Financial Management, (4th Edition), Tata McGraw-Hill Publishers.
4. Maheswari S.N. 2014, Financial Management, (26th Reprint) Vikas Publishing House Pvt. Ltd, New Delhi.

Course Code	Title of the Course
33533	**Principles of Personnel Management**

Learning Objectives:

1. To evaluate and apply theories of social science disciplines to workplace issues.
2. To enhance their effectiveness for optimizing the human resource potential of their organization in order to achieve business and strategic objectives.
3. To examine current issues, trends, practices, and processes in personnel management.

UNIT – I: Personnel Management – Definition – Objectives and functions – Role and structure of personal function in organizations – Personnel principles and policies – Managerial functions – Operative functions.

UNIT – II: Job Evaluation – Job Analysis – Job Design – Job Description – Job Specification – Methods of job evaluation – Ranking method – Advantages and disadvantages of job evaluation.

UNIT – III: Human Resource Planning: Characteristics – Need for planning – Human Resource Planning Process – Factors influencing human resource plan – Limitations of human resource planning – Forecasting the demand for HR.

UNIT – IV: Organising the human resource – Definitions – Process of organization – Importance of organization – Organization Structure – Principles of organization – Theories of organization.

UNIT – V: Recruitment - Selection Process - Placement and induction – Training and development – Need for training – Importance of training – Essentials of good training programme - Promotion – Demotions – Transfers – Separation.

UNIT – VI: Performance appraisal – Meaning – Features of performance appraisal – Methods of performance appraisal - Human relations - approaches to good human relations – Punishment.

UNIT – VII: Wage and Salary Administration: Factors – Principles – Compensation plan – Individual –Group – Incentives – Bonus – Fringe benefits – Job evaluation systems – Wage and salary administration in relation to personal taxation.

UNIT – VIII: Motivation – Definition – Characteristics of motivation – Importance of motivation – Process of motivation – Types of motivation – Theories of motivation.

UNIT – IX: Leadership – Meaning – Need and characteristics of leadership – Importance of leadership – Qualities of a leader – Different kinds of leadership styles – Theories of leadership.

UNIT – X: Morale and job satisfaction – Meaning – Characteristics of morale – Morale and productivity – Measurement of morale – Determinants of job satisfaction – Personal factors – Organizational factors.

UNIT – XI: Absenteeism – Meaning – Causes of absenteeism – Measures to control absenteeism – Discipline and grievance – Objectives of discipline – Characteristics of grievance – Cause of grievance.

UNIT – XII: Employee maintenance and integration – Welfare and society – Accident prevention – Administration of discipline – Employee motivation – Need and measures.

UNIT –XIII: Industrial relations - Trade unionism - Collective bargaining and worker's participation in management – Quality of work life.

UNIT – XIV: Personnel Records / Reports: Personnel Research and Personnel Audit: Objectives – Scope and importance.

REFERENCE BOOKS:

1. Tripathi P.C., 2013, Personnel Management and Industrial Relations, Sultan Chand and Sons, New Delhi.
2. Aswathappa K., 2013, Human Resource Management: Text and Cases, McGraw Hill Education, New Delhi.
3. Memoria C. B. & Rao V. S. P., 2014, Personnel Management - Text & Cases, Himalaya Publishing house, New Delhi.
4. Khanka S. S., 2007, Human Resource Management - Text & Cases, S. Chand & Company Ltd., New Delhi.

Course Code	Title of the Course
33534	**Corporate Accounting**

Learning Objectives:

1. To give a clear understanding and knowledge to the students in the area of corporate accounting and other related matters.
2. To provide the knowledge based on corporate need to have a global perspective and grow accordingly.

UNIT – I: Issue of shares: Par, Premium and Discount - Forfeiture - Reissue - Surrender of Shares - Right Issue – Underwriting.

UNIT – II: Redemption of Preference Shares – Provisions of the companies Act - Debentures - Issue – Treatment of different items relating to debenture in final accounts – Redemption – Methods of redemption of Debenture - Sinking Fund Method – Insurance Policy method.

UNIT – III: Final Accounts of Companies – Trading Account – Profit and Loss Account – Profit and Loss Appropriation Accounts – Balance sheet - Managerial Remuneration – Remuneration payable to different categories of managerial personnel – Calculation of Managerial remuneration.

UNIT – IV: Valuation of Goodwill - Need - Methods of valuation of Goodwill - Average Profit method – super profit method – capitalization method.

UNIT – V: Shares - Methods of valuation of Shares – Net asset method – Yield Method – Fair value Method.

UNIT – VI: Liquidation of Companies –Modes of Winding up - Statement of Affairs - Deficiency account or Surplus Account.

UNIT – VII: Company Final accounts – Schedule VI Part I and Part II – Profit prior to incorporation - Managerial remuneration – Preparation of profit and loss account and Balance Sheet.

UNIT – VIII: Amalgamation – Meaning – Advantages – Purchase consideration – Types of amalgamation – Net present value method - Absorption (Excluding inter – company holdings).

UNIT – IX: External reconstruction and Internal reconstruction – Meaning – Accounting treatment – Alteration of share capital – Capital reduction account.

UNIT – X: Holding company accounts excluding inter-company holdings – Mutual Owings, Contingent Liability, Unrealized Profit, Revaluation of Assets.

UNIT – XI: Liquidation of companies – Meaning – Reasons for winding up – Liquidator – Preferential creditors – Calculation of liquidator's remunerations – Liquidator's final statement of accounts.

UNIT – XII: Accounts for banking companies - Preparation of profit and loss account and balance sheet - Accounts for Insurance Companies - Preparation of profit and loss account and balance sheet.

UNIT – XIII: International Financial Reporting Standards (IFRS) – Meaning – Advantages - Disadvantages.

UNIT – XIV: Human resource accounting - Characteristic, Applications methods - Principles of Government accounting – Principles of Responsibility accounting.

REFERENCE BOOKS:

1. Dr. M. A. Arulanandam, and Dr. K. S. Raman, 2003, "Advanced Accountancy, Part-I" - Himalaya Publications, New Delhi.
2. Jain S. P. & Narang K. L., 2004 - "Advanced Accounting" - Kalyani Publishers, New Delhi.
3. Gupta R. L. & Radhaswamy M., 2006, "Corporate Accounts" – Theory, Method and Application, Sultan Chand & Co., New Delhi.
4. Reddy & Murthy, 2004, "Financial Accounting" - Margham Publicatuions, Chennai.
5. Jain S. P and Narang K. L. 2004, Corporate Accounting, (First Edition) Kalyani Publications, Chennai.
6. Joseph. T. 2009, Corporate Accounting, Vol. 1, (1st Edition), Tata McGraw-Hill Education Pvt. Limited, New Delhi.

FOURTH SEMESTER

Course Code	Title of the Course
33541	**Management Accounting**

Learning Objectives:

1. To enable the students to understand the conceptual framework of Management Accounting.
2. To acquaint the students with the Management Accounting Techniques that facilitates managerial decision making.

UNIT – I: Management Accounting: Definition – Scope – Objectives – Functions – Role – Importance – Limitations – Management accounts Vs Financial accounting – Management and Cost accounting.

UNIT – II: Installation of management accounting system – Tools of management accounting – Reporting.

UNIT - III: Financial Statement Analysis: Financial Statements – Nature and limitations of financial statements – Analysis and Interpretation.

UNIT – IV: Comparative statements – Common size statements – Criticism of published accounts and cost accounting.

UNIT - V: Ratio Analysis – Significance and uses – Important managerial uses of ratio analysis.

UNIT – VI: Types of ratios – Profitability ratios – Turnover ratios – Liquidity ratios – Proprietary ratios – Market earnings ratios – Factors affecting efficiency of ratios.

UNIT – VII: Ratio analysis – Uses and limitations – Construction of profit and loss account and balance sheet with ratios and relevant figures.

UNIT - VIII: Funds Flow Analysis - Need and meaning – Preparation of schedule of change in working capital and the fund flow statement – Projected fund flow statement – Managerial uses and limitations of fund flow analysis

UNIT – IX: Cash Flow Analysis - Need – Meaning – Preparation of cash flow statement – Managerial uses of cash flow statement – Limitations – Differences between fund flow and cash flow analysis.

UNIT - X: Budgeting: Meaning of Budget and Budgetary Control – Importance – Limitations – Classification of budgets and budgets preparation – Cash budget – Sales budget – Production budget.

UNIT – XI: Materials purchase budget - Fixed and flexible budgeting – Performance budgeting – Master budget - Zero-base budgeting.

UNIT - XII: Marginal Costing and Break Even Analysis: Definition – Marginal costing Vs Absorption costing – Justification for marginal costing – Marginal cost sheet – Segregation of semi-variable costs – Contribution – Key factor.

UNIT – XIII: Managerial uses of marginal costing – Pricing decisions – Level of activity planning – Mix of sale – Profit planning techniques – Make or buy decisions.

UNIT – XIV: Break-even Analysis – Break-even meaning, chart and graph – P/V ratio – Margin of Safety – Assumptions of break – Even analysis – Limitations of break-even analysis – Advantages and limitations of marginal costing – Differential costing.

REFERENCE BOOKS:

1. Sharma., and Gupta, S.K. (2006), Management Accounting, Kalyani Publishers, New Delhi.

2. Reddy, T.S., and Hari Prased Reddy, Y. (2010), Management Accounting, Margham publications. Chennai.

3. Maheswari, S.N. (2004), Management Accounting, Sultan Chand & Sons, New Delhi.

4. Jain, S.P., and Narang. K.L. (2001), Cost and Management Accounting, Kalyani Publishers, New Delhi.

Course Code	Title of the Course
33542	**E-Commerce**

Learning Objectives:

1. To enable the students to understand the technology of E–Commerce for Business Application.
2. To enable awareness on the Application of E–Commerce.

UNIT - I: E-Commerce: Defining Commerce; Main Activities of Electronic Commerce; Benefits of E-Commerce; Broad Goals of Electronic Commerce; Main Components of E-Commerce; Functions of Electronic Commerce – Communication, Process Management, Service Management, Transaction Capabilities; Process of E-Commerce; Types of E-Commerce; Role of Internet and Web in E-Commerce; Technologies Used; E-Commerce Systems; Pre-requisites of E-Commerce; Scope of E-Commerce; E-Business Models.

UNIT - II: E-Commerce Activities: Various Activities of E-Commerce; Various Modes of Operation Associated with E-Commerce; Matrix of E-Commerce Types; Elements and Resources Impacting E-Commerce and Changes; Types of E-Commerce Providers and Vendors; Man Power Associated with E-Commerce Activities; Opportunity Development for E-Commerce Stages; Development of E-Commerce Business Case; Components and Factors for the Development of the Business Case; Steps to Design and Develop an E-Commerce Website.

UNIT - III: Internet – The Backbone for E-Commerce: Early Ages of Internet; Networking Categories; Characteristics of Internet; Components of Internet – Internet Services, Elements of Internet, Uniform Resource Locators, Internet Protocol; Shopping Cart, Cookies and E-Commerce; Web Site Communication; Strategic Capabilities of Internet.

UNIT – IV: ISP, WWW and Portals: Internet Service Provider (ISP); World Wide Web (WWW); Portals – Steps to build homepage, Metadata; Advantages of Portal; Enterprise Information Portal (EIP).

UNIT - V: Reference Models: Open Systems Interconnection (OSI) Model – Physical layer, Data link layer, Network layer, Transport layer, Session layer, Presentation layer, Application layer; Transmission Control Protocol (TCP) / Internet Protocol (IP) Model;

Protocol – Internet Protocol (IP), User Datagram Protocol (UDP), Transmission Control Protocol (TCP), Dynamic Host Configuration Protocol (DHCP), Hyper Text Transfer Protocol (HTTP), File Transfer Protocol (FTP), Telnet, Post Office Protocol (POP), Simple Mail Transfer Protocol (SMTP).

UNIT - VI: XML and Data Warehousing: Definition of eXtensible Markup Language (XML); XML Development Goals; Comparison between HTML and XML; Business importance in using XML Based Technology; Advantages, Disadvantages and Applications of XML; Structure of an XML Document; XHTML and X/Secure; Data Warehousing; Data Marts and Operational Data Stores.

UNIT - VII: E-Marketing: Traditional Marketing; E-Marketing; Identifying Web Presence Goals – Achieving web presence goals, Uniqueness of the web, Meeting the needs of website visitors, Site Adhesion: Content, format and access; Maintaining a Website; Metrics Defining Internet Units of Measurement; Online Marketing; Advantages of Online Marketing.

UNIT - VIII: E-Security: Security on the Internet; Network and Website Security Risks – Denial-of-Service attacks, Viruses, Unauthorized access to a computer network; Vulnerability of Internet Sites; Network and Website Security – Transaction security and data protection, Security audits and penetration testing; E-Business Risk Management Issues; Firewall – Network policy, Advanced authentication mechanism, Packet filtering, Application gateways; Defining Enterprise Wide Security Framework.

UNIT - IX: E-Payment Systems: Electronic Funds Transfer; Digital Token Based E-Payment Systems; Modern Payment Systems; Steps for Electronic Payment; Payment Security; Net Banking.

UNIT - X: E-Customer Relationship Management: Customer Relationship Management (CRM) – Marketing automation, Enterprise customer management; Customer Relationship Management Areas; CRM Processes; Architectural Components of a CRM Solution – Customer's information repository, Campaign management, Event triggers, business logic and rules repository, Decision support tools, Higher level statistical analysis, Forecasting and planning tools, True channel management, Workflow management, Collateral management; Electronic Customer Relationship Management; Need, Architecture and Applications of Electronic CRM.

UNIT - XI: Supply Chain Management: Supply Chain Management (SCM); Goals of SCM; Functions of SCM; Strategies of SCM; Electronic SCM and its benefits; Components of Electronic SCM; Electronic Logistics and its Implementation.

UNIT - XII: Wireless Application Protocol: Wireless Application Protocol (WAP); Architecture of WAP; Working of WAP; Wireless Technologies; Generations in Wireless Communications; Security Issues related to Wireless Communications; Mobile Computing in Four Dimensions; Wireless Millennium.

UNIT -XIII: Knowledge Management: Knowledge Management and its Goals; Collaborative Computing and Knowledge Management; Knowledge Management Tools; Features of Knowledge Management Tools; Knowledge Creating Process; Knowledge Management Strategies for Different Organizations; Knowledge Management in Research and Development Organizations.

UNIT - XIV: Implementation of E-Commerce: WWW.EBAY.COM - B2C Website – Registration, Time factor, Bidding process, Growth of eBay; PayPal – New Trend in Making Payments Online; National Electronic Funds Transfer.

REFERENCE BOOKS:

1. Bharat Bhasker**.** 2009. **Electronic Commerce** [Third Edition]. Tata Mc Graw Hill Publishing Co Ltd., New Delhi.

2. RaviKalakota., and Andrew B.Whinston. 2013. **Frontiers of Electronic Commerce** [Fourteenth Edition]. Dorling Kindersley (India) Pvt Ltd,.

3. Daniel Minoli., and Emma Minoli**.** 2007. **Web Commerce Technology Handbook**. [Thirteenth Edition]. Tata McGraw Hill Publishing, New Delhi.

4. Elias,M. and Awa. .2009. **E–Commerce From vision to Fulfillment [**Third Edition]. PHI Publishing, New Delhi.

Course Code	Title of the Course
33543	**Research Methodology**

Learning Objectives:

1. To provide students with a firm foundation and understanding of business research methods and the research process.

2. To understand the relevance of and be able to apply a range of both quantitative and qualitative research methods.

UNIT – I: Research: Meaning of research; Types of research- Exploratory research, Conclusive research; Process of research; Research applications in social and business sciences; Features of a Good research study.

UNIT – II: Research Problem and Formulation of Research Hypotheses: Defining the Research problem; Management Decision Problem Vs Management Research Problem; Problem identification process; Components of the research problem; Formulating the research hypothesis- Types of Research hypothesis; Writing a research proposal- Contents of a research proposal and types of research proposals.

UNIT – III: Research Design: Meaning of Research Designs; Nature and Classification of Research Designs; Exploratory Research Designs: Secondary Resource analysis, Case study Method, Expert opinion survey, Focus group discussions; Descriptive Research Designs: Cross-sectional studies and Longitudinal studies; Experimental Designs, Errors affecting Research Design.

UNIT – IV: Primary and Secondary Data: Classification of Data; Secondary Data: Uses, Advantages, Disadvantages, Types and sources; Primary Data Collection: Observation method, Focus Group Discussion, Personal Interview method.

UNIT – V: Attitude Measurement and Scaling: Types of Measurement Scales; Attitude; Classification of Scales: Single item Vs Multiple Item scale, Comparative Vs Non-Comparative scales, Measurement Error, Criteria for Good Measurement.

UNIT – VI: Questionnaire Design: Questionnaire method; Types of Questionnaires; Process of Questionnaire Designing; Advantages and Disadvantages of Questionnaire Method.

UNIT – VII: Sampling: Sampling concepts- Sample Vs Census, Sampling Vs Non Sampling error; Sampling Design- Probability and Non Probability Sampling design;

Determination of Sample size- Sample size for estimating population mean, Determination of sample size for estimating the population proportion.

UNIT – VIII: Data Processing: Data Editing - Field Editing, Centralized in house editing; Coding - Coding Closed ended structured Questions, Coding open ended structured Questions; Classification and Tabulation of Data.

UNIT – IX: Univariate and Bivariate Analysis of Data: Descriptive Vs Inferential Analysis, Descriptive Analysis of Univariate data- Analysis of Nominal scale data with only one possible response, Analysis of Nominal scale data with multiple category responses, Analysis of Ordinal Scaled Questions, Measures of Central Tendency, Measures of Dispersion; Descriptive Analysis of Bivariate data.

UNIT – X: Testing of Hypotheses: Concepts in Testing of Hypothesis – Steps in testing of hypothesis, Test Statistic for testing hypothesis about population mean; Tests concerning Means- the case of single population; Tests for Difference between two population means; Tests concerning population proportion- the case of single population; Tests for difference between two population proportions.

UNIT – XI: Chi-square Analysis: Chi square test for the Goodness of Fit; Chi-square test for the independence of variables; Chi-square test for the equality of more than two population proportions.

UNIT – XII: Analysis of Variance: Completely randomized design in a one-way ANOVA; Randomized block design in two-way ANOVA; Factorial design.

UNIT – XIII: Research Report Writing: Types of research reports – Brief reports and Detailed reports; Report writing: Structure of the research report- Preliminary section, Main report, Interpretations of Results and Suggested Recommendations; Report writing: Formulation rules for writing the report: Guidelines for presenting tabular data, Guidelines for visual Representations.

UNIT – XIV: Ethics in Research: Meaning of Research Ethics; Clients Ethical code; Researchers Ethical code; Ethical Codes related to respondents; Responsibility of ethics in research – Uses of library and internet in research.

REFERENCE BOOKS:

1. Gupta S. L and Hitesh Gupta (2015), “Research Methodology: Text and Cases with SPSS Application”, International Book House Private Limited, New Delhi.

2. Peer Mohamed and Shazuli Ibrahim (2013), “Research Methodology”, Pass Publications, Madurai.

Course Code	Title of the Course
33544	**Banking and Insurance**

Learning Objectives:

1. To provide adequate basic understanding about the Banking, Insurance.

2. To give adequate exposure to the operational environment in the field of the Banking and Insurance.

3. To prepare students to exploit opportunities being newly created in the field of Banking & Insurance.

UNIT – I: Banker and Customer – Origin of banking – Banker – Banking and other business – customer – Relationship between banker and customer – General relationship – Special relationship – Banker's lien.

UNIT – II: Deposits – General precautions for opening account – Current deposit account – Fixed deposit account – Savings deposit account – Recurring deposit – Other deposits.

UNIT – III: Pass book – Correct entry – Wrong entry – Entries favourable to the customer – Entries favourable to the bankers.

UNIT – IV: Crossing – General crossing – Special crossing – Double crossing – Who can cross a cheque – Opening of crossing.

UNIT – V: Paying banker – Circumtances under which a cheque can be dishonoured – Answers to dishonor cheques – Payment in due course – Holder in due course – Recovery of money paid by mistake.

UNIT – VI: Collecting banker – Banker as a holder for value – Banker as an agent – Conversion – Statutory protection – Basis of negligence – Duties of collecting banker.

UNIT – VII: Subsidiary services – Agency services – Payment and collection – Purchase and sale of securities – Executor – Administrator and trustee – Attorney – Miscellaneous services.

UNIT – VIII: Insurance and Risk - significance of insurance and risk, general structure of the insurance market, significant aspects of this industry - Reforms in Indian Insurance Industry - importance of the privatization of insurance industry, problems associated with public insurance enterprises, relation between insurance and economic growth.

UNIT – IX: Regulations Relating to Insurance Accounting and Management - framework for IRDA rules and regulations regarding general insurance investment in the country, role of financial reporting in managing insurance operations, significance of determining solvency margins.

UNIT – X: Life Insurance - factors influencing the key functioning of insurance organizations insurable interest, role of riders in insurance policies - Non-life Insurance - elements of fire insurance contract and its ancillary features. Significance of marine insurance and its various policies, the role of rural insurance in making people's lives better in rural India.

UNIT – XI: Non-life Insurance - II - types of motor insurance policies, critical aspects of aviation industry in the country, significance of liability insurance in India - Functions and Organization of Insurers - components of the distribution system of life insurance companies in the country, role of agents in the life insurance sector in India, important activities carried out in a life insurance organization

UNIT – XII: Product Design and Development: Product development in the life and non-life insurance sectors in India, role of risk evaluation in the process of insurance product formation, future trends in the domain of insurance product design and development - Insurance Underwriting - need for insurance underwriting, factors that affect the activities performed by the underwriter, steps involved in the process of insurance underwriting.

UNIT – XIII: Claims Management: factors affecting the insurance claim management system, types of documents needed in various types of claims, meaning of 'Causa Proxima' in insurance claim settlement - Insurance Pricing and Marketing - principles of insurance pricing and marketing, tools and techniques used in pricing individual life and health insurance.

UNIT – XIV: Financial Management in Insurance Companies and Insurance Ombudsman: importance of financial management in insurance companies, tools of managing expenses in the insurance companies, modes used by the insurance companies in channelizing their funds - Reinsurance: reinsurance in the insurance sector. Areas of the application of reinsurance - Information Technology in Insurance - application of information technology in the insurance sector, role of insurance companies in insurance security, contours of the future of insurance in rural areas.

REFERENCE BOOKS:

1. Gordon E and Natarajan K, (2010), "Banking Theory, Law and Practice", Himalaya Publishing House, Mumbai.
2. Dr. Sunilkumar (2016), "Insurance and Risk Management", Galgotia Publishing Company.
3. Dr. P. Periasamy (2011), "Principles and Practice of Insurance", Himalaya Publishing House, Mumbai.

Duration of the Programme: The programme for the degree of Master of Commerce (Finance and Control) consists of two academic / calendar years divided into four semesters. Each semester is having four core papers carries four credits each. The duration of course covers 64 credits which are divided equally sixteen credits into four semesters.

Faculty and Support Staff Requirements: The programme for the degree of **Master of Commerce (F & C)** requires the following faculty and supporting staff:

Staff Category	Required
Core Faculty	3
Faculty for Specialization	2
Clerical Assistant	1

Instructional Delivery Mechanisms:

- Self Learning Materials
- Web based Teaching and Learning
- Electronic study materials

6. Student Support Services:

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like counselling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issue of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact programme for theory courses will be held at the university campus. The conduct of end-semester examinations, evaluation and issue of certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

Procedure for Admission, curriculum transaction and evaluation:

f. 1. Procedure for Admission:

A candidate who has passed any UG Degree is eligible B.Com / B.Com (CA) / BCS / B.Com (CS) / BBM / BBA / B.Com. (C.S and C.A) shall be given preference, as per the norms set by the Government of Tamil Nadu or an Examination accepted as equivalent thereto by the Academic Council, subject to such conditions as may be prescribed thereto are permitted to appear and qualify for the Master of Commerce (Finance and Control) Degree Examination of this university after a course of study of two academic / calendar years.

f. 2. Curriculum Transactions:

The classroom teaching would be lecture method by using electronic devices. The face to face contact classes cover group discussion, seminars and paper presentation.

The face to face contact classes will be conducted in following durations;

Course Type	**Face to Face Contact Classes Per Semester (in Hours)**
Theory Courses (4 courses with 4 credits each)	**64**

3. Evaluation: The evaluation method which comprises both internal assessment and end semester examination. The internal assessment shall contain maximum 25 marks for each paper. The end semester examination shall contain 3 hours duration to each course at the end of each semester. The end semester examination shall comprise maximum 75 marks for each paper. The candidate who failed in any papers are permitted to appear in subsequent examinations to clear the failed papers.

Question Paper Pattern:

Answer all questions (one question from each unit with internal choices

Time: 3 Hours; Maximum Marks: 75

Part A - 10 x 2 Marks = 20 Marks

Part B - 5 x 5 Marks = 25 Marks

Part C - 3 x 10 Marks = 30 Marks

Distribution of Marks in Continuous Internal Assessments

The following procedure shall be followed for awarding internal marks for the courses

Component	Marks
Assignments (per course)	25

Passing Minimum:

The students will be evaluated by examinations. There shall be no passing minimum for internal. The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50% marks (50% out of 75 marks) in the University end semester examination. Then the marks secured by the candidate in the end semester examination will be taken and added with his/her internal marks (Maximum marks 25). In aggregate (External + Internal) the passing minimum shall be 50% for each course. Grading shall be based on overall marks obtained.

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

Marks and Grade:

The following table gives the marks, grade points, letter grades and classification to indicate the performance of the candidate.

Conversion of marks to grade points & Letter grade (Performance in a paper / Course)

Range of Marks	Grade Points	Letter Grade	Description
90 – 100	9.0 – 10.0	O	Outstanding
80 – 89	8.0 – 8.9	D+	Excellent
75 – 79	7.5 – 7.9	D	Distinction
70 – 74	7.0 – 7.4	A+	Very Good
60 – 69	6.0 – 6.9	A	Good
50 – 59	5.0 – 5.9	B	Average
00 – 49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

Ci = Credits earned for course i in any semester

Gi = Grade point obtained for course i in any semester

n = refers to the semester in which such courses were credited.

Grade Point Average (GPA): Grade point average (GPA) is calculated for each part taking into account all the courses studied under each part.

$$GPA = \frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$$

$$GPA = \frac{\sum_i (C_i \times G_i)}{\sum_i C_i}$$

Where Ci = Credits earned for course i in any semester.

G_i= Grade points obtained for course i in any semester

Cumulative Grade Point Average (CGPA): For the entire program CGPA is calculated in the following manner.

$$\textbf{CGPA} = \frac{\sum_n \sum_I C_{ni} \times G_{ni}}{\sum_n \sum_i C_{ni}}$$

$$\text{CGPA} = \frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses of the entire programme}}$$

Classification of CGPA: Based on the Cumulative Grade Points Average the performance of the students is classified as follows.

CGPA	Grade	Classification of Final Result
9.5 - 10.0	O+	First Class – Exemplary
9.0 and above but below 9.5	O	
8.5 and above but below 9.0	D++	First Class with Distinction
8.0 and above but below 8.5	D+	
7.5 and above but below 8.0	D	
7.0 and above but below 7.5	A++	First Class
6.5 and above but below 7.0	A+	
6.0 and above but below 6.5	A	
5.5 and above but below 6.0	B+	Second Class
5.0 and above but below 5.5	B	
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

Maximum duration for the completion of the course: The maximum duration for completion of M.Com., (F & C) Degree programme shall not exceed ten semesters from their fourth semester.

Commencement of this Regulation: These regulations shall come into force from the academic year 2018 - 2019 (June session) i.e., for students who are to be admitted to the

first year of the course during the academic year 2018 - 2019 (June session) and thereafter.

Fee Structure: The programme has the following Fee Structure:

Sl. No.	Fees Detail	Amount in Rs.	
		First Year	Second Year
1.	Admission Processing Fees	300.00	-
2.	Course Fees	4400.00	4400.00
3.	ICT Fees	150.00	150.00
	Total	**4850.00**	**4550.00**

The above mentioned fee structure is exclusive of Exam fees.

Requirement of the laboratory support and Library Resources: Alagappa University, Karaikudi having well established Library facility with adequate number of copies of books in relevant titles for Commerce (Finance and Control) programme.

Cost estimate of the programme and the provisions:

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with M.Com., (F & C) Programme are as follows:

Sl. No.	Expenditure Heads	Approx. Amount in Rs.
1.	Programme Development (Single Time investment)	10,00,000/-
2.	Programme Delivery (Per Year)	20,00,000/-
3.	Programme Maintenance	3,00,000/-

Quality Assurance Mechanism and Expected Programme Outcomes: Separate Internal Quality Assurance Cell (IQAC) has been established with certain parameters in the Directorate of Distance Education in order to provide quality teaching to the stakeholders.

University Motto: ‘Excellence in Action’

University Vision Statement: Achieving Excellence in all spheres of Education, with particular emphasis on **“PEARL”-** Pedagogy, Extension, Administration, Research and Learning.

University Objectives: Providing for Instructions and Training in such Branches of Learning as the University may determine. Fostering Research for the Advancement and Dissemination of Knowledge.

University Quality Policy: Attaining Benchmark Quality in every domain of **‘PEARL’** to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

University Quality Quote: Quality Unleashes Opportunities towards Excellence **(QUOTE).**

Minutes of the Meeting of the Board of Studies in Commerce [For M.Com., (Finance & Control) Programme to be offered through ODL Mode] held at the Directorate of Distance Education, Alagappa University, Karaikudi – 630 004. on 04.09.2017 (Monday), at 3:15 p.m.

Members Present:

1. Dr. T. R. Gurumoorthy	-	Chairman
2. Dr. S. Ganapathy	-	Member
3. Dr. A. Muthusamy	-	Member
4. Dr. R. Ganapathi	-	Member

After the deliberation and discussion the board resolved the following:

1. The board considered the curriculum design and detailed syllabi of M.Com., (Finance & Control) programme, prepared as per the norms by the Chairman and the Board Members, scrutinized and suitably modified the same.

2. The board resolved to approve curriculum design, detailed syllabi and other regulations for the M.Com., (Finance & Control) programme to be offered by the Directorate of Distance Education of Alagappa University are given in Annexure – I.

Dr. T. R. Gurumoorthy Dr. S. Ganapathy Dr. A. Muthusamy

Dr. R. Ganapathi

ALAGAPPA UNIVERSITY

Accredited with A+ Grade by NAAC (CGPA 3.64) in the Third Cycle
Karaikudi – 630003. Tamilnadu , INDIA

Directorate of Distance Education

PROGRAMME PROJECT REPORT

for

Master of Computer Applications (M.C.A)

submitted to

UGC, Distance Education Bureau (DEB),
New Delhi

for seeking approval to introduce programme through Distance Education Mode

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ALAGAPPA UNIVERSITY, KARAIKUDI

DIRECTORATE OF DISTANCE EDUCATION

MASTER OF COMPUTER APPLICATIONS (M.C.A)
Credit Based System (CBS)
(With effect from June 2018-2019 Onwards)

(a) Programme's Mission and Objectives

Mission

Mission is to offer excellent career opportunities in various industries including software development companies in the areas of System analysis/design/developments/supports, mobile application programming, game programming, web and e- commerce development, database administration, software testing, education and training etc.

Programme Objectives:

- ✓ Discover, investigate the requirements of a problem and find the solution to them using computing principles.
- ✓ Create and evaluate a computer based system, components and process to meet the specific needs of applications.
- ✓ Utilize current techniques and tools necessary for complex computing practices.
- ✓ Develop and integrate effectively system based components into user environment.
- ✓ Identify the need and develop the skill to employ in learning as a computing professional.
- ✓ Execute effectively in a team environment to achieve a common goal.
- ✓ Classify opportunities and use innovative ideas to create value and wealth for the betterment of the individual and society.
- ✓ Proficiency in developing application with required domain knowledge.
- ✓ To facilitate learners to nurture skills to practice their professions competently to meet the ever-changing needs of society such as Digital India, Safety and Privacy.

Programme Outcome:

- ✓ To widen the ability to plan, analyze, design, code, test, implement & maintain a software product for real time system
- ✓ To support learners capability to set up their own enterprise in various sectors of Computer applications
- ✓ To improve the knowledge of the learners in finding solutions and developing system based applications for real time problems in various domains involving technical, managerial, economical & social constraints
- ✓ To develop in problem solving and programming skills in the various computing fields of IT industries.
- ✓ To prepare the learners to pursue higher studies in computing or related disciplines and to work in the fields of teaching and research.

(b) Relevance of the program with HEI's and Alagappa University Mission and Goals

This programme is aligned with HEI's and Alagappa University mission and goals to be offered through distance mode to reach quality higher education to the unreachable and/or rural learners. Higher education in Computer Science offered through distance mode meets the mission of HEI's like digital India and e-cash transaction will enrich the Human resources for the uplift of the nation.

(c) Nature of prospective target group of learners

The nature of prospective target group of learners is graduates from various disciplines like Commerce, Mathematics, Physics, Chemistry, Biology, Electronics, and Engineering etc. It also includes the learners who want to become entrepreneurs like Web Designers, Software Developers, BPO's, KPO's etc.,

(d) Appropriateness of programme to be conducted in Open and Distance Learning mode to acquire specific skills and competence;

M.C.A Programme through Distance Learning mode is developed in order to give subject-specific skills including i) Knowledge about various kinds of programming languages ii) Digital Logic fundamentals, Operating systems, RDBMS, Data Structure and Software Engineering iii) inter-disciplinary knowledge like Discrete Mathematics and Accounting and Financial Management iv) Concepts like Object Oriented Analysis and Design Compiler Design Data Mining and Ware Housing v) Cutting Edge Technologies like Web Technology and .Net Framework.

(e) Instructional Design

e.1 Revisions of Regulation and Curriculum Design

1. The University reserves the right to amend or change the regulations, schemes of examinations and syllabi from time to time based on recent market dynamics, industrial developments, research and feedback from stakeholders and learners.
2. Each student should secure 96 credits to complete M.C.A. programme.
3. Each theory and practical course carries 4 credits with 75 marks in the University End Semester Examination (ESE) and 25 marks in the Continuous Internal Assessment (CIA).

Programme code

M.C.A	315	**M.C.A(Lateral Entry LE)**	340

M.C.A Credit Based Curriculum and Evaluation System

Courses of Study

S.No	Course Code M.C.A	Course Code M.C.A (LE)	Title of the Course	CIA Marks Max	ESE Marks Max	TOT Marks Max	C Max.
	FIRST YEAR						
	I Semester						
1	31511		Digital Computer Organization	25	75	100	4
2	31512		Object Oriented Programming and C++	25	75	100	4
3	31513		Data Structure and Algorithms	25	75	100	4
4	31514		Data Structure using C++ Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
	II Semester						
5	31521		Software Engineering	25	75	100	4
6	31522		Relational Database Management Systems(RDBMS)	25	75	100	4
7	31523		Computer Graphics	25	75	100	4
8	31524		RDBMS Lab	25	75	100	4
				100	**300**	**400**	**16**
	SECOND YEAR						
	III Semester						
9	31531	34031	Discrete Mathematics	25	75	100	4
10	31532	34032	Operating System	25	75	100	4
11	31533	34033	Object Oriented Analysis and Design	25	75	100	4
12	31534	34034	Operating System Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
	IV Semester						
13	31541	34041	Accounting and Financial Management	25	75	100	4
14	31542	34042	Communication Skills	25	75	100	4
15	31543	34043	Internet and Java Programming	25	75	100	4
16	31544	34044	Internet and Java Programming Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
	THIRD YEAR						
	V Semester						
17	31551	34051	Computer Networks	25	75	100	4
18	31552	34052	Data Mining and Warehousing	25	75	100	4
19	31553	34053	Visual Programming with •NET	25	75	100	4
20	31554	34054	VB•NET Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
	VI Semester						
21	31561	34061	Cloud Computing	25	75	100	4
22	31562	34062	Soft Computing	25	75	100	4
23	31563	34063	Big Data Analytics	25	75	100	4
24	31564	34064	Mini Project	25	75	100	4
			Total	**100**	**300**	**400**	**16**
			Grand Total	**600**	**1800**	**2400**	**96**

CIA : Continuous Internal Assessment **ESE :** End semester Examination **Max**. Maximum Marks; **C :** Credits

Course Code Legend:

3	1	5	S	C

315 – Programme code for Master of Computer Applications (M.C.A).
S -- Semester Number; C – Course Number in the Semester

e.2 Detailed Syllabi

The detailed Syllabi of study and shall be as shown in Appendix.

e.3 Duration of the Programme: (Minimum 3 Years, Maximum – 5 Years)

The M.C.A programme shall consist of a period of three years (Six Semesters).

e.3.1 Medium of Instruction

The medium of instruction is only in **English**.
The course material is also in **English.**

e.4 Faculty and Support Staff Requirements:

The following faculty and support staff is required for this programme.

Staff Category	Required
Core Faculty	3
Faculty for Specialization	2
Laboratory Assistant	1
Clerical Assistant	1

*Faculty at least in Assistant Professor level

e.5 Instructional Delivery mechanisms

The instructional delivery mechanisms of the programme includes SLM- Study materials, Lab instruction manual, Personal contact session for both theory and practical courses of the programme, e-version of the course materials in the form of CD, e-book, e-tutorials, Massive Open Online Courses (MOOC) courses, Open Educational Resources(OER) and virtual lab.

e.6 Identification of media

The printed version of SLM – study material shall be given to the learners in addition to MOOC, e-tutorial and virtual lab.

e.7 Student Support Services

The student support services will be facilitated by the Directorate of Distance Education, Alagappa University, Karaikudi and its approved learning centres located in various parts of Tamilnadu.

The pre-admission student support services like counseling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at Directorate of Distance Education or Learning centres. The post-admission student support services like issuing Identity card, study materials will be provided thru Directorate or Learning centres. The face to face contact sessions of the programme for both theory and practical's will be held at the Directorate or Learning centres. The student support regarding the conduct of examinations, evaluations, publication of results and certificates done by the Office of the Controller of Examinations, Alagappa University, Karaikudi

(f) Procedure for Admissions, curriculum transaction and evaluation

f.1 Minimum qualification for admission

Candidates for admission to the first year of the Master of Computer Applications (M.C.A) programme shall be required to have passed the following examinations. Candidates who have passed any degree with Mathematics at +2 level of any Recognized University or authority accepted by the Syndicate of the Alagappa University as equivalent thereto shall be eligible.

f.1.1 Lateral Entry(LE)

Candidates who have passed any Bachelor degree with P.G.D.C.A (Post Graduate Diploma in Computer Applications) are eligible for admission into the Second Year of M.C.A programme, provided the first year subjects of M.C.A. were studies in P.G.D.C.A. The candidates should appear for examination in subjects which are not covered in P.G.D.C.A.

f.2 Curriculum transaction

- The face to face contact sessions in class room teaching with the support of SLM, Power Point Presentations, web based tools, audio and animated videos.

- The practical classes are based on the respective subject study materials containing requirement for the laboratory experiments.

- Face to face contact sessions will be conducted for both theory and practical courses in the following manner.

Course Type	Face to face contact session per semester (in Hours)
Theory courses (3 Courses with 4 credits each)	48
Practical course (1 Course with 4 credits)	120
Total	**168**

f.3 Evaluation

There shall be two types of evaluation systems; Continuous internal assessment and end semester examination will be conducted by the University according to the following scheme. The internal assessment for both theory and practical's is maximum of 25 marks for each course. The end semester examination for theory and practical is maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

Internal assessment

- Internal assessment of theory courses is through home assignment with workbook, case studies, review questions, quiz, multiple choice questions etc., for 25 marks.
- The internal assessment for the practical courses shall be through home assignment which includes workbook designing algorithm, preparing source code, PL/SQL coding etc., for 25 marks.
- Student should submit assignment for theory and practical courses of every course and semester.

Division of Internal Marks (Assignment)

Theory		Practical	
Assignment	**Marks**	**Assignment**	**Marks**
Review questions	15	Algorithm Design	15
Workbook, case studies, quiz, multiple choice questions etc	10	Workbook for preparing source code, PL/SQL coding , results etc	10
TOTAL	**25**	**TOTAL**	**25**

End Semester Examination (ESE)

The university end Semester Examinations shall be of three hours duration with maximum of 75 Marks for both theory and practical courses.

f.3.1 Minimum for a pass:

To pass in each course, a candidate is required to secure 50% marks in the End Semester examination and 50% marks in the aggregate (marks in End Semester Examination + marks in Internal Assessment).

The students who does not secure required minimum marks for pass in a course(s) shall be required to reappear and pass the same in the subsequent examination,

f.3.2 Question Paper Pattern - Theory

The end semester examination will be conducted in the duration of 3 Hours and maximum of 75 Marks.

All the Blocks Should Be Given Equal Importance

Part – A (10 x 2 Marks: 20 Marks) Answer all questions

Part – B (5 x 5 Marks: 25 Marks) Answer all questions choosing either (a) or (b)

Part – C (3 x 10 Marks: 30 Marks) (Answer any 3 out of 5 questions)

End Semester Examination (ESE) - Practical

Students are required to prepare a separate lab record for each lab course. The practical counsellor should duly sign this lab record after each session.

Students shall prepare practical record note book which includes aim, algorithm, source code, input, expected output and result of the experiment and submit during end semester practical examination.

Division of marks in ESE – Practical (Maximum 75 marks)

The end semester practical examination will be conducted in the duration of 3 Hours and maximum of 75 Marks.

Practical details	**Max. Marks**
Algorithm / Flowchart	10
Source Code	20
Debugging	10
Execution	10
Results	10
Viva-Voce	5
Record	10
Total	**75**

f.3.3 Procedure for Completing the Course:

A student shall be permitted to continue the programme from I to VI semester irrespective of failure(s) in the courses of the earlier semesters. The candidate will qualify for the MCA degree only if he/she passes all the (including arrears) courses with in a period of FIVE years from the date of admission.

f.3.3.1 P.G.D.C.A. Certification

Candidates admitted to M.C.A Degree programme after completing the first year of the MCA course, can opt for discontinuing the programme, will be issued a P.G.D.C.A. (Post Graduate Diploma in Computer Applications) provided he/she passed all the subjects in the First and Second semesters of the M.C.A. programme.

f.3.4 Results and Classification:

Results will be declared at the end of each semester of the University examination and the marks/grade obtained by the candidate will be forwarded to them by the Controller of Examinations, Alagappa University.

f.3.4.1 Marks and grades

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
96 - 100	10.00	S+	First class – Exemplary
91 - 95	9.5	S	
86 - 90	9.0	D++	First class – Distinction
81 - 85	8.5	D+	
76 - 80	8.0	D	
71 - 75	7.5	A++	First Class
66 - 70	7.0	A+	
61 - 65	6.5	A	
56 - 60	6.0	B	Second Class
50 - 55	5.5	C	
Below 50	0.00	F	Fail
ABSENT	0.00	AAA	Absent

For a semester

Grade Point Average[GPA] = $\sum C_i G_i / \sum C_i$

M.C.A Credit Based Curriculum and Evaluation System

$$GPA = \frac{\text{Sum of the multiplication of Grade points by the credit of the courses}}{\text{Sum of the credit of the courses in the semester}}$$

$$= \frac{\text{Sum of [Credit earned x Grade Points]}}{\text{Sum of the credits earned in the semester}}$$

For the entire programme

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni}\ G_{ni} / \sum_n \sum_i C_{ni}$

$$= \frac{\text{sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme}}$$

Where

C_i - Credits earned for the course i in any semester
G_i – Grade Point earned for course i in any semester
n - is number of all Courses successfully cleared during the particular semester in the case of GPA and during all the semesters (programme) in the case of CGPA.

CGPA	Grade	Classification of Final Result
9.6 – 10.00	S+	First class – Exemplary*
9.1 – 9.5	S	
8.6 – 9.0	D++	First class with Distinction*
8.1 – 8.5	D+	
7.6 – 8.0	D	
7.1 – 7.5	A++	First Class
6.6 – 7.0	A+	
6.1 – 6.5	A	
5.6 – 6.0	B+	Second Class
5.0 – 5.5	C	
Below 5.0	U	Reappear

* The candidates who have passed in the first appearance and within the prescribed semester

f.4 Fees Structure

Fee Particulars	Amount in Rs.		
	First Year	Second Year	Third Year
Admission Processing Fees	300	--	--
Course Fees	15400	15400	15400
ICT fees	150	150	150
Total Fees	**15,850**	**15,550**	**15,550**

The above mentioned fees structure is exclusive of examination fees.

(g)Requirement of the laboratory support and library resources

g.1 Laboratory Support

A well- equipment Computer Laboratory was established in the Alagappa University, Karaikudi with necessary software's as per the practical's syllabi for conducting face to face contact sessions for practical courses of this programme.

g.2 Library Resources

The Directorate of Distance Education, Alagappa University provides library facility with number of books and Self Learning materials for Computer Science programmes. The Central library of Alagappa University provides the collection of volumes of Self Learning Materials, Printed books, Subscriptions to printed periodicals and Non-book materials in print form for the learner's references. All these library resources are meant for learner's reference purpose only.

(h)Cost estimate of the programme and the provisions:

Expense details	**Amount in (Rs.) Approx.**
Programme development (Single time Investment)	20,00,000/-
Programme delivery (per year)	24,00,000/-
Programme maintenance (per year)	5,00,000/-

(i)Quality assurance mechanism and expected programme outcomes:

i.1 University's Moto:

' Excellence in Action'

i.2 University's Vision and Mission

Vision

Achieving Excellence in all spheres of Education, with particular emphasis on ' PEARL' - Pedagogy, Extension, Administration, Research and Learning.

Mission

Affording a High Quality Higher Education to the learners so that they are transformed into intellectually competent human resources that will help in the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

i.3 University Objectives

1. Providing for instructions and training in such branches of Learning at the university may determine.
2. Fostering Research for the Advancement and Dissemination of Knowledge and Application.

i.4 Quality Policy

Attaining Benchmark Quality in every domain of 'PEARL' to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

i.5 Quality Quote

Quality Unleashes Opportunities Towards Excellence (QUOTE).

i.6. Course benchmarks

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations and number of enrolments of students. Feedback from the alumni, students, parents, stakeholders and employers will be received to analyze the benchmark qualities for the further improvement of the programme.

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Appendix

Detailed Syllabi
FIRST YEAR
SEMESTER I

Course Code	Title of the Course
31511	DIGITAL COMPUTER ORGANIZATION

Course Objectives:

To impart the knowledge in the field of digital electronics
To impart knowledge about the various components of a computer and its internals.

Course Requirements:

- Before studying this course, the student has knowledge about
- Basic principles of number system
- Concepts of digital, Boolean and instruction

Course Outcome:

After the completion of this course, the student will be able to :

- Design and realize the functionality of the computer hardware with basic gates and other components using combinational and sequential logic.
- Understand the importance of the hardware-software interface

Unit No	Description
	BLOCK 1 :NUMBER SYSTEMS
1	**Number Systems :** Binary, Octal, Decimal and Hexadecimal number systems – Conversion from one base to another base – Use of complements – binary arithmetic – Numeric and Character codes.
2	**Boolean algebra and Combinational Circuits:** Fundamental concepts of Boolean Algebra – De Morgan's theorems
3	**Simplification of expressions** – Sum of products and products of sums – Karnaugh map simplification – Quine - McKluskey method – two level implementation of Combinational Circuits.
	BLOCK 2 COMBINATIONAL CIRCUITS AND SEQUENTIAL CIRCUITS
4	**Combinational Circuits:** Half Adder – Full Adder – Subtractors – Decoders – Encoders – Multiplexers – Demultiplexer.
5	**Sequential Circuits:** Flip flops – Registers – Shift Registers – Binary Counters –

	BCD Counters – Memory Unit.
6	**Data Representation :** Data Types – Complements – Fixed Point Representations – Floating Point Representations – Other Binary Codes – Error detection codes.
	BLOCK 3 : BASIC COMPUTER ORGANIZATION AND DESIGN
7	**Instruction Codes :** Instruction Codes – Computer Registers – Computer Instructions – Timing and Control
8	**Instruction cycle:** – Memory reference instructions – Input output and Interrupt – Complete Computer Description – Design on Basic Computer – Design of Accumulator logic
	BLOCK : 4 CENTRAL PROCESSING UNIT
9	Introduction – General Register organization – Stack organization
10	**Instruction formats:** – Addressing modes – Data transfer and manipulation – Program control.
11	**Input – output organization:** Peripheral devices – Input output interface – Asynchronous data transfer – Modes of transfer
12	**Priority interrupt:** – DMA – IOP – Serial Communication.
	BLOCK : 5 MEMORY ORGANIZATION
13	Memory Hierarchy – Main memory – Auxiliary memory – Associative memory
14	**Memory organization:** Cache memory – Virtual memory – Memory management hardware.

Text Books:

1. Digital Computer Fundamentals, 6th Edition, Thomas C. Bartee, Tata McGraw Hill, 2008.
2. Digital Logic and Computer Design, M. Morris Mano, Pearson Education, 2008.

Reference Books:

1. Digital Computer Electronics, 3rd Edition, Albert Paul Malvino and Jerald A. Brown, Tata McGraw Hill, 2008.
2. Computer Organization, 5th Edition, V.C. Hamacher et al, Tata Mcgraw Hill.

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Course Code	Title of the Course
31512	**OBJECT ORIENTED PROGRAMMING and C++**

Course Objectives:

- To provide an overview of working principles of object oriented paradigm
- To understand and apply the OOPs fundamentals
- To implement the features of OOP in real world applications

Course Outcome:

- Able to understand the object oriented programming techniques

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	**Introduction and Features**: Evolution of Object Oriented Language, Object oriented Paradigm, Basic concept of object-oriented programming- objects, classes, encapsulation and data abstraction, inheritance, polymorphism, dynamic binding, message passing
2	**Popular OOP languages.** Moving from C to C++ Introduction – Predefined console streams, hierarchy of console stream classes,
3	**I/O operations;** Unformatted I/O operations, formatted console I/O operations, manipulators, custom/user-defined manipulators.
	BLOCK 2 : CLASSES AND OBJECTS
4	**Classes and Objects:** Introduction, class specification, class objects, accessing class members, defining member functions, accessing member functions within a class, outside member functions as inline, private member function,
5	**Memory allocation for objects:** array of objects, function prototype, call by reference, return by reference, objects as function arguments, inline function, friend function, constant parameter and member function.
6	**Object Initialization:** Introduction - constructors, default constructor, parameterized constructors, multiple constructors in a class, dynamic initialization through constructors, copy constructor, dynamic constructor, destructor. Dynamic Objects: Introduction, pointers to objects, array of pointers to objects, this pointer.
	BLOCK 3 : INHERITANCE, POLYMORPHISM AND DATA CONVERSION
7	**Inheritance:** Introduction, derived class declaration, forms of inheritance, inheritance and member accessibility, multilevel inheritance, multiple inheritance, hierarchical inheritance, hybrid inheritance.
8	**Polymorphism:** Introduction, Function overloading, Operator overloading introduction, unary operator overloading, binary operator overloading, assignment operator overloading, overloading with friend functions.
9	**Data conversion:** conversion between basic data types, conversion between objects and basic types, conversion between objects of different classes. Virtual function: Introduction, need for virtual functions, pure virtual functions, abstract classes.
	BLOCK 4 : TEMPLATES AND FILES
10	**Generic Programming with Templates:** Introduction - class templates – class template with multiple arguments
11	**Function template:** function template with multiple arguments. inheritance of class template.

12	**Streams with Files :** Introduction, hierarchy of file stream classes, opening and closing of files, file pointers and their manipulators, sequential access to a file, file input/output with stream class, random access to a file.
	BLOCK 5 : EXCEPTION HANDLING
13	**Exception Handling:** Introduction – Basics of exception handling, exception handling mechanism, throwing mechanism, catching mechanism. Exceptions in constructors and destructors
14	**Other Exception Handling methods:** Handling uncaught exceptions, exceptions in operator overloaded functions, exception in inheritance tree, exceptions in class templates, memory allocation failure exception.

TEXT BOOK:

1. E.Balagurusamy, Object oriented programming in C++, Third Edition, Tata McGraw Hill Publications, 2007.
2. Mastering C++, K.R Venugopal and Rajkumar, T.Ravishankar, Tata McGraw Hill Publishing Company Ltd., 2006.

REFERENCE BOOK:

1. Object Oriented Programming in C++, Fourth Edition, Rober Lafore, Galgotia Publications Pvt. Ltd., New Delhi. 2010.

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Course Code	Title of the Course
31513	**DATA STRUCTURE AND ALGORITHMS**

Course Objectives:

- The learner should be well versed with the fundamentals of Algorithms, learn various data structures, should be able to use them appropriately as per need during development of programs.
- Also, the learner should know different sorting and searching techniques so that correct techniques can be used in different programs so that the complexity of the program does not increase due the sorting/ search technique employed.

Course Outcome

After the completion of this course, the student will able to

- To write programs using structures, strings, arrays, pointers and strings for solving complex computational problem.
- Using the data structures real time applications
- Able to analyze the efficiency of Data Structures

Unit No	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction to Data Structure :** Types of Data Structure , Primitive data types
	Algorithms: –Time and space Complexity of algorithms
2	**Arrays:** Array initialization, Definition of Array, Characteristic of Array ,One-dimensional Array, Two-dimensional array and Multi dimensional array
	BLOCK 2 : LINEAR DATA STRUCTURE
3	**Stack :** Stack related terms, Operations on a stack,
4	**Representation of Stack:** Implementation of a stack – application of Stack. Expression Evaluation Polish notation.
5	**Queues:** Operations on queue Circular Queue, Representation of Queues, Application of Queues
6	**List:** Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list
7	**Operation on Linked List :** Insertion and Deletion of linked list
8	**Traversal:** Traversing a linked list , Representation of linked list.
	BLOCK : 3 NON-LINEAR DATA STRUCTURE
9	**Trees:** Binary Trees, Types of Binary trees, Binary Tree Representation
10	**Binary Tree operations / Applications :** Traversing Binary Trees, Binary Search tree,
11	**Operations on Binary Tree:** Insertion and Deletion operations, Hashing Techniques.
	BLOCK 4 : SEARCHING TECHNIQUES
12	**Searching :** Introduction, Searching, Linear Search, Binary Search
	BLOCK 5 : SORTING TECHNIQUES
13	**Sorting:** Bubble sort, Insertion sort, Radix sort
14	**Other sorting Techniques:** Selection sort, Quick sort, Tree sort.

Text Books:

1. Fundamentals of Data structures , Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
2. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.

Reference Books:

1. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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Course Code	Title of the Course
31514	DATA STRUCTURE USING C++ - LAB

Course Objectives

- To be able to solve data structure problems using C++ language
- To learn and implement C++ language programming techniques
- To introduce the efficiency of the algorithm

Course Outcome

- Students can develop programming knowledge/
- Students can solve any kind of problems using C++ language
- Data Structure based problems can be solved

Experiments based on c++ programming and Data Structures

Unit No.	Contents
	BLOCK 1 : SIMPLE C++ PROGRAMS
1	**Introduction** Simple C++ Programs
2	**Control Structures:** Using if and switch constructs Programs
3	**Looping , Arrays ,Structure statements**: for, while, do-while, Strings and Matrices Programs Problems
	BLOCK 2 : OOPs CONCEPTS
4	**Functions:** static function, friend function ,constructor , destructor and operator overloading and Recursive programs
5	**Inheritance and polymorphism:** Inheritance types and polymorphism types, Virtual function
6	**File:** File Handling C++ Programs, opening and closing a data file - creating a data file, processing a data file.
7	**Pointers :** Pointers and Pointers with Arrays Programs
	BLOCK 3: LINEAR DATA STRUCUTURE
8	**Stacks :** Stack Implementation, expression evaluation, Polish notation
9	**Queues:** Queue Implementation, Applications of Queue
10	**Linked List programs:** List, Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list, Insertion and Deletion of linked list, Traversing a linked list.
	BLOCK 4 : NON LINEAR DATA STRUCTURE
11	**Tree Programs :** Trees, Binary Trees, Types of Binary trees, Binary Tree Representation, Traversing Binary Trees, Binary Search tree, Insertion and Deletion operations,

12	**Graphs:** Shortest Path Algorithms o Dijkstra's Algorithm o Graphs with Negative Edge costs o Acyclic Graphs o All Pairs Shortest Paths Algorithm **Minimum cost Spanning Trees** o Kruskal's Algorithm o Prims's Algorithm o Applications Breadth First Search
	BLOCK 5 : SEARCHING AND SORTING ALGORITHMS
13	**Searching Techniques:** Linear and Binary search Programs
14	**Sorting techniques**: Bubble sort, Quick sort, Insertion sort, Merge sort

Reference Books:

1. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.
2. Fundamentals of Data structures in C, Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
3. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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SEMESTER II

Course Code	Title of the Course
31521	**SOFTWARE ENGINEERING**

Course Objective:

- To know of how to do project planning for the software process.
- To learn the cost estimation techniques during the analysis of the project.
- To understand the quality concepts for ensuring the functionality of the software

Course Requirement:

- Fundamental concepts of Software Engineering

Course Outcome:

- Understand the activities during the project scheduling of any software application.
- Learn the risk management activities and the resource allocation for the projects.
- Able to create reliable, replicable cost estimation that links to the requirements of project

planning and managing.

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Software:** Role of software, Software myths. Generic view of process: A layered technology, a process framework, The Capability Maturity Model Integration (CMMI)
2	**Process patterns**, Process assessment, Personal and Team process models.
3	**Process model:** The waterfall model, Incremental process models, Evolutionary process models, The Unified process.
	BLOCK 2 : REQUIREMENT ENGINEERING:
4	Design and Construction, Requirement Engineering Tasks, Requirements Engineering Process, Validating Requirements.
5	**Building the Analysis Model:** Requirement analysis, Data Modeling concepts, Object-Oriented Analysis
6	**Modeling:** Scenario-Based Modeling, Flow-Oriented Modeling Class-Based Modeling, Creating a Behavioral Model.
	BLOCK 3 : SYSTEM DESIGN
7	**Design Engineering:** Design process and quality, Design concepts, the design model.
8	**Architectural Design:** Software architecture, Data design, Architectural styles and patterns, Architectural Design.
9	**User interface design:** The Golden rules, User interface analysis and design, Interface analysis, Interface design steps, Design evaluation.
	BLOCK 4 : SYSTEM TESTING
10	**Testing Strategies:** Approach to Software Testing, Unit Testing, Integration Testing, Test strategies for Object-Oriented Software, Validation Testing, System Testing, the art of Debugging, Black-Box and White-Box testing.
11	**Product Metrics:** Software Quality, Product Metrics, Metrics for Analysis Model, Design Model, Source code and Metrics for testing, Metrics for maintenance. Metrics for Process and Projects Domains: Software Measurement, Metrics for Software Quality and Software Process.
	BLOCK 5 : RISK and QUALITY MANAGEMENT
12	Risk Strategies: Reactive vs. Proactive Risk strategies, software risks, Risk identification
13	**Risk Protection and refinement:** Risk projection, Risk refinement, Risk Mitigation, Monitoring and Management, RMMM Plan.
14	**Quality Management:** Quality concepts, Software quality assurance, Software Reviews, Formal Technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.

TEXT BOOK:

1. Roger S. Pressman Software Engineering - A practitioner's Approach McGraw-Hill 6th Edition (2010)

REFERENCE BOOKS:

1. Richard Fairlay Software Engineering Concepts McGraw Hill Book Company (2005)
2. Pankaj Jalote An Integrated Approach to Software Engineering Narosa Publishing House 3rd Edition (2005)
3. Software Engineering, Somzerville, 8th Edition, Pearson Education 2007.
4. Software Engineering K.K. Agarwal & Yogesh Singh, 3rd Edition New Age International Publishers 2007.
5. Software Engineering an Engineering Approach James F. Peters, Witold Pedrycz - John Wiley & Sons 2000.
6. Software Engineering Principles and Practice Waman S Jawadekar, , Tata McGraw-Hill 2004.

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Course Code	Title of the Course
31522	**RELATIONAL DATABASE MANAGEMENT SYSTEMS (RDBMS)**

Course Objectives:

- To understand the fundamentals of data models
- To make a study of SQL and relational database design.
- To know about data storage techniques an query processing.
- To impart knowledge in transaction processing, concurrency control techniques and External storage

Course Requirements:

- Knowledge about the basic concepts of the database.

Course Outcome:

- Design a database using ER diagrams and map ER into Relations and normalize the relations
- Acquire the knowledge of query evaluation to monitor the performance of the DBMS.
- Develop a simple database applications using normalization.

Unit No	Contents
	BLOCK 1 INTRODUCTION
1	**Data base System Applications**, data base System VS file System – View of Data – Data Abstraction –Instances and Schemas – data Models – the ER Model
2	**Model :**Relational Model – Other Models – Database Languages – DDL – DML – database Access for applications Programs – data base Users and Administrator – Transaction Management – data base System Structure – Storage Manager – the Query Processor.

3	**History of Data base Systems -** Data base design and ER diagrams – Beyond ER Design Entities, Attributes and Entity sets – Relationships and Relationship sets – Additional features of ER Model – Concept Design with the ER Model – Conceptual Design for Large enterprises.
	BLOCK 2 : RELATIONAL MODEL
4	**Introduction**– Integrity Constraint Over relations – Enforcing Integrity constraints – Querying relational data – Logical data base Design – Introduction to Views – Destroying / altering Tables and Views.
5	**Relational Algebra** – Selection and projection set operations – renaming – Joins – Division – Examples of Algebra overviews –
6	**Relational calculus** – Tuple relational Calculus – Domain relational calculus – Expressive Power of Algebra and calculus.
	BLOCK 3 : SQL QUERY
7	**Form of Basic SQL Query** – Examples of Basic SQL Queries – Introduction to Nested Queries – Correlated Nested Queries Set – Comparison Operators – Aggregative Operators – NULL values – Comparison using Null values – Logical connectivity's – AND, OR and NOT – Impact on SQL Constructs – Outer Joins – Disallowing NULL values – Complex Integrity Constraints in SQL Triggers and Active Data bases. Schema refinement
8	**Normal forms :**Problems Caused by redundancy – Decompositions – Problem related to decomposition – reasoning about FDS – FIRST, SECOND, THIRD Normal forms – BCNF–
9	**Join:** Lossless join Decomposition – Dependency preserving Decomposition – Schema refinement in Data base Design – Multi valued Dependencies – FORTH Normal Form.
	BLOCK 4 TRANSACTION
10	**Introduction :**Transaction Concept- Transaction State- Implementation of Atomicity and Durability – Concurrent – Executions – Serializability- Recoverability – Implementation of Isolation – Testing for serializability
11	**Protocols :** Lock Based Protocols – Timestamp Based Protocols- Validation-Based Protocols – Multiple Granularity.
12	**Recovery and Atomicity** – Log – Based Recovery – Recovery with Concurrent Transactions – Buffer Management – Failure with loss of nonvolatile storage- Advance Recovery systems- Remote Backup systems
	BLOCK 5 STORAGE
13	**Data on External Storage** – File Organization and Indexing – Cluster Indexes, Primary and Secondary Indexes – Index data Structures – Hash Based Indexing – Tree base Indexing – Comparison of File Organizations – Indexes and
14	**Performance Tuning-** Intuitions for tree Indexes – Indexed Sequential Access Methods (ISAM) – B+ Trees: A Dynamic Index Structure.

Text Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.

Reference Books:

1. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
2. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
3. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
4. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

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Course Code	Title of the Course
31523	COMPUTER GRAPHICS

Course Objectives:

- To understand computational development of graphics
- To provide in-depth knowledge of display systems, image synthesis, shape modeling of 3D application.

Course Outcome:

- Enhance the perspective of modern computer system with modeling, analysis and interpretation of 2D and 3D visual information.
- Able to develop interactive animations.

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction** Application areas of Computer Graphics, overview of graphics systems, video-display devices, raster-scan systems, random scan systems, graphics monitors and work stations and input devices.
2	**Output primitives:** Points and lines, line drawing algorithms, mid-point circle and ellipse algorithms.
3	**Filled area primitives:** Scan line polygon fill algorithm, boundary-fill and flood-fill algorithms.
	BLOCK 2 : 2 D TRANSFORM AND CLIPPING
4	**2-D geometrical transform:** Translation, scaling, rotation, reflection and shear transformations
5	**2D Matrix representations:** homogeneous coordinates, composite transforms, transformations between coordinate systems.
6	**2-D viewing:** The viewing pipeline, viewing coordinate reference frame, window to view-port coordinate transformation, viewing functions,
7	**Clipping Algorithms:** Cohen-Sutherland and Cyrus-beck line clipping

	algorithms, Sutherland –Hodgeman polygon clipping algorithm.
	BLOCK 3 : 3D OBJECT REPRESENTATION
8	**Introduction:** Polygon surfaces, quadric surfaces, spline representation,
9	**Curve and surfaces:** Hermite curve, Bezier curve and B-Spline curves, Bezier and B-Spline surfaces. Basic illumination models, polygon rendering methods.
	BLOCK 4 : 3D GEMETRIC TRANSFORMATION
10	**3-D Geometric transformations:** Translation, rotation, scaling, reflection and shear transformations, composite transformations.
11	**3-D viewing:** Viewing pipeline, viewing coordinates, view volume and general projection transforms and clipping.
	BLOCK 5 : VISIBLE SURFACE DETECTION METHODS AND ANIMATION
12	**Classification**, back-face detection, depth-buffer, scan-line, depth sorting, BSP-tree methods, area sub-division and octree methods.
13	**Computer animation:** Design of animation sequence, general computer animation functions, raster animation,
14	**Other Animation Techniques:** Computer animation languages, key frame systems, motion specifications.

Text Books

1. Donald Hearn and M.Pauline Baker, Computer Graphics C version, Pearson Education, 2007.

Reference Books:

1. M. Newman and F. Sproull, Interactive Computer Graphics, McGraw Hill 2004
2. Foley, VanDam, Feiner and Hughes, Computer Graphics Principles and Practice, 2nd Edition in C, Pearson Education, 2004.
3. Plastok and Gordon Kalley, Computer, McGraw Hill 2000.

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Course Code	Title of the Course
31524	**RELATIONAL DATABASE MANAGEMENT SYSTEMS (RDBMS) LAB**

Course Objectives:

- To understand the fundamentals of data models
- To make a study of SQL and relational database design.
- To know about data storage techniques an query processing.
- To impart knowledge in transaction processing, concurrency control techniques and External storage

Course Requirements:

- Knowledge about the basic concepts of the database.

Course Outcome:

- Design a database using ER diagrams and map ER into Relations and normalize the relations
- Acquire the knowledge of query evaluation to monitor the performance of the DBMS.
- Develop a simple database applications using normalization.

Use the concepts like data normalization, link between table by means of foreign keys and other relevant database concepts for the following applications. The implementation of each should have necessary input screen (forms) Menu-driven query processing and reports. Necessary validations should be made for each table;

Unit No.	Contents
	BLOCK 1 : TABLE MANIPULATION
1	**Table** creation, Renaming a Table, Copying another table, Dropping a Table
2	**Table Description:** Describing Table Definitions, Modifying Tables, Joining tables, Number and Date functions.
	BLOCK 2 : SQL QUERIES AND SUB QUERIES
3	**SQL Queries:** Queries, Sub Queries, and aggregate functions
4	**DDL:** Experiments using database DDL SQL statements
5	**DML:** Experiment using database DML SQL statements
6	**DCL:** Experiment using database DCL SQL statements
	BLOCK 3 : INDEX AND VIEW
7	**Index :** Experiment using database index creation, Renaming a index, Copying another index, Dropping a index
8	**Views:** Create Views, Partition and locks
	BLOCK 4 : EXCEPTION HANDLING AND PL/SQL
9	**Exception Handling:** PL/SQL Procedure for application using exception handling
10	**Cursor:** PL/SQL Procedure for application using cursors
11	**Trigger:** PL/SQL Procedure for application using triggers
12	**Package:** PL/SQL Procedure for application using package
13	**Reports:** DBMS programs to prepare report using functions
	BLOCK 5 : APPLICATION DEVELOPMENT
14	**Design and Develop Application:** Library information system, Students mark sheet processing, Telephone directory maintenance, Gas booking and delivering, Electricity bill processing, Bank Transaction, Pay roll processing. Personal information system, Question database and conducting Quiz and Personal diary

Reference Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.

2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.
3. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
4. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
5. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
6. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

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SECOND YEAR
SEMESTER III

Course Code	Title of the Course
31531 /34031	DISCRETE MATHEMATICS

Course Objectives:

- To understand the concepts and operations Set theory, Graph Theory
- To understand and apply the Mathematical Logic in computer science.

Course Requirements:

- Knowledge about Logics and graphs

Course Outcome:

- Acquire the basic knowledge of matrix, set theory, functions and relations concepts needed for designing and solving problems
- Acquire the knowledge of logical operations and predicate calculus needed for computing skill
- Able to design and solve Boolean functions for defined problems

Unit No.	Contents
	BLOCK 1 : MATHEMATICAL LOGIC
1	**Mathematical Logic:** Statements and Notation - connectives -normal forms – The theory of inference for the statement calculus -
2	**Predicate Calculus**: The predicate calculus - Inference theory and predicate calculus.
3	**Set theory:** Sets – Basic concepts – notation - inclusion and equality of sets - the power set
	BLOCK 2 : RELATIONS
4	**Relations and ordering** properties – relation matrix and graph of a relation
5	**Relations** Partition – equivalence and compatibility relations
6	**Composition and partial ordering**:Composition – partial ordering – partially

	ordered set
	BLOCK 3 : FUNCTIONS
7	**Functions** – definition – composition – inverse – binary and n-ary operations
8	**Other Functions** : Characteristic function – hashing function.
	BLOCK 4 : ALGEBRAIC STRUCTURES
9	**Algebraic Structures:** Algebraic Systems: Examples and General Properties
10	**Semigroups and Monoids:** Definitions and Examples - Homomorphism of Semigroups and Monoids - Subsemigroups and Submonoids
11	**Groups:** Definitions and Examples - Cosets and Lagrange's Theorem
12	**Normal Subgroups** – Algebraic Systems with two Binary Operations.
	BLOCK 5 : GRAPH AND FINITE PROBABILITY
13	**Graph theory:** Basic concepts – definition – paths - reach - ability and connectedness – matrix representation of graphs - trees.
14	**Finite Probability** – Probability Distributions – Conditional Probability Independence – Bayes' Theorem – Mathematical Expectation

TEXT BOOKS:

1. J.P. Tremblay and R. Manohar Discrete mathematical structures with applications to Computer Science TMH Publishing Company 2003.
2. Judith L. Gersting, Mathematical Structures for Computer Science, 5thEdition, W.H. Freeman and Company, 2003.

REFERENCE BOOKS:

1. Venkatraman M K, Sridharan N and Chandrasekaran N, Discrete Mathematics, The National Publishing Company, 2004.
2. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science PHI,2003.
3. Ramasamy, Discrete Mathematical Structures with application to combinatorics, Univeristies Press, 2006.
4. Bernand Kolman, Roberty C. Busby, Sharn Cutter Ross, Discrete Mathematical Structures, Pearson Education, 2006.
5. Richard Johnsonbaugh, Discrete Mathematics, Fifth Edition, Pearson Education. 2001.
6. Garry Haggard and others, Discrete Mathematics for Computer science, Thomson.

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Course Code	Title of the Course
31532 /34032	**OPERATING SYSTEMS**

Course Objective

- Able to understand the operating system principles
- Able to know the Principles of Deadlock, processor scheduling and memory management.

Course Requirements:

- To be aware of the evolution and fundamental principles of operating system, processes and their communication

Course Outcome

- Students have acquired the knowledge about the types of operating systems
- Students have acquired the knowledge about the functions of operating system

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction:** Definition of Operating Systems – Computer System Organization
2	**Computer System Architecture** – Operating System Structure – Operating System Operations
3	**System Structures:** Operating System Services – System Calls – System Programs – Operating System Design and Implementation.
	BLOCK 2 : PROCESS CONCEPT
4	**Process Concept:** Process Scheduling – Operations on Processes – Inter Process Communication
5	**Process Scheduling:** Scheduling Concepts – Scheduling Criteria – Scheduling Algorithms – Multiple Processor Scheduling
	BLOCK 3 : SYNCHRONIZATION
6	**Synchronization:** The Critical Section Problem – Synchronization Hardware – Semaphores – Classic Problems of Synchronization – Monitors
7	**Deadlocks:** Deadlocks Characterization – Methods for Handling Deadlocks
8	**Deadlock** Prevention – Avoidance – Detection –Recovery from Deadlock.
	BLOCK 4 : MEMORY MANAGEMENT
9	**Memory Management Strategies:** Swapping – Contiguous Memory Allocation – Paging – Segmentation
	BLOCK 5 : FILE SYSTEM
10	**File Concept** – Access Methods – Directory
11	**Structure** – File System Mounting – File Sharing – Protection.
12	**Implementing File Systems:** File System Structure – File System Implementation
13	**Directory Implementation** – Allocation Methods – Free Space Management
14	**Secondary Storage Structure:** Overview of Mass Storage Structure – Disk Structure – Disk Attachment – Disk Scheduling – Disk Management.

TEXT BOOK:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Principles", 7th Edition, Wiley India Edition, 2008.

REFERENCE BOOKS:

1. Andrew S.Tanenbaum, Operating Systems Design and Implementation, 3rd edition, Prentice Hall, 2006.
2. Harvey M. Deitel, An Introduction to Operating Systems, 3rd Edition, Addison Wesley 2007.

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Course Code	Title of the Course
31533 /34033	**OBJECT ORIENTED ANALYSIS AND DESIGN**

Course Objective:

- To demonstrate and apply basic object oriented techniques to create and modify object oriented analysis and design models.
- To understand and apply testing techniques for object oriented software

Course Requirement:

- Basic knowledge about oops concepts

Course Outcome:

- Able to learn the various object oriented methodologies and choose the appropriate one for solving the problem
- Understand the concept of analysis, design & testing to develop a document for the project

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Object Oriented System Development :** Introduction – Object basics - The Object Model : Evolution – Elements
2	**Classes and Objects:** Object nature – Relationship among objects – Class nature – Relationships among classes
3	**Classes and objects:**Building quality classes and objects – System Development Life Cycle
	BLOCK 2 : OBJECT ORIENTED METHODOLOGIES
4	**Methodologies :** - Rumbaugh Object Modeling Technique, Booch – Jacobson – Shaler/Mellor – Coad/Yardon
5	**Patterns :** Patterns – Frame Works

6	**The Unified Approach** – UML – Static and Dynamic Model – UML diagrams
	BLOCK 3 : OBJECT ORIENTED ANALYSIS
7	**Object Oriented Analysis -** Identifying Use Cases – Use Case Model – Documentation
8	**Classification:** Identifying Classes – Noun Phrases Approach – Common Class Pattern Approach
9	**Use Case Driven Approach** – Identifying Object Relationship Attributes and Methods.
	BLOCK 5 : OBJECT ORIENTED DESIGN
10	**Object Oriented Design :** Introduction – Design Process – Design Axioms – Designing Classes – Visibility – Refining Attributes –
11	**Designing Methods -** Access Layer Design – View Layer Design
12	**Managing Analysis And Design** – Evaluation Testing – Impact of object oriented testing -
	BLOCK 5 : CODING AND MAINTENANCE
13	**Coding and Maintenance:** Coding – Maintenance – Metrics
14	**Case Study** Foundation Class Library – Client/Server Computing.

Text Books:

1. Grady Booch, Robert A.Maksimchuk et.al, Object Oriented Analysis and Design with applications, Pearson Education, 3rd Edition, 2009.
2. Ali Bahrami, Object Oriented System Development, Tata McGraw Hill Edition, 2008.

Reference Books:

1. James Rumbaugh et.al, Object Oriented Modeling and Design, Addison Wesley, 2006.
2. Larman, Applying UML & Patterns, An Introduction to Object Oriented Analysis and Design, Pearson Education, 2nd Edition, 2003.
3. Martin Fowler, Kendall Scott , UML, Distilled Addison Wesley, 2004.
4. Ivar Jacobson Object Oriented Software Engineering: A Use Case Driven Approach Addison Wesley, 2004.

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Course Code	Title of the Course
31534 /34034	**OPERATING SYSTEM LAB**

Syllabi based on operating system theory

Objective of the course

- To learn and get familiar with unix/ubundo operating system
- To understand the functions unix/ubundo operating system through basic commands

Course outcome

On completion of the course

- Students will attain hands on experience in linux operating system.
- Basic unix commands their syntax and purpose can be well understood by the students
- Shell commands and their syntax and purpose can be well understood by the students

Unit No.	Contents
	BLOCK 1 INTRODUCTION
1	**Introduction** Operating system ,objective, History, Features of Unix
2	**Kernal and shell**
3	**Unix file system:** File and common commands-Shell-More about files-Directories-Unix system-Basics of file-Directories and filenames-
	BLOCK 2 : UNIX PERMISSIONS
4	**Permissions**- Inodes-Directory hierarchy-Devices-the grep family-Other filters
5	**Stream editor** sed - awk pattern scanning and processing language-files and good filters.
6	**Wild card characters**
	BLOCK 3 : UNIX COMMANDS
7	**Unix commands with syntax:** Syntax and unix commands
8	**Unix shells:** History of unix shells
9	**Deciding on a shell**
	BLOCK 4 : SHELL COMMADS - FILES
10	**Shell Command files**
11	**Bourne shell**: Bourne shell programming
12	**Shell programming files:** Shell programming on files
	BLOCK 5 : MENU DRIVEN SHELL PROGRAM:
13	**Menu Driven File handling**
14	**Menu Driven shell program – file:** edit, create and delete file using menu

REFERENCE BOOKS:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Principles”, 7th Edition, Wiley India Edition, 2008.
2. Andrew S.Tanenbaum, Operating Systems Design and Implementation, 3rd edition, Prentice Hall, 2006.
3. Harvey M. Deitel, An Introduction to Operating Systems, 3rd Edition, Addison Wesley 2007.

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SEMESTER IV

Course Code	Title of the Course
31541 /34041	ACCOUNTING AND FINANCIAL MANAGEMENT

Course Objectives:

- To understand the process of estimating the cost of a particular product.
- To Prepare the estimate for various business activities such as purchase, sale, production and cash budgets

Course Requirements:

- Basic principles of Accounting

Course Outcome:

- Able to do balance sheet preparation and do analysis
- Able to do the budget preparation and control of a company

Unit No	Contents
	BLOCK 1 : INTRODUCTION - Financial Accounting:
1	**Financial Accounting:** Meaning and Scope – Principles – Concepts – Conventions
2	**Accounting process:** Journal - Ledger – Trail Balance – Trading Account – Profit and Loss Account – Balance Sheet
3	**Accounting Ratio Analysis** – Funds Flow Analysis –Cash Flow Analysis – Computerized account.
	BLOCK 2 : COST AND MANAGEMENT ACCOUNTING
4	**Introduction:** Meaning Scope and uses of cost and management accounting – Elements of Cost
5	**Cost Sheet** – Marginal Costing and Cost Volume Profit Analysis
6	**Break Even Analysis:** Concept, Applications and Limitations
	BLOCK 3 : STANDARD COSTING AND BUDGETING:
7	**Introduction :** Concept and importance standard costing - Variance Analysis – Material – Labor – Overhead – Sales – Profit Variances -
8	**Budgets and Budgetary Control** – Meaning and Types of budgets – Sales Budget – Production Budget
9	**Budgets:**Cash Budget – Master Budget – Flexible budgeting – Zero Base Budgeting.

	BLOCK 4 : FINANCIAL MANAGEMENT
10	**Introduction:** Objectives and Functions of Financial Management – Risk – Return Relationship –Time Value of Money
11	**Capital Budgeting:** Basic Methods of Appraisal of investments –
12	**Working Capital:** Concepts of working Capital ,Factors Affecting working Capital – Estimation of working capital requirements
	BLOCK 5 : COST OF CAPITAL
13	**Cost of Capital Capital Structure and Dividend**: Meaning and types of Cost of Capital – computation of cost for debt and equity sources of capital and weighted average cost of capital
14	**Capitial Structure** Meaning and types of capital structure – determinants of capital structure – types of Dividend Policy – Types of Dividend decision.

TEXT BOOKS:

1. Maheswari S N, Financial and Management Accounting, Sultan Chand & Sons, 2003.
2. Pandey I M, Financial Management, 4th Edition, Vikas Publications, 2002.

REFERENCE BOOKS:

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, 2005
2. S.P. Iyengar, “Cost and Management Accounting”, Sultan Chand & Sons.
3. I. M. Pandey, “Elements of Management Accounting”, Vikas Publishing House.
4. Shula and T.S.Grewal, “Advanced Accounting”, S.Chand and Company, 2010
5. S.N.Maheswari, “Management Accounting & Financial Accounting”, Vikas Publications 2010
6. Ravi M Kishore, Management Accounting and Financial Analysis”, Taxmons 2010
7. Khan and Jain, Management accounting: Text, problems and cases, 2011

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Course Code	Title of the Course
31542 /34042	**COMMUNICATION SKILLS**

Course Objectives:

- To enable the students to learn the basics of communication skills, soft skills and quantitative aptitude.
- To improve the written communication skills so as to write reports and letters.
- To enable the students to speak English with correct accent and pronunciation.
- Interact efficiently in real life situation in work place

Course Requirements:

- Basic English knowledge in grammar

Course Outcome:

- Understood the basics of communication skills and soft skills
- Acquired knowledge in quantitative aptitude
- Take part in professional and social communication
- Make oral presentations
- Engage in debates
- Face Interviews
- Participate in Group Discussion

Unit No	Contents
	BLOCK 1 : IMPORTANCE OF COMMUNICATION
1	**Communication:** Importance of Communication: – Principles of effective communication
2	**Levels of communication** (Extrapersonal, Interpersonal, Intrapersonal and Mass Communication)
3	**Channels of communication** (Downward communication, Upward communication, Horizontal communication and Diagonal communication) Barriers to communication
	BLOCK 2 : CONVERSATIONS SKILLS
4	**Soft Skills** (Importance, Definition and attributes) – Verbal and Non-Verbal Communications
5	**Telephone conversations:** Do's and Don'ts – Modes of conversation – Greeting, Requesting, Thanking, Congratulating, Introducing, Apologizing, Agreeing/Disagreeing, permitting/Not permitting
	BLOCK 3 PRESENTATION SKILLS
6	**Introduction:**Planning (Analyzing audience and locale)
7	**Preparing** (Introduction, Middle and End) Practice and presentation
8	(**Nuances and Delivery** – Kinesics, Proximics, paraliquistics, chronemics) Audio visual Aids.
	BLOCK 4 : GROUP COMMUNICATION
9	**Group Communication:** Group discussion – Purpose - Process of Group Discussion - Presentation - Getting Started - Art of Guiding and Controlling Discussion - Personality test through Group Discussion - Lateral thinking - Participation techniques - Mock Group Discussion.
10	**Interviews and job interviews** What and Why? - Types of Interviews -

	Understanding the intricacies - Planning for Interviews - Answering Skills - Effective Communication during Interviews - IPS - Mock Interview.
11	**Meetings** Meaning - Importance - Objectives - Leading and participating in meetings - Communication skills for meetings - Mock Meetings - Seminars. Skills Needed for Team Work.
	BLOCK 5 : WRITING SKILLS
12	**Writing Letters** (Job Application, resume, curriculum vitae) Effective writing - Report writing - Speech writing
13	**E-mail** and Advertising – Writing of Agenda, Agenda writing - Letters - Articles writing - Improving English language writing - When to write and when not to write.
14	**Minutes** and memos – report writing.

TEXT BOOK:

1. Green Level (1984) Building English Skills USA: MC Dougal, Little.
   S.R.Sharma (2006) Communication Skills in English Jaiput: Mark Publishers

REFERENCE BOOKS:

1. Green Rajeevan and P.Kiranmani Dutt. (2006) Basic Communication Skills New Delhi: Foundation Books
2. Martin Hewings(2004) Pronunciation Practice Activities United kingdom: The Cambridge University Press.

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Course Code	Title of the Course
31543 /34043	**INTERNET AND JAVA PROGRAMMING**

Course Objectives:

- To provide an overview of working principles of internet, web related functionalities
- To understand and apply the fundamentals core java, packages, database connectivity for computing

Course Requirements:

- Basic knowledge of internet and programming principles

Course Outcome:

- Able to understand the internet standards and recent web Technologies
- Able to implement, compile, test and run Java program,
- Able to make use of hierarchy of Java classes to provide a solution to a given set of requirements found in the Java API

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Basic Internet Concepts:** Connecting to the Internet – Domain Name System – E-mail
2	**The World Wide Web** – Internet Search Engines – Web Browsers – Chatting and conferencing on the Internet
3	**Online Chatting** –Messaging – Usenet Newsgroup – Internet Relay chat (IRC) – FTP – Telnet.
	BLOCK 2 : FUNDAMENTALS OF OBJECT-ORIENTED PROGRAMMING
4	**Basic concepts** of OOP – Benefits – Applications . Java Evolution: Features – how java differs from C and C++ - java and internet- java support system – java environment
5	**Overview of Java Language** –Introduction – Simple Java Program – Comments – Java Program Structure – Tokens – Java Statements – Implementing a Java Program – JVM – Command Line Arguments. Constants – Variables – Data Types – Type Casting..
6	**Operators and Expressions:** Arithmetic Operators – Relational, Logical, Assignment, Increment and Decrement, Conditional, Bitwise, Special Operators – Arithmetic Expressions, Evaluation of Expression – Precedence of Arithmetic Operators – Type Conversions – Operator Precedence and associativity – Mathematical Functions. **Decision Making and Branching:** If –if…..else –Nesting of if……. Else – else if– switch. Decision Making and Looping: While – do – for – jump in loops – labeled loops.
	BLOCK 3 : CLASSES, OBJECTS AND METHODS
7	**class:** Defining a class –fields –methods –creating objects – accessing class members – constructors – methods overloading –static members –nesting of methods – Inheritance –overriding methods –final variables-classes –methods
8	**Arrays, Strings and Vectors** :One dimensional Arrays –creating of array – Two dimensional arrays- strings –vectors –Wrapper classes – Enumerated Types - Interfaces: Multiple Inheritance
9	**Packages:** Defining interface –Extending interfaces – Implementing Interfaces - Putting Classes Together
	BLOCK 4: MULTITHEREADING , EXCEPTION AND APPLETS
10	**Multithreaded Programming** – Creating Threads –Extending the thread class – Stopping and Blocking a thread –Life cycle of a thread –using thread methods – Thread Exceptions –Priority –Synchronization –Implementing the 'Runnable' Interface
11	**Managing Error and Exceptions:** Types of errors –Exceptions –Syntax of

	Exception Handling code – Multiple Catch statements –using finally statement – Throwing our own Exceptions – using exceptions for Debugging - Graphics Programming: The Graphics Class – Lines and Rectangles – Circles and Ellipses – Drawing Arcs – Drawing Polygons – Line Graphs – Using Control Loops in Applets – Drawing Bar Charts.
12	**Applet Programming:** How applets differ from Applications – preparing to write applets – Building Applet Code – Applet life cycle – creating an Executable Applet – Designing a Web Page – Applet Tag – Adding Applet to HTML file – Running the Applet – Passing parameters to Applets – Displaying Numerical values – Getting input from the user
	BLOCK 5 : MANAGING INPUT/OUTPUT FILES IN JAVA
13	**Introduction** – concept of streams –stream classes – byte stream classes – character Stream
14	**I/O classes**: –using stream –using the file class –Input / output Exceptions – creation of files – Reading / writing characters – reading writing bytes Random access files- Interactive input and output –Other stream classes

Text Books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004 (Unit I).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.

Reference books:

1. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
2. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
3. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
4. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.
5. Cay S. Horstmann, Gary Cornell, Core Java, Volume I and II, 5th Edition, Pearson Education, 2003.
6. Topley, J2ME in A Nutshell, O'Reilly Publishers, 2002.
7. Hunt, Guide to J2EE Enterprise Java, Springer Publications, 2004.
8. Ed Roman, Enterprise Java Beans, Wiley Publishers, 1998.

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Course Code	Title of the Course
31544 /34044	**INTERNET AND JAVA PROGRAMMING LAB**

Course Objective:

- To understand and practice markup languages
- To understand and practice embedded dynamic scripting on client side Internet Programming
- To understand and practice web development techniques on client-side

Course Requirement:

- Basic concepts of Web and Java programming

Course Outcome:

- Explore markup languages features and create interactive web pages using them
- Learn and design Client side validation using scripting languages
- Acquire knowledge about Open source JavaScript libraries
- Able to design front end web page and connect to the back end databases.

Experiments based on Internet Programming Theory

Unit No.	Contents
	BLOCK 1 : JAVA FUNDAMENTAL PROBLEMS:
1	Simple Java Problems
2	class and objects
3	Conditional control using java
4	Looping using java
	BLOCK 2 : OOP CONCEPTS
5	Function overloading programs
6	Operator overloading programs
7	Inheritance programs, Packages
8	Polymorphism programs Message passing programs
	BLOCK 3 : THREAD & VIRTUAL FUNCTION
9	Threads
10	Virtual function programs
	BLOCK 4 : I/O AND EXCEPTION HANDLING
11	Exception handling programs
12	I/O manipulation programs,
	BLOCK 4 :NETWORK PROGRAMMING
13	Applet programs
14	Implementation of simple network programs using java

Reference books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004 (Unit I).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.

3. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
4. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
5. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
6. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.
7. Cay S. Horstmann, Gary Cornell, Core Java, Volume I and II, 5th Edition, Pearson Education, 2003.
8. Topley, J2ME in A Nutshell, O'Reilly Publishers, 2002.
9. Hunt, Guide to J2EE Enterprise Java, Springer Publications, 2004.
10. Ed Roman, Enterprise Java Beans, Wiley Publishers, 1998.

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THIRD YEAR
SEMESTER V

Course Code	Title of the Course
31551 /34051	**COMPUTER NETWORKS**

Course Objectives:

•To understand networking concepts and basic communication model
•To understand network architectures and components required for data communication.

Course Requirements:

•Basic knowledge of networking

Course Outcome:

•Able to understand the working principles of various application protocols
•Acquire knowledge about security issues and services available

Unit No.	Contents
	BLOCK 1 : INTRODUCTION & PHYSICAL LAYER
1	**Introduction;** Computer Networks - Applications - Line configuration - Topology - Transmission Modes
2	**Categories of Network:** LAN, MAN, WAN - OSI Layer.
3	**Physical Layer:** Analog and Digital Signals Performance - Transmission Media
	BLOCK 2 : DATA LINK LAYER
4	**Data Link Layer:** Error Detection and correction – Introduction – Block Coding – Cyclic Redundancy Check – Framing – Flow and error Control –
5	**Data link layer protocols:** stop - wait protocol and sliding window protocol - ARQ, Go-back-n ARQ, selective - repeat ARQ.

6	**Multiple Access Protocols:** ALOHA – CSMA – CSMA/CD – CSMA/CA.
	BLOCK 3 : NETWORK LAYER
7	**Introduction:** Circuit switching - packet switching - message switching - Virtual circuit and Datagram subnets
8	**Routing algorithm :** Static routing -shortest path routing, Flooding, Flow based routing - Dynamic routing - distance vector routing, link state routing
9	**Other Routing Algorithms:** Hierarchical routing, Broad cast, Multi cast routing - Congestion, Control Algorithms
	BLOCK 4 : TRANSPORT LAYER
10	**Introduction:** Process to process delivery – UDP – TCP - Connection oriented Vs connectionless services.
11	**Applications and services:** Domain name system - Remote Logon – Mail Exchange - File Transfer
12	**Remote** Procedure Call - Remote File Access – WWW and HTTP – SNMP.
	BLOCK 5 : NETWORK SECURITY
13	**Introduction:** Cryptography – Encryption model – Transposition and Substitution Chipers – Cryptographic principles
14	**Symmetric key cryptography:** DES – AES – Asymmetric key cryptography: RSA – Security services.

Text Books:

1.Computer Networks, 3rd Edition, Andrew S Tanenbaum, Pearson Education, 2010.
2.Data Communications and Networking, 4th Edition, Behrouz A. Forouzan, TMH, 2009.

Reference Books:

1. Data and Computer Communications, 8th Edition, William Stallings, Prentice Hall.
2. An Engineering Approach to Computer Networks, 2nd Edition, S.Keshav, Pearson Education, 2008

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Course Code	Title of the Course
31552 /34052	**DATA MINING AND WAREHOUSING**

Course Objective:

- This course presents on depth of to data mining techniques; association rule, clustering, classification, web mining, temporal and sequential data mining and provide a practical exposure using data mining tool orange.
- To enable the students to learn the basic functions, principles and concepts of Data Mining
- To understand the fundamentals of Big Data Analytics

Course Requirements:

- Basic Concepts of Database

Course Outcome:

On successful completion of the course the students should have:

- Understand the data mining techniques, classification and web mining

Unit No.	Contents
	BLOCK 1 : DATA MINING and WAREHOUSING INTRODUCTION
1	**Data Warehousing Introduction** – Definition-Architecture-Warehouse Schema-Warehouse server-OLAP operations. Data Warehouse technology – Hardware and operating system
2	**Data Mining** - Definition – DM Techniques – current trends in data mining - Different forms of Knowledge – Data selection, cleaning, Integration, Transformation, Reduction and Enrichment.
3	**Data:** Types of data - Data Quality - Data Preprocessing - Measures of similarity and dissimilarity. **Exploration:** Summary statistics – Visualization.
	BLOCK 2 : ASSOICATION RULE MINING AND CLASSIFICATION
4	**Association rules:** Introduction – Methods to discover association rule – Apriori algorithm Partition Algorithm
5	**AR Algorithms:** Pincher search algorithm – Dynamic Item set algorithm – FP Tree growth algorithm.
6	**Classification:** Decision Tree classification – Bayesian Classification – Classification by Back Propagation.
	BLOCK 3 : CLUSTERING TECHNIQUES AND MACHINE LEARNING
7	**Introduction** – Clustering Paradigms – Partitioning Algorithms – K means & K Mediod algorithms – CLARA – CLARANS – Hierarchical clustering – DBSCAN – BIRCH – Categorical Clustering algorithms – STIRR – ROCK – CACTUS.
8	**Introduction to machine learning** – Supervised learning – Unsupervised learning – Machine learning and data mining.
9	**Neural Networks:** Introduction – Use of NN – Working of NN Genetic Algorithm: Introduction –Working of GA.
	BLOCK 4 : WEB MINING AND VISUAL DATA MINING
10	**Introduction** –Web content mining – Web structure mining –Web usage mining –Text mining –Text clustering, Temporal mining -Spatial mining
11	**Visual data mining** – Knowledge mining – Various tools and techniques for implementation using weka, Rapidminer and Matlab.
	BLOCK 5 : INTRODUCTION TO BIG DATA ANALYTICS
12	**Big Data Characteristics**- Types of Big Data- Traditional Versus Big Data Approach
13	**Technologies** Available for Big Data
14	**Hadoop** – Introduction - What is Hadoop? - Core Hadoop Components - Hadoop Ecosystem - Physical Architecture - Hadoop Limitations

Text Books:

1. Arun K Pujari, "Data Mining Techniques", University press, 2008.
2. C S R Prabhu, "Data Warehousing – concepts, techniques and applications ", 2nd Edition, Prentice Hall of India, 2002.
3. Radha Shankarmani, M Vijayalakshmi, "Big Data Analytics", Wiley Publications, first Edition, 2016

Reference Books:

1. Jaiwei Han, Michelinne Kamber, "Data Mining: Concepts and Techniques", Harcourt India / Morgan Kauffman publishers, 2008.
2. Alex Berson, Stephen J.Smith , "Data Warehousing , Data Mining & OLAP", Tata McGraw Hill, 2004.
3. Seema Acharya, Subhashini Chellappan, "Big Data and Analytics", Wiley Publication, first edition. Reprint in 2016
4. DT Editorial Services, "Black Book- Big Data (Covers Hadoop 2, MapReduce, Hive, Yarn, PIG, R, Data visualization)", Dream tech Press edition 2016.

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Course Code	Title of the Course
31553 /34053	**VISUAL PROGRAMMING WITH •NET**

Course Objective:

To develop an understanding of Visual Basic .Net
To develop the skills necessary to create software solutions using VB with .Net
To learn how to analyze certain types of problems with a software solution in mind

Course Requirements:

- Basic knowledge of Visual Basic

Course Outcome:

- Able to understand and design the solution to a problem using VB. Net
- Understand and implement the features of .Net for providing programmed solutions to complex problems

Unit No	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction** - What Is Visual Studio ? - Navigating the Visual Studio - The Menu

	– Toolbar - Work Area
2	**Toolbox** - Solution Explorer - Status Bar - Managing VS Windows
3	**Visual Studio Project Types** - Windows Projects - Web Projects - Office Projects - SharePoint Projects - Database Projects
	BLOCK 2 : C# AND VB.NET
4	**Basic Syntax** - Code Skeleton - The Main Method - The Program Class - The First Program Namespace - VS Code Editor - Class and Member Locators – Bookmarks - Running Programs - Primitive Types and Expressions - Enums - Branching Statements - Loops
5	**Creating Classes** - Class Inheritance – Class Snippet - Writing Methods - Parameters Passing - Returning Data
6	**Method Snippets** - Coding Fields and Properties - Declaring and Using Properties - The Property Snippet
	BLOCK 3 : UNDERSTANDING DELEGATES AND EVENTS
7	Events - Delegates - Handler Code - Implementing Interfaces - The interface Snippet - Applying Arrays and Generics -
8	**Creating and Building Projects** - Constructing Solutions and Projects - Navigating the Solution Explorer - Examining Property Settings - Assembly Name - Default Namespace - Target Framework - Output Type
9	**Building Projects** :Startup Object - Icon and Manifest - Compiling Applications - Rebuilding Solutions/Projects - Cleaning Solutions/Projects - Managing Dependencies, Compilation Settings - Navigating with Class View - Using the Class Designer - Class Designer Code Generation
	BLOCK 4 : DEBUGGING WITH VISUAL STUDIO
10	**Debugging methods:** Breakpoints - Stepping Through Code – Inspecting Application State - Locals and Autos Windows - Watch Windows - The Immediate Window - The Call Stack Window - The Quick Watch Window - Watching Variables with Pin To Source - Working with IntelliTrace
11	**Working with Databases** - Server Explorer - Creating a Database - Adding Tables - Relating Tables with Foreign Keys - Adding Stored Procedures - Configuring Database Options
	BLOCK 5 : BUILDING PROGRAMS WITH VS 2010
12	Building Desktop Applications with WPF - Starting a WPF Project - Understanding Layout - Grid Layout - StackPanel Layout - DockPanel Layout - WrapPanel Layout - Canvas Layout
13	**Using WPF Controls** - Managing Windows for Controls - Setting Properties - Handling Events - Coding Event Handlers - Working with Data in WPF - Data Source - Configuring a ComboBox
14	**Reading and Saving Data** - Using the DataGrid - Summary -Creating Web Applications with ASP.NET MVC - Designing Silverlight Applications - Deploying Web Services with WCF

Text Book:

1. Joe Mayo, Visual Studio 2010 - A Beginner's Guide, Tata Mc Graw Hill Edition, 2010.

Reference Books:

1. Nick Randolph, David Gardner, Professional Visual Studio 2010, Wiley Publishing 2010.
2. Andrew Moore, Visual Studio 2010 All-in-One For Dummies, Weiley Publishing, 2010.

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Course Code	Title of the Course
31554 /34054	**VISUAL BASIC •NET LAB**

Course Objective:

To develop an understanding of Visual Basic .Net
To develop the skills necessary to create software solutions using VB with .Net
To learn how to analyze certain types of problems with a software solution in mind

Course Requirements:

- Basic knowledge of Visual Basic

Course Outcome:

- Able to understand and design the solution to a problem using VB. Net
- Understand and implement the features of .Net for providing programmed solutions to complex problems

Experiments based on Visual Programming with .NET Theory

Unit No.	Contents
	BLOCK 1 : SIMPLE APPLICATIONS
1	**Simple Applications:** Developing simple applications using VB.NET a. Finding factorial Value b. Money Conversion c. Quadratic Equatin d. Temperature Conversion e. Login control
2	**Login form:** Create and Validate Login Form, Program to design Class, Program to demonstrate Inheritance, Polymorphism and Interfaces.
	BLOCK 2 : CONTROLS
3	**Controls:** Advance Controls, Common Dialog Controls. 2. Adrotator Control 3. Calendar control

	a. Display messages in a calendar control b. Display vacation in a calendar control c. Selected day in a calendar control using style d. Difference between two calendar dates 4. Treeview control a) Treeview control and datalist b) Treeview operations 5. Validation controls
4	**Active X Controls:** Working with intrinsic controls and ActiveX controls
	BLOCK 3 : MDI AND DATA CONTROLS
5	**MDI:** Application with multiple forms
6	**Data controls:** Application using data controls
	BLOCK 3 : DIALOGS AND MENU
7	**Dialogs:** Application with dialogs
8	**Common Dialogs:** Application using Common Dialogs
9	**Menus:** Application with Menus
	BLOCK 4 : EVENTS AND DATABASE
10	**Events and Database:** Drag and Drop Events Database Management Creating ActiveX Controls
11	**DataGridView:**ADO.NET Code to show records in DataGridView Control. 1. Databinding using datalist control 2. Datalist control templates 3. Databinding using datagrid 4. Datagrid control template 5. Datagrid hyperlink 6. Datagrid button column 7. Datalist event 8. Datagrid paging
12	**Database operations:**ADO.NET Code to perform Insert, Delete, Update and Select operations.
	BLOCK 5 : CRYSTAL REPORTS AND WEB APPLICATION
13	**Crystal Reports**
14	**Web Application** using ASP.NET that uses validation controls.

Reference Books:

1. Joe Mayo, Visual Studio 2010 - A Beginner's Guide, Tata Mc Graw Hill Edition, 2010.
2. Nick Randolph, David Gardner, Professional Visual Studio 2010, Wiley Publishing 2010.
3. Andrew Moore, Visual Studio 2010 All-in-One For Dummies, Weiley Publishing, 2010.

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SEMESTER VI

Course Code	Title of the Course
31561 /34061	CLOUD COMPUTING

Course Objective:

Lets learner to understand how to access all applications and documents from everywhere in the world, freeing from the confines of the desktop and making it easier for group members in different locations to collaborate.

Course Requirements:

Basic knowledge about internet and its application.

Course Outcome:

Understood the importance of cloud computing and its services.

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Fundamentals** :Cloud Computing – History – Working of cloud computing – Cloud computing today – Pros and cons of Cloud Computing – Benefits of cloud computing
2	**Non users of Cloud computing** – Developing cloud services – Pros and Cons of Cloud service Development
3	Types of Cloud Service Development – Discovering Cloud Services development services and tools.
	BLOCK 2 : CLOUD COMPUTING FOR EVERYONE
4	**Centralizing Email Communications** – Collaborating of Grocery lists – Collaborating on To-Do lists –
5	**Collaborating** on Household budgets – Collaborating on Contact lists – Communicating across the community – Collaborating on Schedules
6	**Collaborating on group projects** and events – Cloud computing for corporation.
	BLOCK 3 : CLOUD SERVICES
7	Exploring online calendar applications – Exploring online scheduling applications – Exploring online planning and task management – Collaboration on event management –
8	Collaboration on Contact Management – Collaboration on Project Management –
9	Collaborating on Word Processing and Databases – Storing and Sharing files and other online content.
	BLOCK 4 : ISSUES IN CLOUD
10	**Federation in cloud** – Four levels of federation – Privacy in cloud

11	**Security in Cloud** –Software as a security service – Case Study: Aneka – service level agreements
12	**Cloud Storage:** Over view of cloud storage – Cloud storage providers – Amazon S3 – Cloud file system – Map Reduce – Hadoop
	BLOCK 5 : CLOUD DEPLOYMENT TOOLS:
13	Study of open source cloud platforms – Eucalyptus
14	Nimbus – Open Nebula

Text Books:

1. Michael Miller, "Cloud computing – Web based applications that change the way you work and collaborate online", Pearson Education Inc., 2008
2. John W.Rittinghous, James F.Ransome, "Cloud Computing: Implementation, Management and Security", CRC Press 2010.

Books for Reference:

1. Danielle Ruest and Nelson Ruest, "Virtualization: A Beginners's Guide", McGraw Hill,2009.
2. Tom White, "Hadoop: The Definitive Guide", O'RIELLY Media 2009.
3. Rajkumar Buyya, James Broberg, Andrezj Goscinski, "Cloud computing – Principles and Paradigms", John Wiley and Sons, 2011.

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Course Code	Title of the Course
31562 /34062	**SOFT COMPUTING**

Course Objective:

- To learn the key aspects of Soft computing
- To know about the components and building block hypothesis of Genetic algorithm.
- To study the fuzzy logic components

Course Requirements:

- Basic concepts of Fuzzy Set and Neural Network

Course Outcome:

- Write Genetic Algorithm to solve the optimization problem
- Understand fuzzy concepts and develop a Fuzzy expert system to derive decisions.

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction:** Soft Computing Constituents – Soft Computing Vs Hard Computing – Characteristics - Applications

2	**Artificial Neural Network (ANN):** Fundamental Concept – Application Scope - Basic Terminologies – Neural Network Architecture – Learning Process
3	**Fundamental Models of ANN:** McCulloch-Pitts Model – Hebb Network – Linear Separability.
	BLOCK 2 : SUPERVISED AND UNSUPERSIVED LEARNING NETWORKS
4	**Perceptron Network** – Adaline and Madaline Networks – Back Propagation Network
5	**Radial Basis Function Network.** Associative Memory Networks – BAM - Hopfield Network - Boltzmann Machine.
6	**Unsupervised Learning Networks:** Kohonen Self Organizing Network – Counter Propagation Network – ART Network.
	BLOCK 3: FUZZY SETS
7	**Fuzzy Sets:** Basic Concept – Crisp Set Vs Fuzzy Set - Operations on Fuzzy Set – Properties of Fuzzy Sets
8	**Fuzzy Relations:** Concept – Fuzzy Composition – Fuzzy Equivalence and Tolerance Relation
9	**Membership Functions:** Features – Fuzzification – Methods of Membership value assignments – Defuzzification – Methods.
	BLOCK 4 : FYZZY ARITHMETIC
10	**Fuzzy Arithmetic** – Extension Principle – Fuzzy Measures
11	**Fuzzy Rules and Fuzzy Reasoning:** Fuzzy Propositions – Formation of Rules – Decomposition of Rules – Aggregation of Rules – Approximate Reasoning
12	**Fuzzy Inference and Expert Systems** – Fuzzy Decision Making – Fuzzy Logic Control Systems.
	BLOCK 5 : GENETIC ALGORITHM
13	**Genetic Algorithm:** Fundamental Concept – Basic Terminologies – Traditional Vs Genetic Algorithm - Elements of GA - Encoding - Fitness Function
14	**Genetic Operators:** Selection – Cross Over - Inversion and Deletion - Mutation – Simple and General GA - The Schema Theorem - Classification of Genetic Algorithm – Genetic Programming – Applications of GA.

Text Book:

1. S.N. Sivanandam, S.N. Deepa, “Principles of Soft Computing”, Wiley India, 2007.

Reference Books:

1. J.S.R. Jang, C.T. Sun, E. Mizutani, “Neuro-Fuzzy and Soft Computing”, Prentice Hall India, 2004
2. S. Rajasekaran, G.A.V. Pai, “Neural Networks, Fuzzy Logic, Genetic Algorithms”, Prentice Hall India, 2004.

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Course Code	Title of the Course
31563 /34063	BIG DATA ANALYTICS

Course Objective:

Able to understand the characteristics of Big Data.
To know about the link analysis, mapreduce and social networks

Course Requirement:

Basic Knowledge about data mining

Course Outcome:

Understood the importance of Big Data Analytics and social networks

Unit No.	Contents
	BLOCK 1 : INTRODUCTION TO BIG DATA
1	**Big Data Characteristics**- Types of Big Data- Traditional Versus Big Data Approach
2	**Technologies** Available for Big Data
3	**Hadoop** – Introduction - What is Hadoop? - Core Hadoop Components - Hadoop Ecosystem - Physical Architecture - Hadoop Limitations
	BLOCK 2 : MapReduce
4	MapReduce and The New Software Stack- MapReduce- Algorithms Using MapReduce -
5	Finding Similar Items – Introduction - Nearest Neighbor Search - Applications of Nearest Neighbor Search- Similarity of Documents
6	Collaborative Filtering as a Similar-Sets Problem - Recommendation Based on User Ratings- Distance Measures.
	BLOCK 3 : MINING DATA STREAMS
7	**Introduction**- Data Stream Management Systems- Data Stream Mining - Examples of Data Stream Applications -
8	**Stream Queries**- Issues in Data Stream Query Processing - Sampling in Data Streams - Filtering Streams - Counting Distinct Elements in a Stream
9	**Querying on Windows** − Counting Ones in a Window -Decaying Windows.
	BLOCK 4 : LINK ANALYSIS
10	**Introduction**- History of Search Engines and Spam –
11	**PageRank** - Efficient Computation of PageRank - Topic-Sensitive PageRank- Link Spam- Hubs and Authorities
12	**Recommendation Systems** : A Model for Recommendation Systems - Collaborative-Filtering System - Content-Based Recommendations - Mining Social Network Graphs .
	BLOCK 5 : SOCIAL NETWORKS

13	**Introduction** - Applications of Social Network Mining - Social Networks as a Graph - Types of Social Networks
14	**Social Graphs:** Clustering of Social Graphs - Direct Discovery of Communities in a Social Graph - SimRank-Counting Triangles in a Social Graph

Text Book:

1.“Big Data Analytics”, Radha Shankarmani, M Vijayalakshmi, Wiley Publications, first Edition, 2016

Books for Reference:

1. Seema Acharya, Subhashini Chellappan, “Big Data and Analytics”, Wiley Publication, first edition. Reprint in 2016
2. DT Editorial Services, “Black Book- Big Data (Covers Hadoop 2, MapReduce, Hive, Yarn, PIG, R, Data visualization)”, Dream tech Press edition 2016.

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Course Code	Title of the Course
31564 /34064	**MINI PROJECT**

Course Objective:

- To help the student to develop his/her ability to apply multi-disciplinary concepts, cutting edge technologies, software tools and techniques to solve programming problems.

Course Requirement:

- Basic concepts of Web and C/ Java /VB /ASP .NET programming

Course Outcome:

- Explore markup languages features and create interactive web pages
- Able to design front end web page and connect to the back end databases.
- Able to study, design, develop , implement and maintain software applications

Syllabi

Student should be able to design, develop and implement mini project based on latest cutting edge technologies using latest software.

Unit No	Contents
	BLOCK 1 : INTRODUCTION AND STUDY PHASE

1.	Define the Abstract of the project
2.	System Analysis :Identify the Problem (Aim & Objectives)
3.	Problem description
4.	Scope of the Problem
5.	Motivation
6.	Identify the existing system and its Limitations, Feasibility study
7.	Proposed system to overcome limitations
	BLOCK 2 : SYSTEM DESIGN
8.	System Design : Prepare Input / Output Design, Prepare Form Design
9.	Prepare software design (UML Diagram, Data Dictionary, Use case, Activity diagram, E-R diagram)
	BLOCK 3 : SYSTEM DEVELOPMENT
10.	System Development and Implementation the project
	BLOCK 4 : SYSTEM TESTING
11.	System Testing and maintenance of the project
	BLOCK 5 : DOCUMENTATION
12.	Prepare the mini project documentation.
13.	Sample Source Code
14.	Reports/Screen Layouts

Reference Books:

1. Grady Booch, Robert A.Maksimchuk et.al, Object Oriented Analysis and Design with applications, Pearson Education, 3rd Edition, 2009.
2. Ali Bahrami, Object Oriented System Development, Tata McGraw Hill Edition, 2008.
3. James Rumbaugh et.al, Object Oriented Modeling and Design, Addison Wesley, 2006.
4. Larman, Applying UML & Patterns, An Introduction to Object Oriented Analysis and Design, Pearson Education, 2nd Edition, 2003.
5. Martin Fowler, Kendall Scott , UML, Distilled Addison Wesley, 2004.
6. Ivar Jacobson Object Oriented Software Engineering: A Use Case Driven Approach, Addison wesley, 2004.

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MINUTES OF THE MEETING OF THE BOARD OF STUDIES FOR MASTER OF COMPUTER APPLICATIONS (M.C.A) PROGRAMME

Minutes of the Meeting of the Board of Studies in Computer Science for the Master of Computer Applications (M.C.A), M.Sc(Information Technology), M.Sc. (Computer Science), Post Graduate Diploma in Computer Applications (P.G.D.C.A), Bachelor of Computer Applications (B.C.A), B.Sc (Information Technology), B.Sc. (Computer Science) Programmes to be offered through Open Distance Learning (ODL) Mode held at The Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 04.09.2017, (11.00 A.M).

Members Present

1.	Dr. V. Palanisamy	-	Chairman
2.	Dr. E.Ramaraj	-	Member
3.	Dr. K.Kuppusamy	-	Member
4.	Dr. T.Meyyappan	-	Member
5.	Dr. S.S.Dhenakaran	-	Member
6.	Dr. K.Mahesh	-	Special Invitee
7.	Dr. A. Padmapriya	-	Special Invitee
8.	Dr. P. Prabhu	-	Member
9.	Mr.S.Balasubramanian	-	Member

After the deliberation and discussion the board resolved the following:

1. The Board considered the curriculum design and detailed syllabi of Computer Science programmes, prepared as per the norms and the Board scrutinized and necessary modifications are specified.

2. The Board resolved to approve curriculum design, detailed syllabi and other regulations for the Master of Computer Applications (M.C.A), M.Sc(Information Technology), M.Sc (Computer Science), Post Graduate Diploma in Computer Applications (P.G.D.C.A), Bachelor of Computer Applications (B.C.A), B.Sc (Information Technology), B.Sc. (Computer Science) programmes to be offered from 2018-2019 academic year onwards by the Directorate of Distance Education of Alagappa University, Karaikudi.

Dr. V. Palanisamy | Dr. E.Ramaraj | Dr. K.Kuppusamy

Dr. T.Meyyappan | Dr. S.S.Dhenakaran | Dr. K.Mahesh

Dr. A. Padmapriya | Dr. P. Prabhu | Mr.S.Balasubramanian

MASTER OF ARTS

(JOURNALISM AND MASS COMMUNICATION)

(DISTANCE EDUCATION PROGRAMME)

Curriculum (2017 onwards)

With

Credit Based System (CBS)

DIRECTORATE OF DISTANCE EDUCATION

Alagappa University

(A State University Accredited with A^{+} Grade among the Universities by NAAC (CGPA:3.64) in The Third Cycle)

Karaikudi- 630 003

2017

Master of Arts (M.A) – Journalism and Mass Communication
Credit Based System (CBS)
(With effective from June 2018 – 2019 onwards)

Sl.No.	Course Code No.	Name of the Course	CAI max	ESE max	Total max	C
I SEMESTER						
1	30911	Introduction to Journalism & Mass Communication	25	75	100	4
2	30912	Evolution of Media	25	75	100	4
3	30913	Photography	25	75	100	4
4	30914	Reporting & Editing	25	75	100	4
		TOTAL	**100**	**300**	**400**	**16**
II SEMESTER						
5	30921	Theories of Communication	25	75	100	4
6	30922	Advertising & Public Relations	25	75	100	4
7	30923	Audio Production	25	75	100	4
8	30924	Video Production	25	75	100	4
		TOTAL	**100**	**300**	**400**	**16**
III SEMESTER						
9	30931	Graphic Communication	25	75	100	4
10	39032	Communication Research Methods	25	75	100	4
11	39033	Media Laws and Ethics	25	75	100	4
12	39034	Internship(One Month)	25	75	100	4
		TOTAL	**100**	**300**	**400**	**16**
IV SEMESTER						
13	39041	Development Communication	25	75	100	4
14	39042	New Media Communication	25	75	100	4
15	39043	Corporate Communication	25	75	100	4
16	39044	Project Work / Dissertation	25	75	100	4
		TOTAL	**100**	**300**	**400**	**16**
		TOTAL CREDITS	**400**	**1200**	**1600**	**64**

Course Code Legend:

3	9	0	Y	Z

390-M.A. Journalism and Mass Communication

Y-Semester Number

Z- Course Number in the Semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C:Credit Points, MAX :Maximum.

a. Objectives of the Course

The focus of the course is to develop competent and qualified media professionals who will be an asset to the media industry. The programme provides students with both theoretical and practical exposure to media. The main areas that are covered in the course are Evolution of Media, Reporting and Editing, Advertising and Public Relations, Media Laws and Ethics, Audio and Video Production, Graphic Communication, New Media Communication. In addition, students are encouraged to contribute articles to, or report major events for, various media, during the course. Print journalism is the key focus of the course, along with electronic media. With this noble objective we have launched Master of Arts in Journalism and Mass Communication programme through distance mode.

The objectives of the programme include;

- To give an expanded knowledge about various kind of media.
- To gain vivid knowledge on Audio and Video Production.
- To understand concept of media and industry.
- To study about radio, television, satellite communication and internet.
- To understand about modern concepts like advertisement, public relation and culture.
- To educate the undergraduate students in the fascinating field.

b. Relevance of the Programme with Algappa University's Mission and Goals:

In order to align with the mission and goals of Alagappa University, Master of Arts (M.A) Journalism and Mass Communication Programme is planned to deliver in Distance Learning mode which may reach the maximum number of student aspirants who are unable to thrive to spend non-elastic timings of formal conventional class room education. Such a higher education in arts subject will enrich the human resources for the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

c. Nature of Prospective Target Group of Learners:

Master of Arts (M.A) Journalism and Mass Communication through Distance Learning mode is developed by keeping in mind to give an opportunity to economically and socially excluded people includes graduates of various socio-economic status viz., unemployed youths, employed with marginalized salary due to lack of sufficient knowledge in the subject mass communication. Also, the target group of learners includes various level employees of Media including Print and Electronic, Media related aspirants and Persons interested in Journalism.

d. Appropriateness of programme to be conducted in Distance Learning mode to acquire specific skills and competence:

Master of Arts (M.A) Journalism and Mass Communication through Distance Learning mode is developed in order to give subject-specific skills including i) knowledge about various kinds of media, their classification, i) Radio, Television, Advertisement, Function of mass communication, New Media communication, Verbal and Non- verbal communication ii) Inter-disciplinary knowledge like media law and ethics, iii) Modern concepts like reporting skills,

print journalism, iv) Application part of art of typography v) Development of electronic journalism. The programme is developed to give detailed exposure theoretically and free-hands experience on practical parts of the study in order to impart skills of mass communication and journalism to the learners.

FIRST SEMESTER

Course Code	Title of the Course
30911	**INTRODUCTION TO JOURNALISM AND MASS COMMUNICATION**

OBJECTIVES OF THE PAPER

- To explain the definitions of communication.
- To discuss the concept of Mass Media.
- To explain the concept of Journalism.
- To Know the various Indian news agencies

Possible Outcomes of the course:

- The course will give a clear ideas about Journalism and Mass Communication
- The Study of Mass media and various concept of Journalism.
- The course will make the learners knowing about communication, human communication skill and handling the various media device.

UNIT I

Communication: Definitions

Communication and Society

Types of Communication

UNIT II

Basic models of Communication

Linear model

Non-linear model

UNIT III

Communication, Culture and Media Literacy

Internet and Society

Impact on Politics

Education

Business and Culture.

UNIT IV

The human communication process

A review of some basic models and the ingredients.

UNIT V

The concept of Journalism

The functions of press

Press freedom

Responsibility and the theories of Press

UNIT VI

Current trends in Journalism

Press codes and ethics of Journalism

A code of ethics for the Indian Press.

UNIT VII

Press Commissions and Committees

The First and Second Press Commission reports

UNIT VIII

Reports of Chanda committee

Varghese Committee

Joshi committee

Karanth working group etc.

UNIT IX

Media for Mass Communication

Print media

Electronic media

Radio

Television and New Media

UNIT X

Prasar Baharati Bill

FM and Community Radio

DTH

Cable Revolution.

UNIT XI

Mass Media institutions in India

Government Media Units – Akashvani, Doordarshan, PIB, DAVP etc.

Press Registrar of India,

Press council of India

UNIT XII

Indian News agencies

Professional organizations as INS, AINEC, IFWJ, PRST, AAAI, ILNA etc.,

Media educational institutions.

UNIT XIII

Introduction to mass media

Meaning of mass media

The concept of Mass Media

The influence of technology on the means of communication

UNIT XIV

A discussion of the characteristics of individual mass medium

Mass Media in India

Present status of mass media

Advantages of mass media

Limitations of mass media

REFERENCE BOOKS:

1. Baran, Stanley J. Introduction to Mass Communication: Media Literacy and Culture. Boston: McGraw Hill, 2002.
2. McQuail, Denis. (1994). Mass Communication: An Introduction (2nd). Newbury Park, CA: Sage.
3. David K. Berlo, The Process of Communication, Holt Rhinehart and Winston, 1960. 4. Sean Macbride "Many Voices, One World".
4. Keval J.Kumar, Mass Communication in India, Vikas Publication,s New Delhi, 1994
5. John Comer and et. al, Communication Studies, Longman, London, 1981.

Course Code	Title of the Course
30912	**EVOLUTION OF MEDIA**

OBJECTIVES OF THE PAPER

- To understand the growth and functions of press in India after Independence
- To discuss the growth and development of Radio
- To learn the importance of folk media.
- To state and explain the origin and development of television

Possible Outcomes of the course:

- The course will give a clear idea about the history of press in India.
- The Study of folk media growth and its development.
- The course will make the learners knowing about the importance of News papers and magazines

UNIT I

Early communication systems in India

Development of printing

Early efforts to publish newspaper in different parts of India.

UNIT II

History of the print media and evolution and development of printing technology in India and World; various types of print media, nature, concepts, scope; Content making for print media, reach, advantages and importance.

UNIT III

Newspaper and Magazines in the nineteenth century

First war of Indian Independence and the press

Issues of freedom,

Political and press freedom.

UNIT IV

Invention and Development of Radio medium,

Radio Broadcasting since its inception in world and India,

Radio Technology,

Establishment of radio stations.

UNIT V

Television – origin and development, nature, scope, audience, genre, functions of television; Ownership pattern, debate on autonomy; BBC model of ownership, control and programming – commercial and public service, state and private sectors' expansion; potential for future development.

UNIT VI

The press in India after Independence

Social, political and economic issues

The role of Indian press problems

Prospects of Indian press

UNIT VII

Introduction to Radio

Growth and Development of Radio

AM and FM,

Vividh Bharati, Gyanvani, Community Radio

Satellite Ratio, Ham Operators

Evaluation of Content in Radio.

UNIT VIII

Introduction to Television

Growth and development of Television

Terrestrial and Satellite

Digital divide

UNIT IX

Introduction to Television

Evolution of content in Television,

Entertainment News

Public interaction

Commercialism

UNIT X

Introduction to Films

Early efforts

Film as a mass medium

Historical development of Indian films

Silent era of films

Talkies as a new approach in films

UNIT XI

Indian cinema after Independence

Parallel cinema and commercial cinema

Meaning and concept of documentaries

Issues and problems of Indian cinema.

UNIT XII

Folk media – Traditional media in India – regional diversity – content – form – character – utility – evaluation – future.

UNIT XIII

Exploring the New media

Internet

Growth and development of Internet communication

Online journalism

E-Publishing.

UNIT XIV

Definitions of Media and Mass Media;

Traditional Forms of Media

Signs, wood carving, Sound, drawings, sculptures

Folk Media – Songs, Drama, Puppet Shows, Story Telling etc.

REFERENCE BOOKS:

1. R.K.Ravindran, "Media in Development Arena", Indian Pub & Distributors 2000.
2. Straubhar, Larsoe, "Media Now", Thomson Wordsworth, 4th Edition,2004
3. KevalJ.Kumar, "Mass Communication in India", Jaico Publishing Co.2003.
4. J.K.Sharma, "Print Media and Electronic Media – Implications for the future", Authors Press, New Delhi, 2003.
5. The Evolution of Media Kindle Edition by Michael A. Noll (Author), Rowman & Littlefield Publishers, 2013

Course Code	Title of the Course
30913	PHOTOGRAPHY

OBJECTIVES OF THE PAPER

- To understand the different tools of photography
- To explain the reflection and refraction of life
- To define the various techniques in composition
- To understand the concept of photo journalism and colour photography

Possible Outcomes of the course:

- The course will give a clear ideas about different tools of photography
- The Study of the concept of photo journalism and colour photography
- The course will make the learners knowing about various techniques in composition

UNIT I

Photography- Meaning, Definition

History of Photography

Tools of Photography

Camera- Parts of Camera (Shutter, Apertures, Lens, Films)

Types of Camera.

UNIT II

Composition: Need for composing a picture, rule/ conventions of composition, elements of composition and their role/ relevance in communication a message.

UNIT III

Pin-hole, box, folding and DSLR, Large and Medium format cameras, Single lens reflex (SLR) and twin lens reflex (TLR).

UNIT IV

Principal focus and focal length, size of the image, speed and power of the lens, depth of field, angle of view and perspective.

UNIT V

Photographic Optics: Reflection and refraction of light, Dispersion of Light through a glass prism, Lenses, Different kinds of image formation

UNIT VI

Miniature, Sub-miniature and instant camera, Choice of camera and sizes, rising , falling, cross movements and swing back

UNIT VII

Lens, Aperture, Shutters, Various types and their functions, view-finders and focusing system.

UNIT VIII

Film chamber: Exposure counter, self timer, tripod stand, panning tilt head, lens hood, cable release, extension tubes and bellows, tele-converter and changing bag.

UNIT XI

Natural source- Natural Source, the sun, Nature and intensity of the sunlight at different times of the day, different weather condition types of light sources used

UNIT X

Artificial light sources- nature, intensity of different types of light sources used : Photo flood lamp, Halogen lamp, Barn doors and shoot, flash unit : Bulb flash and electronic flash.

UNIT XI

Photographing people, Portrait and still, wildlife, environment, sports, landscape, Industrial disasters.

UNIT XII

Photography for advertising

Conflicts war political

Social photography.

UNIT XIII

News values for pictures, photo essays- photo features: qualities for photo-journalism, picture magazines – colour photography,

UNIT XIV

Impact of technology

Practical

Field assignment and their evaluation.

REFERENCE BOOKS:

1. The encyclopedia of photography (3rd edn.) (1993) by Richard Zakia, LeatieStroebel, Focal Press, London.
2. The Manual of Photography (2000) by Ralph E Jacobson/Geoffrey G Attridge/Sidney F Ray, Focal Press
3. Ninth Edition. Understanding Digital Photography by Joseph A .Ippolito, Thomson Press, New Delhi, 2005.
4. The Manual of Photography (2000) by Ralph E Jacobson/Geoffrey G Attridge/Sidney F Ray, Focal Press,
5. The Photography Book, Jeffrey, Ian, Phaidon Press, London 200

Course Code	Title of the Course
30914	REPORTING & EDITING

OBJECTIVES OF THE PAPER

- To impart knowledge on news and reporting.
- To describe about the different types of reporting.
- To develop the knowledge on editing of news paper and magazine.
- To enhance the knowledge on the organization of the editorial department.

Possible Outcomes of the course:

- The course will give a clear ideas about different types of reporting.
- The Study of impart knowledge on news and reporting.
- The course will make the learners knowing about on the organization of the editorial department.

UNIT I

Meaning of Journalism

Definitions of Journalism

Functions and role of Journalism

UNIT II

Journalist,

Role of a Journalist,

Journalism ethics and standards,

Qualities of a Journalist.

UNIT III

Concept of news,

Definition of news,

Types of news.

UNIT IV

News Values,

Elements of news,

Functions of news,

News Sources.

UNIT V

Concept of reporting,

Types of reporting,

Reporting Skills

UNIT VI

Covering News beat,

Political reporting,

Business Reporting,

Sports Reporting,

Science & Technology reporting,

Education Reporting.

UNIT VII

Investigative and Interpretative reporting,

Development reporting,

Reporting with new technologies.

UNIT VIII

Journalistic Writing Techniques,

Writing Book review,

Film Review and Sports Review.

UNIT IX

Editing – meaning,

concept and significance,

contemporary trends in print journalism.

UNIT X

Introduction

Contemporary presentation styles

Editing of news paper and magazines.

UNIT XI

Structure of News Paper organization – Editorial, Management,

Production,

Circulation,

Marketing.

Human resource development

UNIT XII

Editing – principles,

Tools & techniques,

Role and function of copy desk,

Art of copy editing,

Steps and precautions in editing.

UNIT XIII

Difference between editing of newspapers,

Magazines & web editing,

Computerized editing.

UNIT XIV

Function and organization of the editorial department of a news paper, functions of Editor, Resident editor, Asst.editor, News editor, Chief sub- editor, Sub – editor, and Chief reporter etc.

REFERENCE BOOKS:

1. D.bruse, Newspaper writing and reporting for today's media(2001) Italy
2. Susan, News paper journalism (2004) Pape
3. Keeble,Richeard, News paper handbook(2001) Routledge, london
4. Sourin, Banergji, News Editing in Theory and Practice (2001) K.P Bagchi and Company, Calcutta.
5. Aggarwal, Vir Bala, Essentials of Practical Journalism,(2006) Concept Publishing Company , New Delhi.
6. Verma M.K News Reporting and Edition

SECOND SEMESTER

Course Code	Title of the Course
30921	THEORIES OF COMMUNICATION

OBJECTIVES OF THE PAPER

- To understand the Historical development of Media Theories
- To develop and understanding of Strength and Limitation of basic Theories of Communication
- To comprehend how Mass Communication theories are accepted the society.
- To critically evaluate theories as applied to practical Mass Communication problems.

Possible Outcomes of the course:

- The course will give a clear ideas about Historical development of Media Theories
- The Study of understanding of Strength and Limitation of basic Theories of Communication
- The course will make the learners knowing about critically evaluate theories as applied to practical Mass Communication problems.

UNIT I:

Introduction to Communication

Meaning of Communication

Importance of Communication

Features of Communication

Scope of Communication

UNIT II

Functionalist

Marxist

Critical Political Economy

Feministic perspectives

UNIT III

White's Gate – Keeping model,

Socialist, Culturalist,

Psychoanalytical

Behavioral Theories.

UNIT IV

Normative Theories: Authoritarian Media Theory,

Libertarian or free press Media theory

UNIT V

Psychological and Sociological Communication theories: Cultivation theory,

` Uses and Gratification theory,

Gestalt theory of motivation

UNIT VI

Cultural dimensions, mass media as cultural industry, cultural norms theory, effects theory, Source Theories, Connectional background, Agenta setting (Mc Comband Shaw) Merits and demerits.

UNIT VII

Message theories

Diffusion of Innovation,

Propaganda theory

UNIT VIII

Channel theories:

Gate keeping,

Technological determinism,

Social shaping of technology

UNIT IX

Audience Theories

Social Categories theory
Social Learning theory

UNIT X

Scope and functions of communication in the society-social aspects in shaping communication behavior.

UNIT XI

Media audiences

Public and public opinion
Mass media and public opinion
Media in society

UNIT XII

Introduction
Political Communication
Political Socialization
Communication Politics.

UNIT XIII

Introduction

Western and Eastern Perspectives
Differences
Western Theories in the Indian Context

UNIT XIV

Evolution of Communication

Theories in developing countries
Alternative communication
Indian experience
Participatory

REFERENCE BOOKS:

1. Theories of Communication A Short Introduction Armand Mattelart - University of Paris,France, Michèle Mattelart - University of Haute-Bretagne Sage, 1998
2. Blunder, J. and E. Katz, The Uses of Mass Communication. Thousand Oaks, CA: Sage, 1974.
3. Richard Collins, Media, Culture and Society, Sage Publication, New Delhi 1986
4. Keval J. Kumar, Mass Communication in India, Jaico Publishing Housing, Bombay, 1991
5. Duai R.& Manonmani .T, Culture and Communication: New Perspectives, Galgotia Publication, New Delhi, 1997.

Course Code	Title of the Course
30922	ADVERTISING & PUBLIC RELATIONS

OBJECTIVES OF THE PAPER

- To explain the perception of advertising in the marketing process.
- To understand the functions of advertising.
- To acquire the knowledge about structure of an advertisement.
- To realize the importance of advertising campaigns and
- To recognize the professional organizations in advertising.

Possible Outcomes of the course:

- The course will give a clear ideas about the perception of advertising in the marketing process.
- The Study of the knowledge about structure of an advertisement.
- The course will make the learners knowing about the professional organizations in advertising.

UNIT I

Introducation

Evolution and growth of advertising

Definitions of advertising

UNIT II

Introduction

Structure of an Advertisement

Advertisement Design

UNIT III

Introduction

Types of advertising

Advertising media.

UNIT IV

Visualization,

Headlines,

Body copy,

Visuals copy appeal etc.

UNIT V

Introduction

Copy writing techniques

Fundamentals of Arts in the layout

design

UNIT VI

Introduction

Professional organizations in advertising

The code for commercial advertising on AIR

UNIT VII

Introduction

The code of commercial advertising on Doordarshan

The code of advertising practice of the Advertising Standard Council of India.

UNIT VIII

Public Relations:

Definitions

Public Relations and publicity

UNIT IX

Nature and Scope of Public Relations.

Qualifications

Responsibilities of a Public Relations Officer.

UNIT X

Introduction

PR and Public Opinion

History and Development of PR in India

UNIT XI

Introduction

Role and functions of PR in management

PR Policy.

UNIT XII

Structure and functions of a PR,

Department in Government,

Public

Private Sectors

UNIT XIII

Introduction

Functions of a PRO

Media relations

Employee relations.

UNIT XIIV

PR Professional Organizations

New trends in PR

Ethics of Public relations

REFERENCE BOOKS:

1. Melvin L. Defluer, Everette .L Dennis : Understanding Mass Communication – A Liberal arts perspective – (Publishers – Houghton Mifflin Company – Boston, Toronto – year – 1994).
2. Keval J. Kumar : Mass Communication in India – (Publishers - Jaico Publishing house New Delhi – year -2002 – III Edition).
3. Reubean Ray : Communication Today – Understanding Creative skill (Publishers - Himalaya Publishing House, Mumbai – year – 1997).
4. Angela Wadia : Communication and Media – Studies in Ideas, Initiatives and institutions – (Publishers - Kanishka Publishers New Delhi – year – 1999).

5. Y.K.D'souza, : Communication today and tomorrow – (Publishers -Discovery Publishing House New Delhi. Year – 1999).
6. C.S. Rayadu : Media and Communication Management – (Publishers Himalaya Publishing House, Mumbai – year – 1998 – III Edition).
7. Larry L. Barker, Deborah L. Barker, : Communication –(Publishers - Allyn and Bacon, Massachutes – Year – 1993 Sixth Edition)

Course Code	Title of the Course
30923	AUDIO PRODUCTION

OBJECTIVES OF THE PAPER

- To Understand the developments and advances in radio journalism
- To acquire Knowledge in radio formats
- To gain vivid knowledge in news writing and presentation
- To be Competent in production management.

Possible Outcomes of the course:

- The course will give a clear ideas about Knowledge in radio formats
- The Study of vivid knowledge in news writing and presentation
- The course will make the learners knowing about Competent in production management.

UNIT I

History of Radio - Developments and Advances in Radio Journalism and techniques since inception – Radio in today's Media Scenario; Future of Radio. Introduction to acoustics; different kinds of studios vis-à-vis programme formats; varieties of Microphones; the broadcast chain;

UNIT II

Recording & Transmission systems; Modulation(Am & FM) Antennas, Receivers Amplifiers, High Fidelity systems; Multi-track recording technique; Stereo; Recording & Editing Consoles.

UNIT III

Radio Formats- Writing & Production skills vis-avis Diverse Formats; The spoken word/ Interviews/Discussions /Symposia – Radio plays / Radio Features & Documentaries/Music on Radio,

UNIT IV

Special Audience programmes on Radio- Programmes on Radio- Programme for Children, Women, Youth Senior citizens, Rural Folk, Industrial workers, Defense personnel.

UNIT V

News Writing and Presentation- Principals of News writing in a public service broadcasting organization, as contrasted with News in private Radio

UNIT VI

Principals of News presentation; News features; Professionals and Freelance stringers reporting for Radio; Disaster coverage News Bulletins.

UNIT VII

Production Management- Economic Production Management; Principles of Production Planning and Course of Production; Pre-production, Production and Post-production; Management of personnel Financial and Technical resources;

UNIT VIII

Budgetary planning and control-Direct and Indirect costs; Human resource Development; fixed variable factors in planning subject- research; conducive production conditions.

UNIT IX

Introduction

Innovative Developments In Radio Communication

Information Service Programmes on Radio

UNIT X

Introduction

Community Radio;

Local Radio;

Campus Radio;

Private FM Radio stations.

UNIT XI

Introduction

Application of Sound in Studio formats

Introduction to various Recording Softwares in the industry

UNIT XII

Introduction

Application of Sound recording in Radio Stations.

UNIT XIII

Sound spectrum and frequencies

History of recording and formats

UNIT XIV

Digital technology and advancements - Creating files and naming them – Recording – Using EQ, DYNMS, SFX, and Reverb – Source Selection – Cleaning and Editing - Elements of music recording - Mixing Pre and Post Mastering

REFERENCE BOOKS:

1. Louie Tabing, “How to do community radio” UNESCO, 2002.
2. Carole Fleming, “The Radio Handbook”, 2nd edition, Routledge, 2002.
3. Jan Maes and March Vereammen “Digital Audio Technology”, 4th Edition Focal Press, 2001.
4. Ministry of Information and Broadcasting, Govt. of India, Annual Report 2002-03.
5. All India Radio, Audience Research Unit, Prasar Bharat,2002

Course Code	Title of the Course
30924	VIDEO PRODUCTION

OBJECTIVES OF THE PAPER

- To Understand the preproduction planning stage
- To acquire Knowledge in shooting
- To Understand the importance of post production stage
- To be Competent in technical areas.

Possible Outcomes of the course:

- The course will give a clear ideas about Knowledge in shooting
- The Study of Understand the importance of post production stage
- The course will make the learners knowing about Competent in technical areas.

UNIT I

Preproduction Planning Stage - Concept –content – research-the basic script – the role of writer for television-budget – logistics- Crew – Location survey- Talents,

UNIT II

Roles of the production crew like the Producer, Production assistant, Cameraman, and the Studio crew both production and technical and other outdoor crew- who is who,

UNIT III

For studio and outdoor shows the set design backdrop and properties to be used, the role of the art director or the set designer. Props-Wardrobe-Makeup.

UNIT IV

Shooting Stage- Camera equipment and accessories – Shooting techniques – composition and framing types of shots- taking notes – writing the dope sheets- Shooting techniques used for News, Sports and business reporting,

UNIT V

Creative affairs programmes. Documentaries, features, Live-shows, event shows, Creative productions like serials, audio plays and outdoor serials, advertisement films, music videos, new formats etc.

UNIT VI

Technical inputs equipment required for various shows including lightning etc and Crew required for the various shows. Shooting techniques for the mega and the live shows and live News.

UNIT VII

Post Production Stage- Editing –linear-Nonlinear Equipment – preparation of edit-list, Use of the Dope sheet, editing schedule – Editing of the programme identify the editor, special effects, writing for the programme, recording the audio,

UNIT VIII

Use of Voice-over for the documentary- musical score recording and using the music lying of the tracks. Computer graphics (titling etc) – meet the deadlines.

UNIT IX

Management

Professional practice

Management

Legal issues

Ethics

UNIT X

Technology -Technical areas- Equipment-Formats-Maintenance-Satellite TV- DTH-Outside Broadcast- Live links-Uplink, Downlink, Latest Technology-HDTV

UNIT XI

Single-camera shooting, Continuous single –camera shooting, Discontinuous single camera shooting, segmented shooting. Multi-camera treatment – Visual variety, Shot organisation, Subjective and objective approaches, Focusing audience attention, Creating tension, Pace, Timing and Visual clarity.

UNIT XII

Visual effects-Mirror effects, Electronic effects, Chromo key techniques and Digital video effects. Timecode sync and color sync, remote controlling.

UNIT XIII

Studio Floor, Cameras on Pedestals, Microphones, Lighting rigs and the controlling Equipments, the role of Floor Manager, Production Control Room, Vision Control and Sound Control, Video Monitor, Vision Mixer, Audio Mixing console, Character Generator Digital Video Effects, VTR, and Talk Back.

UNIT XIV

Different video recording formats – Tapes: (VHS,Beta, Digi-Beta,HDV, DVCam, U-Matic) Tapeless: DVD, Optical and Blue ray Disks, Compact Flash Cards and Solid State cards, Hard-disk based recording etc.

REFERENCE BOOKS:

1. Gerald Millerson, Video Production hand Book, 3rd Edition, Focal Press, 2002.
2. Peter W.Rea& David K.Irving, Producing & Directing the Short Film and Video, 2nd Edition, Focal Press, 2001.
3. Deslyver& Graham Swainson, Basics of Video Production, 2nd Edition, Focal Press, 2001
4. Deslyver& Graham Swainson, Basics of Video Lighting, Focal Press, 2003.
5. Vasuki Belavadi, Video Production Second Edition, Published by Oxford university, 2017

SECOND YEAR

THIRD SEMESTER

Course Code	Title of the Course
30931	**GRAPHIC COMMUNICATION**

OBJECTIVES OF THE PAPER

- To Understand the concepts and significance of graphic communication
- To Understand the functions of a good design;
- To realize the opportunities and challenges of Graphic Communication.
- To Acquire Sound Knowledge in Publication design

Possible Outcomes of the course:

- The course will give a clear ideas about the concepts and significance of graphic communication
- The Study of the opportunities and challenges of Graphic Communication.
- The course will make the learners knowing about Acquire Sound Knowledge in Publication design

UNIT I

Design – definition & fundamentals – purpose & functions of a good design – principles of design – design decisions – graphic communication –

UNIT II

definition, nature & scope, design process – layout stages & types – appropriate visual structure – shaping media architecture – modern design – opportunities and challenges.

UNIT III

Introduction

Basic components of design

visuals,

text,

graphics and

colour, typography

UNIT IV

Definition,

principles & significance,

visuals

categories, c

riteria for selection,

editing pictures,

UNIT V

Photography & designing

ethical issues

colour basics, color theories, colour psychology,

importance of colour in designing.

UNIT VI

Publication design

name plate,

master pages, templates, style sheets

dummying process role of computers in designing – quark xpress

UNIT VII

Architectural components of newspapers and magazines,

formats & page make – up,

front page, inside page,

editorial & opinion pages,

UNIT VIII

Life styles & feature pages,

food & fashion, entertainment,

business & classifieds, designing special & regular sections,

book design.

UNIT IX

Designing for public relations – newsletters.

Letterhead & logo design,

identify & collateral materials, product & packaging,

UNIT X

Hospitality materials & branding,

business correspondence material,

promotional material, advertising design, poster design.

UNIT XI

Graphics input - output devices:

Direct input devices - Cursor devices

Direct screen interaction - logical input.

Line drawing displays - raster scan displays.

UNIT XII

Two dimensional graphics. Raster graphics - Scan conversion of polygons - region filling - algorithms. File formats – GIF, JPEG, TIFF, Graphics Animation Files, Postscript/Encapsulated Postscript files.

UNIT XIII

Curves and surfaces: Parametric representation of curves - parametric representation of surfaces - planes - curved surfaces - ruled surfaces.

UNIT XIV

Three dimensional graphics: 3D transformations - normal, oblique central projections - 3D algorithms - hidden lines and hidden surfaces removal. Lighting, perception and depth of field.

REFERENCE BOOKS:

1. William Ryan, Thonsar Delma Learning, “Graphic Communication Today”, IV Edition, 2009.
2. Yolanda Zappaperra, Rotovision,2002 , “Editorial Design for Print & Electronic Media”
3. Chris Forst , II Edition, “Designing for Newspapers & Magazines”, Rouledge 2003
4. Traditional and Digital Techniques for Graphic Communication , Paul Lase, 2000
5. Notes on Graphic Design and Visual Communication, Gregg Berryman, Crisp Publication, 1990

Course Code	Title of the Course
30932	COMMUNICATION RESEARCH METHODS

OBJECTIVES OF THE PAPER

- To Understand the concept and significance of Research
- To acquire the knowledge about various Research Designs
- To explain the different methods of Communication Research
- To master the students in Research Report writing

Possible Outcomes of the course:

- The course will give a clear ideas about the concept and significance of Research
- The Study of the different methods of Communication Research
- The course will make the learners knowing about various Research Designs

UNIT I

Definition

elements of research

scientific approach

research and communication theories

UNIT II

Role and function

scope and importance of communication research

basic and applied research.

UNIT III

Research design components

experimental,

quasi-experimental, bench mark,

longitudinal studies – simulation

panel studies –co-relational designs.

UNIT IV

Methods of communication research

census method, survey method,

observation method – clinical studies

case studies – content analysis.

UNIT V

Tools of data collection: sources, media source books, questionnaire and schedules, people's meter, diary method, field studies, logistic groups, focus groups, telephone, surveys, and online polls.

UNIT VI

Random sampling methods and representativeness of the samples,

sampling errors

distributions in the findings.

UNIT VII

Report writing

data analysis techniques

coding and tabulation – non-statistical methods

descriptive – historical

UNIT VIII

Statistical analysis

parametric and non-parametric

uni - variate – bi-variate – multi-variate

UNIT IX

Tests of significance

levels of measurement

central tendency – tests of reliability and validity

SPSS and other statistical packages.

UNIT X

Media research as a tool of reporting.

Readership and / audience surveys,

preparation of research reports / project reports / dissertations / theses.

Ethical perspectives of mass media research.

UNIT XI

Sampling in communications Research, Types, their applications and limitations. Methods of data Collection: Interview, questionnaire, observation and case study applications and limitations of different methods.

UNIT XII

Use of statistics in communication research, Basic statistical tools: measures of central tendency (mean mode and medium): measures of dispersion (standard deviation), correlation and chi square.

UNIT XIII

Data processing, Analysis,
presentation and interpretation of data.
Use of graphics in data presentation.

UNIT XIV

Writing a research proposal
writing research report
Components and style.

REFERENCE BOOKS:

1. Arthur Asa Berger, "Media Research Techniques", Sage Publications, New Delhi
2. Roger D.Wimmer, Mass Media Research
3. Wrench.et al. Qualitative Research Methods for Communication, Oxford University Press.
4. Media and Communication Research methods, Arthur asa Bergur, San Feansisco State University , USA, 2016
5. Mass Communication Research Methods, Volume 1 Anders Hansen, SAGE, 2009

Course Code	Title of the Course
30933	MEDIA LAWS AND ETHICS

OBJECTIVES OF THE PAPER

- To Understand the concept and significance of Media Laws in India
- To Acquire Sound Knowledge on Press Laws In India.
- To Understand the Architectural components of Civil And Criminal Laws:
- To attain the Knowledge about Role and responsibilities of professional bodies

Possible Outcomes of the course:

- The course will give a clear ideas about the concept and significance of Media Laws in India
- The Study of the Knowledge about Role and responsibilities of professional bodies
- The course will make the learners knowing about the Architectural components of Civil And Criminal Laws

UNIT I

History Perspective of the Media Laws in India- Constitution and Media: Fundamentals Rights, Freedom of Speech and Expression, Directive principles of state policy; Powers and Privileges of the Parliament / State legislative assemblies.

UNIT II

Press Laws in India

Definition of contempt of court

Intellectual property rights

Trademark

Patently and copy right

UNIT III

Parliamentary privileges- books and registration act- working journalist act- press council of India- press commissions of India- official secrets act.

UNIT IV

Press censorship, Right to information, Code of ethics for media professionals, the role of press council of India and other professional councils.

UNIT V

Civil And Criminal Laws: Civil and Criminal Law of Defamation; Relevant provisions of the Indian Penal Code with reference to sedition, obscenity, crime against women, children etc.,;

UNIT VI

Laws dealing with obscenity,

Laws and constitutional provisions pertaining to Human Rights in India.

UNIT VII

Cinematograph Act, 1952; Prasar Bharathi Act; Committees on Broadcasting and Information Media; Broadcasting regulations – key issues; GATT and Intellectual property right legislations;

UNIT VIII

Cyber Laws in India: Need for cyber-laws, nature and scope of cyber laws, approaches to cyber laws, cyber – crimes, piracy, Convergence bill, Information Technology Legislation.

UNIT IX

Ethics**,**

Role and responsibilities of professional bodies;

Themes and issues in Media Ethics:

Principles and ethics of Journalistic conduct;

UNIT X

Comparison of ethical norms;

Codes for radio, television and advertising;

Case studies in media ethics and major ethical violations.

UNIT XI

Domestic violence act- tabloid and yellow journalism – violence and brutality – reporting during special sensitive situations- ethical construes in investigative journalism.

UNIT XII

Law of copyright,

major copyright issues/cases,

WIPO,

piracy of media products and the IT Act 2001.

UNIT XIII

The limits of the right to know – journalism ethics and patriotism- new roles of journalism and public opinion – journalist code of conduct – broadcast content complaints council.

UNIT XIV

Codes of ethics for media professionals, role of press council of India and other professional councils and case studies of major ethical violations by the Indian media.

REFERENCE BOOKS:

1. Basu, "Law of the Press in India", Prentice Hall of India,2003
2. Basu, "Introduction to Indian Constitution", Prentice Hall of India,2003
3. Hameling, Cess, "Ethics of Cyber-space", Sage Publications,2001
4. Day, E Ethics in Media Communications: Cases and Controversies, Thomson Learning 2000
5. Leslie, "Mass Communication Ethics", Thomson Learning, 2000.

Course Code	Title of the Course
30934	**INTERNSHIP (ONE MONTH)**

Students should go for an Internship for 4-5 weeks (one month), after the second semester before the end of third semester to any media organization with the dual approval of the University and submit the report with the work diary in the III semester

Scheme of Marks

Work Diary	- 25 Marks
Report	- 50 Marks
Viva Voce	- 25 Marks
	100 Marks

Viva Voce Examination will be conducted at the end of III Semester

FOURTH SEMESTER

Course Code	Title of the Course
30941	**DEVELOPMENT COMMUNICATION**

OBJECTIVES OF THE PAPER

- Understand the concepts and Communication – Definition, scope, forms and purpose; Types of Communication
- Knowledgeable in New communication technologies
- Thorough in Theories of Communication
- Competent in Media systems and theories,

Possible Outcomes of the course:

- The course will give a clear ideas about Knowledgeable in New communication technologies
- The Study of Theories of Communication
- The course will make the learners knowing about Competent in Media systems and theories

UNIT I

Development: Concept – Dynamics of development – Development issues – Development indicators – Dysfunctions of development – Communication perspective on development

UNIT II

Role of Communication in Development:

Development motivation,

Development participation

Approaches to Development Communication.

UNIT III

Dominant paradigm of Development:

Evolutionary model

Psychological variable model

UNIT IV

Cultural factors model

Economic growth model

Industrialization approach –

The critique of the above models.

UNIT V

Introduction

Communication approaches of Dominant paradigm:

Powerful effects model of mass media

UNIT VI

Diffusion of Innovations

Mass media in modernization

The critique of above models.

UNIT VII

Alternative paradigms of Development and development communication:

Basic needs programme

Integrated Development

Intermediate technology

UNIT VIII

Self Development – Self reliance – Popular participation – New communication technologies – Traditional media use – Development support communication.

UNIT IX

Historical analysis of India's Development:

Gandhi Metha model,

Elawath experiment,

Nilokheri experiment

UNIT X

Five Year Plans, Models of Experimental Project: Rural Television – SITE, Kheda, Communications Project, Radio Rural Forum.

UNIT XI

Role of mass media organizations in Development Communication, Newspaper, Radio, TV, Traditional Media, PIB, DAVP, Song and Drama Division etc., Strategies of Development Communication, Role of NGOs in Development.

UNIT XII

` Development support Communication in Agriculture, Health and Family Welfare, Education and Literacy, Environment, Women Empowerment, Poverty and Employment.

UNIT XIII

Case Studies On:

a) Development Communication Experiences
b) Role of NGOs in Development

UNIT XIV

Case Studies On:
c) Application of Development support Communication in Agriculture / Health and Family Welfare / Literacy.

REFERENCE BOOKS:

1. Mozammel, Mazud. "Development Communication: Challenges in an Empowered Information Environment". Retrieved 28 August 2012.
2. Mefalopulos, Paolo (2008). Development Communication Sourcebook: Broadening the Boundaries of Communication. Washington ".
3. McPhail, Thomas. (2009). Development communication: Reframing the role of the media. London, UK: Wiley-Blackwell
4. Flor, Alexander G. (1995). Development Communication Praxis. University of the Philippines Open University.
5. Flor, Alexander; Ongkiko, Ila Virginia (2006). Introduction to Development Communication.

Course Code	Title of the Course
30942	**NEW MEDIA COMMUNICATION**

OBJECTIVES OF THE PAPER

- To explain the meaning of New Media Communication
- To understand the various trends in New Media.
- To acquire the knowledge about Cyber Journalism.
- To realize the importance of Online Editing

Possible Outcomes of the course:

- The course will give a clear ideas about the knowledge about Cyber Journalism.

- The Study of the various trends in New Media.
- The course will make the learners knowing about the importance of Online Editing

UNIT I

Communication Technology (CT): concept and scope, CT and IT: similarities and differences – telephony – electronic digital exchange, C-Dot - Pagers, Cellular Telephone.

UNIT II

Internet: LAN, MAN, WAN, E-mail, Web, Ownership and administration of Internet, ISPs, WAP, types of Internet connections: Dial-up, ISDN, lease-line.

UNIT III

Optical fibre:
structure,
advantage and application;
protocols of Internet: SLIP, CSLIP, TCP/ IP, PPP

UNIT IV

WEB PAGE, Websites, Homepages.
Introduction to HTTP, HTML, ELP, DNS, JAVA;
browsing and browsers, bookmarks

UNIT V

Searching: through directory search engine, s

earch resources; video conferencing and telephony,

e-commerce: m-commerce, buying, selling, banking, and advertising on Internet.

UNIT VI

Web page development, inserting, linking; editing,

publishing, locating,

promoting and maintaining a website

UNIT VII

Cyber Journalism: On-line editions of newspapers
management and economics;
cyber newspapers-creation, feed, marketing, revenue and expenditure

UNIT VIII

Online editing,

e-publishing; security issues on Internet;

social, political, legal and ethical issues related IT and CT.

UNIT IX

Social and Cultural effects of new Media:

Social Networking,

Information Overload,

Information Rich and Information Poor,

UNIT X

Knowledge Gap and Cultural Alienation New media impact on old media

ICTs for Development

Empowerment,

right to information

UNIT XI

Connotation, Denotation, Reading Comprehension - Reading between Lines – Listening for cues – Arguing skills – Negotiating skills – Introducing a chief guest- Introducing a Programme – Summarizing - Evaluative & Analytical Writing

UNIT XII

Foreign words in English – Tense forms – Participle, Perfect – Reading Reviews – Listening to interpret & analyze – Presenting and marketing a product – Scene description – Writing recommendations - Writing a news report – Group created written reports giving instruction on various aspects of target vocabulary.

UNIT XIII

British/American English – Media related Vocabulary – Understanding Schedules – Listening to interviews & Dialogues – Role plays in various authentic situations – Conducting interviews – Organizing a programme – Job Application with CV (with Cover letter)

UNIT XIV

Prepositional phrases – Active & Passive – Extensive Reading – Novels & Plays – Listening for data collection – Evaluating problems and giving suggestions – Giving Directions – Oral & Written - Creative Writing – Using online resources to extract authentic materials on specific areas of interest.

REFERENCE BOOKS:

1. Global Communication in Transition: The end of diversity (1996), HamidMowlana, Sage Publication, Newbury Park.
2. Global information and World Communication (2nd edition) – (1997) Hamid Mowlana Sage Publications, New Delhi.
3. World Communication Report: The media and the challenge of the new technologies (1997) - Ed. Alaine Modouz, UNESCO Publishing.
4. New Media – Ronald Rice, (1984) Sage Publications
5. Public Communication Campaigns, Ronald E.Rice, Charles K.Atkin, Sage Publications,2012

Course Code	Title of the Course
30943	**CORPORATE COMMUNICATION**

OBJECTIVES OF THE PAPER

- To explore the scope and functions of Corporate Communication
- To understand the Crisis and Disaster Communication Management
- To acquire the Knowledge on Corporate Social Responsibility
- To understand the Importance of Organizational Communication, Business Communication

Possible Outcomes of the course:

- The course will give a clear ideas about scope and functions of Corporate Communication
- The Study of the Importance of Organizational Communication, Business Communication
- The course will make the learners knowing about the Crisis and Disaster Communication Management

UNIT – I

Definition,

scope

functions of Corporate Communication

UNIT – II

Corporate Culture

Citizenship

Philanthropy

International Communication

UNIT – III

Corporate Identity Philosophy/Image Building – Event & Conference Organization & management

UNIT – IV

Introduction of Image,

Event

Conference Management

UNIT – V

Counselling,

Crisis and Disaster Communication Management

Functions

UNIT – VI

Art of persuasion,

feedback,

campaign planning and strategies

UNIT – VII

Advertising and the marketing communication environment,

customer care,

strategic planning and campaign management

UNIT – VIII

Communication Strategy – 4 P's

UNIT – IX

Corporate Social Responsibility

UNIT – X

Corporate Crisis and Conflict Situation

UNIT – XI

Social Media and Corporate Communication

UNIT – XII

Employee Communication & Media Relations,

UNIT – XIII

Organizational Communication, Business Communication

UNIT– XIV

Case study of Corporate Crisis Management

REFERENCE BOOKS:

1. Ananthakrishnan, M. (2011) Impact of Corporate Communication on Internal Public – A Case Study of Hindustan Aeronautics Ltd., Unpublished Ph.D., Thesis, University of Mysore, Karnataka, India.

2. Andrews, P. H. and Bird, J. E. (1989) Communication for Business and the Professions, Dubuque, Wim C. Brown Publishers, Iowa, U.S.A.

3. Banerjee, A. K. (2009) Art of Corporate Communication, Lotus Press.

4. Prakash, Jagadeesh (2007) Corporate Communication Practices in Public Sector: A Case Study of Karnataka Power Corporation Limited, Ph.D., Thesis, Bangalore University, Bangalore, Karnataka, India.

5. Cornelissen, J., Bekkum, T., Van, and Ruler, B., Van, (2006) Corporate Communication: A Practice-Based Theoretical Conceptualization, Corporate Reputation

Course Code	Title of the Course
30944	PROJECT WORK

Guidelines for Project work / Dissertation

1. Conceptualization of subject And Research Problem	:	15 Marks
Review of Literature	:	10 Marks
2. 8Presentation of Methodology	:	20 Marks
3. Data Analysis & Dissertation	:	20 Marks
4. Final Draft & Presentation	:	10 Marks
5. Viva –voce	:	25 Marks
Total		100 Marks

Chapterisation

1. Introduction
2. Review & Related Literature
3. Research Methodology
4. Analysis and Interpretation
5. Discussion & Conclusion
6. Bibliography
7. Appendices

Duration of the Programme:

The programme for the degree of Master of Journalism and Mass Communication (MJMC) shall consist of two academic years divided into four semesters. Each semester consists of four papers including internship in third semester and fourth semester one project work. Each course carry 4 credits each. Each semester consists of 16 credits and the programme consists of 64 credits in total.

Faculty and Support Staff Requirements:

PG - Non-Science Programmes

Staff Category	Required
Core Faculty	3
Faculty - Specialization	2
Clerical Assistant	1

Instructional Delivery Mechanisms:

The instructional delivery mechanisms of the programme includes SLM – study materials, face to face contact session and e-content of the study materials in the form of CD will be supplied.

Student Support Services:

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like counseling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issuance of Identity Card, Study materials, etc. will be routed through the Learning Centres. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter only. The conduct of end semester examinations, evaluation and issuance of certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

Procedure for Admission, curriculum transaction and evaluation:

1. Procedure for Admission:

A candidate who has passed a bachelor, degree in any discipline (10 + 2+ 3 system) of the University or an Examination of any other University accepted by the Syndicate as equivalent thereto shall be eligible for the Master of Journalism and Mass Communication (MJMC) of this University.

2. Curriculum Transactions:

The classroom teaching would be through chalk and talk method, use of OHP, Power Point presentations, web-based lessons, animated videos, etc. The face to face contact sessions would be such that the student should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

For practical courses exclusive study materials containing the requirements, procedure for the experiments will be issued to the learners. In the laboratory, instruction would be given for the experiments followed by demonstration and finally the students have to do the experiments individually.

The face to face contact sessions will be conducted in following durations;

Course Type	Face to Face Contact Session per Semester (in Hours)
Theory Courses (4 courses with 4 credits each)	64
Total	**64**

3. Evaluation:

The evaluation shall be conducted separately for theory and practical to assess the knowledge acquired during the course of study. There shall be two systems of evaluation viz., internal assessment and end semester examinations.

In the case of theory courses the internal evaluation shall be conducted as Continuous Internal Assessment via. Student's assignment preparation, Seminar, etc. The internal assessment shall comprise of maximum 25 marks for each subject. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

In the case of Practical courses for Master of Arts (M.A) Journalism and Mass Communication Programme should be conducted at end semester. The internal evaluation will be based on Internship/Work diary, Project work/ Dissertation and viva voce. The external evaluation consists of an end semester practical examinations which comprise of 75 marks for

each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

3.1. Question Paper Pattern:

Answer all questions (one question from each unit with internal choices Time: 3 Hours Max. Marks: 75

Part A- 10 x 2 Marks = 20 Marks

Part B -5 x 5 Marks = 25 Marks

Part C- 3 x 10 Marks = 30 Marks

3.2. Passing Minimum:

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50% (50 marks out of 100 marks) in the University end semester examination. Then the total marks secured by the candidate will be taken and add with his/her internal marks (Maximum marks 25).

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

3.3. Marks and Grades:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester

G_i = Grade Point obtained for course i in any semester.

n refers to the semester in which such courses were credited

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

Grade Point Average = $\frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = $\frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme.}}$

CGPA	Grad	Classification of Final Result
9.5-10.0	O+	First Class- Exemplary*
9.0 and above but below 9.5	O	
8.5 and above but below 9.0	D++	First Class with Distinction*
8.0 and above but below 8.5	D+	
7.5 and above but below 8.0	D	
7.0 and above but below 7.5	A++	First Class
6.5 and above but below 7.0	A+	
6.0 and above but below 6.5	A	
5.5 and above but below 6.0	B+	Second Class
5.0 and above but below 5.5	B	
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

4. Maximum duration for the completion of the course:

The maximum duration for completion of Degree in Master of Journalism and Mass Communication (MJMC) programme shall not exceed ten semesters from their fourth semester.

5. Commencement of this Regulation:

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter.

6. Fee Structure:

The programme has the following Fee Structure:

Sl. No.	Fees Detail	Amount in Rs.	
		First Year	Second Year
1	Admission Processing Fees	**300.00**	**-**
2	Course Fees	**5000.00**	**5000.00**
5	ICT Fees	**150.00**	**150.00**
	TOTAL	**5,450.00**	**5,150.00**

The above mentioned fee structure is exclusive of Exam fees.

Requirement of the laboratory support and Library Resources:

Directorate of Distance Education, Alagappa University, Karaikudi housing an exclusive Library facility with adequate number of copies of books in relevant titles for Master of Journalism and Mass Communication (MJMC). The Central Library of Alagappa University also having good source of reference books. The books available at both the libraries are only for reference purpose and not for lending services.

Cost estimate of the programme and the provisions:

The cost estimate of the programme and provisions for the fund to meet out the expenditure will be as per the university norms.

Quality assurance mechanism and expected programme outcomes:

The quality of the programme depends on scientific construction of the curriculum, strong-enough syllabi, sincere efforts leading to skilful execution of the course of the study. The ultimate achievement of M.A programme of study may reflect the gaining of knowledge and skill in the subject. And all these gaining of knowledge may in journalism may helps the students to get new job opportunities, upgrading in their position not only in employment but also in the society, make students feel thirsty to achieve in research in the fields associated with the discipline.

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations. Apart from the end semester examination-based review feedback from the alumni, students, parents and employers will be received and analyzed for the further improvement of the quality of the Master of Arts (M.A) Journalism and Mass Communication Programme.

Minutes of the Meeting of the Board of Studies of M.A in Journalism and Mass Communication Programme to be offered through ODL Mode held at the Directorate of Distance Education , Alagappa Universitry, Kraikudi – 630003 on 04.09.2017 at 11.00 A.M

Members Present

1. Dr.S.Nagarathinam - Chairman
2. Dr.B.Radha - Member
3. Dr.C.Jaisanker - Member
4. Dr.R.Rajan - Member
5. Dr.N.Johnson - Member

The modified syllabi of M.A in Journalism and Mass Communication programme was scrutinized and discussed in the meeting. The board has resolved the following:

The board has unanimously accepted the curriculum design and modified syllabi of M.A in Journalism and Mass Communication programme are prepared as per the norms

The board has approved the Regulations, Curriculum and Syllabi for M.A in Journalism and Mass Communication programme to be offered by the Directorate of Distance Education, Alagappa University

The Approved Syllabi are provided in the Annexure- I (M.A in Journalism and Mass Communication)

4/9/17 (S.NAGARATHINAM) (B.RADHA) 4/9/17 (C.JAISANKAER)

4/9/17 (R.RAJAN) 4/9/17 (N.JOHNSON)

PROGRAM PROJECT REPORT

MASTER OF LIBRARY AND INFORMATION SCIENCE (M.Lib.I.Sc)

Directorate of Distance Education

Alagappa University

Karaikudi

(a) Programme Mission and Objectives

- To create awareness of the evolution of knowledge society and its role in the social transformation and economic prosperity of the nation
- To analyse the complex issues of the access and use of knowledge and its productive utility in the social development
- Special coaching classes are arranged periodically for the benefit of rural students
- To create an understanding about the methods, techniques, skills as well as approaches in the information processing and management
- To give the students an understanding of the basic principles of fundamental laws of Library and Information Science and to enable them to understand, appreciate and develop professionalism to work in contemporary "Information Age".
- to impart the students a thorough understanding of patterns of knowledge development and its organization;
- to train the students in the basics of professional skills for information or knowledge management, so that they serve the society through an institution of library and information centre
- to train the students in the analysis, repackaging, marketing,planning and management of the systems of library and information centers;
- to provide the students thorough understanding of I T applications in information environment including networks and communication systems;
- to make students fully aware of various sources of information and to train them in techniques of dissemination of information in the context of different user groups.
- to train the students in the advanced skills of information/ knowledge, gathering, processing, organization and retrieval;
- to train students in the techniques of Information Management and equip them with the application of Information Technologies (IT) in libraries and information centers
- to provide an understanding of research methods and activities of research organizations.
- to acquaint the students with the development of the Universe of knowledge and methods of its organization in a library system

(b) Relevance of the Programme with HEI's Mission and Goals

TheVision of the HEI is "Achieving Excellence in all spheres of Education, with particular emphasis on PEARL - Pedagogy, Extension, Administration, Research and Learning". This course M.Lib.I.Sc is offered in relation to the Vision.

Also, the Mission, namely, "Affording a High Quality Higher Education to the learners so that they are transformed into intellectually competent human resources that will help in the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM)", is adhered to in the course.

(c) Nature of Prospective target group of learners

A Candidate with B.L.I.Sc degree or a Equivalent Degree from a recognised University shall be eligible for M.L.I.Sc. Also, the target group of learners includes various level employees of educational institutions, secondary –level school teachers, research aspirants, women taking care of family –the important unit of the community, etc.,

(d) Appropriateness of programme to be conducted in Open and Distance Learning modeto acquire specific skills and competence

It makes the students fully aware of various sources of information and trains them in techniques of dissemination of information in the context of different user groups.

It gives the students an understanding of the basic principles of fundamental laws of Library and Information Science and enables them to understand, appreciate and develop professionalism to work in contemporary "Information Age".

It trains the students in the analysis, planning and management of the systems of library and information centers.

It provides the students thorough understanding of IT applications in information environment including networks and communication systems.

(e) Instructional Design

i. Curriculum Design

The Open University system is more learner-oriented, and is geared to cater to the needs of motivated students assuming that the students is an active participant in the teaching-learning process.

Face to Face Contact Programme by University Professors/ experienced professionals.

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
FIRST YEAR						
I Semester						
1.	**32311**	Information Processing and Retrieval	25	75	100	4
2.	**32312**	Library and Information System Management	25	75	100	4
3.	**32313**	Information Technology and information Systems	25	75	100	4
4.	**32314**	Information Technology (Practice)	25	75	100	4
		Total	**100**	**300**	**400**	**16**

II Semester						
5.	**32321**	Academic Library System	25	75	100	4
6.	**32322**	Technical Writing	25	75	100	4
7.	**32323**	Research Methodology	25	75	100	4
8.	**32324**	Information Processing and Retrieval (Practice)	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		Grand Total	**200**	**600**	**800**	**32**

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

No. of Credits per Course (Theory) -4

No. of Credits per Course (Practical) - 4

Total No. of Credits per Semester- 16

Total No. of Credits per Programme- 16 X 2 = 32

ii. Detailed Syllabi

Paper 1: INFORMATION PROCESSING & RETRIEVAL

Unit I

Concepts of Information transfer – Universe of subjects

Unit II

Structure & development – Impact on the schemes for classification - CC, DDC, UDC, & LC

Unit III

Indexing Languages – Vocabulary Control – Thesaurus

Unit IV

Design of indexing languages, general theory of subject indexing languages.

Unit V

Indexing Systems & Techniques – Pre coordinate indexing – PRECIS, POPSI, Chain indexing – Relational indexing,

Unit VI

Post Coordinate Indexing Systems, Uniterm Indexing, Citation Indexing, KWIC and KWOC, Evaluative Studies – Crane field. I.

Unit VII

Bibliographic Standards – ISBD, (G), AACR 2R, ISBN, ISDN, ISSN, ISO 2709

Unit VIII

Bibliographic Formats - Bibliographic Standards : MARC, CCF, UNIMARC, MARC21, MARC XML, Dublin Core Z39.5.

Unit IX

Information Retrieval System – Structure, Functions and Components

Unit X

Search strategy – Criteria for evaluation – Recall, Precision – Relevance and failure analysis.

Unit XI

Automatic Indexing, Web Ontology

Unit XII

Sequential file, structure of a sequential file, inverted file, structure of an index file, matching criteria,

Unit XIII

Boolean logic, limitations of Boolean logic, processing query expression: rules for operations **Unit XIV**

Recent Trends in IRS - Internet information retrieval - Web-based information retrieval

SUGGESTED READINGS:

1. Alberico, R. and Micco M.(1990). Expert systems for reference and Information retrieval. West Port : Meckler.
2. Atchison, J. & Gilchrist, A.(1972). Thesaurus construction: a practical manual. London: Aslib.
3. Austin, D.(1984). PRECIS: A manual of concept analysis and subject Indexing. 2nd ed.
4. Chowdhruy, G.G.(2003). Introduction to modern Information retrieval. 2nd Ed. London, Facet Publishing.
5. Cleaveland, D. B.(2001). Introduction to Indexing and abstracting. 3rd Ed. Englewood, Colo. : Libraries Unlimited.
6. Ghosh, S.B. and Biswas, S.C. (1998). Subject Indexing systems: Concepts, methods and techniques. Rev. ed. Calcutta: IASLIC.

7. Lancaster, F. W. (1968). Information retrieval systems, characteristics, testing and evaluation. London: Facet Publishing.
8. Pandey, S.K. Ed.(2000).Library Information retrieval. New Delhi: Anmol.
9. Seetharama, S. (1997).Information consolidation and repackaging. New Delhi: ESS ESS.
10. Vickery, B.C.(1970). Techniques of Information retrieval. London: Butterworths.

Paper 2: LIBRARY AND INFORMATION SYSTEM MANAGEMENT

Unit I

Concept of management and organization – Definition – Library and information system as Non Profit Organizations – Library as a system - Organisational Structure of different types of library

Unit II

Various Schools of Management Thought: Classical, Human relations

Unit III

Behavioral schools of thought – Management theories: Taylor, Fayol, Gantts, McGregor, Maslow.

Unit IV

Concept and principles of Scientific Management – Definition and scope

– Methodology – Advantages and limitations. Application of Scientific Management principles to Library and Information Centres

Unit VI

Systems approach – Systems analysis in library and information systems – Contingency approach – Decision making approach, MBO, POSDCORB

Unit VII

Collection Management: Policy and procedures for print and non-print resources including print and ejournals - Selection criteria and tools - Barriers of acquisition including licensing of electronic resources

-Library security.

Unit VIII

Management Information System (MIS) – Designing – Work Analysis – Flow process chart – Decision flow charts, Block diagram, Gantt chart, network analysis, PERT and CPM.

Unit IX

Housekeeping Operations: Book / Information Resource selection and acquisition section, License negotiation and relevant rights issues - Technical processing section- Serial control and circulation control Policy, procedures and methods of maintenance and stock verification -- Collection evaluation and weeding out

Unit X

Personnel management – Human resources planning – Recruitment – Selection – Training and Development - Performance appraisal promotion – Motivation.

Unit XI

Financial Management - Sources of Library Finance in different types of libraries - Budget techniques and method, budgetary control - Costing library process, functions and services - Cost effectiveness and Cost benefit analysis Report writing and Library Statistics

Unit XII

Building and space management of library and information centres - Safety issues - Equipments and furniture- in addition for differently able people - Library standards - Indian and International

Unit XIII

Management of Electronic libraries - Job descriptions of IT manager - Evaluation of IT - Technology Assessment –Equipment, Infrastructure, Service, Staff, Self - Technology development –updation

Unit XIV

Total Quality Management : Concept, Definition, Elements - Operations Management Systems - Tools and techniques for improving quality-Inventory planning and control, - Inventory control model - Quality Audit, LIS related Standards - Resource mobilization, Outsourcing, Library Consortia, Open Access - Technology Management

SUGGESTED READINGS:

1. Evans, G Edward. Developing Library and Information centre Collections. New York, Libraries Unlimited, 2005
2. Evans,GEdward: Management techniques for librarians,2ndEd.,NewYork,Academic Press,1983.
3. Gorman, G.E. International yearbook of Library and Information management 2003-2004 metadata applications and management. London, L.A., 2003
4. Kishan Kumar. Management of libraries in Electronic environment. Delhi, Har-Anand Publications, 2007
5. Kishore, Jugal. Personal Management in Libraries. Delhi, EssEss, 1981

6. Krishna Kumar. Library Administration and Management. Delhi, Vikas, 2004
7. Kumar, P.S.G. Management of Library and Information Centres (paper V of UGC Model Curriculum). Delhi, B.R.Pub., 2003
8. Lahiri, Ramansu. Management of Libraries concepts and practices. New Delhi, EssEss, 1996
9. Lancaster, F.W. Technology and Management in Library and Information Services. London, Lib. Assoc., 1997
10. Bavakutty, M &Majeed, Abdul. Methods for Measuring Quality of Libraries. ISBN : 81-7000-439-X , 2005

Paper 3: INFORMATION TECHNOLOGY & INFORMATION SYSTEMS

Unit I

Computer basics – Computer generations and classification - Understanding IT and components of IT Computers and Communication Technologies (Data Process Cycles and Operations)

Unit II

Role of Computers in information transfer. Block Diagram of Computer - Classification of computers – Analog and Digital - Generation of computers Stand alone systems including Note Books and Servers

Unit III

Input / Output Devices: Understanding Personal Computer: CPU, Storage and Input/Output Devices, RAM and ROM, USB, Hard Discs, Scanners and Digital Camera, Joysticks & Printers

Unit IV

Data presentation in Computers: Binary Number System, Overview of Character

Coding Standards- ASCII and UNICODE

Unit V

Computer Software : Windows, LINUX - System and Application Software - Programming Concepts - Open Source and Proprietary Library Software.

Unit VI

File Organisation: Files and Databases, Data Elements, Fields, Records, DBMS and RDBM Packages - Database models – Hierarchical, network, relational.

Unit VII

Planning of Information System. Database system – Definition, scope, need and purpose - Basic features of WINISIS and MS Access

Unit VIII

Information System analysis and Design – Overview, components, System Development Lifecycle

Unit IX

Hardware and software management : Server configuration - Managing the servers - Backups - RAID application - Software licensing- AMC issues

Unit X

Networking: Technological development in communication: Transmission media - Digital Networks – LAN, WAN, PSTN, ISDN

Unit XI

Library Automation : Basic: Retrospective Conversion Techniques , Library Automation Software – OPAC - Automation Identification Methods: Bar coding, RFID - Selection criteria - for hardware and software - Library Automation Software – Open source / Commercial

Unit XII

Communication : Land line and Mobile networks - Data transmission in telephone networks with Major Telecommunication –Networks - Motivation for ISDN and ISDN channels - User interfaces - Broadband ISDNO ptical Communication systems, FAX, Modem, Teletext, Videotext, email, Internet, and Intranet.

Unit XIII

National Information Systems – : NISCAIR (formerly INSDOC), DESIDOC,

SENDOC, INFLIBNET, DELNET

Unit XIV

International Information Systems - – INIS, AGRIS, BIOSYS – Open Office Tools.

National Information Systems

SUGGESTED READINGS:

1. Balakrishanan, Shyama&Paliwal, P.K. Current Scenario of Information Technology.Delhi, Anmol, 2001

2. Brophy, Rowley. The basics of information systems. London, Library Association, 1996.
3. Dhiman, A.K. Basics of Information Technology for Library and Information Scientists. 2 Vols., Delhi, EssEss, 2003
4. Kumar, P.S.G. Information Technology: Basics: (Paper IV of UGC Model Curriculum). Delhi, B.R.Pub., 2003
5. Mohamed Acly and Gill, Needham, Eds. M- Libraries 3: Transforming libraries with Mobil technology. Chennai: Allied, 2012.
6. Microsoft Corporation. Microsoft Visual C++ 6.0 RUN - TIME Library Reference – Vol.4. Washington, Microsoft Press, 1998.
7. Kumar, P.S.G. Information Technology: Basics: (Paper IV of UGC Model Curriculum). Delhi, B.R.Pub., 2003

Paper 4: INFORMATION TECHNOLOGY- PRACTICE

Practice

1. Creating a database using MS – Access, My SQL
2. Thorough Knowledge of MS-Word, MS-EXCEL & and Power Point.
3. Installing and searching CD-ROM Database and Online Databases
4. Formulating Queries and searching using Boolean Operators.

Paper -5 ACADEMIC LIBRARY SYSTEM

Unit I

Types of Libraries – Role of Academic libraries and functions of higher education

Unit II

Growth of University and College libraries in India and the role of UGC and other

Unit III

National Bodies in promoting Academic Libraries.

Unit IV

Authorities in University/college libraries – Budgeting – Collection Building – problems and methods

Unit V

Centralization & Decentralization of University Libraries – Merits and Demerits

Unit VI

Resource Sharing and – Networking – Role of INFLIBNET

Unit VII

Academic libraries – Types of users & their information needs

Unit VIII

User education and services – User behavior and user studies.

Unit IX

Staffing Pattern – staff Formula – standards for Academic Libraries

Unit X

Automation in academic libraries – Impact of information technology on academic library services

Unit XI

Electronic Library, Digital Library, Virtual Library.

Unit XII

Library Building – Furniture's and equipment's

Unit XIII

Preservation and Conservation of Library Materials – Methods and Techniques.

Unit XIV

Recent developments of academic libraries and its Services

SUGGESTED READINGS:

1. American Association of School Librarians. Standards for school library programmes. 1969. ALA, Chicago (Latest).
2. Baker, David, Ed. Resource management in academic libraries. 1997. L.A.London.
3. Balakrishanan, Shyama&Paliwal, P.K. Academic Library automation
4. Bavakuty, M. Libraries in Higher Education. ESS ESS Pub., 1988
5. BhaskaraRao, P. Information Networks and Resource sharing. Delhi, Reliance, 1998
6. Brophy, Peter. The academic library. 2000. Library Association, London.
7. Chapman, Liz. Managing acquisitions in library and information services 2001. Library Association, London.
8. Jordon, Peter. The academic library and its users.1998. Gower, London.
9. Lyle, G R. Administration of the college library. Ed. 4. 1974. Wilson, New York.
10. Metcalf, K D. Planning academic and research library building. 1965. McGraw Hill, New York.

Paper 6: TECHNICAL WRITING

Unit I

Communication Process – Characteristic features of technical writing - Reader-writer relationship.

Unit II

Types of Communication – Verbal, Non-Verbal, Written - Effective Communication Skills, Oral and Written Communication Skills

Unit III

Language as a medium for communication of thought - Body language and common gestures - Meeting, Telephonic Communication and Presentation Skills - Good Questioning and Listening Skills

Unit IV

Characteristics and features of technical writing - Target groups in written communication- Level of technicality in Scientific Communication - Readability and text – Aberrations in technical writing.

Unit V

Organization and Presentation of data in abstracts, textual manner, references

Unit VI

Preparation of popular articles, technical reports, monographs, house journals.

Unit VII

Repackaging of information: Preparation of Review, Trend report, Progress report.

Unit VIII

Editorial Process: Editorial tools, use of style manuals - proof reading – Role of Editor – Publication

Unit IX

Use of style manuals - APA, MLA and Chicago style manuals using MS Word and Zotero

Unit X

Office Communication: Report Writing : Annual Report, Daily Progress Report, Event Report, Promotion - Report, Confidential Report, User Satisfaction Report - Office Writing: Notice Writing, Memo Writing, -Letter Writing – Publisher, Book - Seller, Binders, Users-Patrons-Clienteles, = □ Presentation: Body language, Book review, At the time of Library

Committee meeting, Staff meeting, Condolence meeting, Business meeting, Orientation, Conference, Seminars or Workshop – Training Programme

Unit XI Categories of Technical Communication : Structure, function and types of Technical communication - Definition, purpose, characteristics of Technical Communication

Unit XII

Technical papers / Articles, Review articles, Technical Reports, Monographs, Dissertations, In-House bulletins - Information analysis, Consolidation and Repackaging Products- Technical Digest, - Trend Reports, State-of-the-art Reports, Annual Reports, Manuals, Handbooks

and Directories

Unit XIII

Business Writing: Business Plan and Mission Writing - Terms and Condition with Book Sellers, Publishers, Venders, Service Providers - MOUs – Licensing, Contract Writing - Effective Covering Letters - Publisher, Book Seller, Binders, Users-Patrons- Clienteles

Unit XIV

Legal Issues : Freedom of information and privacy- Intellectual property in media - Database rights - Patents and Tread Marks - Quality issues and liabilities of

Ethics – Pre-publication and post-publication process.

SUGGESTED READINGS:

1. Gordon, H. M. and Walter J. A. *Technical Writing.* 5th ed. London: Holt, 1986.
2. James, H. S. *Handbook of Technical Writing*. NTC Business Books, 2010.
3. Richard, W. S. *Technical Writing*. New York: Barnes and Noble, 2008.
4. Krishnan Kumar, Research Libraries in Developing Countries.
5. Santhosh Gupta, Research Methodology and Statistical Techniques, New Delhi: Deep & Deep, 2000.
6. Lancaster FW, information Retrieval Syste,s, Ed- 2, 1976.

Paper 7: RESEARCH METHODOLOGY

Unit I

Foundations of research; Nature, definition and objectives - Types of research, Basis Concepts of research; Scientific Method; Ethical consideration of research.

Unit II

Library and Information Science (LIS) as an interdisciplinary subject, Significance of research in LIS; Areas of research in Library and Information Science.

Unit II

Scientific method – Nature of research in library & information science.

Unit III

Research methods – Definitions – Sources – Advantages – Limitations historical method, case study method, survey method, experimental method and other methods (Field investigation Research, Evaluation research, Action research, Ex post Facto)

Unit IV

Research Problem: Sources of research problem – Locating the problem – Formulation of the research problem – Criteria in selecting a problem – Defining and delimiting problems

Unit V

Literature search – Importance of surveying related literature – Library sources, research reviews, catalogue, indices, abstracts, bibliographies, microforms, computerized information retrieval systems.

Unit VI

Hypothesis – Meaning, Importance, types, sources, characteristics- Formulation of Hypothesis, Different forms of hypothesis – Difficulties in formulation – Testing the hypothesis.

Unit VII

Planning of research; Planning Process; Review of literature, Selection of a

problem-problems, process of identification, criteria of selection, formulation of

problem - Research design-Essentials of good research design & its importance, preparation of the research design/writing the research proposal

Unit VIII

Types: Descriptive, Diagnostic, Exploratory; and Experimental.

Unit IX

Data collection, primary and secondary data, methods of data collection, Research Technique and Tools; Questionnaire, Schedule, Interview, Sampling,

Scale and Check list, Library Records and Reports

Unit X

questionnaire construction & design, types of questionnaire – secondary data sources and precautions in the use of secondary data

Unit XI

Data analysis, interpretation and presentation – Research reporting.

Unit XII

Statistical analysis software SPSS, PSP and SOFA

Unit XIII

Report Writing – Structure and Components – Style manuals – APA, MLA, Chicago. MS Word and Zotero;

Unit XIV

Bibliometrics, Scientometrics, and Informetrics, Webometrics

SUGGESTED READINGS:

1. Busha, C.Hand Harter, S.S. (1980). Research methods in librarianship: Techniques and interpretation. Orlando, Academic press.
2. Charles, H. et.al.(1993). Research methods in librarianship: Techniques and interpretations, New Delhi: Sage.
3. Fowler, F.J. (1993).Survey research methods. New Delhi: Sage.
4. Goode, W.J. and Hatt, P.K. (1986). Methods in social Science research. New Delhi: McGraw Hill.
5. KrishanKumar(1992). Research methods in Library and Information Science. New Delhi: Vikas
6. Krishnaswami, O.R. (1993). Methodology of Research in Social Sciences. Bombay:Himalaya.
7. Leddy, P. D. (1980).Practical research: Planning design. London: Clive-Bingley.
8. RavichandraRao, I.K. (1985). Quantitative methods for Library and Information Science. New Delhi: Wiley Eastern.
9. Slater, M. (1990). Research methods in Library and Information studies. London: L.A.
10. Stevens, R.E. Ed.(1971). Research methods in librarianship. London: Clive Bingley.

Paper 8. INFORMATION PROCESSING & RETRIEVAL –PRACTICE

<u>Practice</u>

1. Classification of Documents according to abridged English Edition of UDC and CC-6th edition.
2. Cataloguing of books, Serials and Non - Book material according to AACR 2R and Sears List of Subject Headings.

iii. Duration of the Programme

It is One Year.

iv. Faculty requirement

The Department Professors and the Professors from the other Universities are employed for classes.

Staff Category	Required
Core Faculty	2
Faculty - Specialization	1
Clerical Assistant	1

v. Instructional Delivery Mechanisms

The instructional delivery mechanisms of the programme includes SLM – study materials, face to face contact session for both theory and practical courses of the programme, e-content of the study materials in the form of CD, and virtual laboratory wherever applicable.

vi. Identification of Media:

The SLM – designed study materials will be provided in print media as well is in the form of CD which carries electronic version of the study material in addition to virtual laboratory courses.

vii. Student Support Services

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like counselling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issuance of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter and Learning Centres. The conduct of end semester examinations, evaluation and issuance of certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

(f) Procedure for admission, curriculum transaction and evaluation

A Candidate with B.L.I.Sc degree or a Equivalent Degree from a recognised University shall be eligible for M.L.I.Sc.

i. Fee Structure

Sl No.	Fee Details	Amount in Rs.
1	Admission Processing Fee	300
2	Course Fee	5000
3	ICT	150
	Total	5450

ii. Admission Policy

Admission under Distance Education Stream

The candidate may seek admission either directly to the University or through any one of the approved Learning Centres.

If the candidates enroll directly with the University, the Course Materials shall be directly handed over to them or sent to them and they have to attend the Contact Classes at Karaikudi only.

If candidates enroll through Learning Centres, the Course Materials shall be sent through the Learning Centres and they have to attend the Contact Classes arranged by the Learning Centres at their places.

Direct admission with the University

The following documents are to be enclosed along with the filled-in application:

- The original entry qualification of HSC/ Diploma Certificate or Degree or Provisional Certificate and a copy attested by the Gazetted Officer. (Original Certificates will be returned immediately after verification).
- Student Index Card with stamp size photo and signature affixed.
- Demand Draft for the prescribed total fee drawn in favour of the Director, DDE, Alagappa University payable on any bank at Karaikudi.
- No Transfer Certificate is required for admission to any of the Courses. However, a 'Course Completion Certificate' may be issued for those candidates who apply for it.

- The Transfer Certificate can be issued only on production of the following documents: by post (or) in person (candidate only)
    - Previous original T.C of the candidate.
    - Fees Rs-100/- Through Demand Draft ("Favouring the Director, DDE, Alagappa University, Karaikudi") (OR) payment through cash Counter at DDE campus.
    - Copy of Student I.D card.

The filled-in application along with the required documents should be sent to the following address by Registered Post before the last date specified:

The Director
Directorate of Distance Education
Alagappa University
Karaikudi - 630 003

Tamil Nadu.

Admission through Learning Centre

The filled-in application has to be submitted through the Learning Centres along with the following documents:

- The original entry qualification of HSC/ Diploma Certificate or Degree or Provisional Certificate and a copy attested by the Gazetted Officer. (Original Certificates will be returned immediately after verification).
- Student Index Card with a stamp size photo and signature affixed.
- The candidate has to submit two Demand Drafts separately towards Course Fee
- One Demand Draft in favour of the Director, DDE, Alagappa University, payable at Karaikudi towards the University Share, and the others in favour of the Learning Centre concerned through whom the candidate seeks admission.
- The above Two Demand Drafts should be submitted to the Learning Centre along with the filled-in Application.
- No Transfer Certificate is required for admission to any of the Courses. However, a 'Course Completion Certificate' may be issued for those candidates who apply for it.
- The candidates are advised not to pay any fees in the form of Cash to the Learning Centre under any circumstances.

iii. Curriculum transaction

PCP Classes are conducted at regular intervals. The classroom teaching would be through chalk and talk method, use of OHP, Power Point presentations, web-based lessons, animated videos, etc. The face to face contact sessions would be such that the student should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

For practical courses, the procedure will be issued to the learners. In the concerned areas, instruction would be given for the practical activities followed by showing the records and the documents. And finally the students have to do the activities individually.

The face to face contact sessions will be conducted in following durations;

Course Type	Face to Face Contact Session per Semester (in Hours)
Theory Courses (3 courses with 4 credits each)	48
Practical Courses (1 course with 4 credits)	120
Total	**168**

iv. Evaluation

The evaluation shall be conducted separately for theory and practical to assess the knowledge acquired during the course of study. There shall be two systems of evaluation viz., internal assessment and end semester examinations.

In the case of theory courses the internal evaluation shall be conducted as Continuous Internal Assessment via Student assignment preparation. The internal assessment shall comprise of maximum 25 marks for each subject. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

For Practical courses, the internal evaluation will be done through continuous assessment of skill in collecting the data and record note preparation. The external evaluation consists of an end semester practical examinations which comprise of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

Question Paper Pattern:

Answer all questions (one question from each unit with internal choices Time: 3 Hours Max. Marks: 75

Part A- 10 x 2 Marks = 20 Marks

Part B -5 x 5 Marks = 25 Marks

Part C- 3 x 10 Marks = 30 Marks

Distribution of Marks in Continuous Internal Assessments:
The following procedure shall be followed for awarding internal marks for **theory** courses

Component	Marks
Assignments (5 questions per course)	25
Total	**25**

The following procedure shall be followed for awarding internal marks for **Practical**courses

Internal –Practical	Marks
Record	25
Total	**25**

Passing Minimum:

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50%. There shall be no passing minimum for internal. For external examination, passing minimum will be 50%. An aggregate of 50% is needed (External + Internal) for a pass in each theory.

For a pass in the Practical paper, a candidate has to secure a minimum of 50% marks. There shall be no passing minimum for internal. For external examination, passing minimum will be 50%. An aggregate of 50% is needed (External + Internal) for a pass in each practical. However submission of all record note book is a must.

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.
Marks and Grades:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester
G_i = Grade Point obtained for course i in any semester.
n refers to the semester in which such courses were credited

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

Grade Point Average = $\frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = $\frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme}}$

CGPA	Grad	Classification of Final Result
9.5-10.0 9.0 and above but below 9.5	O+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D++ D+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A++ A+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

Maximum duration for the completion of the course:

The maximum duration for completion of MLISc programme shall not exceed ten semesters from the completion of the course.

Commencement of this Regulation:

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter.

(g) Requirement of the laboratory and Library Resources

The course contains two papers which are meant for practice in the laboratory. Computer laboratory available in the Library will be utilised for these kinds of practicing.

(h) Cost estimate of the programme and the provisions

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with M.L.I.Sc Programme is as follows:

Sl. No.	Expenditure Heads	Approx. Amount in Rs.
1	Programme Development	7,00,000/-
2	Programme Delivery	14,00,000/-
3	Programme Maintenance	2,10,000/-

(i) Quality assurance mechanism and expected programme outcomes

1.University's Moto:

'Excellence in Action'

2. University's Vision Statement:

Achieving Excellence in all spheres of Education, with particular emphasis on "PEARL"- Pedagogy, Extension, Administration, Research and Learning.

3. University's Objectives:

1. Providing for Instructions and Training in such Branches of Learning as the University may determine.
2. Fostering Research for the Advancement and Dissemination of Knowledge

4. University's Quality Policy:

Attaining Benchmark Quality in every domain of 'PEARL' to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

5. University's Quality Quote:

Quality Unleashes Opportunities towards Excellence (QUOTE)

6. Programme's Review Mechanism:

The quality of the programme depends on scientific construction of the curriculum, strong-enough syllabi, sincere efforts leading to skilful execution of the course of the study. The ultimate achievement of this programme of study may reflect the gaining of knowledge and skill in the subject. And all these gaining of knowledge may help the students to get new job opportunities, upgrading in their position not only in employment but also in the society, make students feel thirsty to achieve in research in the fields associated with the discipline-Library Sciences achieving in competitive examinations on the subject.

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations. Apart from the end semester examination-based review feedback from the alumni, students, parents and employers will be received and analyzed for the further improvement of the quality of the MLISc Programme.

ALAGAPPA UNIVERSITY, KARAIKUDI
(Accredited with A+ Grade by NAAC (CGPA: 3.64) in the Third Cycle)
Directorate of Distance Education

Minutes of the Board of Studies Meeting of Library and Information Science Courses (DDE) held on 30th May, 2017 at 10.30 a.m. in the Department of Library and Information Science, Alagappa University.

Members Present:

Dr. S. Thanuskodi Professor & Head, DLIS, Alagappa University	:	Chairperson
Dr. A. Thirunavukkarasu Librarian, Central Library, Alagappa University	:	Member
Dr. S. Ravi Professor, DDE, Annamalai University	:	Member
Dr. N. Radhakrihnan Associate Professor, DLIS, Periyar University	:	Member
Dr. R. Sevukan Associate Professor & Head, DLIS, Pondicherry University	:	Member

At the outset, the chairman extended a very warm welcome to all the members of the Board of Studies and briefly narrated the need for revision of syllabi for Certificate, UG and PG Courses of Library and Information Science offered through Directorate of Distance Education, Alagappa University.

The Board of studies thoroughly scrutinized the curriculum structure of the above mentioned programmes of Library and Information Science and made necessary changes in the curriculum structure incorporating the nascent developments in the fields. In tune with the changes and updations it is recommended to write new lessons for all the papers with Self Learning Materials (SLM) pattern. The board also recommended to offer the revised curriculum at the earliest.

Dr. S. Ravi | Dr. A. Thirunavukkarasu | Dr. R. Sevukan

Dr. N. Radhakrihnan | Dr. S. Thanuskodi

ALAGAPPA UNIVERSITY

(Accredited with A+ Grade by NAAC (CGPA:3.64) in the Third Cycle)

Karaikudi-63 003, Tamil Nadu, India.

Directorate of Distance Education

PROGRAMME PROJECT REPORT

for M. Sc., BOTANY

to be submitted to

UGC, Distance Education Bureau (DEB), New Delhi

for seeking approval to introduce new programme through Distance Education Mode

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Degree of Master of Science (M.Sc.,) in Botany
Credit Based System (CBS)

a. Programme's Mission & Objectives:

To afford a High Quality Post Graduate Degree M. Sc., Botany through Distance Learning mode to the graduate-aspirant in order to nurture the natural science- Botany among the young minds. The pupil nurtured with botanical knowledge is the need of the hour to save the natural environment, educate the human resources towards conservation and save planet thereon.

The objectives of the programme include;

- To give an expanded knowledge about various kind of life forms of plant kingdom.
- To teach about naming and classification of plants.
- To understand about anatomical, embryological, cellular and molecular level approach of science in studying plants.
- To study about microorganisms, their impact on plants and various kind of plant diseases.
- To understand about modern concepts like plant molecular biology, plant genetic engineering and plant tissue culture.
- To introduce inter-disciplinary approaches like biophysics, biostatistics and bioinformatics.
- To study about application part of botany viz., algal technology, mushroom technology, economic botany, and horticulture.
- To enlighten the students about biodiversity, conservation and Intellectual Property Rights.

b. Relevance of the Programme with Alagappa University's Mission and Goals:

In order to align with the mission and goals of Alagappa University the M. Sc., Botany Programme is planned to deliver in Distance Learning mode which may reach the maximum number of student aspirants who are unable to thrive to spend non-elastic timings of formal conventional class room education. Such a higher education in science subject with appropriate laboratory experiences will enrich the human resources for the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

c. Nature of Prospective Target Group of Learners:

This Master of Science in Botany programme through Distance Learning mode is developed by keeping in mind to give opportunity to economically and socially excluded people includes graduates of various socio-economic status viz., unemployed youths, employed with marginalized salary due to lack of sufficient knowledge in the subject Botany. Also, the target group of learners includes various level employees of agro-based companies, secondary – level school teachers, research aspirants, women taking care of family –the important unit of the community, etc.,

d. Appropriateness of programme to be conducted in Distance learning mode to acquire specific skills and competence:

M. Sc., Botany programme through Distance Learning mode is developed in order to give subject-specific skills including i) knowledge about various kind of life forms of plant kingdom, their classification, ii) anatomical, embryological, cellular, genetical, molecular approach and plant metabolisms, iii) inter-disciplinary knowledge like biophysics, biostatistics and bioinformatics, iv) modern concepts like plant biotechnology, plant tissue culture, v) application part of botany viz., algal technology, herbal technology, economic botany, horticulture and forestry, and vi) biodiversity, conservation and intellectual property rights. The programme is developed to give detailed exposure theoretically and free-hands experience on practical parts of the study in order to impart skills of Botany to the learners.

e. Instructional Design:

e. 1. Curriculum Design:

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
FIRST YEAR						
I Semester						
1.	34611	**Plant Diversity**	25	75	100	4
2.	34612	**Plant Taxonomy**	25	75	100	4
3.	34613	**Biological Techniques in Botany**	25	75	100	4
4.	34614	**Practical – Lab I: Plant Diversity, Plant Taxonomy and Biological Techniques in Botany**	25	75	100	4
		Total	**100**	**300**	**400**	**16**
II Semester						
5.	34621	**Cell Biology, Genetics and Plant Breeding**	25	75	100	4
6.	34622	**Plant Anatomy and Embryology**	25	75	100	4
7.	34623	**Plant Physiology and Biochemistry**	25	75	100	4
8.	34624	**Practical – Lab II: Cell Biology, Genetics & Plant Breeding, Plant Anatomy & Embryology, Plant Physiology and Biochemistry**	25	75	100	4
		Total	**100**	**300**	**400**	**16**
SECOND YEAR						
III Semester						
9.	34631	**Microbiology and Plant Pathology**	25	75	100	4
10.	34632	**Ecology, Biodiversity Conservation and Economic Botany**	25	75	100	4
11.	34633	**Algal Technology and Mushroom Technology**	25	75	100	4
12.	34634	**Practical – Lab III: Microbiology & Plant Pathology, Ecology, Biodiversity Conservation, Economic Botany, Algal Technology and Mushroom Technology**	25	75	100	4
		Total	**100**	**300**	**400**	**16**
IV Semester						
13.	34641	**Plant Molecular Biology**	25	75	100	4
14.	34642	**Biostatistics, Biophysics and Bioinformatics**	25	75	100	4
15.	34643	**Horticulture and Plant Tissue Culture**	25	75	100	4
16	34644	**Practical - Lab IV: Plant Molecular Biology, Biostatistics, Biophysics and Bioinformatics, Horticulture and Plant Tissue Cu (Paper Sl. Nos.13, 14, and 15)**	25	75	100	4
		Total	**100**	**300**	**400**	**16**

Course Code Legend:

3	**4**	**6**	**X**	**Y**

346- M. Sc., Botany Programme
X -Semester Number
Y - Course Number in the Semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

No. of Credits per Course (Theory) - 4

No. of Credits per Course (Practical) - 4

Total No. of Credits per Semester- 16

Total No. of Credits the Programme- 16 X 4 = 64

e. 2. Detailed Syllabi:

FIRST SEMESTER

Course Code	Title of the Course
34611	**Plant Diversity**

Objectives:

- To understand diversity of algae, fungi, lichens and bryophytes.
- To define and characterize diversity of lower vascular plants.
- To understand the dynamics of diversity
- To realize the significance of diversity.

Possible Outcomes of the course:

- The course will enrich the knowledge on algae, fungi, lichens and bryophytes.
- The course will impart the adequate knowledge about lower vascular plants.
- The course will help the students to have an insight on evolutionary approach with the plant kingdom.
- The course will help the students to answer the questions in competitive exams viz., SET, NET, GATE, etc. successfully.

Block 1: **Algae:**

Unit-1: Definitions and concepts of plant diversity- Thallus organization in Algae.

Unit -2: Reproduction and life cycle patterns in algae- Classification of algae (Smith).

Unit-3: Reproduction and life cycles in *Cyanophyceae, Chlorophyceae, Phaeophyceae* and *Rhodophyceae* (Comparative study only).

Block 2: Fungi; Lichens:

Unit-4: Classification of fungi (Alexopoulos and Mims, 1979).

Unit-5: Thallus organization and Reproductive pattern in fungi.

Unit-6: Life cycle patterns in fungi (Phycomycetes to Deuteromycetes) - Spores and spore dispersal mechanisms exhibited by fungi.

Unit-7: General account on Lichen- Classification of lichens (Miller, 1984) - Structure and reproduction of lichen.

Block 3: Bryophytes:

Unit-8: Classification of bryophytes (Watson, 1964).

Unit-9: Structural variations in the gametophytes and sporophytes of *Marchantiales, Sphaerocarpales, Jungermanniales,* and *Calobryales* (Comparative study only).

Unit-10: Structural variations in the gametophytes and sporophytes of *Anthocerotales, Sphagnales, Andreales* and *Bryales* (Comparative study only).

Block 4: The Lower Vascular Plants (Pteridophytes and Gymnosperms) :

Unit- 11: Classification of Pteridophytes (Reiner's).

Unit- 12: General characteristics and life cycle patterns in *Psilopsida, Lycopsida, Sphenopsida* and *Pteropsida*.- Study of fossils in Pteridophytes (*Sphenophyllum, Lepidodendron*).

Unit- 13: Classification (K.R.Sporne, 1967) of Gymnosperms.

Unit- 14: General characteristics of *Cycadales, Ginkgoales, Coniferales* and *Gnetales* (Comparative study only). - Study of fossil gymnosperms (*Medullosa, Cycadeoidea* and *Caytonia).*

Reference Books:

1. Robert Edwardlee, 2008, Phycology, 4th Edition, Cambridge University Press, London, ISBN: 0521682770.
2. O. P. Sharma, 2011, Algae, 1st Edition, Cambridge University Press, Mcgraw Hill Education, ISBN: 0070681945.
3. Steven L. Stephenson, 2010, The Kingdom Fungi: An Introduction to Mushrooms, Molds and Lichens, 1st Edition, Timber Press, ISBN: 0881928917.
4. Himanshu Rai, Dalip K. Upreti, 2014, Terricolous Lichens in India: Volume 2: Morphotaxonomic Studies, Springer NY, ISBN: 1493903594
5. W.B Schofield, 2001, Introduction to Bryology, 1st Edition, The Blackburn Press, ISBN: 1930665261.
6. S. S. Sambamurty, 2005, A Text Book of Bryophytes, Pteridophytes, Gymnosperms and Palobotany, I K International Publishing House Pvt. Ltd, ISBN: 8188237450.
7. Pteridophytes and Gymnosperms, 2010, 1st edition, Springer, ISBN: 3642080804.
8. C. J. Alexopoulos C. W. Mims M. Blackwell, 2002, Introductory Mycology, Wiley India Pvt Ltd. ISBN: 9788126511082.

9. K. R. Sporne, 2014, The Morphology of Pteridophytes; The Structure of Ferns and Allied Plants - Primary Source Edition, Nabu Press, ISBN: 9781294843078.

Course Code	Title of the Course
34612	**Plant Taxonomy**

Objectives:

- To acquire the fundamental knowledge of plant systematics.
- To know about the basic concepts and principles of plant systematics.
- To know how to identify the plants.
- To create awareness of the f taxonomic relationships in plant systematic studies.

Possible Outcomes of the course:

- The course will nurture the knowledge on classification of plants.
- The course will give an expertise in understanding characteristic features of various plant families.

Block-1: Scope and Applications of Plant Taxonomy:

Unit-1: Scope and applications- Species concept, Biotype, Ecad, Ecotype- Binomial System of Nomenclature.

Unit-2: Theories of Biological Classification- Structural, Biological and Molecular systematics.

Unit -3: Historical Background, Plant classification- Plant classification systems: Bentham and Hooker, Engler and Prantl, Takhtajan and Hutchinson.

Block-2: Taxonomic Structure:

Unit-4: Taxonomic structure: Biosystematics, Chemotaxonomy, Numerical taxonomy-. Modern inter-disciplinary approaches to Taxonomy.

Unit -5: Botanical Nomenclature- Need for scientific names- Principles of ICBN. Type method, author citation, Publication of names, rejection of names.

Unit-6: Principle of priority, limitations, conservation of names of species-Draft Biocode.

Block-3: Salient features of plant families:

Unit-7: Study of the Monocotyledons: *Hydrocharitaceae* and *Dioscoreaceae*

Unit-8*:* Study of the Monocotyledons: *Arecaceae* and *Cyperaceae.*

Unit-9: Study of the Monochlamydeae families*: Polygonaceae* and *Amaranthaceae.*

Unit-10*:* Study of the Monochlamydeae families: *Aristolochiaceae* and *Loranthaceae*.

Block- 4: Salient features of plant families:

Unit-11: Study of the Gamopetalae families: *Sapotaceae, Rubiaceae, Asteraceae, Apocynaceae.*

Unit-12: Study of the Gamopetalae families: *Convolvulaceae, Bignoniaceae, Scrophulariaceae* and *Verbenaceae.*

Unit-13: Study of the Polypetalae families: *Magnoliaceae, Menispermaceae, Papaveraceae, Polygalaceae, and Tiliaceae.*

Unit-14: Study of the Polypetalae families: *Geramiaceae, Mimosaceae, Myrtaceae, Meliaceae* and *Sapindaceae*.

Reference Books:

1. Anupam Dikshit, M.O. Siddiqui, Ashutosh Pathak, 2016, Taxonomy of Angiosperms: Basic Concepts, Molecular Aspects & Future Prospects, 1st Edition, Studera Press, ISBN: 9385883070
2. O. P. Sharma, 2012, Plant Taxonomy, 2nd Edition, Mcgraw Hill Education, ISBN: 0070141592
3. Yilma Dessalegn, Getachew Mekonnen, 2012, Plant Taxonomy and Systematics, LAP Lambert Academic Publishing, ISBN: 3659164739.
4. Jones and Luchsinger, 1987, Plant Systematics, 2nd Edition, McGraw Hill International Editions. New York.
5. Neeru Mathur, 2012, Taxonomy of Angiosperms, Sonali Publications, ISBN: 8184114494.
6. K.V. Krishnamurthy, 2004, An Advanced Text Book on Biodiversity – Principles and Practice, Oxford and IBH Publishing Co. Pvt. Ltd. New Delhi. ISBN: 9788120416062.

Course Code	Title of the Course
34613	**Biological Techniques in Botany**

Objectives:

- To acquire the knowledge about biological techniques.
- To know about the basic concepts, principles and significance of various analytical and molecular techniques.
- To understand the various anatomical techniques.

Possible Outcomes of the course:

- The course will nurture the knowledge on biological samples especially plant samples.
- The course will give an expertise in understanding the various important biological techniques to be employed in the field of botany.

Block-1: Microscopy:

Unit-1: Principles, image formation and applications of Light, Polarizing Microscpy.

Unit-2: Transmission and Scanning electron microscopes - Confocal Microscopy, Phase Contrast Microscopy. Photomicrography- Digital imaging.

Unit-3: Microscopic measurements- Micrometers – Ocular, Stage, Haemocytometer and Camera lucida.

Block-2: Microtomy:

Unit-4: Microtomes- Rocking, Rotary, Sledge and Ultra microtomes and their uses.

Unit-5: Material preparation techniques for microtome sectioning - fixatives, dehydrating agents, killing, fixing methods.

Unit-6: Stains and their uses and staining of plant tissues- Clearing, Mounting and mountants.

Block-3: Histochemical studies:

Unit-7: Histochemical techniques – staining of Proteins, Carbohydrates, Lipids and enzymes.

Unit-8: Microslide preparation—Whole mounts, Smears, Squashes.

Unit-9: Sectioning of Biological specimens- Free hand, Hand microtome, sludge and rotary microtome sectioning, Embedding methods.

Unit-10: Dewaxing and staining of the sections, fixing coverslips and ringing.

Block-4: Maceration, Separation Tracer and Blotting Techniques:

Unit-11: Maceration technique- PAGE, SDS – PAGE and Agarose gel electrophoresis. Isoelectric focusing. 2D Electrophoresis.

Unit-12: Separation technique -Ultracentrifugation- TLC, HPLC, HPTLC, FPLC, GC, MS, MALDI Tof.

Unit-13: Tracer techniques- Principles and applications of radioactive isotopes, Autoradiography and Liquid scintillation spectrometry.

Unit-14: Blotting techniques -Principles and techniques of Southern, Northern and Western blotting techniques and hybridization. Principles and applications of PCR, RFLP, RAPD, AFLP and DNA fingerprinting.

Reference Books

1. D.K. Singh, 2013, Principles and Techniques in Histology, Microscopy and Photomicrography, CBS Publishers.
   ISBN: 9788123909509.
2. R. Mortin, 1996, Gel Electrophoresis: Nucleic Acids (Introduction to Biotechniques), Garland/ BIOS Scientific Publishers. ISBN: 9781872748283.
3. A. S. Sameer, 2011, Molecular Biology and Biotechniques, VDM Verlag, ISBN: 9783639324990.
4. S.K. Dewab, 2012, Organic Spectrosopy (NMR IR MASS and UV), CBS Publishers, ISBN: 9788123919065.

Course Code	Title of the Course
34614	**Practical – Lab I: Plant Diversity, Plant Taxonomy and Biological Techniques in Botany**

Suggested Laboratory Exercises:

1. Morphological study of representative member of Algae: *Microcystis, Oocystis, Pediastrum, Hydrodityon, Ulva, Pithophora, Stigeoclonium, Drapranaldiopsis, Closterium, Cosmarium, Chara.*

2. Morphological study of representative member of fungi: *Albugo, Mucor, Pilobolus, yeast, Polyporus, PeniCillium, Aspergillus, Colletotrichum.*

3. Identification of fungal cultures: *Rhizopus, Mucor, Aspergillus, Penicillium, Chaetomium, Curvularia, Fusarium.*

4. Study of morphological and internal structures of representative genera of fruticose, foliose, and crustose lichen.

5. Morphological study of representative member of bryophytes and pteridophytes: *Marchantia, Anthoceros, Polytrichum, Psilotum, Lycopodium, Selaginella, Equisetum, Ophioglossum, Isoates, Lygodium, Regnellidium* and *Marsilia*

6. Study of morphology, anatomy and reproductive structures *of* bryophytes.

7. Study of complex tissues *viz*. Xylem and Phloem, Tracheids, Vessels and Sieve tubes and Companion cells.

8. Comparative study of the wood anatomy and vegetative and reproductive parts of *Cycas, Ginkgo, Cedrus, Abies, Picea, Cupressus, Araucaria, Cryptomeria, Taxodium, Podocarpus, Agathis, Taxux, Ephedra* and *Gnetum*.

9. Study of the living gymnosperms in the botanical garden of the university.

10. Study of important fossil gymnosperms from prepared slides and specimens.

11. Monographic study of the living gymnopserms.

12. Description of a specimen from representative, locally available families.

13. Description of a species based on various specimens to study intraspecific variation a collective exercise.

14. Description of various species of a genus; location of key characters and preparation of keys at generic level.

15. Location of key characters and use of keys at family level.

16. Field trips; compilation of field notes and preparation of herbarium sheets.

17. Training in using floras and herbaria for identification of specimens described in the class.

18. Demonstration of the utility of secondary metabolites in the taxonomy of some appropriate genera.

19. Comparison of different species of a genus and different genera of a family to calculate similarity coefficients and preparation of dendrograms.

20. Microscopic measurement practices using Micrometers, Haemocytometer and Camera lucida.

21. Microprepartaions of plant materials.

22. Microtomy, microslide preparations.

23. Staining for Proteins, Carbohydrates, Lipids and enzymes.

24. Demonstration of PAGE, Agarose gel electrophoresis.

25. Demonstration of PCR and blotting techniques.

SECOND SEMESTER

Course Code	Title of the Course
34621	**Cell Biology, Genetics and Plant Breeding**

Objectives:

- To explore the cellular structure and inclusions of the cell.
- To understand about genetics and inheritance of traits.
- To realize the significance of plant breeding techniques in improving the plant productivity and strain improvement.

Possible Outcomes of the course:

- The course will impart the wisdom about structure and function of the cells, cell organelles.
- The course will give a detailed account on fundamentals and application of genetics
- The course will give a scientific approach to plant breeding techniques and their significances.

Block-1: The Cell:

Unit-1: Structure of Prokaryotic and Eukaryotic cell.

Unit-2: Structure and function of Nucleus, Endoplasmic Reticulum, Golgi complex, Mitochondria, Chloroplast and Lysosomes.

Unit-3: Architecture of Nucleus and nuclear transport.

Unit-4: Architecture and functions of Cytoskeleton networks (Microfilaments, Intermediate filaments and Microtubules).

Block-2: Assembly and Functions of Cellular Membranes:

Unit-5: Biological Membrane – Structure (lipid bilayer, membrane proteins) - Assembly and basic functions, Transport of ions and molecules across the membranes.

Unit-6: Protein sorting in mitochondria, chloroplast, endoplasmic reticulum and nucleus.

Unit-7: Protein processing and trafficking from Endoplasmic reticulum to Golgi- Cell division and Cell cycle.

Block-3: Genetics:

Unit-8: Basic account on Mendelian Genetics and Gene interaction, Linkage, crossing over, Gene mapping.

Unit-9: Sex linkage, Cytoplasmic inheritance, male sterility, Origin, induction and applications of prions.

Unit-10: Polyploidy – Types, and their origin, Significance of polyploids.

Unit-11: Basic account on mutation (Causative agents, induction and types)- Basic account of population genetics (Hardey Weinberg's Law).

Block-4: Plant Breeding:

Unit-12: Objectives of plant breeding, Genetic variability and its role in plant breeding.

Unit-13: Breeding methods in self pollinated, cross pollinated, vegetatively propagated and apomictic plants.

Unit-14: Inbreeding depression theories- Hybrid vigour in plant breeding- Mutation breeding and breeding for disease resistance and stress tolerance.

Reference Books:

1. S. C. Rastogi, 2012, Cell Biology, 3rd Edition, New Age International, ISBN: 8122416888.
2. S. P. Vyas, A. Mehta, 2011, Cell and Molecular Biology, 1st Edition, CBS Publisher, ISBN: 8123919654.
3. P. S. Verma and A. K. Agarval, 2004, Cell Biology, Genetics, Molecular Biology, Evolution and Ecology, S. Chand Publishing, ISBN: 8121924421.
4. William S. Klug, Michael R. Cumming, Charlotte A. Spencer, Michael A. Palladino, 2016, Concepts of Genetics, 10th Edition, Pearson Education, ISBN: 9332577463.
5. J. E. Bradshaw, *et al.*, Root and Tuber Crops (Handbook of Plant Breeding), 2010, 1st Edition, Springer-verlag Gmbh, ISBN: 0387927646.
6. Mahabal Ram, 2014, Plant Breeding Methods, PHI Publishers, ISBN: 8120348508.
7. E.D.P. De Roberties and E. M. F. De Roberties Jr., 2010, Cell and Molecular Biology, Lippincot Williams & Wilkins, ISBN: 9788184734508.
8. J.E. Krebs, T. Kilpatrick, and E.S. Goldstein, 2014, Lewins Genes IX, Viva Books Pvt Ltd., ISBN: 9789380853710.

Course Code	Title of the Course
34622	**Plant Anatomy and Embryology**

Objectives of the Course:

- To study the plant anatomy which facilitate the process of understanding the internal structures of various plant parts and their significance.
- To study the development of male and female gametes, pollination and fertilization reveals the various steps involved in development of new plant.

Possible Outcomes of the Course:

- The course will illustrate anatomy of various plant parts.
- The course will make the learners understanding about various stages of development.

Block-1: Meristem, Connective tissues:

Unit-1: General account and theories of organization of meristems.

Unit-2: Light and Electron microscopic structure of cell walls.

Unit-3: Structural diversity and phylogenetic specialization of xylem and phloem.

Unit-4: Distribution, structure and significance of transfer cells, Vascular cambium – storied, non-storied and the mode of activity.

Block-2: Vascular Tissues:

Unit-5: Vascular differentiation in the primary body of stem, root and leaf.

Unit-6: Root stem transition- Molecular aspects of developing vegetative organs.

Unit-7: Cambial variants and floral vasculature.

Block-3: The Wood:

Unit-8: Structure, identification, classification and uses of woods.

Unit-9: Physical, chemical and mechanical properties of wood.

Unit-10: Natural defects, knots, reaction wood, compression wood tension wood.

Unit-11: Molecular aspects on wood differentiation, Commercial woods of South India.

Block-4: The Developmental Biology:

Unit-12: Anther development- Pollen morphology, Pollen stigma compatibility.

Unit-13: Megasporogenesis female gametopohyte- Nutrition of embryo sac-Endosperm types.

Unit-14: Apomixis- Vegetative reproduction- Agamospermy and apospory- Exploitation of polyembryony and apomixis in plant improvement progrmmes- Molecular aspects of higher plant reproduction.

Reference Books:

1. Pandey, B.P. 2011. Plant Anatomy, 21st Edition. S. Chand & Company Ltd., New Delhi. ISBN 8121901456.
2. Katerine Esau. 2010. Anatomy of Seed Plants, 2nd Edition. Wiley India Pvt., Ltd., New Delhi. ISBN 8126508205
3. Bhojwani, S.S. and S.P. Bhatnagar 2010. The Embryology of Angiosperms, 5th Revised Edition. Vikas Publishing House Pvt Ltd. Noida. ISBN: 8125923466.
4. Sharma, H.P. 2009, Plant Embryology: Classical and Experimental. Narosa Book Distributors Pvt. Ltd., New Delhi. ISBN: 8173199698.

Course Code	Title of the Course
34623	**Plant Physiology and Biochemistry**

Objectives of the Course:

- To study the plant physiological processes.
- To study the bio-chemical architecture of various bio-materials and their synthesis.

Outcomes of the Course:

- The course will enlighten the students about various metabolic pathways of plants.
- The course will make the learners understanding about various enzymes and its kinetics.

Block-1: Plant-Water Relations:

Unit-1: Water transport process, diffusion, osmosis, water potential, Chemical potential, **Unit-2:** Absorption of water, water transport through trachieds and xylem.

Unit-3: Transpiration and its significance, factors affecting transpiration, mechanism of stomatal movement- Water stress on crop production.

Block-2: Photosynthesis:

Unit-4: Ultra structure of photosynthetic apparatus.

Unit-5: Photochemical reaction- electron transport pathway in chloroplast membranes, photophosphorylation.

Unit-6: C4 carbon cycle- Crassulacean acid metabolism- Photorespiration.

Block-3: Respiration and Flow of Energy:

Unit-7: Glycolysis- TCA Cycle- electron transport in mitochondria.

Unit-8: Oxidative phosphorylation- pentose phosphate pathway- cyanide –resistant respiration.

Unit-9: Nutrient uptake and transport mechanism.

Unit-10: Biological nitrogen fixation, Nitrate and ammonia assimilation.

Block-4: Chemistry of Biomolecules

Unit-11: Carbohydrates- Classification, Structure of mono, di and polysaccharides, stereoisomers, enatiomers and epimers.

Unit-12: Amino acids and Proteins- Structure, characteristics and classification- aminoacid synthesis- peptide bond and polypeptide chain- primary, secondary, tertiary and quaternary structure of proteins.

Unit-13: Enzymes- General aspects (Classification and structure), allosteric mechanism, regulatory and active sites, isoenzymes, enzymatic catalysis- Michaelis-Menton equation and its significance.

Unit-14: Lipids- Classification and structure, biosynthesis of fatty acids, Oxidation of fatty acids- Nucleic acids- Composition of nucleic acids and nucleotide synthesis.

Reference Books:

1. Salisbury and Ross. 2006. Plant Physiology. CBS Publications ISBN: 9788123910437.

2. Taiz and Zeiger, 2006, Plant Physiology, MAC Publishers, ISBN: 9780878938568.

3. S.N. Pandey and Sinha, 2005, Plant Physiology, Vikas Publishing House Pvt. Ltd.. ISBN: 9788125918790.

4 . David Hames and Nigel Hooper, 2008, Instant Notes on Biochemistry, Tylor & Francis, ISBN: 9780415367783.

5. David L. Nelson and Michael N. Cox, 2013, Lehninger Principles of Biochemistry, Macmillon, ISBN: 9781464109621.

Course Code	Title of the Course
34624	**Practical – Lab II: Cell Biology, Genetics & Plant Breeding, Plant Anatomy & Embryology, Plant Physiology and Biochemistry**

Suggested Laboratory Exercises:

1. Study of different types of chromosomes
2. Preparation of karyograms.
3. Study of different stages of mitotic cell division in suitable material.
4. Study of meiotic cell division.
5. Linear differentiation of chromosomes through banding techniques such as G-banding, C-banding and Q-banding.
6. Application of colchicines in genetics.
7. Studying pea plant as tool for investigating Laws of Inheritance.
8. Demonstration of Mendel's Law of segregation.
9. Demonstration of Mendel's Law of Independent Assortment.
10. Studying deviations from Mendel's laws and applying statistics.
11. Studying *Drosophila* as model organism.
12. Effect of chemicals on mutation.
13. Effect of radiations on mutation.
14. Isolation of DNA from different sources.
15. Study of cell structure from onion leaf peels; demonstration of staining and mounting methods.
16. Comparative study of cell structure in onion cells, *Hydrilla* and *spirogyra.*
17. Study of cyclosis in *Tradescantia* staminal hairs.
18. To study the permeability of plasma membrane using different concentrations of organic solvents.
19. To study the effect of temperature on permeability of plasma membrane.
20. To prepare the standard curve of protein and determine the protein content in unknown samples.
21. Separation of chloroplast pigments by solvent method.
22. Determining the osmotic potential of vacuolar sap by plasmolytic method.
23. Determining the water potential of any tuber.
24. Separation of amino acids in a mixture by paper chromatography and their identification by comparison with standards.

25. Comparison of the rate of respiration of various plant parts
26. Protein, carbohydrate estimation
27. Estimation of Nitrogenase activity.
28. Acid and alkaline acid phosphatase activity
29. Isolation and identification of Rhizobium from leguminous plants.
30. Separation of amino acid through paper and column chromatography.

THIRD SEMESTER

Course Code	Title of the Course
34631	**Microbiology and Plant Pathology**

Objectives of the Course:

- To study microorganisms and scope of microbiology.
- To study about plant pathology and maintaining plant health.

Possible Outcomes of the Course:

- The course will give a detailed introduction of microorganisms and their classification.
- The course will make the learners understanding about which factor is lying behind the plant diseases and how to overcome that.

Block-1: Scope of Microbiology

Unit-1: Introduction to Microbiology – Scope of Microbiology – Evolution into science – Characterization of microorganisms.

Unit-2: Classification of microorganisms – Archea and bacteria – Characteristics of bacteria - Morphology – Ultrastructure.

Unit-3: Nutrition – Growth – Reproduction- Bacterial culture and culture characteristics – Economic importance of bacteria

Block-2: Virus and Mycoplasmas

Unit-4: Virology – General features – Classification of viruses – Characteristics – Ultrastructure

Unit-5: Virus- Isolation – Purification – Chemical nature – Replication – Transmission – Virions.

Unit-6: Prions – Phytoplasma (including mycoplasma).

Unit-7: Microbial products – Antibiotics – Enzymes – Human diseases caused by bacteria and viruses.

Block-3: Plant Pathology

Unit-8: Introduction to Plant Pathology – History - Causal agents responsible for plant diseases.

Unit-9: Types of plant diseases – Methods of studying plant diseases – Koch's postulates – Symptoms of plant diseases.

Unit-10: Etiology - Plant disease epidemics - Plant disease forecasting – Disease triangle – Disease cycle.

Block-4: Host-Pathogen Interactions and Plant Diseases:

Unit-11: Host-pathogen interactions – Defense mechanisms in plants.

Unit-12: Control measures – Cultural practices – Chemical control – Biological control - Integrated plant disease management.

Unit-13: Plant diseases- Tobacco mosaic virus – Bunchy top of banana –Blast and sheath blight of paddy – Citrus canker – Red rot of sugarcane – Downey mildew of grapes

Unit-14: Plant diseases – Late blight of potato – Leaf spot diseases of groundnut - Anthracnose of mango – Wilt of cotton – Rust of wheat.

Reference Books:

1. P. D. Sharma, 2012, Microbiology and Plant Pathology, 3rd Edition, Rastogi Publications, ISBN: 97881-7133-794-5.
2. Michael Pelczar, ECS Chan, R. Krieg, 2001, Microbiology, 5th Edition, Mcgraw Hill Education, ISBN: 074623206.
3. Allan Granoff, Robert G. Webster, 1999, Encyclopedia of Virology, 2nd Edition, Elsevier. ISBN: 0122270304.
4. H. C. Dube, 2007, A text Book of Fungi, Bacteria and Viruses, 3rd edition, Agrobios India, ISBN: 8188826383.
5. A. Bohra and A. Mishra, 2011, Plant Pathology - Disease and Management. AgroBios, Jodhpur, ISBN: 817754246X.
6. N. G. Ravi chandra, 2013, Fundamentals of Plant Pathology, Phi Learning, ISBN: 812034703X.
7. George N. Agrios, 2004, Plant Pathology, Elsevier, ISBN: 0120445654.

Course Code	Title of the Course
34632	**Ecology, Biodiversity Conservation and Economic Botany**

Course Objective:

To introduce various concepts in Ecology.

To introduce various components of biogeography

To create awareness on the biodiversity and importance of its conservation.

To know about intellectual property rights.

To know about economically important plants.

Possible Outcomes of the Course:

- The course will give an introduction about ecosystem ecology and population ecology.
- The course will make the learners understanding about biodiversity and the necessity to conserve them.
- The course will introduce intellectual property rights and legal protection towards it.
- The course will make impart of knowledge about the economically important plants.

Block-1: Ecology:

Unit-1: Ecosystem Ecology- concept and dynamics – Abiotic and biotic components, energy input in ecosystem, Biomass, primary and secondary production.

Unit-2: Concept of food chain and food web – community organization – Concept of habitat, functional role and niche – ecotone – edge effect – ecological succession.

Unit-3: Population biology- Basic concepts –Gause's principle, survivorship curves – self- regulating mechanisms.

Unit-4: Species interaction - evolution of cooperation, inter-specific competition, competition coexistence- Negative interaction- predation, herbivory, parasitism - Positive interaction - commensalisms and mutualism.

Block-2: Biodiversity Conservation:

Unit-5: Introduction to biodiversity- Types of biodiversity, Biodiversity concepts, Centres of diversity, Agro biodiversity, Values and uses of biodiversity, Loss of biodiversity.

Unit-6: Phytogeographic zones, Vegetation types of India and Tamilnadu, Endemism, Red listed plants, Red Data Book, Threatened plants and animals of India.

Unit-7: Biodiversity act of India 2002 and 2004. General overview of plant conservation, Conservation of biodiversity - Wildlife Sanctuaries, National parks and Biosphere Reserves, Hotspot biodiversity areas in India- Sustainable use of plant genetic resources, Biotechnology assisted plant conservation (*in situ* and *ex situ* conservation).

Block-3: Intellectual Property Rights:

Unit-8: General Agreement on Trade and Tariff (GATT) and World Trade Organization, Establishment and functions of GATT and WTO, Physical and Intellectual Property.

Unit-9: Different types of intellectual property rights (IPR) - Patents, Trade mark, Trade secret and Copy right.

Unit-10: Plant biotechnological examples of patents, trademark, trade secret and copyright. Plant breeder's rights.

Unit-11: Flavr Savr™ tomato as a model case for GM food, Case studies on patents (Basmati rice, Neem and Turmeric,), General guidelines for research in transgenic plants.

Block-4: Economic Botany:

Unit-12: General account on economic botany Origin and History, Botanical description, Cultivation and uses of Spices and Condiments: Ginger, Pepper, Cardamom, Clove, Nut-Meg, Chilly, Coriandrum, Turmeric.

Unit-13: Origin and History, Botanical description, Harvesting, Extraction and uses of Fatty oils and Vegetable Fats: Sun flower, Soya bean, Peanut, Palm Oil, Coconut and Gingelly. Fibers and Timber: Cotton, Jute, Sun hemp, Teak, Rosewood, Ebony, Sal and Mahogany.

Unit-14: Ethnobotany -Origin and History, Botanical description, Cultivation, Processing and uses of - Medicinal Plants: Rauvolfia, Aconitum, Jatamansi, Sathavari, Goggul, Basil, Saraca and Neem.

References Books:

1. James P. Kimmins, 2003, Forest Ecology 3rd edition Benjamin Cummings Publisher. ISBN: 0130662585.
2. Peter Stiling, 2001, Ecology – Theories and applications ISBN: 013091102X.
3. Jessica Gurevitch, 2006, The Ecology of Plants Sinauer Associates; 2 nd edition ISBN: 0878932941.
4. D.L. Hawskworth, 2009, Methods and Practice in Biodiversity Conservation, Springer, ISBN: 9789048138487.
5. Biber-Klemm and T. Cottier, 2006, Rights to plant genetic resources and traditional

knowledge: Basic issues and Prespectives, Cabi Publication, ISBN: 9780851990330.

6. Sammbamurthy, 2008,A Textbook of Modern Economic Botany, 1st edition, CAS publishers & Distributors Pvt, Ltd, ISBN: 8123906293.

7. Ashwin Dutt, 2008, Economic Botany, 1st edition, Aahyayan Publishers & Distributors, ISBN: 8184350694.

8. Gary W. Barrett, Murray Barrick and Eugene Odum, 2005, Fundamental of Ecology, Cengage Learning, ISBN: 9788131500200.

Course Code	Title of the Course
34633	**Algal Technology and Mushroom Technology**

Objectives:

- To give introduction about values of algae.
- To understand about micro and macro algal distribution, cultivation, and their economic importance.
- To reveal the importance and cultivation methods of mushroom.

Possible Outcomes of the course:

- The course will give a clear picture on distribution, availability and utilization of various kinds of algae belongs to both fresh water and marine habitats.
- The course will train the students to promote mass cultivation of algae and improve the livelihood of the fishermen community through providing alternative employment.
- The course will make the learners skilled in cultivation, harvesting and processing of mushroom.

Block-1: Algal Technology:

Unit-1: Economic importance of algae: Fresh water and marine algae – Macro and Micro algae – Occurrence – distribution - Cultivation – Importance of cultivation.

Unit-2: Mass cultivation techniques of microalgae- Upstream and downstream processes of algal cultivation - *Spirulina, Dunaliella, Hematococcus* and *Botryococcus*.

Unit-3: Single cell protein – bioactive compounds, industrial enzymes, biofuel and other bioproducts from algae.

Block-2: Micro and Macro Algae:

Unit-4:Micro algae used as biofertilizers – nitrogen fixing forms – free living and symbiotic nitrogen fixers – Azolla – Mass cultivation of blue green algae in field -Importance and selection of carrier materials – Immobilization technique.

Unit-5: Mass cultivation of macro algae: rope cultivation – culturing in the laboratory – Applications of seaweeds in biotechnology.

Unit-6: Seaweed liquid fertilizers preparation and their potential in agriculture and horticulture.

Unit-7: Genetics of algae – *nif* genes – strain improvement – transformation – protoplast fusion technique for macro algae.

Block-3: Mushroom Technology:

Unit-8: Introduction to mushroom cultivation – History – scope of edible Mushroom – Types of edible mushroom available in India – Medicinal and other uses - Poisonous mushroom.

Unit-9: Pure culture – preparation of medium (PDA and Oatmeal Agar medium) Sterilization – preparation of test tube slants.

Unit-10: Spawn preparation: Spwan substrate, Mother spawn in saline bottle – Inoculation, incubation, storage and transportation of spawn – Quality of spawn and contaminants.

Unit-11: Preparation of compost and cultivation of white button mushroom (*Agaricus bisporus*)- Cultivation of paddy straw mushroom (*Volvariella volvacea*) and oyster mushroom (*Pleurotus spp.*) - Low cost mushroom farm design of production.

Block-4: Storage and Values of Mushroom:

Unit-12: -Factor affecting mushroom cultivation (Temperature, pH, air and water management) - Insects and pest attacking mushroom – fungal, bacterial, viral diseases.

Unit-13: Packing and preservation techniques for mushroom - Storage -short-term storages, long term storages, drying, storages in salt solution,

Unit-14: Nutrition- nutritive value – amino acids, mineral elements – carbohydrates, crude fiber – vitamins-Cost benefit ratio – Marketing in India and abroad, export value.

Reference Books:

1. Faizal, B. and Yusuf, C. 2016. Algal biotechnology: Products and processes. Springer. ISBN: 3319123335.
2. Sharma, O.P, 2011. Algae. McGraw Hill Education (India) Private Limited. ISBN: 0070681945.
3. Demirbas, 2010, Algae Energy, 1st Edition, Newnes Publisher, ISBN: 1849960496
4. Pathak. V. N, 2011. Mushroom Production and Processing Technology. 1st Edition. Agrobios (India). ISBN: 8177540068.
5. Suman, B.C, Sharma, V.P, 2007. Mushroom Cultivation in India. Daya Publishing House, ISBN: 817035479X.

Course Code	Title of the Course
34634	**Practical – Lab III: Microbiology & Plant Pathology, Ecology, Biodiversity Conservation, Economic Botany, Algal Technology and Mushroom Technology**

Suggested Laboratory Exercises:

1. Gram staining of bacteria.
2. Sterilization methods, preparation of media and stains.
3. Symptomology of some diseased specimens: White rust, downy mildew, powdery smuts, ergot, groundnut leaf spot, red rot of sugarcane, wilts, paddy blast, bacterial blight of paddy, angular leaf spot of cotton, tobacco mosaic, little leaf of brinjal, seasame phyllody, mango malformation (based on availability).
4. Preparation of different media for the isolation and culture of fungi from soil and diseased materials
5. Isolation of bacteria from diseased plant specimen.
6. Collection and study of the crop diseases from the local and out station fields.
7. Demonstration of the production of cell wall degrading enzymes.
8. Demonstration of the production of mycotoxins.
9. Culture and study of some common plant pathogenic fungi.

11. Study of minimal size of the quadrat by Species- Area Curve method for studying the forest and grassland vegetation.

12. To determine minimal number of quardats by Species –Area Curve method in forest and grassland.

13. Sampling of the grassland vegetation by quadrat method for determining the density, frequency and basal cover of different species.

14. To determine diversity indices (Shannon-Wiener, concentration of dominance, species richness, equitability and B-diversity) for protected and unprotected grassland stands.

15. To estimate IVI of the species in a woodland using point centred quarter method.

16. To determine gross and net phytoplankton productivity by light and dark bottle method.

17. To determine soil moisture content, porosity and bulk density of soils collected for varying depths at different locations.

18. To determine the water holding capacity of soils collected from different locations.

19. To determine percent organic carbon and organic matter in the soils of cropland, grassland and forest.

20. To estimate the dissolved oxygen content in fresh waters by azide modification of Winkler's method.

21. To estimate rate of carbon dioxide evolution from different soils using soda lime or alkali absorption method.

22. Isolation and cultivation of fresh and marine water algae.

23. Cultivation of Spirulina, Azola.

24. Immobilization of algae and use of algae as biocarrier.

25. Preparation of seaweed liquid biofertilizers.

26. Survey of mushrooms.

27. Isolation and pure culture of edible mushrooms.

28. Spawn preparation for the growth of mushrooms.

29. Mass cultivation of edible mushrooms.

30. Value added food preparation from mushroom.

FOURTH SEMESTER

Course Code	Title of the Course
34641	**Plant Molecular Biology**

Objectives of the Course:

- To study about molecular aspects of plants.
- To study about manipulation at molecular level for the strain improvement of the plants.
- To study about molecular pharming.

Possible Outcomes of the Course:

- The course will give knowledge about plant genome organization.
- The course will make the learners understanding about various methods of molecular manipulation.
- The course will make the leaners understanding about the production of transgenic plants and molecular pharming.

Block-1: Plant Genome Organization:

Unit-1: Plant genome organization – Nucleus, Chloroplast and Mitochondria, Structural features of a typical plant gene- Chromatin organization in plants.

Unit-2: Nucleus-encoded and chloroplast-encoded genes for chloroplast proteins- Targeting of nuclear encoded cytoplasmic proteins to chloroplast compartments.

Unit-3: Mitochondrial genome and Cytoplasmic male sterility- Seed storage proteins – Classification and functions.

Unit-4: Regulation of gene expression in plant development- Plant hormones- Plant transposons.

Block-2: Plant Genetic Engineering:

Unit-5: Introduction to Plant Genetic Engineering- Tools of genetic engineering- General Methodology.

Unit-6: Selectable markers – Types and their role in plant transformation- Antibiotic sensitivity assay- Reporter genes – Types and role in optimizing transformation- Promoters used in plant vectors.

Unit-7: Plant transformation techniques – Direct (chemical, mechanical and electrical) and Indirect methods (Agrobacterium mediated)-Mechanism of T-DNA transfer to plants-

Unit-8: Types of Ti-plasmid based vectors (Co integrate and binary vectors) for plant transformation, Agro infection, Symbiotic nitrogen fixation in legumes by Rhizobia.

Block- 3: Transgenic Plants

Unit-9: Plant genetic engineering for herbicide resistance- Abiotic stress tolerance.

Unit-10: Insect pest resistance (Bt and proteinase inhibitor)- Cytoplasmic male-sterility- Virus resistance (Antisense RNA approach, Cross protection Satellite RNA, Ribozymes and Coat protein mediated protection)- delays of fruit ripening and resistance to fungi and bacteria.

Unit-11: Case studies - Golden rice, Flavr Savr®, Chloroplast engineering and Transplastomic plants.

Block- 4: Molecular Markers; Molecular pharming:

Unit-12: Molecular markers – STS, Microsatellites, RAPD, SCAR (Sequence Characterized Amplified Region) and AFLP for genetic diversity.

Unit-13: Molecular pharming – Introduction, Transgenic plant derived products for commercial applications,

Unit-14: Bioremediation through plants. Tagging, mapping and cloning of plant genes.

Reference Books:

1. H. S. Chawla, 2009, Introduction to Plant Biotechnology, 3rd Edition, Oxford and IBH Publishing Co. Pvt. Ltd. ISBN: 978-81-204-1732-8.
2. Peter J. Lea, Richard. C. Leegood, 1999, Plant Biochemistry and Molecular Biology, 2nd Edition, John Wiley and Sons.
3. J. Hammond, P. McGarvey and V. Yusibov (Eds), 2000, Plant Biotechnology-New Products and Applications. Springer-Verlog, ISBN: 978-3-540-66265-5.
4. Maarten J. Chrispeels and David E. Sadava, 2000, Plants, Genes and Agriculture, Jones and Barlett Publishers. ISBN: 0763715967.
5. Adrian Slater, Nigel W. Scott and Mark R. Fowler, 2003, Plant Biotechnology: The Genetic Manipulation of Plants, Oxford University Press.

Course Code	Title of the Course
34642	**Biostatistics, Biophysics and Bioinformatics**

Objectives of the Course:

- To study about basics of statistics and utilization of statistical method in biology.
- To study about physical properties of biomaterials.
- To study about modern tools of bioinfromatics an inter-disciplinary subject to help the biologists in research perspectives.

Possible Outcomes of the Course:

- The course will give knowledge about data collection, processing and interpretation of biological samples through statistical methods.
- The course will make the learners understanding about basic physical property of biological materials.
- The course will give knowledge about modern tools of bioinformatics.

Block-1: Biostatistics:

Unit-1: Data collection and interpretation- Types of population - sample – non probability sampling techniques - random sampling techniques.

Unit-2: Choice of sampling methods-sampling and non-sampling errors- Diagrammatic and graphical representation of data.

Unit-3: Measures of central tendency: Mean - median - mode. Measures of dispersion: Range - mean deviation - standard deviation.

Unit-4: Test of significance: Null hypothesis - alternate hypothesis - confidence interval -level of significance - p value - S.E of mean - S.E of standard deviation - Z test - t test - chi square test.

Block-2: Biophysics

Unit-5: Bioenergetics- Energy and work- Laws of Thermodynamics-Energy transductions in biological systems.

Unit-6: Redox potential, Redox couples, ATP bioenergetics, Order of reactions.

Unit-7: Photobiology: Dual nature of light, characteristics of solar radiation, solar energy.

Unit-8: Efficiency of atoms - Absorption spectra in molecules, energy states, De-excitation.

Block-3: Bioinformatics-Data Retrieval:

Unit-9: Bioinformatics - introduction - biological data bases - nucleotide sequence data

bases, protein sequence data bases, specialized sequence data bases.

Unit-10: Data retrieval and analysis, sequence and retrieval system.

Unit-11: Sequence alignment - sequence similarity searches, amino acid substitution matrices,

Block-4: Data base search; Sequence alignment; Phylogenetic analysis:

Unit-12: Data base searches - FASTA, BLAST - PSI BLAST.

Unit-13: Mutiple sequence alignment – Clustered W:

Unit-14: Phylogenetic analysis, PHYLODRAW- Phylogenetic tree.

Reference Books:

1. Bernard Rosner, 2010, Fundamentals of Biostatistics, 7th Edition, Cengage Learning, Inc, ISBN: 0538733497.
2. Rodney Cotterill, 2014, Biophysics: An Introduction, Wiley India Private Limited, ISBN: 8126551607.
3. Rastogi, 2008, Bioinformatics: Methods and Applications : Genomics, Proteomics and Drug Discovery, 4th eition, PHI Learning Private Limited, New Delhi, ISBN: 8120347854.
4. Pranav Kumar, 2014, Biophysics and Molecular Biology, Patnfinder Publisher, ISBN: 938047315X.
5. R. Perl, 2014, Building Bioinformatics Solutions, 2nd edition, Oxford university, ISBN: 9780199658558.

Course Code	Title of the Course
34643	**Horticulture and Plant Tissue Culture**

Objectives of the Course:

- To learn about horticulture.
- To acquire the fundamental of plant tissue culture techniques.
- To know about the basic concepts and principles of plant tissue culture.
- To learn about various types of gardening and maintenance of the same.

Possible Outcomes of the Course:

- The study of horticulture gives knowledge about the art, science, technology and business of growing plants.
- The course will make the learners knowing about micro propagation technique towards development of disease-free plants.

Block-1: Horticulture:

Unit-1: Scope of horticulture - relation to agriculture, agronomy and forestry - Divisions of horticulture.

Unit-2: Climate, soil and nutritional needs- Importance of macro and micronutrients,- different types of soil- Sterile soil mixtures (vermiculite, perlite, etc.).

Unit-3: Different types of organic manure's and inorganic fertilizers - Water irrigation; advanced irrigation system such as drip, microtube and sprinkler systems.

Block-2: Biological Components of Horticulture:

Unit-4: Growth and Development- Seed dormancy, viability and germination-Vegetative and reproductive growth of plants.

Unit-5: Native and synthetic hormones and other growth regulators- their importance in horticulture, gardening and landscaping.

Unit-6: Vegetative propagation using stem, leaf and root cuttings- Propagation by division and layering, bulbs, corms, tubers and rhizomes-budding and grafting.

Unit-7: Production of seeds, their certification, storage and germplasm collection.

Block- 3: Gardening:

Unit-8: Outdoor Gardening: Principles and methods of designing outdoor garden - hedges, edges, fences, trees, climbers, rockeries, arches, terrace garden - Lawn making and maintenance.

Unit-9: Water garden - cultivation of water plants- Layout for a model college garden.

Unit-10: Indoor gardening: Foliage plants, flowering plants, hanging basket, Bonsai plants - Training and pruning.

Block-4: Plant Tissue Culture:

Unit-11: Introduction to plant cell and tissue culture- Totipotency- Sterilization techniques- Nutrient media composition and preparation of solid and liquid cultures- Establishment and maintenance of callus and suspension cultures from representative monocot and dicot plants.

Unit-12: Micropropagation – Introduction, stages and types of explants for commercial propagation- Virus elimination- Commercial importance and applications of micropropagation.

Unit-13: Plant regeneration-Organogenesis and Somatic embryogenesis. Role of hormones in regeneration- Artificial seeds-Principle and method involved in the production- Somoclonal variations-Significance, mechanism and applications.

Unit-14: Protoplast isolation - Principles and protocols, protoplast culture and fusion- Importance of protoplast fusion and applications- In vitro production of secondary plant products.

Reference Books:

1. Chadha, K. L. 2012. Hand Book of Horticulture. ICAR, New Delhi. ISBN: 89171640065.
2. Simson, S. P., and M. C. Straus 2009. Basics of Horticulture, Oxford Book Company, New Delhi. ISBN: 9380179189
3. Misra, Kumar, K and R. Kumar, 2014. Fundamentals of Horticulture, Biotech, New Delhi. ISBN: 9380179189.
4. S. S. Bhojwani and M. K. Razdan, 2004, Plant Tissue Culture: Theory and Practice, Elsevier, ISBN: 0444-816232.
5. O. L. Gamborg and G. C. Philips, 2013, Plant Cell, Tissue and Organ Culture, Springer. ISBN: 9783642489747.
6. H. S. Chawla, 2009, Introduction to Plant Biotechnology, 3rd Edition, Oxford and IBH Publishing Co. Pvt. Ltd. ISBN: 9788120417328.

Course Code	Title of the Course
34644	**Practical - Lab IV: Plant Molecular Biology, Biostatistics, Biophysics and Bioinformatics, Horticulture and Plant Tissue Culture.**

Suggested Laboratory Exercises:

1. Isolation of plant genomic DNA.
2. Isolation of total RNA.
3. Isolation of plasmid DNA from bacteria.
4. Quantification of DNA.
5. Examples of positive and negative control of gene expression.
6. Restriction and modification of DNA by using restriction enzymes and ligases.
7. Cloning vectors - pBR322, pUC18 and Col E1.
8. Biology and molecular basis of *Agrobacterium* mediated plant transformation and its applications.
9. Gene transfer methods – Biolistics ®, PEG and Electroporation.
10. Genetic Markers – RFLP and ISSR.
11. Blotting techniques: Southern, Northern and Western blotting.
12. PCR- Types of PCR- DNA foot printing, finger printing.
13. Genetically Engineered Foods.
14. Sampling of biological data.
15. Measures of central tendency.
16. Test of significance- Chi square, T-test
17. FASTA, BLAST - PSI BLAST.
18. Clustered W, PHYLODRAW
19. Sterilization of Plant Tissue Culture Media/Glassware with the help of autoclave and hot air oven.
20. Media components and composition of important culture media (MS medium).
21. Micropropagation of given plant materials – shoot and root formation.
22. Plant Growth Regulators – IAA, IBA, NAA, GA_3, 2, 4-D.
23. Somatic embryogenesis – Embryo development.
24. Another culture, pollen culture, ovule and ovary culture.
25. Cryopreservation technique.
26. Importance of macro and micro nutrients in horticulture.

27. Types of irrigation methods – drip, microtube and sprinkler systems.
28. Seed dormancy.
29. Seed viability test by Tetrazolium chloride.
30. Role of growth hormones on horticultural propagating plant.
31. Propagation methods – Layering, budding and grafting.
32. Bonsai plants.

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e. 3. Duration of the Programme:

The programme for the degree of Master of Science in Botany shall consist of two academic years divided in to four semesters. Each semester consists of Three Theory course and One Practical course. Theory course and Practical course carry 4 credits each. Each semester consists of 16 credits and the programme consists of 64 credits in total.

e. 4. Faculty and Support Staff Requirements:

The programme for the degree of Master of Science in Botany requires following faculty and supporting staff:

Staff Category	Required
Core Faculty #	3
Faculty- Specialization #	2
Laboratory Assistant	1
Clerical Assistant	1

Faculty may be belongs to at least Assistant Professor level either permanent or part-time.

e. 5. Instructional Delivery Mechanisms:

The instructional delivery mechanisms of the programme includes SLM – study materials, face to face contact session for both theory and practical courses of the programme, e-content of the study materials in the form of CD, MOOC courses and virtual laboratory wherever applicable.

e. 6. Identification of Media:

The SLM – designed study materials will be provided in print media as well is in the form of CD which carries electronic version of the study material in addition to MOOC and virtual laboratory courses.

e. 7. Student Support Services:

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like counseling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres.

The post-admission student support services like issuance of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter only. The conduct of end semester examinations, evaluation and issuance of certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

f. Procedure for Admission, curriculum transaction and evaluation:

f. 1. Procedure for Admission:

A candidate who has passed the B.Sc., degree (10 + 2+ 3 system) examination in Botany/ Plant Science/ and Plant Biology & Plant Biotechnology of the University or an Examination of any other University accepted by the Syndicate as equivalent thereto shall be eligible to appear and qualify for the M.Sc. Degree in Botany of this University after a course of study of two academic years.

f. 2. Curriculum Transactions:

The classroom teaching would be through chalk and talk method, use of OHP, Power Point presentations, web-based lessons, animated videos, etc. The face to face contact sessions would be such that the student should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

For practical courses exclusive study materials containing the requirements, procedure for the experiments will be issued to the learners. In the laboratory, instruction would be given for the experiments followed by demonstration and finally the students have to do the experiments individually.
The face to face contact sessions will be conducted in following durations;

Course Type	Face to Face Contact Session per Semester (in Hours)
Theory Courses (3 courses with 4 credits each)	48
Practical Courses (1 course with 4 credits)	120
Total	**168**

f. 3. Evaluation:

The evaluation shall be conducted separately for theory and practical to assess the knowledge acquired during the course of study. There shall be two systems of evaluation viz., internal assessment and end semester examinations.

In the case of theory courses the internal evaluation shall be conducted as Continuous Internal Assessment via. Student's assignment preparation, seminar, etc. The internal assessment shall comprise of maximum 25 marks for each subject. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

In the case of Practical courses for M.Sc. programme in Botany should be conducted at every semester. The internal evaluation of practical courses will be done through continuous assessment of skill in demonstrating the experiments, record note preparation, and *viva voce*. The external evaluation consists of an end semester practical examinations which comprise of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

f. 3.1. Question Paper Pattern:

Answer all questions (one question from each Block with internal choices Time: 3 Hours Max. Marks: 75
Part A- 10 x 2 Marks = 20 Marks
Part B -5 x 5 Marks = 25 Marks
Part C- 3 x 10 Marks = 30 Marks

f. 3.2. Distribution of Marks in Continuous Internal Assessments:

The following procedure shall be followed for awarding internal marks for theory courses

Component	Marks
Student's Assignments	25
Total	**25**

The following procedure shall be followed for awarding internal marks for theory courses

Internal –Practical	Marks
Continuous Assessment (demonstration and observation of experiments)	10
Record Note Book	10
Viva	5
Total	**25**

f. 3.3. Passing Minimum:

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50% (50 marks out of 100 marks) in the University end semester examination. Then the total marks secured by the candidate will be taken and add with his/her internal marks (Maximum marks 25).

For a pass in the Practical paper, a candidate has to secure a minimum of 50 % (37.5 marks) marks in the University end semester practical examination (75 marks) examination. Then the total marks secured by the candidate will be taken and add with his/her internal marks (Maximum marks 25).

There is no passing minimum for the record notebook. However submission of a record notebook is a must.

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

f. 3.4. Marks and Grades:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester
G_i = Grade Point obtained for course i in any semester.
n refers to the semester in which such courses were credite

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

Grade Point Average = $\frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = $\frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme.}}$

CGPA	Grad	Classification of Final Result
9.5-10.0 9.0 and above but below 9.5	O+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D++ D+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A++ A+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

f. 4. Maximum duration for the completion of the course:

The maximum duration for completion of M.Sc., Degree in Botany programme shall not exceed ten semesters from their fourth semester.

f. 5. Commencement of this Regulation:

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter

f. 6. Fee Structure:

The programme has the following Fee Structure:

Sl. No.	Fees Detail	Amount in Rs.		Nature of Fees
		First Year	Second Year	
1	Admission Processing Fee	**300.00**	**-**	Non- Refundable
2	Course Fee	**20000.00**	**20000.00**	Non- Refundable
3	ICT Fees	**150.00**	**150.00**	Non- Refundable
	TOTAL	**20450.00**	**20150.00**	Non- Refundable

The above mentioned fee structure is exclusive of Exam fees.

g. Requirement of the laboratory support and Library Resources:

A well-equipped Botany Laboratory exclusive for Distance Education Programme was established at the Science Block, Alagappa University, Karaikudi with all the necessary equipments for conducting face to face contact sessions for Practical courses of M. Sc., Botany programme. The students who have enrolled themselves in M. Sc., Botany Programme shall attend the face to face contact session for Practical Courses **only** at Botany Laboratory, Directorate of Distance Education, Located at Science Block, Alagappa University, **Karaikudi**.

Directorate of Distance Education, Alagappa University, Karaikudi housing an exclusive Library facility with adequate number of copies of books in relevant titles for M. Sc., Botany programme. The Central Library of Alagappa University also having good source of reference books. The books available at both the libraries are only for reference purpose and not for lending services.

h. Cost estimate of the programme and the provisions-Approximate:

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with M. Sc., Botany Programme as follows

Sl. No.	Expenditure Heads	Approx. Amount in Rs.
1	Programme Development	**20,00,000.00**
2	Programme Delivery	**24,00,000.00**
3	Programme Maintenance	**5,00,000.00**

i. Quality assurance mechanism and expected programme outcomes:

The quality of the programme depends on scientific construction of the curriculum, strong-enough syllabi, sincere efforts leading to skilful execution of the course of the study. The ultimate achievement of M. Sc., Botany programme of study may reflect the gaining of knowledge and skill in the subject Botany. And all these gaining of knowledge may in Botany may helps the students to get new job opportunities, upgrading in their position not only in employment but also in the society, make students feel thirsty to achieve in research in the fields associated with the discipline- Botany, achieving in competitive examinations on the subject- Botany, etc.

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations. Apart from the end semester examination-based review feedback from the alumni, students, parents and employers will be received and analyzed for the further improvement of the quality of the M. Sc., Botany Programme.

Minutes of the Meeting of the Board of Studies in Botany (For M.Sc. Botany Programme to be offered through ODL Mode) held at The Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 01.09.2017, at 12.05 P.M.

Members Present:

1.	Dr. S. Murugesh	-	Member
2.	Dr. S. Chandrasekaran	-	Member
3.	Dr. M. Ramesh	-	Member
4.	Dr. A. Arumugam	-	Chairman
5.	Dr. M. Jothi Basu	-	Member
6.	Dr. G. Selvakumar	-	Member
7.	Dr. N. Anusuya	-	Member

After the deliberation and discussion the board resolved the following:

1. The board considered the curriculum design and detailed syllabi of M.Sc., Botany programme, prepared as per the norms by the Chairman and the Board Members, scrutinized and suitably modified the same.
2. The board resolved to approve curriculum design, detailed syllabi and other regulations for the M.Sc., Botany programme to be offered by the Directorate of Distance Education of Alagappa University are given in Annexure I.

Dr. S. Murugesh

Dr. S. Chandrasekaran

Dr. M. Ramesh

Dr. A. Arumugam

Dr. M. Jothi Basu 01/09/17

Dr. G. Selvakumar

Dr. N. Anusuya

ALAGAPPA UNIVERSITY

[Accredited with 'A+' Grade by NAAC (CGPA:3.64) in the Third Cycle]
(A State University Established by the Government of Tamilnadu)

KARAIKUDI – 630 003

DIRECTORATE OF DISTANCE EDUCATION

website: alagappauniversity.ac.in
Phone: 04565-226 250

e-Mail ID: dde@alagappauniversity.ac.in
Fax: 04565-225216

13.10.2017

Dr. P. Sivakumar
Director i/c

WRONG ENTRY

The programme M.Sc., Botany - Specialization with Plant Biotechnology has been entered mistakenly. We also sent the mail to the technical team for correction. They informed that, reverse action is not possible. We are unable to delete the same, therefore this Programme may not be considered.

13/10/17

Director i/c

Director
Distance Education
Alagappa University
Karaikudi - 630 004

ALAGAPPA UNIVERSITY

(State University Accredited with 'A+' Grade, 3.64 CGPA on 4.0 point scale by NAAC)

KARAIKUDI

DIRECTORATE OF DISTANCE EDUCATION

PROGRAMME PROJECT REPORT

M.Sc. DEGREE in CHEMISTRY

REGULATIONS AND SYLLABUS

[For the Candidates admitted from the academic year 2018 – 2019 onwards]

Credit Based System

CONTENTS

a. Programmer's mission & objectives

b. Relevance of the programme with Alagappa University's mission and goals:

c. Nature of prospective target group of learners

d. Appropriateness of programme to be conducted in distance learning mode to acquire specific skills and competence

e. Instructional design

- e. 1. Curriculum Design
- e. 2. Detailed Syllabi:
- e. 3. Duration of the Programme:
- e. 4. Faculty and Support Staff Requirements:
- e. 5. Instructional Delivery Mechanisms:
- e. 6. Identification of Media:
- e. 7. Student Support Services:

f. Procedure for Admission, curriculum transaction and evaluation

- f. 1. Procedure for Admission
- f. 2. Curriculum Transactions
- f. 3. Evaluation
  - f. 3.1. Question Paper Pattern
  - f. 3.2. Distribution of Marks in Continuous Internal Assessments
  - f. 3.3. Passing Minimum
  - f. 3.4. Marks and Grades
  - f. 3.5. Maximum duration for the completion of the course
  - f. 3.7 Transfer of credits
  - f. 3.8. Revision of regulations and curriculum
  - f. 3.9. Commencement of this regulation
- f. 4. Fee Structure

g. Requirement of the laboratory support and Library Resources

h. Cost estimate of the programme and the provisions

i. Quality assurance mechanism and expected programme outcomes

a. Programme's Mission & Objectives:

To afford a High Quality Post Graduate Degree (M.Sc.,) Chemistry through Distance Learning mode to the graduate-aspirant in order to to educate the undergraduate students in the fascinating fields of chemistry in an effective manner. M.Sc. Chemistry is a unique kind of course dealing with all aspects of chemistry such as preparation, properties, structure elucidation, kinetics and mechanism of the reaction, techniques of analysis for different kinds of materials, which are very essential for the human society

OBJECTIVES OF THE COURSE

All the changes in life in one-way or other involve CHEMISTRY. Chemistry is central to the current revolutions in Science. No educated person today can understand the modern world without a basic knowledge of chemistry. The existence of a large number of chemical factories, mines and related industries necessitates chemistry education. The major objectives of M.Sc. Chemistry course are:

- To impart knowledge in fundamental aspects of the three branches of chemistry viz. Organic chemistry, Inorganic Chemistry and Physical Chemistry.
- To acquire deep knowledge in the study of physical, chemical, electrochemical and magnetic properties, structure elucidation using various techniques and applications of various organic and inorganic materials and
- To acquire basic knowledge in the specialized areas such as Polymer chemistry and Instrumental methods of analysis.

b. Relevance of the Programme with Alagappa University's Mission and Goals:

In order to align with the mission and goals of Alagappa University the M.Sc., Chemistry Programme is planned to deliver in Distance Learning mode which may reach the maximum number of student aspirants who are unable to thrive to spend non-elastic timings of formal conventional class room education. Such a higher education in science subject with appropriate practical experiences will enrich the human resources for the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

c. Nature of Prospective Target Group of Learners:

This M.Sc., Chemistry programme through Distance Learning mode is developed by keeping in mind to give opportunity to economically and socially excluded people includes graduates of various socio-economic status viz., unemployed youths, employed with

marginalized salary due to lack of sufficient knowledge in the subject chemistry. Also, the target group of learners includes various level employees of industrial employees, research and development company worker, secondary –level school teachers, research aspirants, women taking care of family –the important unit of the community, etc.,

d. Appropriateness of programme to be conducted in Distance learning mode to acquire specific skills and competence:

M.Sc., Chemistry programme through Distance Learning mode is developed in order to give subject-specific skills including i) Apply knowledge obtained in Chemistry lecture to problem solving and critical thinking in the laboratory, ii) Engage in safe laboratory practices by handling laboratory glassware, equipment, and chemical reagents appropriately, using general guidelines and basic knowledge about the common hazards associated with them in an organic chemistry laboratory, iii) Maintain an appropriate scientific notebook using notational and descriptive content containing information on relevant chemical reagents, experimental procedure followed, data collected, and observations made during the experimental process, iv) Assemble glassware and perform the following techniques as a part of synthetic procedures: aqueous workup, distillation, reflux, separation, isolation, and crystallization, v) Predict the outcome of several common organic reaction types through a basic understanding of starting materials, functional groups, mechanism, and typical reaction conditions, vi) Characterize prepared substances by physical and spectroscopic means, vii) Develop the skill set necessary to continue on to higher studies such as M.Phil and Ph.D. in Chemistry, viii) Can confidently attend and clear competitive examinations especially CSIR NET, ix) Become Chemistry teachers in educational institutes and scientist in research laboratories.

e. Instructional design
e. 1. Curriculum Design

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	Credit.
FIRST YEAR						
I Semester						
1.	34411	Inorganic Chemistry -I	25	75	100	4
2.	34412	Organic Chemistry - I	25	75	100	4
3.	34413	Physical Chemistry - I	25	75	100	4
4.	34414	Practical : Analytical Chemistry Practical	25	75	100	4
		Total	**100**	**300**	**400**	**16**
II Semester						
5.	34421	Inorganic Chemistry -II	25	75	100	4
6.	34422	Organic Chemistry - II	25	75	100	4
7.	34423	Physical Chemistry - II	25	75	100	4
8.	34424	Practical : Organic Chemistry Practical	25	75	100	4
		Total	**100**	**300**	**400**	**16**
SECOND YEAR						
III Semester						
9.	34431	Advanced Inorganic Chemistry	25	75	100	4
10.	34432	Advanced Organic Chemistry	25	75	100	4
11.	34433	Spectroscopy – Applications in Organic and Inorganic Chemistry	25	75	100	4
12.	34434	Practical : Inorganic Chemistry Practical	25	75	100	4
		Total	**100**	**300**	**400**	**16**
IV Semester						
13.	34441	Analytical Chemistry	25	75	100	4
14.	34442	Applied Chemistry	25	75	100	4
15.	34443	Advanced Physical Chemistry	25	75	100	4
16	34444	Practical : Physical Chemistry Practical	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		GRAND TOTAL	**400**	**1200**	**1600**	**64**

Course Code Legend:

3	**4**	**4**	**Y**	**Z**

344- M. Sc., Chemistry Programme
Y -Semester No
Z- Serial of Course number in the semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

No. of Credits per Course (Theory)	- 4
No. of Credits per Course (Practical)	- 4
Total No. of Credits per Semester	- 16
Total No. of Credits the Programme- 16 X 4	= 64

e. 2. Detailed Syllabi:

FIRST SEMESTER

Course Code	Title of the Course
34411	INORGANIC CHEMISTRY – I

Course Objectives: The major objectives of this course are to understand the concepts of

- ✓ Chemical periodicity, structure and bonding of atoms
- ✓ Acids and basis, solid state structures and its determination
- ✓ To distinguish isopolyacids from heteropolyacids.
- ✓ To know about various types of silicates.

Learning Outcomes: The student would be able to

- ✓ Predict the shape of atoms and chemical bonding
- ✓ The apply the Bronsted and Lewis concept of acids and bases for different explanations
- ✓ Understand the structure of solids having different ratio of atoms
- ✓ Know about the chemistry of pH and Buffer solution

Block -1: Basic concepts in inorganic chemistry

Unit –1: Chemical periodicity

Chemical periodicity – ionic radii, ionization potential, electron affinity, electro negativity, concept of hybridization - Molecular orbitals and electronic configuration of homonuclear and heteronuclear diatomic molecules - Shapes of polyatomic molecules.

Unit – 2: VSEPR theory

VSEPR theory - shapes of molecules. The concept of multi centre bonding. Structure and bonding in fluorine and oxygen compounds of xenon. Bonding in simple triatomic molecules/ions.

Unit – 3: Molecular Orbital theory of covalent bonding

Bond length, bond order, bond angle, bond energy and magnetism – ionic character in a covalent bond. MO approach to covalent bonding - symmetry and overlap of atomic orbitals -symmetry of molecular orbitals - sigma-pi-and delta-bonding - energy levels in homo- and hetero nuclear diatomic molecules.

Unit – 4: Intermolecular forces and Lattice energy

Intermolecular forces -Lattice energy and its calculations by Born-lande and Born-Meyer equations- Determination by Born-Haber cycle - Kapustinski equation. Dipolemoment.

Unit – 5: Properties of ionic compounds

Properties of ionic compounds-hardness and electrical conductivity. Energetic of dissolution of ionic compounds in polar solvents.

Block -2: Acids, and Base

Unit -6: Bronsted concept of acids and bases

Bronsted concept - relative strength in aqueous medium-levelling and differentiating solvents - periodic trends in Bronsted acidity – acid strength of oxy-acids- advantages and limitations of Bronsted concept.

Unit -7: Lewis concept of acids and bases

Lewis acids and bases – relative order of acid strength of boron halides and basic strength of hydrides. Comparison between Bronsted and Lewis acids and bases

Unit -8: HSAB

Hard and Soft Acid and Bases (HSAB) principle– applications-limitations. pH and Buffer solutions.

Block -3: Polyacids and Silicates

Unit -9: Polyacids

Isopoly and heteropoly acids, and their anions, Anderson structure. Keggin structure.

Unit -10: Silicates

Types of silicates - Ortho and meta silicates, pyrosilicates, ring silicates, chain silicates, double chain silicates, sheet silicates, three dimensional silicates.

Unit -11: Silicate with frame work structures

Feldspar, zeolites - molecular sieves - clay minerals

Block -4: Solid State Chemistry

Unit -12: Crystal structure of solids

Close packing of atoms and ions –HCP, FCC and BCC types of solids- calculation of packing voids – radius ratio rule –its influence on structures. Classification of ionic structures - AX, AX_2, AX_3 types – AX type (ZnS, NaCl, CsCl) structures only - AX_2 type (fluorite, rutile, beta-cristobolite) structure only - layer structure – CdI_2- Nickel arsenite structure.

Unit -13: Defects in crystal

Schottky and Frenkel defects -explanation and calculation of number defects per cm^3 – metal excess defect - F-centers and interstitial ions - metal deficiency defect - positive ions absent - extra interstial negative ions.

Unit -14 : Electrical properties of solids

Electrical properties of conductor, semiconductors and insulators - band theory of solids. Structure of graphite and diamond

Text books

1. **Modern aspects of Inorganic chemistry,** H.J. Emelius and Sharpe, Universal book Stall, New Delhi, 1989.

2. **Inorganic Chemistry- Principles of structure and reactivity**, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th edition, Pearson-Education, 2002.
3. **Advanced Inorganic Chemistry** - F.A. Cotton and G. Wilkinson, Wiley Eastern, 5th edition, 1998.
4. **Inorganic Solids,** D. M. Adams, John Wiley Sons, 1974.

Reference books

1**. Inorganic Chemistry**, D. F. Shriver and P. W. Atkins, Oxford U.K., 1999.
2. Concise Co-ordination Chemistry, R. Gopalan, 1E 2nd reprint, VPH (P) Ltd., 2009.
3. Inorganic Chemistry, G. S.Sodhi; Ist Edition, VB (P) Ltd, 2006.
4. Solid state chemistry and its applications, A.R. West, Wiley, New York, 1984.
5. Structural methods in Inorganic Chemistry, E.A.V. Ebsworth, D.WH. Rankine and S. Craddock, Black well Scientific Publ., 1987.

Course Code	Title of the Course
34412	ORGANIC CHEMISTRY – I

Course Objectives: The primary objective of this course is to introduce the student to the concepts of organic chemistry and to develop critical thinking skills. The objectives are

- To learn the about the nucleophilic and electrophilic substitutions reaction
- To be able to interpret the reaction pathways
- To learn the stereochemistry of organic compounds

Learning Outcomes: The students shall be able to:

- ✓ Understand and give the IUPAC name of all organic compounds, Reaction Mechanism, Aromaticity nature of the compounds.
- ✓ Efficient knowledge in the reaction mechanism of electrophilic and Nucleophilic reaction and naming reactions.
- ✓ Create a valuable understanding of the main and important concepts in this course.

Block -1: Fundamentals of organic chemistry

Unit – 1: IUPAC nomenclature of organic compounds

Bicyclic, polycyclic and Heterocyclic compounds.

Unit – 2: Electron Displacement in molecules

Inductive and field effects – mesomeric effect – steric inhibition of resonance – steric enhancement of resonance – hyperconjugation - time variable effects -hydrogen bonding – effect of structure on the dissociation constants of acids and bases.

Unit – 3: Aromaticity

Electron delocalization and resonance -Aromatic, antiaromatic, homoaromatic and non-aromatic compounds -Molecular orbital picture of Aromaticity- HMO theory

Unit – 4: Alternate and non-alternate hydrocarbons

Aromaticity on cyclopentadienyl anion, fulvene, ferrocene, azulene, tropolones, annulens and tropylium cations. Aromaticity on larger annulenes, hetero annulenes and fullerenes (C_{60}).

Block -2: Stereochemistry of Organic compounds

Unit – 5: Introduction to stereochemistry

Introduction to molecular symmetry and chirality – examples from common objects to molecules – axis, plane, center, alternating axis of symmetry. Stereoisomerism – definition based on symmetry and energy criteria – configuration and conformational stereoisomers.Center of chirality – molecules with C, N, S based chiral centers – absolute configuration - enantiomers – racemic modifications.

Unit – 6: Nomenclature

R and S nomenclature using Cahn-Ingold-Prelog rules – molecules with a chiral center and Cn – molecules with more than one center of chirality – definition of diastereoisomers – constitutionally symmetrical and unsymmetrical chiral molecules - erythro, threo nomenclature – E and Z nomenclature – out/in isomers.

Unit – 7: Stereochemistry and absolute configuration

Axial, planar and helical chirality – examples – stereochemistry and absolute configuration of allenes, biphenyls and binaphthyls, ansa and cyclophanic compounds, spiranes, exo-cyclic alkylidenecycloalkanes.

Block -3: Reaction mechanism and molecular rearrangement

Unit – 8: Kinetics of reaction mechanisms

Classification of organic reactions - Principle of microscopic reversibility - Hammond postulate - Kinetic and thermodynamic control of chemical reactions - Kinetic and non-kinetic methods for determining organic reaction mechanisms

Unit – 9: Carbocation

Structure and stability of carbocations, Classical and non-classical carbocations, Neighbouring group participation and rearrangements including Wagner-Meerwein, Pinacol-pinacolone, semi-pinacol rearrangement

Unit – 10: Molecular rearrangements

Mechanisms of Wagner – Meerwein, Demzonev, Wolff, Baeyer-Villiger, Stern, Beckmann and Favorskii rearrangements.

Block -4: Substitutions reaction

Unit – 11: Aliphatic nucleophilic substitutions

Nucleophiles and nucleofuge – S_N1, $S_N2$and S_Ni mechanisms with examples - Factors influencing the Aliphatic nucleophilic substitutions– kinetics of Aliphatic nucleophilic substitutions– stereochemistry- competition between$S_N1$and S_N2

Unit – 12:Aliphatic electrophilic substitutions

S_E1, S_E2 and S_Ei reaction and mechanism.

Unit – 13:Aromatic electrophilic substitutions

Aromatic electrophilic substitution reaction, O/P ratio, ring activator, deactivator, arenium ion mechanism, typical reaction and Mechanisms of nitration, diazonium coupling, sulphonation, halogenation, Friedel craft alkylation and acylation - Gatermann Koch formylation - VilsmierHaak reaction.

Unit – 14:Aromatic Nucleophilic substitutions

Addition-elimination reaction, elimination –addition reaction and mechanism- benzyne mechanism - Von-Richter reaction.

Text books

1. **Advanced Organic Chemistry – Reactions, Mechanisms and Structure,** Jerry March, IV Edn., John Wiley & Sons, 1992.
2. **A Guide Book to Mechanisms in Organic Chemistry**, P. Sykes, VI Edn., Longmans Scientifics and Technical, Essex 1986.
3. **Reaction Mechanism in Organic Chemistry**, S.M. Mukherji and S.P. Singh, III Edn. MacMillan.1984.
4. **Organic Chemistry, Vol. I & II**, I.L. Finar**,** V Edn. First Indian reprint, Pearson Education Asia Pvt. Ltd. 2000.

Reference books

1. **Advanced Organic Chemistry, Part A& B**, F.A. Carey and Sundberg, III edition, Plenum Press, 1990.
2. **Organic Chemistry,** S.H. Pine, J.B. Hendrickson, D.J. Cram and G.S. Hammond, IV Edn. McGraw-Hill Company 1980.
3. **Organic Reaction Mechanisms**, V.K. Ahluwalia and R.K. Prashar, 4th edition, Alpha Science International, UK, 2011.
4. **Organic Reactions and Mechanisms**, P.S. Kalsi, II Edn., New Age International Publishers, 2000.
5. **Fundamentals of Organic Reaction Mechanisms**- J.M. Harris and C.C. Wamser, John Wiley & Sons, Inc. 1976.
6. **Organic Reaction Mechanisms**- R.K. Bansel, Tata McGraw Hill, 1975.
7. **Organic Chemistry,** P. Mehta & M. Mehta, Prentice Hall India, New Delhi, 2005.
8. **Organic Chemistry,** StevanA.Fleming, 4th ed., W.W. Norton & Compound, London, 2010.
9. **Organic Chemistry**, R.T. Morrison and R. N. Boyd's, 6th edition, Spring, 2008.
10. **Fundamentals of Reaction Mechanisms in Organic Chemistry**, R.P. Narain, PHI Learning Private Limited, New Delhi, 2011.

Course Code	Title of the Course
34413	PHYSICAL CHEMISTRY – I

Course Objectives: To make the students:

- Familiarity with basic concepts in thermodynamics and to relate the characteristics
- This unit covers the principles of chemical kinetics, theory of kinetics
- Quantum Chemistry will be applied to understanding the basic energetics of atoms and molecules.

Course Outcomes: The students shall be able to:

- ✓ Recognize the importance of quantum chemistry and of its applications.
- ✓ Describe the fundamental chemical and physical properties that determine chemical reaction rates.
- ✓ To study the solution and gas phase kinetics and some fast reaction kinetics

Block -1: Classical thermodynamics

Unit – 1: Law of thermodynamics

First and Second law of thermodynamics – Need, Statements. Entropy - Definition, entropy changes in reversible and irreversible processes, Carnot's cycle clausius inequality, entropy changes in ideal gases, entropy of mixing, entropy changes in phase changes - Degradation of energy.

Unit – 2: Gibb's and Helmholtz free energies

Gibb's and Helmholtz free energies - Criteria for spontaneity and conditions of equilibrium Maxwell relations - Thermodynamic equations of state Free energy changes in ideal gases. Gibbs Helmholtz equation, applications.

Unit – 3: Nernst Heat theorem

Nernst Heat theorem - Third law of thermodynamics, apparent exceptions to third law - Partial molar quantities - chemical potential Gibb's Duhem equation - Duhem Margules equation - determination of partial molar quantities. Zeroth law of thermodynamics.

Unit – 4: Fugacity

Fugacity and its determination - Activity and activity co-efficient - determination of mean activity co - efficient of electrolytes - Reaction isotherm - equilibrium constant and its dependence on temperature and pressure.

Block -2: Electrochemistry

Unit – 5: Transport number

Transport number and ionic mobilities (only definition and not determination) - Debye Huckel theory of interionic attraction.

Unit – 6: Debye Huckel Onsagar equation

Debye Huckel Onsagar equation - Validity and extension of the theory - Activity of ions in solution - Debye Huckel limiting law

Unit – 7: Applications of conductivity measurements

Applications of conductivity measurements - Electrode potential and Nernst equation - types of electrodes and electrochemical cells.

Unit – 8: EMF

EMF Applications of cell EMF - Electrode Kinetics over voltage and its determination - Butler-Volmer equation and approximation of the equation.

Block -3: Quantum Chemistry

Unit – 9: Fundamental of quantum chemistry

Inadequacy of classical mechanics, Block body radiation, and photoelectric effect-wave-particle dualism - Heisenberg's uncertainty principle.

Unit – 10: Mathematical preparation for quantum chemistry

Mathematical preparation for quantum chemistry: functions, operators, matrices, vectors – Eigen value and Eigen functions.

Unit – 11: Postulates of quantum mechanics

Postulates of quantum mechanics-Schrodinger wave equation - Application of quantum chemistry to particle in one and three dimensional boxes – degeneracy.

Block -4: Chemical Kinetics

Unit – 12: Theories of reaction rates

Theories of reaction rates: Absolute reaction rate theory(ARRT) - thermodynamic and statistical treatment - comparison to simple collision theory - Application of ARRT to unimolecular (Lindemann, Hinshelwood and KRRM and Slater) bimolecular and third order reactions

Unit – 13: Isotopic effect

Potential energy surfaces, Kinetic isotopic effect (qualitative approach only) - Principles of microscopic reversibility - steady state approximation –

Unit – 14: Kinetics of complex reactions

Parallel consecutive and opposing or reversible reactions, Branched chain and explosive reactions - Fast reactions - Flow, relaxation and NMR methods.

Text Books

1. **Thermodynamics for Students of Chemistry**, J. Rajaram and J.C. Kuriacose, Lal Nagin Chand, New Delhi, 1986.
2. **Physical Chemistry,** P.W. Atkins, Oxford University Press, Oxford, 1990.

3. **Text Book of Physical Chemistry**, D.A. McQuarrie, University Science Books, Mill Valley, California, 1983.
4. **Quantum Chemistry,** Henry Eyring, John Walter, George E. Kimball, BiblioBazaar, California, 2011.
5. **Chemical Kinetics**, Farrington Daniels, BiblioBazaar, Cornell University Press LONDON, 2011.
6. **Kinetics and Mechanism of Chemical Transformations**, J. Rajaram and J.C. Kuriacose, MacMillan India Ltd. 1993.

Reference Books

1. **Molecular Quantum Mechanics**, P.W. Atkins, Oxford University Press, Oxford, 1983.
2. **Quantum Mechanics in Chemistry**, M.W. Hanna, W.A. Benjamin Inc. London 1965
3. **Thermodynamics for Chemists**, S. Glasstone, Affiliated East West Press, New Delhi 1960.
4. **Chemical Kinetics,** K.J. Laidler, **3rd ed.,** Pearson Education Inc, New Delhi, 2008.
5. **Kinetics and Mechanism**, R.G. Frost and Pearson, Wiley New York, 1961.
6. **Quantum Chemistry**, R.K. Prasad, Wiley Eastern, New Delhi, 1992.
7. **Thermodynamics,** Enrico Fermi, Create Space, Dover Publications USA, 2011.

Course Code	Title of the Course
34414	ANALYTICAL CHEMISTRY PRACTICAL

Course Objectives

- To understand the principle behind the estimation of organic compounds and get practical knowledge.
- To improve the skill in semi-micro qualitative analysis of inorganic mixtures containing two less familiar cations and two familiar cations.

Learning Outcomes: Students will get the experience in the

- ✓ Estimation of various organic compounds
- ✓ Semi-micro qualitative analysis of cations.

This will make the students to work well in the research institutes and quality control laboratories.

Block -1: Quantitative Estimation -1

Unit 1: Quantitative estimation of aniline

Estimate the amount of aniline present in the whole of the given solution.

Unit 2: Quantitative estimation of phenol

Estimate the amount of phenol present in the whole of the given solution

Unit 3: Quantitative estimation of ethylmethylketone

Estimate the amount of ethylmethylketone present in the whole of the given solution

Unit 4: Quantitative estimation of glucose

Estimate the amount of glucose present in the whole of the given solution

Block -2: Semi-micro qualitative analysis -1

: Unit 5: Semi-micro qualitative analysis of mixture -1

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Unit 6: Semi-micro qualitative analysis of mixture -2

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Unit 7: Semi-micro qualitative analysis of mixture -3

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Block -3: Semi-micro qualitative analysis -2

Unit 8: Semi-micro qualitative analysis of mixture -4

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Unit 9: Semi-micro qualitative analysis of mixture -5

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Unit 10: Semi-micro qualitative analysis of mixture -6

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Block -4: Semi-micro qualitative analysis -3

Unit 11: Semi-micro qualitative analysis of mixture -7

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Unit 12: Semi-micro qualitative analysis of mixture -8

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Unit 13: Semi-micro qualitative analysis of mixture -9

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Unit 14: Semi-micro qualitative analysis of mixture -10

Analysis of mixtures containing two less familiar cations like W, Tl, Mo, Se, Te, Ce, Zr, Th, Ti, V, U and Li and two familiar cations like Pb, Cu, Bi, Cd, Mn, Ni, Co, Zn,Ca, Ba, Sr and Mg.

Reference Books

1. **A Laboratory Manual of Inorganic Chemistry,** John Bernard Ekeley, BiblioLife, 2010.
2. **Laboratory Manual of Organic Chemistry**, Raj K. Bansal, III Edition, New Age International (P) Ltd., 1996.
3. **Vogel's qualitative Inorganic analysis**, G. Svehla, VI Edition, Orient Longman, 1987.
4. **Inorganic Semimicro Qualitative analysis,** V.V. Ramanujam, National Publishing Co., 1971.

SECOND SEMESTER

Course Code	Title of the Course
34421	INORGANIC CHEMISTRY – II

Course Objectives: The objectives are to understand the concepts of

- The Nomenclature of coordination compounds
- The formation of coordination compounds by CFT and MOT
- Nuclear structure and nuclear reactions
- Extraction of lanthanides, spectral and magnetic properties of lanthanides and actinides.

Learning outcomes: The students will have knowledge in

- ✓ The formation of coordination complexes based on the various theories
- ✓ Magnetic properties of coordination complexes
- ✓ Basic mechanism of nuclear reactions and nuclear reactor
- ✓ Uses of lanthanide and actinides

Block -1: Coordination chemistry

Unit- 1: Fundamentals of coordination chemistry

Nomenclature of coordination compounds, Geometrical and optical isomerisms in octahedral, square planar and tetrahedral complexes. Theory on coordination compounds – valence bond theory, limitation of VBT

Unit-2: Crystal field Theory in octahedral and tetrahedral complexes

CFT – Splitting in octahedral filed – CFSE - Strong field and weak field splitting-calculation of CFSE for d^n systems - splitting in tetrahedral complexes - only weak field splitting – reason, spectrochemical series.

Unit-3: Crystal field Theory in tetragonal and square planar complexes

Tetragonal symmetry - differences between tetrahedral and tetragonal symmetries - Jahn-Teller distortion - theorem - square planar symmetry - factors affecting 10Dq - Jorgensen relation - evidences for CFSE.

Unit-4: Molecular orbital theory of Coordination complexes

MOT - Octahedral, tetrahedral, square planar complexes-pi bonding and MOT ligands having empty and filled π orbitals – effect on 10Dq, comparision of VBT and CFT

Unit-5: Magnetic properties of complexes.

Para, dia, ferro, ferri, antiferro magnetisms - calculation of μ_{eff} values for complexes.

Block -2: Nuclear Chemistry

Unit -6: Basics of nuclear structure

Nuclear structure - composition of nuclei,– nuclear forces-its characteristics - meson field theory nuclear models - liquid drop, shell and collective models- Properties of nucleus.

Unit -7: Nuclear stability

Nuclear stability, factors affecting the nuclear stability; Mode of decay - alpha, beta, gamma and orbital electron capture; Q value - threshold energy- reaction cross section; isobars- nuclear isomerism

Unit -8 : Radioactive decay and detection

Radioactive decay - theories of decay processes – Laws of radioactivity, series of radioactivity. Detection and measurements of radiations –Half life period, Geiger Muller counter, Scintillation counters.

Block -3: Artificial radioactivity

Unit -9: Classification of nuclear reactions and Artificial radioactivity

Nuclear reactions - transmutation, stripping and pick up, fission, fusion, spallation and fragmentation reactions - nuclear cross-section.

Unit -10: Particle accelerators

Charged particle accelerators, Cyclotron and synchrotron, Uses of accelerator.

Unit -11: Application of nuclear Chemistry

Application C^{14} dating – agriculture - biology – neutron activation and isotopic dilution analysis.

Block -4: Lanthanides and Actinides

Unit -12: Position of Lanthanides and Actinides

Lanthanides and Actinides -position in the periodic table, electronic configuration and oxidation states

Unit -13: Lanthanides and Actinides - occurrence, extraction and separation techniques

Lanthanides - occurrence, extraction and separation techniques -fractional crystallization, precipitation, ion exchange, solvent extraction and thermal decomposition, selective reduction and oxidation

Unit -14: Properties and uses of lanthanides and Actinides

Lanthanides and Actinides contraction – Causes of Lanthanides contraction - spectral and magnetic properties - coordination compounds of lanthanides. Comparative account of lanthanides and actinides, Uses of lanthanides and Actinides and their compounds

Text books

1. **Advanced Inorganic Chemistry**, F.A. Cotton and G. Wilkinson, Wiley Eastern (P) Ltd., 1988.
2. **Essentials of nuclear chemistry**, H.J. Arniker, 2nd edition Wiley eastern Co., 1987.
3. **Concise Inorganic Chemistry,** J.D. Lee, Fifth edition Oxford, 2008.

Reference books

1. **Co-ordination Chemistry,** D. Bannerjea, Tata-McGraw Hill, 1993.
2. **Inorganic Chemistry- Principles of structure and reactivity**, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th edition, Pearson-Education, 2002.
3. **The Magneto Chemistry of Complex Compounds** in Modern Coordination Chemistry, B. N. Figgeis and J. Lewis, Ed: Lewis & Wilkins, Interscience. N.Y., 1967.
4. **Nuclear and radiochemistry,** G. Friedlander, J.W. Kennedy and J.M. Miller, Wiley, 1964.
5. **Elements of Nuclear Chemistry,** A.K. Srivatsava and P.C. Jain, S. Chand and Co., 1989.

Course Code	Title of the Course
34422	ORGANIC CHEMISTRY – II

Course Objectives: To make the students:

- To understand on the basic concepts on how an organic compound undergoes photochemical or pericyclic reactions.
- Emphasis the stereoisomerism of organic compound
- Emphasis is on the construction of organic compounds through the reactive intermediates.
- To understand the elimination and addition reaction.

Course Outcomes: The students shall be able to:

- ✓ Increase in ability of isomerism and stereochemistry of organic compounds
- ✓ Understand the importance of photochemistry and pericyclic reaction.
- ✓ Recognize the mechanism of addition and elimination reaction
- ✓ Recognize the mechanism of oxidation and reduction reactions in organic synthesis.

Block -1: Elimination and Addition reaction

Unit – 1: Elimination reactions

E_1, E_2 and E_1cB mechanisms - orientation of the double bond - effect of substrate, base, leaving group and reaction medium

Unit – 2: Rules of Elimination reactions

Hofmann and Saytzeff rules - elimination versus substitution - pyrolytic cis elimination - Bredt's rule.

Unit – 3: Addition reaction

Electrophilic, Nucleophilic and free radical additions - stereochemistry of additions - addition to conjugated systems - regioselectivity and chemoselectivity in additions

Block -2: Addition to multiple bond and Conformational analysis

Unit – 4: Addition to carbon-carbon multiple bond

Hydration of olefins – hydroboration - Michael addition - and lithium dimethyl Cuprate - Diels-Alder reaction.

Unit – 5: Addition to carbon-hetero multiple bond

Mechanisms of Aldol condensation, Perkin reaction, Knovenagal reaction, Mannich reaction, Claisen ester condensation, Dickmann condensation, Darzen reaction. Wittig reaction, Cannizzaro reaction, Benzoin condensation and Reformatsky reaction-Addition of Grignard reagents

Unit-6: Conformational analysis of acyclic and cyclic systems

Conformational analysis of acyclic and cyclic systems – substituted n-butanes – cyclohexane and its derivatives – decalins –fused and bridged bicyclic systems – conformation and reactivity some examples

Unit – 7: Topicity and NMR distinction of organic compounds

Topicity and prostereoisomerism – topicity of ligands and faces, and their nomenclature – NMR distinction of enantiotopic/diastereotopic ligands.

Block -3: Reaction intermediates

Unit – 8: Carbenes

Structure of carbenes, generation of carbenes, addition and insertion reactions, rearrangement reactions of carbenes - Wolff rearrangement.

Unit – 9: Nitrenes

Structure of nitrene, generation and reactions of nitrene and related electron deficient nitrogen intermediates, Curtius, Hoffmann, Schmidt, Beckmann rearrangement reactions.

Unit – 10: Free radical reactions

Formation, detection, stability and reactions of free radicals – radicals chain reactions - polymerization, substitution, additions and rearrangements – Barton, Gomberg, Sandmeyer, Ullmann, Pschorr and Hunsdiecker reactions.

Block -4: Photochemistry

Unit – 11: Fundamentals of Photochemistry

Principles - excited states - Energy transfers - Jablonski diagram - sensitization, quenching and quantum efficiency

Unit – 12: Photochemical reaction

Norrish type I and type II reactions – Paterno-Buchi reaction – photoreduction – photooxidation - photochemical reactions of olefins - cis-trans isomerisation - di-π methane rearrangement.

Unit – 13:Pericyclic reactions

Woodward-Hoffmann rules - Frontier molecular orbital theory -perturbation theory - electrocyclic reactions - cycloaddition reactions

Unit – 14: chemotropic reaction

Sigmatropic rearrangements. Cope and Claisen rearrangements; 1,3-dipolar additions - Diels - Alder reaction.

Text books

1. **Advanced Organic Chemistry – Reactions, Mechanisms and Structure,** Jerry March, IV Edn., John Wiley & Sons, 1992.
2. **A Guide Book to Mechanisms in Organic Chemistry**, P. Sykes, VI Edn., Longmans Scientifics and Technical, Essex 1986.
3. **Reaction Mechanism in Organic Chemistry**, S.M. Mukherji and S.P. Singh, III Edn. MacMillan.1984.
4. **Stereochemistry of carbon compounds**, Ernest L. Eliel, T.M.H. Edn., Tata McGraw-Hill Publishing Company, 1962.
5. **Organic Photochemistry**, J.M.Coxon and B. Halton, Cambridge University Press, 2011.
6. **Molecular Rearrangements, Vol.I, Vol. II**, Paul de Mayo, Interscience, NY, 1963.

Reference books

1. **Advanced Organic Chemistry, Part A& B**, F.A. Carey and Sundberg, III edition, Plenum Press, 1990.
2. **Organic Chemistry,** S.H. Pine, J.B. Hendrickson, D.J. Cram and G.S. Hammond, IV Edn. McGraw-Hill Company 1980.
3. **Organic Reaction Mechanisms**, V.K. Ahluwalia and R.K. Prashar, 4th edition, Alpha Science International, UK, 2011.
4. **Organic Reactions and Mechanisms**, P.S. Kalsi, II Edn., New Age International Publishers, 2000.
5. **Organic Reaction Mechanisms**- R.K. Bansel, Tata McGraw Hill, 1975.
6. **Organic Chemistry**, R.T. Morrison and R. N. Boyd's, 6th edition, Spring, 2008.
7. **Fundamentals of Reaction Mechanisms in Organic Chemistry**, R.P. Narain, PHI Learning Private Limited, New Delhi, 2011.

Course Code	Title of the Course
34423	PHYSICAL CHEMISTRY – II

Course Objectives: To make the students

- To study adsorption isotherm for adsorption on to solid surfaces and to understand heterogeneous catalysis.
- To acquire basic knowledge in area of Fundamental concepts of polymer chemistry, Polymerization reactions Polymerization techniques
- To know about photochemistry and radiation chemistry

Learning Outcomes: After completion of the course, the students shall be able to

- ✓ Get deep knowledge about various methods of polymerization and speciality polymers
- ✓ Get knowledge about nomenclature of polymer, degree, types, mechanism and kinetics of polymerization and Characterization of polymers
- ✓ To describe adsorption isotherm for adsorption on to solid surfaces and to understand heterogeneous catalysis

Block -1: Surface Chemistry

Unit – 1: Adsorption of gases

Adsorption of gases on solids - Physical and Chemical adsorption –Freundlich, Langmuir, Temkin and BET isotherms

Unit – 2: Surface area reaction

Surface area determination - Mechanisms of uni and bimolecular surface reactions - Langmuir-Hinshelwood and Langmuir-Riedal mechanisms

Unit – 3: Surface excess

Surface excess - Gibbs adsorption isotherm - spreading of a liquid on another - contact angle – surfactants - micelles and detergents.

Block -2: Photochemistry and radiation chemistry

Unit – 4: photophysical processes

Absorption of light by atoms and molecules - photophysical processes of the electronically excited states - fluorescence and phosphorescence

Unit – 5: Energy transfer mechanisms

Energy transfer mechanisms - photosensitization and Chemiluminescence - actinometers and quantum yield determination

Unit – 6: Flash photolysis

Flash photolysis. Study of photochemical reactions - Hydrogen-Halogen reaction - decomposition of carbonyl compounds

Unit – 7: Radiation chemistry

Radiation chemistry of aqueous solutions-hydrated electron - radiolysis of water – Pulsed radiolysis.

Block -3: Polymer chemistry

Unit – 8: Fundamentals of Polymer

Definition, classification of polymers- addition polymerization – type of initiators – initiator efficiency

Unit – 9: Stepwise polymerization

Stepwise polymerization – Functionality of monomers and its significance – Molar masses - Degree of polymerization

Unit – 10: Kinetics and Mechanism polymerization

Kinetics and Mechanism of free radical, cationic and anionic polymerization.

Unit – 11: Polymerization techniques

Various methods of polymerization – solution, bulk, emulsion and suspension polymerization.

Block -4: Colloids and solar energy

Unit – 12: Colloids

Definition, classification, Stability and properties of colloids.

Unit – 13: Solar energy conversion

Conducting polymers, polymer electrolyte, fire retardant, thermally stable and bio-degradable polymers. Atom radical polymerization- Basic, mechanism and applicaiton. Dendrimer.

Unit -14: Solar energy conversion

Principle, solar panel, photovoltaic cell – dye sensitized solar cell, solar water splitting, artificial photosynthesis, water oxidation and hydrogen evolution reaction.

Text Books

1. **Fundamentals of Photochemistry,** K.K. Rohatgi-Mukherjee, New Age International (P) Ltd, Publishers, New Delhi, 2008.
2. **Physical Chemistry,** P.W. Atkins, Oxford University Press, Oxford, 1990.
3. **Physical Chemistry of surfaces**, A.W.Adamson, 4th edn., Wiley - Interscience, Newyork, 1982.
4. **Text Book of Polymer Science**, F.W.Billmeyer Jr. 3rd edn., John Wiley & Sons, New York, 2003.
5. **A Textbook of Polymers**, . M.S. Bhatnagar, Vol I, S.Chand & Company Ltd., 2004.

Reference Books

1. **Physical Chemistry of Surfaces**, A.W. Anderson, Wiley-Interscience, Newyork, 1990.
2. **Physical Chemistry**, Paul Monk, John Wiley and Sons Limited, England, 2004.
3. **Polymer Science**, V.R. Gowariker, N.V. Viswanathan and J. Sreedhar, New Age International, New Delhi, 2003.
4. **Contemporary Polymer Chemistry**, H.R. Alcock and F.W. Lamber, Prentice Hall,1981.
5. **Principles of polymer chemistry**, P.J. Flory, Cornell University press, New York, 1953.

Course Code	Title of the Course
34424	ORGANIC CHEMISTRY PRACTICAL

Course Objectives
The objectives of this course are to:

- Develop practical skill with reference to organic qualitative analysis and organic preparations
- Have expertise in the chromatographic separations
- Understand how to solve the structure of organic compounds using spectroscopies
-

Learning Outcomes
The student would have through practical knowledge in the

- ✓ Separation of organic mixture and identification of organic compounds
- ✓ Chromatographic separations
- ✓ Confirmation of structure of organic compounds using spectroscopes

Block -1. Qualitative analysis: Separation and Identification of components in a two component mixture and preparation of their derivatives. Determinations of boiling point/melting point for components and melting point for their derivatives.

Unit 1: Analysis of two-component mixtures
Separation and characterization of components -1

Unit 2: Analysis of two-component mixtures
Separation and characterization of components -2

Unit 3: Analysis of two-component mixtures
Separation and characterization of components – 3

Block -2. Qualitative analysis

Unit 4: Analysis of two-component mixtures
Separation and characterization of components – 4

Unit 5: Analysis of two-component mixtures
Separation and characterization of components -5

Unit 6: Analysis of two-component mixtures
Separation and characterization of components -5

Block -3. About a dozen single stage preparation of organic compounds-1

Unit 7: single stage preparation of organic compounds
Preparations of organic compounds illustrating N-acylation

Unit 8: Single stage preparation of organic compounds
Preparations of organic compounds illustrating O-acylation

Unit 9: Single stage preparation of organic compounds
Preparations of organic compounds illustrating bromination

Unit 10: Single stage preparation of organic compounds
Preparations of organic compounds illustrating nitration

Block -4. About a dozen single stage preparation of organic compounds -2

Unit 11: Single stage preparation of organic compounds
Preparations of organic compounds illustrating benzoylation

Unit 12: Single stage preparation of organic compounds
Preparations of organic compounds illustrating diazotization

Unit 13: Single stage preparation of organic compounds
Preparations of organic compounds illustrating rearrangements

Unit 14: Single stage preparation of organic compounds
Preparations of organic compounds illustrating hydrolysis

Reference books

1. **Laboratory Manual of Organic Chemistry**, Raj K. Bansal, III Edition, New Age International (P) Ltd.1996.
2. **Elementary practical organic chemistry: Quantitative organic analysis Part-III, 2e(pb),** A.I.Vogel**,** Pearson Education Asia, 2011
3. **Elementary practical organic chemistry: Qualitative organic analysis Part-II, 2e(pb),** A.I.Vogel, Pearson Education Asia, 2011

THIRD SEMESTER

Course Code	Title of the Course
34431	ADVANCED INORGANIC CHEMISTRY

Course Objectives: To make the students:

- To know synthetic procedure of metal alkyl, alkene, alkyne, and arene complxes
- To describe the various organometallic reaction mechanisms
- To appreciate the uses of organometallic complexes
- To predict their structures and bonding found in inorganic rings
- To distinguish substitution reactions in octahedral and square planar complexes
- To understand spectral and magnetic properties of octahedral complexes
- Distribution of metal ions in bioligands
- Role of metals in medicine and their structure and properties.

Learning Outcomes: After completion of the course the student are able to:

- ✓ Predict the reaction mechanisms of organometallic complexes and catalysis
  They will have expertise in
- ✓ Know the electron transitions in complexes and its effect on magnetic properties
- ✓ Understand the chemistry of cages and clusters

Block -1: Coordination Chemistry

Unit-1: Stability of coordination compounds

Stability constants, stepwise and overall formation constants - pH metric, polarographic and Spectrophotometric methods of determining stability constants - chelate effect.

Unit-2: Kinetics and mechanisms of coordination compounds

Kinetics and mechanisms of reactions in solution - labile and inert complexes - ligand displacement reactions - hydrolysis, anation, aquation in octahedral complexes - substitution reactions in square planar complexes – trans effect – electron transfer reactions

Unit-3: complementary and non-complementary reactions

Inner sphere and outer sphere processes – isomerisation and racemisation - template effect and synthesis of macrocyclic ligands.

Block -2: Spectral properties of complexes, Spinels, Cages, clusters and boranes

Unit -4 : Spectral properties of complexes

Electronic spectra of coordination compounds - selection rules, band intensities and band widths; Term state for d ions in Octahedral complexes, energy level diagrams of Orgel and Tanabe - Sugano diagram.

Unit -5 : Spinels

Spinels – normal and inverse types, site preferences in spinnels.and perovskite structures.

Unit -6: Cages and metal clusters

Inorganic chains - rings – cages, Metal clusters - dinuclear clusters - trinuclear clusters - tetranuclear clusters - hexanuclear cluster.

Unit -7: Boranes

Boranes: Structure and bonding in polyhedral boranes and carboranes, styx notation; Wade's rule; electron count in polyhedral boranes; isolobal analogy.

Block -3: Organometallic chemistry

Unit-8: Definition of Organometallics

M-C – bond - Low oxidation state of metal – explanation – synthesis and structure of metal alkyls and aryls - Olefin and acetylene complexes - Dewar-Chatt approach to bonding in olefins

Unit-9: Metallocenes

Metallocenes – structure — comparison of ferrocene with other metallocenes with respect to their reactivity, stability etc – preparation of ferrocene –properties - fluxional molecules.

Unit-10: Metal carbonyl

Metal carbonyl complexes - synthesis - structure and reactions of metal carbonyls -metal carbonyl anions - metal carbonyl hydrides - metal carbonyl halides - metal carbonyl clusters - metal nitrosyls.

Unit -11: Catalysis involving organometallics

Oxidative addition and reductive elimination, hydrogenation, hydroformylation, Monsanto process, isomerisation and Ziegler-Natta polymerization.

Block -4: Bioinorganic chemistry

Unit-12: Metalloporphyrins and Metalloenzymes

Chlorophyll-hemoglobin and myoglobin structure and function of hemoglobin - cytochromes, enzyme action-inhibition and restoration. Carboxy peptidase-A, Vitamin B_{12} and B_{12} coenzymes. Copper containing oxidases

Unit-13: non-heme iron proteins

Rubridoxin – ferrodoxins – HIPIP, fixation of nitrogen - in vivo systems

Unit-14: Metal ions in biology

Alkali and alkaline earth metal ions in biology - sodium ion pump. Metal poisons and chelating agents in medicine.

Text books

1. Modern aspects of Inorganic chemistry, H.J. Emelius and Sharpe, Universal book Stall, New Delhi, 1989.
2. Inorganic Chemistry- Principles of structure and reactivity, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th edition, Pearson-Education, 2002.
3. Advanced Inorganic Chemistry - F.A. Cotton and G. Wilkinson, Wiley Eastern, 5th edition, 1998.
4. Modern aspects of Inorganic chemistry, H.J. Emelius and Sharpe, Universal book Stall, New Delhi, 1989.
5. Inorganic Chemistry- Principles of structure and reactivity, J.E. Huheey, E.A. Keiter and R.L. Keiter, 4th edition, Pearson-Education, 2002.
6. Advanced Inorganic Chemistry - F.A. Cotton and G. Wilkinson, Wiley Eastern, 5th edition, 1998.

Reference books

1. **Inorganic Chemistry**, D. F. Shriver and P. W. Atkins, Oxford U.K., 1999.
2. **Physical Methods in Inorganic Chemistry**, R. S. Drago, Van Nostrand Reinhold, 2nd Edn., 1968.
3. **Chemistry & Chemical Reactivity**, John C. Kotz, Paul M. Treichel, John Townsend, 8th ed. Cengage Learning, USA, 2012.
4. **A Text book of Quantitative Inorganic Analysis**, A. I. Vogel, ELBS, 3rd Edn, 1969.
5. **Source book of atomic Energy**, S. Glasstone, Van Nonstrand Co., 1969.
6. **Inorganic Chemistry**,. G. S.Sodhi; Ist Edition, VB (P) Ltd, 2006.

Course Code	Title of the Course
34432	ADVANCED ORGANIC CHEMISTRY

Course Objectives: To make the students:

- To learn the about the oxidizing and reducing reagents in organic synthesis
- Emphasis is on the construction of organic compounds through the reactive intermediates.
- To understand the importance and structural characterization of nucleic acid, vitamins, carbohydrate
- Able to synthesis the alkaloids and terpenes
- To understand the importance of target molecules and their synthesis.

Learning Outcomes: The students shall be able to:

- ✓ Create a valuable understanding of the main and important concepts in this course.
- ✓ Understand how systematic the advanced organic syntheses are carried out.
- ✓ Know about the importance and usefulness of protecting groups in organic synthesis.
- ✓ primarily with the principles to understand the oxidation and reduction reaction

Block -1: Oxidation and reduction

Unit-1: Oxidizing reagents in organic synthesis -1

Metal based and non-metal based oxidations of (i) alcohols to carbonyls (Cr, Mn, hypervalent iodine and TEMPO based reagents). (ii) phenols (Fremy's salt, silver carbonate) (iii) alkenes to epoxides (peroxides/per acids based), Sharpless asymmetric epoxidation,

Unit-2: Oxidizing reagents in organic synthesis -2

(i) alkenes to diols (Mn, Os based), Sharpless asymmetric dihydroxylation, Prevost reaction and Woodward modification, (ii) alkenes to carbonyls with bond cleavage (Os and Ru, ozonolysis) (iii) alkenes to alcohols/carbonyls without bond cleavage (hydroboration-oxidation, Wacker oxidation, Se) (iv) ketones to ester/lactones (Baeyer-Villiger).

Unit-3: Reducing reagents in organic synthesis -1

Catalytic hydrogenation- Heterogeneous: Pd/Pt/Rh/Ni, Homogeneous, Wilkinson, Li/Na/Ca in liquid ammonia - Birch, Pinacol formation, McMurry, Acyloin formation, dehalgenation and deoxygenations,

Unit-4: Reducing reagents in organic synthesis -2

Hydride transfer reagents from Group III and Group IV in reductions – $LiBH_4$, $NaBH_4$, triacetoxyborohydride, L-selectride, K-selectride, Luche reduction; $LiAlH_4$, DIBAL-H; Trialkylsilanes, Meerwein-Pondorff-Verley reduction - Stereo/enantioselectiviey reductions -Chiral Boranes, Corey-Bakshi-Shibata

Block -2: Retrosynthetic Analysis and Heterocyclic compounds

Unit-5: Retrosynthetic Analysis

Basic principles and terminology of retrosynthesis, synthesis of aromatic compounds, one group and two group C-X disconnections, one group C-C and two group C-C disconnections, amine and alkene synthesis, important strategies of retrosynthesis, functional group transposition, important functional group interconversions

Unit-6: Functional group protection

Protection and deprotection of hydroxy, carboxyl, carbonyl, carboxy amino groups and carbon-carbon multiple bonds; chemo- and regioselective protection and deprotection; illustration of protection and deprotection in synthesis

Unit – 7: Heterocyclic compounds

Synthesis, structure and reactivity of Indole, Oxazole, Flavone and Anthocyanin.

Block -3: Carbohydrates, Protein and Ezzymes

Unit – 8: Carbohydrates

Configuration and conformation of disaccharides - Maltose and cellobiose – Polysaccharides - starch and cellulose**.**

Unit – 9: Proteins and enzyme

Aspects of structure and classification of proteins.- Primary, secondary and tertiary structure- end group analysis -Solid phase peptide synthesis. Enzyme- coenzyme.

Unit – 10:Nucleic acids

Aspects of structure and classification DNA and RNA. DNA replication and RNA transcription and translation

Block -4: Natural products

Unit – 11: Alkaloids

Structure and synthesis of Morphine and Atropine, Biosynthesis of Alkaloids.

Unit –12: Terpenes

Structure and synthesis of α-Pinene, Camphor and Zingiberene. Biosynthesis of terpenes.

Unit – 13: Vitamins

Chemistry and physiological action of ascorbic acid, thiamin, riboflavin and pyridoxine – Elementary aspect of vitamin A, E, K and B12.

Unit – 14:Cholesterol and steroid

Structural elucidation of cholesterol– biosynthesis of Cholesterol. Structural elucidation and synthesis of Progesterone.

Text books

1. **Advanced Organic Chemistry** – **Reactions, Mechanisms and Structure,** Jerry March, IV Edn., John Wiley & Sons, 1992.
2. **A Guide Book to Mechanisms in Organic Chemistry**, P. Sykes, VI Edn., Longmans Scientifics and Technical, Essex 1986.
3. **Reaction Mechanism in Organic Chemistry**, S.M. Mukherji and S.P. Singh, III Edn. MacMillan.1984.
4. **Organic Chemistry, Vol. I & II**, I.L. Finar**,** V Edn. First Indian reprint, Pearson Education Asia Pvt. Ltd. 2000.

Reference books

1. **Organic Reaction Mechanisms**- R.K. Bansel, Tata McGraw Hill, 1975.
2. **Chemistry of Alkaloids,** S.W. Pelletier, Van Nostrand Reinhold, 1970.
3. **Chemistry of Terpenesand Terpenoids**, A.A. Newman (editor), Academic Press, London, 1972.

4. **Organic Chemistry,** P. Mehta & M. Mehta, Prentice Hall India, New Delhi, 2005.
5. **Chemistry of Terpenoids**, P. De Mayo, Interscience publishers, 1959.
6. **Biochemistry**, A.L. Lehninger, Nath publishers, 2000.
7. **Organic Chemistry,** StevanA.Fleming, 4th ed., W.W. Norton & Compound, London, 2010.
8. **Organic Chemistry**, R.T. Morrison and R. N. Boyd's, 6th edition, Spring, 2008.
9. **March's Advanced Organic Chemistry Reactions, Mechanishms and Structure**, Micheal B. Smith and Jerry March, 6th edition, John Wiley & Sons Inc., New Jersey, 2007.
10. **Fundamentals of Reaction Mechanisms in Organic Chemistry**, R.P. Narain, PHI Learning Private Limited, New Delhi, 2011.

Course Code	Title of the Course
34433	Spectroscopy – Applications in Organic and Inorganic Chemistry

Course Objectives

The primary objective of this course is to introduce the student to the advanced concepts of applications of spectroscopy in organic and inorganic chemistry. The objectives are:

- To learn the about the theory and applications of UV-VIS, and FT IR spectroscopies
- To be familiar with the principles and applications of NMR, ESR, and Mass spectroscopies
- To be able to interpret the spectra and work out conjoined problems in spectroscopies.

Learning Outcomes

Students will be able to

- ✓ Understand how different spectroscopes work and their applications in structure elucidations.
- ✓ Recognize and distinguish the different molecules by applying the spectroscopies
- ✓ Solve spectral problems
- ✓ Know about the importance and usefulness of various spectroscopies in organic and inorganic chemistry.

Block -1: UV-Visible and IR spectroscopy

Unit 1: UV-Visible spectroscopy

Basic Principles – electronic excitations-solvent effects - factors affecting position and intensity of absorption bands - instrumentation

Unit 2: Application of UV-Visible spectroscopy

Applications – Qualitative analysis - Quantitative analysis - spectra of dienes - α,β-unsaturated ketones and aromatic carbonyl compounds – Woodward –Fieser rules - charge transfer complexes.

Unit 3: IR-Spectroscopy

Basic principles-stretching vibrations - Hook's law - Bending vibrations –Overtone and combination bands - Fermi resonance – Instrumentation

Unit 4: Application of IR-Spectroscopy

Applications to organic compounds - characteristic frequencies - effects of substitution, conjugation, bond angle and hydrogen bond - vibrational frequencies.

Block -2: NMR Spectroscopy

Unit 5: ^{1}H NMR Spectroscopy

Theory of ^{1}H NMR spectroscopy – chemical shift – factors affecting chemical shift – spin –spin coupling Instrumentation - first order and non-first order spectra - shift reagents

Unit 6: ^{1}H NMR Spectral Technique

Double resonance - spin tickling - Nuclear Overhauser Effect - Deuterium exchange reactions – Applications.

Unit 7: ^{13}C NMR Spectroscopy

^{13}C NMR, Theory, instrumentation, Application

Block -3: ESR, Mass Spectroscopy and **ORD and CD**

Unit 8: ESR Spectroscopy

Theory – Instrumentation - Presentation of spectrum - comparison between ESR and NMR - 'g' values - applications to organic and inorganic compounds.

Unit 9: Mass Spectroscopy

Principle - parent ion - Meta stable ion - isotopic ions - Basic peak Nitrogen rule - Instrumentation – general rule of fragmentation - Mclafferty rearrangement. Structural elucidation.

Unit 10: ORD and CD

Principle – Circular birefringence and circular dichromism – Cotton effect - ORD curves

Unit 11 Application of ORD and CD

Application on cotton effect curves - α-haloketone rule - octant rule - Applications for determination of conformation and configuration.

Block -4: Thermal and Spectrometric methods of analysis

Unit 12: Thermal analysis

Thermogravimetry - Differential thermal analysis - Differential scanning calorimetry - Thermometric titrations

Unit 13: Fame photometry

Principle, instrumentation and applications of flame photometry

Unit 14: Turbidimetry and Nephelometry

Principle, instrumentation and applications of turbidimetry and Nephelometry

Text Books

1. **Instrumental Methods of Analysis**, Willard, Merit Dean and Settle CBS Publishers and Distributors, IV edition, 1986.
2. **Principles of Instrumental Analysis,** Schoog, Holler, Nieman, Thomson, Asia Pvt. Ltd., Singapore, 2004.
3. **Spectrometric identification of organic compounds**, R.M Silverstein, C.G. Bassler and Morril, VI Edition, John Wiley & Sons, New York, 2002.
4. **Text Book of Quantitative Inorganic Analysis**, A.I Vogel, ELBS III Edition, 1987.
5. **Instrumental methods of chemical analysis**, Chatwal and Anand, Himalaya publishing House New Delhi, 2000.

Reference Books

1. **Chemical Analysis: Modern Instrumentation Methods and Technique**s, F.Rouessac, A.Rouessac , 2nd Edition, Wiley & sons, USA, 2011.
2. **Analytical Chemistry,** J.G. Dick McGraw Hill Publishers, 1974.
3. **Instrumental Methods of Chemical Analysis,** G.W.Ewing McGraw Hill Pub, 1975.
4. **Instrumental methods in Electrochemistry**, R. Greef, R. Peat, L.M. Peter, D. Pletcher and J. Robinson, Ellis Horwood, Chichester, 1985.
5. **Spectroscopy of organic compounds**, P.S. Kalsi, Wiley Eastern Ltd., Madras, 1995.
6. **Fundamentals of Analytical Chemistry,** D.A. Skoog and D.M. West Holt Rinehart and Winston Publications, IV Edition, 2004.
7. **NMR in Chemistry,** W. Kemp, MacMillan Ltd, 1986.
8. **Spectroscopy in Inorganic Chemistry**, C.N.R. Rao, J.R. Ferraro, Methven Co., London, 1968.
9. **Basic Principles of Spectroscopy,** Raymond Chang, Mc Graw Hill Ltd., New York, 1993.
10. **Instrumental Analysis,** G. D. Christian and J.E.O Reilly, Allyn and Bacon Inc, II Edition, 1986.
11. **Structural methods in Inorganic Chemistry,** E.A.V. Ebsworth, D.WH. Rankine and S. Craddock, Black well Scientific Publ., 1987.
12. **Principles of Instrumental Analysis**, Schoog, Holler, Nieman, Thomson, Asia Pvt. Ltd., Singapore, 2004.
13. **Fundamentals of Analytical Chemistry**, D.A.Skoog and D.M.West, Winston Publications, IV Edition, 2004.
14. **Analytical chemistry: a modern approach to analytical science**, J.M. Mermet, M. Otto, R. Kellner, Wiley-VCH, 2004.

Course Code	Title of the Course
34434	INORGANIC CHEMISTRY PRACTICAL

Course Objectives

This course will help in developing practical skill with reference to separation, estimation by volumetric, gravimetric and complexometric method of analysis of metal ions.

Learning Outcomes

The student would have through practical knowledge in quantitave and qualitative estimation of inorganic cations. with suitable methods.

Block-1: Quantitative analysis of mixtures containing two components system-1

Unit 1: Separation and estimation of mixtures containing two components – 1

Separation and estimation of Cu^{2+} - Ni^{2+} mixtures by volumetric method and the other by gravimetric method.

Unit 2: Separation and estimation of mixtures containing two components – 2

Separation and estimation of Cu^{2+} – Ba^{2+} mixtures by volumetric method and the other by gravimetric method.

Unit 3: Separation and estimation of mixtures containing two components – 3

Separation and estimation of Cu^{2+} – Zn^{2+} mixtures by volumetric method and the other by gravimetric method

Unit 4: Separation and estimation of mixtures containing two components – 4

Separation and estimation of Fe^{2+} – Ni^{2+} mixtures by volumetric method and the other by gravimetric method

Block-2: Quantitative analysis of mixtures containing two components system-2

Unit 5: Separation and estimation of mixtures containing two components – 5

Separation and estimation of Fe^{2+} – Zn^{2+} mixtures by volumetric method and the other by gravimetric method

Unit 6: Separation and estimation of mixtures containing two components – 6

Separation and estimation of Fe^{2+} - Cu^{2+} mixtures by volumetric method and the other by gravimetric method

Unit 7: Separation and estimation of mixtures containing two components – 7

Separation and estimation of Zn^{2+} – Cu^{2+} mixtures by volumetric method and the other by gravimetric method

Block-3: Complexometric estimation of binary mixture of cations -1

Unit 8: Complexometric estimation of binary mixture of cations -1

Estimation of Bi^{2+} - pb^{2+} mixtures using EDTA as a complexing agent by adopting any one of the techniques, like precipitation, pH variation, masking and demasking.

Unit 9: Complexometric estimation of binary mixture of cations -2

Estimation of pb^{2+} – Ca^{2+} mixtures using EDTA as a complexing agent by adopting any one of the techniques, like precipitation, pH variation, masking and demasking.

Unit 10: Complexometric estimation of binary mixture of cations - 3

Estimation of Ni^{2+} – Cu^{2+} mixtures using EDTA as a complexing agent by adopting any one of the techniques, like precipitation, pH variation, masking and demasking.

Unit 11: Complexometric estimation of binary mixture of cations -4

Estimation of Fe^{2+} – Ni^{2+} mixtures using EDTA as a complexing agent by adopting any one of the techniques, like precipitation, pH variation, masking and demasking.

Block-4: Complexometric estimation of binary mixture of cations -2

Unit 12: Complexometric estimation of binary mixture of cations -5

Estimation of Zn^{2+} – Cu^{2+} mixtures using EDTA as a complexing agent by adopting any one of the techniques, like precipitation, pH variation, masking and demasking.

Unit 13: Complexometric estimation of binary mixture of cations -6

Estimation of Co^{2+} - Cu^{2+} mixtures using EDTA as a complexing agent by adopting any one of the techniques, like precipitation, pH variation, masking and demasking.

Unit 14: Complexometric estimation of binary mixture of cations -7

Estimation of Zn^{2+} - Ca^{2+} mixtures using EDTA as a complexing agent by adopting any one of the techniques, like precipitation, pH variation, masking and demasking.

Reference Books

1. **Vogel's qualitative Inorganic analysis,** G. Svehla, VI Edition, Orient Longman, 1987.
2. **Inorganic Semimicro Qualitative analysis,** V.V. Ramanujam, National Publishing Co., 1971.
3. **A Laboratory Manual of Inorganic Chemistry**, John Bernard Ekeley, BiblioLife, 2010.

FOURTH SEMESTER

Course Code	Title of the Course
34441	Analytical Chemistry

Course Objectives

The major objectives of this course are to understand the concepts of:

- ✓ Errors in chemical analysis, Statistical analysis and validation comparison of results
- ✓ Purification techniques
- ✓ Electro analytical methods
- ✓ Separation and identification of compounds by Chromatographic methods

Learning Outcomes

The student would be able to

- ✓ Predict the error
- ✓ To know about electroanalytical methods
- ✓ Understand the gas chromatography and High pressure liquid chromatography
- ✓ Know about the Purification techniques

Block-1: Error analysis

Unit 1: Errors in chemical analysis.

Errors in chemical analysis. Classification of errors- systematic and random, additive and proportional, absolute and relative.

Unit 2: Data analysis.

Accuracy and precision. Mean, median, average deviation and standard deviation. Significant figures and rules to determine significant figures. Calculations involving significant figures.

Unit 3: Comparison of results

Confidence limit, correlation coefficient and regression analysis. Comparison of methods: F-test and T-test. Rejection of data based on Q-test. Least squares method for deriving calibration graph.

Block-2: Electroanalytical methods

Unit 4: Electrodics

Ion selective electrodes - Potentiometric methods – electrogravimetry - Coulometric analysis.

Unit 5: Polarography

Principles and applications of polarography – Instrumentation, Types of cells, advantages of dropping mercury electrode. and applications.

Unit 6: Cyclic voltammetry

AC polarography – Cyclic Voltammetry, Instrumentation, advantages over polarographic techniques – test of reversibility of electron transfer reactions – Application

Unit 7: Chrono techniques

Chronopotentiometry – instrumentation, advantages over polarography – controlled potential coulometry.

Block-3: Chromatography

Unit 8: Chromatographic methods

Definition - Classification - Basic and elementary principle and practice of Paper chromatography. Thin Layer Chromatography

Unit 9: Gas Chromatography

Gas chromatographic techniques – Principle, instrumentation – injection system, column, and detector, application of GC.

Unit 10: GC-MS

Gas chromatographic –mass spectral techniques – Principle, instrumentation – interpretation and application of GC-MS

Unit 11: High Pressure Liquid Chromatography

Principle, types of HPLC, Normal and reversed phase liquid chromatography (NP- & RP LC); Instrumentation, Application

Block-4: Ion Chromatography, Purification Techniques and Ionic separation

Unit 12: Ion chromatography

Theory, application of Ion exchange and gel permission Chromatography

Unit 13 - Purification Techniques

General methods of isolation and purification of chemicals – Solvent extraction – Fractional crystallization – sublimation – distillation – vacuum distillation – purification of solvents.

Unit 14: Ionic Separations

Types of equipment employed for electrophoresis, Di-electrophoresis and electrodialysis Controlling factors, Applications.

Text Books

1. A.I Vogel, **Text Book of Quantitative organic Analysis**, ELBS III Edn, 1987.
2. Chatwal and Anand, **Instrumental methods of chemical analysis**, Himalaya publishing House New Delhi, 2000.
3. D.B.Hibbert and J.J. Gooding, **Data Analysis for chemistry**, Oxford University Press, 2006
4. Allen J.Bard and Faulkner, **Electrochemical Methods, Fundamentals and Applications**, John Wiley and Sons, New York, 1983.

Reference Books

1. Willard, Merit Dean and Settle, **Instrumental Methods of Analysis**, CBS Publishers, IV Edn., 1986.
2. Schoog, Holler, Nieman, Thomson, **Principles of Instrumental Analysis**, Asia Pvt. Ltd., Singapore, 2004.
3. D.A.Skoog and D.M.West, **Fundamentals of Analytical Chemistry**, Winston Publications, IV Edn, 2004.
4. J.M.Mermet, M.Otto, R.Kellner, **Analytical chemistry: a modern approach to analytical science**, Wiley-VCH, 2004.
5. F.Rouessac, A.Rouessac, **Chemical Analysis: Modern Instrumentation Methods and Technique**s, 2nd Edition, Wiley & sons, USA, 2011.
6. J. R. Lakowicz, **Principles of Fluorescence Spectroscopy**, 3rd Ed., Springer, New York, 2006.
7. Lacey, R.E. and S.Loaeb - **Industrial Processing with Membranes**, Wiley -Inter Science, New York, 1972.
8. King, C.J. **Separation Processes** , Tata McGraw - Hill Publishing Co., Ltd., 1982.
9. 4. Ronald W.Roussel - Handbook of Separation Process Technology, John Wiley, New York, 1987.
10. James A. Plam Beck, **Electroanalytical Chemistry – Basic Principles and Applications**, John Wiley & Sons, 1982.

Course Code	Title of the Course
34442	APPLIED CHEMISTRY

Course Objectives

- To educate on the basic terminologies of the environment
- To develop knowledge about air, water and soil
- To create awareness various pollutions and abatements
- To gain an understanding of the principles of nanotechnology; characterization of nano structured materials; and tools and equipment for producing and assembling at the nano scale.
- To acquire advanced knowledge about the equipment used in nanotechnology such as XRD, SEM, TEM, STM, AFM, XPS, AES

Learning Outcomes

- ✓ The students will acquire basic knowledge about environment
- ✓ Environmental awareness about the various types of pollution and their control.
- ✓ This multidisciplinary course provides an in-depth view of the synthesis, characterisation and application of nanostructures using chemical routes. Necessarily, it will incorporate various concepts from colloidal chemistry, supramolecular chemistry, polymer chemistry and electrochemistry.
- ✓ This course will gain knowledge in the most exciting, novel and interdisciplinary issues in nanoscale science and Technology.

Block-1: Environmental Wastewater treatment methods and **Electrochemical power sources**

Unit – 1: Environmental Chemistry

Hazardous materials and their ill effects**.** Acid rain, Ozone hole and green house effect. Types of pollution – air, water, land, pesticide, thermal and radioactive. Physicochemical and biological investigations of water - water quality.

Unit – 2: Wastewater treatment methods

Pretreatment, preliminary treatment, secondary (or biological) methods of treatment and tertiary (or advanced) methods of wastewater treatment.

Unit – 3: Electrochemical power sources

Principle of energy conversion, electrochemical energy conversion, Classification of batteries - primary and secondary systems. Basic electrochemical reactions and performance of primary and secondary systems - Fuel cells - Introduction - Types of fuel cells, Advantages - fuel cell, supercapcitors.

Block-2: Electrochemistry-I

Unit – 4: Corrosion

Basic aspects of corrosion: Importance of corrosion studies – EMF and Galvanic series – classification of corrosion – corrosion kinetics – Pourbaix diagram for Fe- H_2O system – passivity

Unit – 5: Methods of corrosion

High temperature corrosion – Forms of corrosion. Chemical and Electrochemical methods of corrosion rate measurements methods.

Unit – 6: Corrosion control methods

General classification of corrosion control methods – Designing aspects in corrosion control – corrosion inhibitors – Electrochemical methods of protection such as anodic and cathodic protection.

Block-3: Electrochemistry-II

Unit – 7: Electroplating

Principles of electroplating – Metal deposition from solutions of simple salts and complex salts – measurement of current density, throwing power and current efficiency of electroplating bath – surface preparation for electroplating. Electroplating of nickel and copper

Unit – 8: Electroforming

Principle and applications, Alloy plating of Brass, Brush plating, Cladding and Vapour deposition

Unit – 9 Electroless plating

Principles, advantages and limitations of electroless plating – Composite coating - principle, mechanism and their applications. Anodizing – principle, types of anodizing bath – colouring of anodizing aluminium.

Block-4: Nanomaterials, Characterization Techniques and Computer applications in Chemistry

Unit – 10: Nanomaterials

Brief introduction to nanoscience and technology. Preparatory synthesis - Sol-gel thermolysis, combustion method, solvothermal method and microemulsion method.

Unit – 11: Physical and chemical method of nanomaterials

Physical methods – vacuum evaporation, sputtering, pulsed laser deposition. Chemical methods - CVD, chemical solution deposition, electrochemical deposition, spray pyrolysis deposition.

Unit 12: Physical characterization techniques of nanomaterial

FT-IR, XPS and Laser Raman spectroscopy. XRD analysis. Microscopic techniques: SEM, AFM and TEM. Thermal analysis: TG/DTA and DSC.

Unit 13: Computer applications in Chemistry

Calculation of pH, solubility product, calculation of bond energy using Born-Lande equation. Standard deviation and correlation coefficient.

Unit 14: Online resources for chemistry

Introduction - Internet service providers, terms used in E-mail-search engines - chemistry databases- table of contents - source for list of journals – Online courses NPTEL, Swayam, MOOC, virtual lab..

Text books

1. **Environmental Chemistry,** Sharma & Kaur, Krishna Publishers, New Delhi, 2000.
2. **Principles and prevention of corrosion**, D.Jones, Macmillan Publications New York, 1992.
3. **The Chemistry of nanomaterials; Synthesis, properties and applications**, C.N.R. Rao, Wiley-VCH Verlag Gmbh&Co, Weinheim, 2004.
4. **Computers in Chemistry,** K.V. Raman, Tata McGraw Hill, New Delhi, 1993.

Reference Books

1. **Environmental Chemistry**, S.E Manahan CRC press, 2010.
2. **Environmental Chemistry,** S.K. Banerji, Prentice Hall of India, New Delhi, 2003.
3. **Wastewater treatment,** Ed. M. Henze, P. Harremoes, J.C. Jansen and E. Arvin, Springer Verlag, New York, 1995.
4. **Cathodic Protection Theory and practice**, J.J. Meketta, Marcel Dekker Publication, NY, 1993.
5. **An introduction of corrosion and corrosion inhibition,** S.N. Banerjee, Oxonian Press Ltd., New Delhi.

6. **Modern Electroplating,** K.A. Lowenheim, Second Edition, John Wiley & Sons, New York, 1963.
7.**BASIC Programming for Chemists,** P.C. Jurns, T.L. Isenhour and C.C. Wilkins, JW.& Sons 1987.
8. **Computers in Chemistry,** K.V. Raman, Tata McGraw Hill, New Delhi, 1993.
9. **Nanoscale materials in Chemsitry**, Kenneth J. Klabunde, John-Wiley & Sons, 2001.
10. **Environmental Chemistry for a Sustainable World: Remediation of Air and Water Pollution,** E. Lichtfouse, J. Schwarzbauer, D. Robert, Springer, 2011.

Course Code	Title of the Course
34443	ADVANCED PHYSICAL CHEMISTRY

Course Objectives: To make the students

- To understand the statistical thermodynamics
- To familiar with one dimensional harmonic oscillator, rigid rotator
- The basic of group theory and character table
- To study about salt effect and catalysis system.

Learning outcomes

The students will gain knowledge about

- ✓ Advanced concepts in quantum mechanics which make the students to understand the atomic orbitals and their structures.
- ✓ Recognize the importance of quantum chemistry and of its applications.
- ✓ Describe and understand the basic group theory and its applications
- ✓ Get deep knowledge about chemical kinetic of reaction in solution.

Block-1: Statistical thermodynamics

Unit – 1: Statistical thermodynamics

Maxwell - Boltzmann distribution law of molecular energies - Negative absolute temperature - Entropy and probability,

Unit – 2: Partition functions

Partition functions and thermodynamic functions, translational, rotational and vibrational partition functions entropies and energies – Equilibrium constant from partition function

Unit – 3: Statistical interpretation of third law

Statistical interpretation of third law - Bose-Einstein distribution law Application of the law to photon gas - Fermi-Dirac distribution law -Application of the law to electron gas

Unit – 4: Heat capacities of solids

Heat capacities of solids: Einstein and Debye's models - Non equilibrium thermodynamics - Elementary treatment, Onsager reciprocal relations.

Block-2: Quantum chemistry

Unit – 5: Quantum chemistry

Application of wave mechanics to simple systems - One dimensional harmonic oscillator, rigid rotor

Unit – 6: Quantum mechanical treatment

Quantum mechanical treatment for radical and angular wave function and hydrogen atom like atoms - Pauli's exclusion principle and Slater determinant – Approximation methods – variation - time independent perturbation and SCF methods

Unit – 7: HMO method

Application of variation methods to hydrogen atom - Application of perturbation method to helium - HMO method – application to butadiene.

Block-3: Group theory

Unit – 8: Assignment of point groups

Symmetry elements and symmetry operations - Rules for forming a group, group multiplication table, group classification - Point groups and systematic assignment of point groups for molecules

Unit – 9: Matrix representation theory

Matrix representation theory - matrix multiplication, inverse of a matrix, matrix diagonalization and matrix representation for symmetry operations

Unit – 10: Character table

Reducible and irreducible representations.-Character table of C_{2v} and C_{3v} point groups. The great orthogonality theorem and character table - Direct product representation –

Unit – 11: Symmetry oriented spectral methods

Application of group theory to IR and Raman spectra - H_2O and NH_3 molecules - Application of group theory to electronic spectra (HCHO and C_2H_4)

Block-4: Chemical kinetics

Unit – 12: Reactions in solution

Reactions in solution – factors which influence the reaction rates in solution. Application of ARRT to solution kinetics – Bronsted – Bjerrum equation,

Unit – 13: Salt effect

Primary salt effect, secondary salt effect - influence of internal pressure - effect of pressure and volume of activation. Effect of solvent: ion-ion and ion-dipole reactions-dielectric constant – Effect of substituents on reaction rates Hammett and Taft equations

Unit – 14: Acid base catalysis

Acid base catalysis-acidity functions – Bronsted relations - Zucker Hammett hypothesis – Enzyme catalysis – Michaelis – Menton equation- Lineweaver- Burke equation, Edie equation – Effect of pH and temperature on enzyme catalyzed reactions.

Text Books

1. **Statistical Thermodynamics**, M.C. Gupta, Wiley Eastern, New Delhi, 1990.
2. **Quantum Chemistry**, D.A. McQuarrie, University Science Books, Mill Valley, California, 1983.
3. **Group Theory**, A.K. Chandra, Discovery Publishing House, New Delhi, 2010.
4. **Kinetics and Mechanism of Chemical Transformations**, J. Rajaram and J.C. Kuriacose, MacMillan India Ltd. 1993.

Reference Books

1. **Introduction to Statistical Thermodynamics**, R.P.H. Gasser and W.G. Richards, World Scientific, Singapore, 1995.
2. **Quantum Chemistry**, R.K. Prasad, Wiley Eastern, New Delhi,1992.
3. **Quantum Mechanics in Chemistry**, M.W. Hanna, W.A. Benjamin Inc. London, 1965.
4. **Chemical Application of Group Theory**, F.A. Cotton, John Wiley and Sons Inc. New York, 1971.
5. **Group theory and its applications to Chemistry**, K.V. Raman, Tata McGraw-Hill Publishing Company, 2008.
6. **Irreversible Thermodynamics,** J. Rajaram and J.C. Kuriacose, Lal Nagin Chand, New Delhi, 1989.
7. **Chemical Kinetics**, K.J. Laidler, Harper and Row, New York,1987.
8. **Kinetics and Mechanism**, R.G. Frost and Pearson, Wiley New York, 1961
9. **Kinetics and Mechanism**, W.J. Moore and R.G. Pearson, 1981.
10. **Symmetry and spectroscopy of molecules,** K. Veera Raddy, New Age International Publishers; Second edition, 2009.
11. **Group theory in Chemistry**, V. Ramakrishnan and M.S. Gopinathan, Vishal Publications, 1988.

Course Code	Title of the Course
34444	PHYSICAL CHEMISTRY PRACTICAL

Course Objectives

- The Physical Chemistry practical course is designed such that to provide deep knowledge and hands on experimenting the more advanced physical chemistry practicals such as kinetics, distribution studies, conductometry, potentiometry, and spectrophotometric method

Learning outcomes

The students will be able to

- ✓ Carry out electrical experiments such as Conductomerty and Potentiometric Titrations
- ✓ Determine out the kinetic parameters in the ester hydrolysis
- ✓ Understand the equilibrium reactions
- ✓ Determine the concentration by spectrophotometric method

Block-1: Partition Co-efficient

Unit 1: Partition Co-efficient - 1

Determination of Partition Co-efficient of iodine in water/

Unit 2: Partition Co-efficient - 2

Determination of Equilibrium constant of KI

Unit 3: Partition Co-efficient - 3

Determination of Unknown KI

Block-2: Conductometric titration

Unit 3: Conductometric titration of Acid vs Base

Cconductometric titration of (i) strong acid vs strong base, (ii) weak acid vs strong base (iii) mixture of acids vs strong base

Unit 4: Conductometric titration of Mixed halides

Conductometric titrations of mixed halides

Unit 5: Conductometric titration of solubility product

Solubility product by conductivity measurement

Unit 6: Determine the strength of the given salt by conductometric titration

Determine the strength of the given salt solution by conductometric titration

Block-3: Potentiometric titration

Unit 7: Potentiometric titration of Acid vs Base

Potentiometric titration of strong acid vs strong base and weak acid vs strong base

Unit 8: Determine the strength of the given salt by Potentiometric titration

Determine the strength of the given salt solution by potentiometric titration (FAS vs $K_2Cr_2O_7$ and FAS vs $KMnO_4$)

Unit 9: Potentiometric titration of simple halides and Mixed halides

potentiometric titration of simple halide and mixture of halides

Block-4: Chemical Kinetics and Spectrophotometric method

Unit 10: First order and second order kinetics of hydrolysis of ester

Acid catalyzed hydrolysis of an ester, base catalyzed hydrolysis of an ester by titration method

Unit 11: Kinetics of hydrolysis of ester by conductivity method

Base catalyzed hydrolysis of an ester by conductivity method

Unit 12: Determination of metal by colorimeter

Determine the amount of manganese present in the given steel sample.

Unit 13: Determination of Iron by spectrophotometer method

Determine the amount of iron present in the given water sample by spectrophotometeric method.

Unit 14: Determination of Copepr by spectrophotometer method

Determine the amount of copper present in the given sample by spectrophotometeric method.

REFERENCE BOOKS

1. **Findlay's Practical Physical Chemistry,** Revised and edited by 'B.P.Levitt, 9th edn., Longman, London, 1985.
2. **Advanced Experimental Chemistry,** J.N.Gurtu and R. Kapoor, Vol.I, S.Chand & Co. Ltd., New Delhi (1980).
3. **Practical Physical Chemistry**, B. Viswanathan and P.S. Raghavan, ViVa Books, 2009.
4. **Systematic Experimental Physical Chemistry**, S.W. Rajbhoj and T.K.Chondhekar, Anjali Publication, Aurangabad, 2000.

e. 3. Duration of the Programme:

The programme for the degree of Master of Science in Chemistry shall consist of two academic years divided in to four semesters. Each semester consists of Three Theory Papers and One Practical Paper. Theory course carry 4 credits each and Practical course carry 4 credits. Each semester consist of 16 credits.

e. 4. Faculty and Support Staff Requirements:

The programme for the degree of Master of Science in Botany requires following faculty and supporting staff

Staff Category	Required
Faculty for Core Chemistry Subjects #	3
Faculty for Special Subject viz.,Analytical Chemistry, Applied chemistry, etc., #	2
Laboratory Assistant	1
Clerical Assistant	1

Faculty may be belongs to at least Assistant Professor level

e. 5. Instructional Delivery Mechanisms:

The instructional delivery mechanisms of the programme includes SLM – study materials, face to face contact session for both theory and practical courses of the programme, e-content of the study materials in the form of CD and virtual laboratory wherever applicable.

e. 6. Identification of Media:

The SLM – designed study materials will be provided in print media as well is in the form of CD which carries electronic version of the study material in addition to virtual laboratory courses.

e. 7. Student Support Services:

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like councelling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issuance of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter only. The conduct of end semester examinations, evaluation and issuance of certificates will be done by office of the Controller of Examinations, Alagappa University, Karaikudi.

f. Procedure for Admission, curriculum transaction and evaluation

f. 1. Procedure for Admission:

A candidate who has passed the B.Sc., degree (10 + 2+ 3 system) examination in Chemistry of the University or an Examination of any other University accepted by the Syndicate as equivalent thereto shall be eligible to appear and qualify for the M.Sc. Degree in Chemistry of this University after a course of study of two academic years.

f. 2. Curriculum Transactions:

The classroom teaching would be through chalk and talk method, use of OHP, Power Point presentations, web-based lessons, animated videos, etc. The face to face contact sessions would be such that the student should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

For practical courses exclusive study materials containing the requirements, procedure for the experiments will be issued to the learners. In the laboratory, instruction would be given for the experiments followed by demonstration and finally the students have to do the experiments individually.

The face to face contact sessions will be conducted in following durations;

Course Type	**Face to Face Contact Session per Semester (in Hours)**
Theory Courses (3 courses with 4 credits each)	48
Practical Courses (1 course with 4 credits)	120
Total	**168**

f. 3. Evaluation:

The evaluation shall be conducted separately for theory and practical to assess the knowledge acquired during the course of study. There shall be two systems of evaluation viz., internal assessment and end semester examinations.

In the case of theory courses the internal evaluation shall be conducted as Continuous Internal Assessment via. student assignment preparation, seminar, etc. The internal assessment shall comprise of maximum 25 marks for each subject. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

In the case of Practical courses for M.Sc. programme in Chemistry should be conducted at every semester. The internal evaluation of practical courses will be done through continuous assessment of skill in demonstrating the experiments, record note preparation, and viva voce. The external evaluation consists of an end semester practical examinations which comprise of 75

marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

f. 3.1. Question Paper Pattern:

Time: 3 Hours Max. Marks: 75

Part A- 10 x 2 Marks = 20 Marks (Answer all questions) Q. No. 1 – 10

Part B -5 x 5 Marks = 25 Marks Q. No. 11 – 15

11 a) or b)
12 a) or b)
13 a) or b)
14 a) or b)
15 a) or b)

Part C- 3 x 10 Marks = 30 Marks (Answer any three from five question s) Q. No. 16 – 20

f. 3.2. Distribution of Marks in Continuous Internal Assessments:

The performance of a student in each course is evaluated in terms of percentage of marks with a provision for conversion to grade points. Evaluation for each course shall be done by internal assessment during Personal Contact Programme as well as by annual examination and will be consolidated at the end of the programme.

The components for continuous internal assessment are:

Internal Tests /Seminar/Assignment - 25 marks

Attendance need not be taken as a component for continuous assessment, although the student should put in a minimum of 75% attendance in each course.

The annual examination, which will be a written examination of at least 3 hours duration, would also form an integral component of the evaluation. The ratio of marks to be allotted to internal assessment and to end semester examination is 25:75. The evaluation of laboratory component, wherever applicable, will also be based on internal assessment and annual practical examination.

Distribution of marks for practical examinations

(Internal Assessment 25 marks + Annual Examination 75 marks)

ESE mark distribution	
Qualitative/Quantitative analysis	50 marks
Record Note	10 marks
Viva voce	15 marks
Total	75 marks

f. 3.3. Passing Minimum:

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50 % in the University external examination and 50% of the total (CIA+ESE) marks. Candidate who does not obtain the required minimum marks for a pass in a course shall be required to reappear and pass the same at a subsequent appearance.

f. 3.4. Marks and Grades:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester
G_i = Grade Point obtained for course i in any semester.
n refers to the semester in which such courses were credited

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

$$\text{Grade Point Average} = \frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = Sum of the multiplication of grade points by the credits of the entire programme

Sum of the credits of the courses for the entire programme

CGPA	Grad	Classification of Final Result
9.5-10.0	O+	First Class- Exemplary*
9.0 and above but below 9.5	O	
8.5 and above but below 9.0	D++	First Class with Distinction*
8.0 and above but below 8.5	D+	
7.5 and above but below 8.0	D	
7.0 and above but below 7.5	A++	First Class
6.5 and above but below 7.0	A+	
6.0 and above but below 6.5	A	
5.5 and above but below 6.0	B+	Second Class
5.0 and above but below 5.5	B	
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

f. 3.5. Maximum duration for the completion of the course:

The maximum duration for completion of M.Sc., Degree in Chemistry programme shall not exceed ten semesters from their fourth semester.

f. 3.6. Commencement of this Regulation:

The Department shall form a Grievance Redressal Committee for each course with the course Teacher and the HOD as the members. This committee shall solve all grievances relating to the internal Assessment marks of the students.

f. 3.7 Transfer of credits

Students are permitted to transfer their programme credits from Directorate of Distance Education (DDE) of Alagappa University to Regular Stream and Vice-versa, if the PG degree programme is same.

f. 3.8. Revision of regulations and curriculum

The University may from time to time revise, amend and change the regulation and the curriculum, if found necessary.

f. 3.9. Commencement of this regulation

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter.

f. 4. Fee Structure:

The programme has the following Fee Structure:

Sl. No.	Fees Detail	Amount in Rs.	
		First Year	**Second Year**
1	Admission Processing Fees	300.00	-
2	Course Fee	20000.00	20000.00
3	ICT Fees	150.00	150.00
	TOTAL	20450.00	20150.00

The above mentioned fee structure is exclusive of Exam fees.

g. Requirement of the laboratory support and Library Resources:

A well-equipped Chemistry Laboratory exclusive for Distance Education Programme was established at the Department of Industrial Chemistry, Alagappa University, Karaikudi with all the necessary equipments for conducting face to face contact sessions for Practical courses of M. Sc., Chemistry programme. The students who have enrolled themselves in M. Sc Chemistry Programme shall attend the face to face contact session for Practical Courses **only** at Chemistry Laboratory, Directorate of Distance Education, Located at Alagappa University, **Karaikudi**.

Directorate of Distance Education, Alagappa University, Karaikudi housing an exclusive Library facility with adequate number of copies of books in relevant titles for M. Sc., Chemistry programme. The Central Library of Alagappa University also having good source of reference books. The books available at both the libraries are only for reference purpose and not for lending services.

h. Cost estimate of the programme and the provisions:

The cost estimate of the programme and provisions for the fund to meetout the expenditure to be incurred in connection with M. Sc., Chemistry Programme as follows:

S.No.	Expenditure Heads	Approx. Amount in Rs.
1	Programme Development	20,00,000/-
2	Programme Delivery	24,00,000/-
3	Programme Maintenance	5,00,000/-

i. Quality assurance mechanism and expected programme outcomes:

University's Moto:

'Excellence in Action'

University's Vision Statement:

Achieving Excellence in all spheres of Education, with particular emphasis on "PEARL"- Pedagogy, Extension, Administration, Research and Learning.

University's Objectives

1. Providing for Instructions and Training in such Branches of Learning as the University may determine.
2. Fostering Research for the Advancement and Dissemination of Knowledge

University's Quality Policy

Attaining Benchmark Quality in every domain of 'PEARL' to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

University's Quality Quote

Quality Unleashes Opportunities Towards Excellence (QUOTE)

Programme's Review Mechanism

The quality of the programme depends on scientific construction of the curriculum, strong-enough syllabi, sincere efforts leading to skilful execution of the course of the study. The ultimate achievement of M. Sc., Chemistry programme of study may reflect the gaining of knowledge and skill in the subject chemistry. And all these gaining of knowledge may in Chemistry may helps the students to get new job opportunities, upgrading in their position not only in employment but also in the society, make students feel thirsty to achieve in research in the fields associated with the discipline- Chemistry, achieving in competitive examinations on the subject- Chemistry, etc.

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations. Apart from the end semester examination-based review feedback from the alumni, students, parents and employers will be received and analyzed for the further improvement of the quality of the M. Sc., Chemistry Programme.

Minutes of the Meeting of the Board of Studies in Chemistry for M. Sc., Chemistry Directorate of Distance Education (DDE), Alagappa University, Karaikudi held on 16.05.2017 at 10.30 A.M of Department of Industrial Chemistry, Alagappa University Karaikudi.

Members Present

1. **Dr. P. Manisankar** - **Chairman**
   Prof. and Head, Department of Industrial Chemistry
   Alagappa University, Karaikudi
2. **Dr. A. Ramu** - **Member**
   Prof and Head, Department of Inorganic Chemistry
   Madurai Kamaraj University, Madurai – 625 021
3. **Dr. E. Murugan** - **Member**
   Professor, Department of Physical Chemistry
   University of Madras, Guindy Campus, Chennai -25
4. **Dr. M. Anbukulandainathan** - **Member**
   Principal Scientist, CECRI, Karaikudi
5. **Dr. N. Sengottuvelan** - **Member**
   Assistant Professor, (DDE)
   Department of Industrial Chemistry, Alagappa University, Karaikudi,
6. **Dr. P. Muthu Mareeswaran** - **Special Invitee**
   DST INSPIRE Faculty, Dept of Industrial Chemistry,

Dr. G. Paruthimal Kalaignan has gone to USA on personal work and he will be back only on 22.06.2017. Dr. M. Sundharavadivelu has informed his inability to attend the meeting.

The Chairman of the Board Dr. P. Manisankar, welcomed the members.

The committee had detailed discussion on the draft syllabus prepared by the Chairman and Faculty Members of Department of Industrial Chemistry.
After detailed discussion the board has passed the following resolution :

1. Resolved to approve the regulation, course structure and Syllabi of DDE M. Sc., Chemistry

2. Resolved to introduce the the internal mark system (25 marks internal and 75 external).

3. Resolved to authorize the Chairman of the committee, submit the regulation, course structure and syllabi of DDE M. Sc., Chemistry, both Soft and hard copies.

Prof. P. Manisankar P. Manisankar 16/05/17

Prof. A. Ramu

Prof. E. Murugan 16/05/2017

Dr. M. Anbukulandainathan 16/05/2017

Dr. N. Sengottuvelan

Dr. P. Muthu Mareeswaran P. Muthu 16/05/2017.

ALAGAPPA UNIVERSITY

(A State University Accredited with A+ Grade by NAAC (CGPA: 3.64) in the Third Cycle)
Karaikudi, Tamilnadu, India

DIRECTORATE OF DISTANCE EDUCATION

PROGRAMME PROJECT REPORT
for
M.Sc (Computer Science)

Submitted to
UGC, DISTANCE EDUCATION BUREAU (DEB)
New Delhi
For seeking approval to introduce new programme through Distance Education Mode

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ALAGAPPA UNIVERSITY, KARAIKUDI

DIRECTORATE OF DISTANCE EDUCATION

MASTER OF COMPUTER SCIENCE (M.Sc)
Credit Based System (CBS)
(With effect from June 2018-2019 Onwards)

(a) Programme's Mission and Objectives

Mission

The mission for M.Sc. Computer Science, to meet the twin objectives one for research and the other is to improving the potential for suitable placements in the industry. The M.Sc. course provides in-depth understanding of core and advanced topics in the subject. In the final semester, students take up and complete a project in the design and development of software. This course is to offer excellent career opportunities in various industries including software development companies in the areas of System analysis/design/developments/supports.

Programme Objectives:

- To train students for careers in research and in the software industry
- To enable students to work on a project in collaboration with well-established software development organizations and acquire demonstrable experience in the design, development and documentation of a software product of significant size and complexity
- To develop problem-solving skills, and to encourage students to communicate effectively
- Discover, analyze, investigate the requirements of a problem and find the solution using computers.

Programme Outcome:

- The programme will enable the students to develop software solutions to real world enterprises through IT with international standards and facilitate them to be outstanding professionals.
- To widen the ability to plan, analyze, design, code, test, implement & maintain a software product for real time systems.
- To support learners capability to set up their own enterprise in various sectors of Computer science
- To improve the knowledge of the learners in finding solutions and developing system based applications for real time problems in various domains involving technical, managerial, economical & social constraints.
- To develop in problem solving and programming skills in the various computing fields of IT industries.
- To prepare the learners to pursue higher studies in computing or related disciplines and to work in the fields of teaching and research like M.Phil., PhD etc.

(b) Relevance of the program with HEI's and Alagappa University Mission and Goals

This programme is aligned with HEI's and Alagappa University mission and goals to be offered through distance mode to reach quality higher education to the unreachable and/or rural learners. Higher education in Computer Science offered through distance mode meets the mission of HEI's like digital India and cash less transactions will enrich the Human resources for the uplift of the nation.

(c) Nature of prospective target group of learners

The nature of prospective target group of learners is graduates from various disciplines like Commerce, Mathematics, Physics, Chemistry, Biology, Electronics, and Engineering etc. It also includes the learners who want to become entrepreneurs like System Engineer, Web Designers, and Software Developers

(d) Appropriateness of programme to be conducted in Open and Distance Learning mode to acquire specific skills and competence;

M.Sc. Computer Science Programme through Distance Learning mode is developed in order to give subject-specific skills including i) programming languages concepts ii) Operating systems, RDBMS, Data Structure and Software Engineering iii) inter-disciplinary knowledge like Mathematical foundation of computer science iv) Concepts like cloud computing, mobile computing v) Cutting Edge Technologies like Web Technology, .Net Frameworks.

(e) Instructional Design

e.1 Revisions of Regulation and Curriculum Design

1. The University reserves the right to amend or change the regulations, schemes of examinations and syllabi from time to time based on recent software trend, Industrial developments, research and feedback from stakeholders and learners.
2. Each student should secure 64 credits to complete M.Sc CS programme.
3. Each theory course carries 4 credits with 75 marks in the University end semester and 25 marks in the Internal Assessment and each practical (Lab) course carries 4 credits with 75 marks in the university end semester examination and 25 marks in the Internal Assessment.

Programme Code

M.Sc(Computer Science)	**341**

Courses of Study M.Sc Computer Science

S.No	Course code	Title of the Course	CIA Marks	ESE Marks	Total Marks	Credits
	I Semester					
1	34111	Design and Analysis of Algorithms	25	75	100	4
2	34112	Applied Mathematics for Computer Science	25	75	100	4
3	34113	Advanced Java Programming	25	75	100	4
4	34114	Lab – Advanced Java Programming	25	75	100	4
		Total	**100**	**300**	**400**	**16**
	II Semester					
1	34121	Computer System Architecture	25	75	100	4
2	34122	Distributed Operating System	25	75	100	4
3	34123	.Net Programming	25	75	100	4
4	34124	Lab – .Net Programming	25	75	100	4
		Total	**100**	**300**	**400**	**16**
	III Semester					
1	34131	Cryptography and Network Security	25	75	100	4
2	34132	Cloud Computing	25	75	100	4
3	34133	Web Technology	25	75	100	4
4	34134	Lab – Web Technology	25	75	100	4
		Total	**100**	**300**	**400**	**16**
	IV Semester					
1	34141	Data Mining and ware housing	25	75	100	4
2	34142	Mobile Application Development	25	75	100	4
3	34143	Artificial Intelligence and Expert Systems	25	75	100	4
4	34144	Project	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		Grand Total	**400**	**1200**	**1600**	**64**

CIA: Continuous Internal Assessment **ESE:** End semester Examination

Course Code	3	4	1	X	Y

– Programme code for Master of Computer Science (M.Sc).

No. of Credits per course (Theory) : 4
No. of Credits per course (Practical) : 4
Total No. of credits per Semester : 16
Total No. of credits of the Programme : 16 X 4 = 64

e.2 Detailed Syllabi

The detailed Syllabi of study and shall be as shown in Appendix.

e.3 Duration of the Programme:

The M.Sc CS programme shall consist of a period of two years (Four Semesters).

e.4 Faculty and Support Staff Requirements:

The following faculty and support staff is required for this programme.

S.No	Staff Category	Required Numbers
1	Computer Science Subject Core Faculty*	3
2	Inter-disciplinary Subject Faculty* (Mathematics)	1
3	Lab Assistant	1
4	Clerical Assistant	1

* Faculty at least in Assistant Professor Level

e.5 Instructional Delivery mechanisms

The instructional delivery mechanisms of the programme includes SLM- Study materials, Lab instruction manual, Personal contact session for both theory and practical courses of the programme, e-version of the course materials in the form of CD, e-book, e-tutorials, Massive Open Online Courses (MOOC) courses, Open Educational Resources(OER) and virtual lab.

e.6 Identification of media

The printed version of SLM – study material shall be given to the learners in addition to MOOC, e-tutorial and virtual lab.

e.7 Student Support Services

The student support services will be facilitated by the Directorate of Distance Education, Alagappa University, Karaikudi and its approved learning centres located in various parts of Tamilnadu.

The pre-admission student support services like counseling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at Directorate of Distance Education or Learning centres. The post-admission student support services like issuing Identity card, study materials will be provided thru Directorate or Learning centres. The face to face contact sessions of the programme for both theory and practical's will be held at the Directorate or Learning centres. The student support regarding the conduct of examinations, evaluations, publication of results and certificates done by the Office of the Controller of Examinations, Alagappa University, Karaikudi

(f) Procedure for Admissions, curriculum transaction and evaluation

f.1 Minimum qualification for admission

Candidates for admission to the first year of the Master of Computer Science (M.Sc) programme shall be required to have passed the following examinations. Candidates who have passed any degree with Mathematics at +2 level of any Recognized University or authority accepted by the Syndicate of the Alagappa University as equivalent thereto shall be eligible.

f.2 Curriculum transaction

- The face to face contact sessions in class room teaching with the support of SLM, Power Point Presentations, web based tools, audio and animated videos.
- The practical classes are based on the respective subject study materials containing requirement for the laboratory experiments.
- Face to face contact sessions will be conducted for both theory and practical courses in the following manner.

Course Type	**Face to face contact session per semester (in Hours)**
Theory courses (3 Courses with 4 credits each)	48
Practical courses (1 Courses with 4 credits each)	120
Total	**168**

f.3 Evaluation

There shall be two types of evaluation systems; internal assessment and end semester examination will be conducted by the University according to the following scheme. The maximum marks for the internal assessment for both theory and practical's is 25 marks. The maximum marks for end semester examination is 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

Internal assessment

- Internal assessment of theory courses is through home assignment with workbook, case studies, review questions, quiz, multiple choice questions etc., for 25 marks.
- The internal assessment for the practical courses shall be through home assignment which includes workbook designing algorithm, preparing source code etc., for 25 marks.

- The learners should submit home assignment with worksheet for each course (Theory and Practical's) to **The Director, Directorate of Distance Education (DDE), Alagappa University, Karaikudi** Only along with response sheet contains name of the programme, name of the learner, enrolment number, course name and subject code.

- Learners should submit home assignments of each courses both theory and practicals at least one month before the commencement of end semester examination of every semester.

Division of Internal Marks (Assignment)

Theory		**Practical**	
Assignment	**Marks**	**Assignment**	**Marks**
Review questions	15	Algorithm Design	15
Workbook, case studies, quiz, multiple choice questions	10	Workbook for preparing source code, PL/SQL coding , results	10
TOTAL	25	TOTAL	25

End Semester Examination

The university end Semester Examinations shall be of three hours duration with maximum of 75 Marks for both theory and practical courses.

Learners shall prepare practical record note book according to the following guidelines; aim, algorithm, source code, input, expected output and result of the experiment and submit during end semester practical examination.

f.3.1 Minimum for a pass:

To pass in each course, a candidate is required to secure 50% marks in the end Semester examination (maximum marks 75) 50% marks in the aggregate (marks in Semester Examination + marks in Internal Assessment) (maximum marks 100).

The students who does not secure required minimum marks for pass in a course(s) shall be required to reappear and pass the same in the subsequent examination,

f.3.2 Question Paper Pattern - Theory

The end semester examination will be conducted in the duration of 3 Hours and maximum of 75 Marks.

Answer ALL questions
One question from each unit from the course syllabi

Part – A (10 x 2 Marks: 20 Marks)

Part – B (5 x 5 Marks: 25 Marks) (Internal Choice)

Part – C (3 x 10 Marks: 30 Marks) (Internal Choice)

f.3.3 Procedure for Completing the Course:

A student shall be permitted to continue the programme from I to IV semester irrespective of failure(s) in the courses of the earlier semesters. The candidate will qualify for the M.Sc degree only if he/she passes all the (including arrears) courses with in a period of **FIVE** years from the date of admission.

f.3.4 Results and Classification:

Results will be declared at the end of each semester of the University examination and the marks/grade obtained by the candidate will be forwarded to them by the **Controller of Examinations, Alagappa University, Karaikudi.**

f.3.4.1 Marks and grades

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0 - 10.00	O	Outstanding
80-89	8.0 - 8.9	D+	Excellent
75-79	7.5 - 7.9	D	Distinction

70-74	7.0 - 7.4	A+	Very Good
60-69	6.0 - 6.9	A	Good
50-59	5.0 - 5.9	B	Average
00-49	0.00	U	Reappear
ABSENT	0.00	AAA	Absent

For a semester

Grade Point Average[GPA] = $\sum C_i G_i / \sum C_i$

Grade Point Average = Sum of the multiplication of Grade points by the credit of the courses

Sum of the credit of the courses in the semester

= Sum of [Credit earned x Grade Points]
Sum of the credits earned in the semester

For the entire programme

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

= sum of the multiplication of grade points by the credits of the entire Programme

Sum of the credits of the courses for the entire programme

Where

C_i - Credits earned for the course i in any semester

G_i – Grade Point earned for course i in any semester

n - is number of all Courses successfully cleared during the particular semester in the case of GPA and during all the semesters (programme) in the case of CGPA.

CGPA	**Grade**	**Classification of Final Result**
9.5 – 10.00	O+	First class – Exemplary*
9.0 and above but below 9.5	O	
8.5 and above but below 9.0	D++	First class with Distinction*
8.0 and above but below 8.5	D+	
7.5 and above but below 8.0	D	
7.0 and above but below 7.5	A++	First Class
6.5 and above but below 7.0	A+	
6.0 and above but below 6.5	A	
5.5 and above but below 6.0	B+	Second Class
5.0 and above but below 5.5	B	
0.0 and above but below 5.0	U	Reappear

* The candidates who have passed in the first appearance and within the prescribed semester

f.4 Fees Structure

Fee Particulars	Amount in Rs.		
	First Year	**Second Year**	**Nature of Fees**
Admission Processing Fees	300	--	Non-Refundable
Course Fees	14,300	14,300	Non-Refundable
ICT fees	150	150	Non-Refundable
Total Fees	**14,750**	**14,450**	Non-Refundable

The above mentioned fees structure is exclusive of examination fees.

(g) Requirement of the laboratory support and library resources

g.1 Laboratory Support

A well- equipment Computer Laboratory was established in the Alagappa University, Karaikudi with necessary software's as per the practical's syllabi for conducting face to face contact sessions for practical courses of this programme. Model Practical Questions is available to the learners in the university website.

g.2 Library Resources

The Directorate of Distance Education, Alagappa University provides library facility with number of books and Self Learning materials for Computer Science programmes. The Central library of Alagappa University provides the collection of volumes of Self Learning Materials, Printed books, Subscriptions to printed periodicals and Non-book materials in print form for the learner's references. All these library resources are meant for learner's reference purpose only.

(h) Cost estimate of the programme and the provisions:

Expense details	**Amount in (Rs.) Approx.**	**Provisions for the fund**
Programme development (Single time)	20,00,000/-	University's Own fund
Programme delivery (per year)	24,00,000/-	University's Own fund and fees to be collected from the learners
Programme maintenance (per year)	5,00,000/-	University's Own fund

(i) Quality assurance mechanism and expected programme outcomes:

i.1 University's Moto:

'Excellence in Action'

i.2 University's Vision and Mission

Vision

Achieving Excellence in all spheres of Education, with particular emphasis on ' **PEARL**' - Pedagogy, Extension, Administration, Research and Learning.

Mission

Affording a High Quality Higher Education to the learners so that they are transformed into intellectually competent human resources that will help in the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (**ESTEEM**).

i.3 University Objectives

1. Providing for instructions and training in such branches of Learning at the university may determine.
2. Fostering Research for the Advancement and Dissemination of Knowledge and Application.

i.4 Quality Policy

Attaining Benchmark Quality in every domain of 'PEARL' to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

i.5 Quality Quote

Quality Unleashes Opportunities Towards Excellence (QUOTE).

i.6. Course benchmarks

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations and number of enrolments of students. Feedback from the alumni, students, parents, stakeholders and employers will be received to analyze the benchmark qualities for the further improvement of the programme.

Appendix

Detailed Syllabi

SEMESTER I

Course Code	Title of the Course
11	DESIGN AND ANALYSIS OF ALGORITHMS

Course Objectives:

- To provide an overall knowledge in an algorithmic problem solving.
- To understand the concept of sorting
- To understand the concepts of back tracking, travelling salesman problem.

Course Outcome:

- Able to understand the various algorithmic problem solving methods.

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	**Introduction**: notion of algorithm, fundamentals of algorithmic problem solving, important problem types, fundamentals of analysis of algorithm efficiency
2	**Asymptotic notations:** Big-oh notation, omega notation, theta notation
3	**Performance analysis:** space complexity, time complexity, pseudo code for algorithms
	BLOCK 2 : MATHEMATICAL ANALYSIS OF NON RECURSIVE ALGORITHMS
4	**Analysis of Recursive algorithms**: algorithms for computing Fibonacci numbers
5	**Empirical analysis of algorithms:** Brute force, selection sort, Bubble sort, sequential sort
6	**Closet-pair and convex-hull problems:** Divide and conquer, merge sort, quick sort, Binary search, Strassens matrix multiplication
	BLOCK 3 : DYNAMIC PROGRAMMING AND SEARCH BINARY TREES
7	**General method:** computing a Binomial coefficient, warshalls and Floyds algorithms, optimal search Binary trees, knapsack problems
8	**Greedy Technique:** General method
9	**Applications :** prims algorithm, kruskals algorithm, dijikstras algorithm
	BLOCK 4 : SORTING AND OPTIMIZATION PROBLEM
10	**Sort and Searching algorithms:** decrease and conquer, Insertion sort, Depth first search and Breadth first search, Topological sorting
11	**Generating combinatorial objects:** Transform and Conquer, presorting, Heap and Heap sort
12	**Optimization Problems:** Reductions, Reduction to Graph Problems
	BLOCK 5 : BACKTRACKING AND GRAPH TRAVERSALS
13	General method: 8 queens problem, sum of subsets, Graph colouring, Hamiltonian cycle, Branch and Bound, assignment problem, knapsack problem, travelling salesman pobles
14	Graph traversals: connected components, spanning trees, NP hard and NP complete problems

Text books:

- Introduction to Design and Analysis of Algorithms, Anany Levitin, Pearson Education

Reference Books

- Fundamentals of computer algorithms, Ellis Horowitz, sartaj sahani and Rajasekaran, Galgotia publications
- Algorithm Design: Foundations, analysis and Internet examples, M T Goodrich and R. Tomassia, John wiley and sons,2006

Course Code	Title of the Course
12	APPLIED MATHEMATICS FOR COMPUTER SCIENCE

Course objective

- To understand the basics of normal forms
- To understand the concept of graph theory

Course outcome

- Able to understand the graph theory
- Able to know transportation problem and solutions

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	Logic : TF statements, connectives, atomic and compound statements,
2	WFF, truth table of a formula,
3	Tautology, tautological implications and equivalence of formulae
	BLOCK 2 : NORMAL FORMS
4	Principal normal forms
5	Theory of inference, open statements, quantifiers, valid formulae and equivalence,
6	Theory of inference for predicate calculus
	BLOCK 3 : GRAPH THEORY
7	Basics concept of graph theory
8	Matrix representation of graphs
9	Trees : Definition, Spanning trees, Rooted trees, Binary trees
	BLOCK 4 : LINEAR PROGRAMMING PROBLEM
10	Mathematical foundations, graphical solutions
11	Slack of variables, simplex method, two phase method
	BLOCK 5 : TRANSPORTATION PROBLEM
12	Transportation table, solutions of transportation problem
13	Testing for optimality, assignment problem
14	The assignment method, special cases in assignment problems

Text Books:

1. Discrete Mathematics – Dr. M.K.Venkataraman, Dr N.Sridharan, N.Chandrasekaran- The National Publishing Company – Reprint 2003 (Unit I, II and III)
2. Operation Research – Kantiswarap, P.K.Gupta, Man Mohan- Sultan Chand & Sons – Reprint 2011.

Book for Reference:

1. Discrete Mathematical Structures with Applications to Computer science – J.P.Trembley, R.Manohar Tata McGraw Hill.

Course Code	Title of the Course
13	ADVANCED JAVA PROGRAMMING

Course Objective

- To understand the advanced concepts in Java Programming
- To understand Internetworking using Java Programming

Course Outcome

- Able to write Programs using JDBC concept
- Able to write Servlet Programs

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	Introduction : JDBC overview, connection class, meta data function
2	SQL Exception, SQL warning
3	Statement, Result set, Other JDBC classes
	BLOCK 2 : INETADDRESS
4	Inetaddress, TCP/IP client sockets, TCP/IP server sockets
5	URL, URL connection, Data grams
6	Client/Server applications using RMI
	BLOCK 3 : BDK AND DESIGN PATTERNS
7	Bean Development Kit, JAR files, Introspection
8	Design Patterns for properties, Events and methods
9	Constrained Properties, Persistence, Customizers
	BLOCK 4 : SERVLETS
10	Life cycle of servlet, Generic Servlet, HTTP servlet
11	Reading Initialization, Parameters, Reading Servlet Parameters
12	Cookies, Session Tracking
	BLOCK 5 : JAPPLET AND AWT CLASSES
13	JApplet, Buttons, Combo, Trees, Tables, Panes
14	AWT Classes, Working with Graphics, Color and Font

Text Books:

1. Patrick Naughton & Herbert Schildt, "The Complete Reference: Java 2", Tata
2. McGraw Hill, 1999. (Chapter - 18, 21, 24, 25, 26, 27)
3. Joseph Weber, "Using Java 2 Platform", Prentice Hall of India, 2000. (Chapter - 39, 40)

Books for Reference:

1. Deitel & Deitel, "Java How to Program", Prentice Hall, 5th Edition ,2002
2. Peter Haggar, "Practical Java: Programming Language Guide", Addison-Wesley Pub Co, 1st Edition, 2000

3. Bruce Eckel, "Thinking in Java", Pearson Education Asia, 2nd Edition, 2000

Course Code	Title of the Course
14	LAB – ADVANCED JAVA PROGRAMMING

Course Objective:

- To understand to write applet programs
- To understand JDBC application programs

Course Outcome:

- Able to develop Client/Server application programs using RMI
- Develop chat server using Java

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	Simple Java program
2	Program using JDBC with create, insert table data
3	SQL Exception, SQL Warning
	BLOCK 2 : INETADDRESS
4	Programs using TCP/IP client sockets, TCP/IP server sockets
5	Program with URL, URL connection, Data grams connection
6	Client/Server applications using RMI
	BLOCK 3 : BDK AND DESIGN PATTERNS
7	Simple programs using Bean Development Kit, JAR files
8	Program with Design Patterns,
9	Program with Events and methods
	BLOCK 4 : SERVLETS
10	Create a servlet to read the parameters
11	Programs using cookies
12	Programs with session tracking
	BLOCK 5 : JAPPLET AND AWT CLASSES
13	Programs using JApplet, Buttons, Combo, Trees, Tables, Panes
14	Programs with AWT Classes, Working with Graphics, Color and Font

Reference Books

1. Joseph Weber," Using Java 2 Platform" PHI ,2000

II Semester

Course Code	Title of the Course
21	COMPUTER SYSTEM ARCHITECTURE

Course Objective:

- To understand the computer system architecture, design
- To understand the Instruction Level Parallelism, memory optimizations

Course Outcome:

- Able to learn the computer design
- Able to learn the memory optimization, storage systems

Unit No.	Contents
	BLOCK 1: FUNDAMENTALS
1	Introduction : Definition, trends, power in IC, cost
2	Performance : Dependability, measuring, reporting and summarizing performance
3	Quality: Quality principles of computer design, performance
	BLOCK 2 : ILP CONCEPTS
4	Introduction: concepts and challenges, Basic computer techniques for exposing ILP, reducing branch costs with prediction, data hazards
5	Scheduling : dynamic scheduling, hardware based speculation, multiple issue and static scheduling, advanced techniques for instruction delivery and speculation
6	Limitations of ILP: hardware and software speculation, multithreading
	BLOCK 3 : THREAD LEVEL PARALLELISM
7	Multiprocessor and thread level parallelism: Introduction, symmetric shared memory architecture
8	Performance and architectures: performance of symmetric shared memory multiprocessors, Distributed shared memory architectures
9	Synchronization models: synchronization, model of memory consistency, cross cutting issues
	BLOCK 4 : MEMORY HIERARCHY DESIGN
10	Introduction : Optimization of cache performance, memory technology and optimizations
11	Protection: virtual memory and virtual machines
12	Issues : crosscutting issues in the design of memory hierarchies
	BLOCK 5 : STORAGE SYSTEMS
13	Introduction : advanced topics in Disk storage, real faults and failures, I/O performance, reliability measures and benchmarks
14	Issues : a little queuing theory, crosscutting issues, designing and evaluating and I/O system, the Internet Archive Cluster

Text Books:

1. John L.Hennessy & David A Patterson Morgan, Compute architecture a quantitative approach,4th edition kufmann

Reference Books:

1. Kai Hwang and A.Briggs, Computer Architecture and parallel processing, International edition, McGraw Hill
2. Dezso Sima, Terence Fountain, Peter kacsuk, Advanced computer architecture, Addison Wesley,1997

Course Code	Title of the Course
22	**DISTRIBUTED OPERATING SYSTEMS**

Course Objective:

- To understand the issues in designing Distributed system
- To understand the distributed file system and security

Course Outcome:

- Able to learn the distributed file system
- Able to learn the security principles in distributed system

Unit No.	Contents
	BLOCK 1: FUNDAMENTALS
1	Introduction : what is distributed operating system, Evolutions, models
2	Issues in designing distributed computing system
3	Introduction to computer networks: Network types, LAN, WAN, communication protocols, Internetworking, ATM technology
	BLOCK 2 : MESSAGE PASSING
4	Introduction: features, issued in PC message passing, synchronization
5	Buffering, multidatagram messages, Encoding and Decoding
6	Process addressing, failure handling, group communication
	BLOCK 3 : DISTRIBUTED SHARED MEMORY
7	Introduction: General architecture of DSM system, Design and implementation issues of DSM, Granularity, structure of shared memory consistency models, Replacement strategy, Thrasing
8	Other approaches to DSM, Heterogeneous DSM, advantages
9	Synchronization: Introduction, clock synchronization, Event ordering, mutual Exclusion, Deadlock, Election algorithm
	BLOCK 4 : DISTRIBUTED FILE SYSTEM
10	Introduction : Desirable features, file modes, file accessing models
11	File sharing semantics, file caching schemes, file replication
12	Fault Tolerence, atomic transaction, Design principles
	BLOCK 5 : SECURITY
13	Introduction : potential attacks to computer system, cryptography,

	authentication
14	Access control, Digital Signatures, Design Principles

Text Book:

1. Distributed Operating Systems – Concepts and Design, Pradeep K Sinha, PHI,2014

Book for Reference:

1. Distributed Operating Systems 1e, Andrew S Tanenbaum, PHI.

Course Code	Title of the Course
23	**.NET PROGRAMMING**

Course Objective:

- To understand the .NET frameworks
- To understand the object oriented programming concepts in an .Net technology

Course Outcome:

- Able to learn visual basic .Net from basics to file handling
- Able to learn the ADO.Net and security models

Unit No.	Contents
	BLOCK 1: .NET FRAMEWORKS
1	Introduction: CLR, namespace, assemblies, class library
2	Basic Terminology: .Net component, .Net garbage collection
3	Oops concept: class, objects, structures, modules, abstraction, encapsulation, inheritance, polymorphism, overloading, overriding, shadowing
	BLOCK 2 : VISUAL BASIC.NET
4	**Introduction:** Data types, operators, arrays, dynamic arrays, String handling
5	Control statements: Conditional and looping statements, sub procedures and functions
6	Windows Forms: MDI form, events, msgbox, inputbox, Dialogboxes, passing forms, RichTextBoxes, Labels, Link labels
	BLOCK 3 : WINDOWS CONTROLS
7	Introduction: Buttons, checkbox, radio buttons, panel, list boxes, combo boxes, scrollbars, splitters, track bars, pickers, notify icons, timers, menus
8	Tree and list view: toolbars, status bars, progress bars, tab controls
9	Debugging and Error Handling: Types of errors, Exceptions and structured exception handling
	BLOCK 4 : ASP.NET
10	Introduction: file types, Importing namespaces, usage of Global.asax file, The page class, HttpRequest, HttpResponse, Server Utility
11	Basic web controls: List controls, validation and Rich controls, Data controls, custom controls
12	Overview of AJAX controls

	BLOCK 5 : ADO.NET
13	Introduction: Database access in the Internet world, characteristics, Data objects, Data namespace
14	SQL Basics: Data binding controls, Data set, Data table, Data row, data column, data list, data grid

Text Book:

1. Visual Basic .Net programming, Steve Holzner, Dreamtech press.
2. The complete reference for ASP.Net, Mathew macdonald, TMH

Books for Reference:

1. Visual Basic .Net programming Bible, Bill Evjen, JasonBeres, Wiley dreamtech press.

Course Code	Title of the Course
24	**LAB – .NET PROGRAMMING**

Course Objective:

- To understand and practice with visual basic.net controls
- To understand and practice ASP.Net file types, AJAX controls

Course Outcome:

- Able to design visual basic.net programs with databases
- Able to design develop ASP.Net programs with Interactive web pages

Unit No.	Contents
	BLOCK 1: .NET FRAMEWORK
1	Programs using variables, constants and data types
2	Programs using arrays and dynamic arrays
3	Program using control flow statement
	BLOCK 2 : VISUAL BASIC.NET
4	Programs using functions and procedures, MDI forms, events
5	Programs using msgbox, inputbox, dialog boxes, working with multiple forms
6	Anchoring and docking controls, event handling, RichTextBoxes
	BLOCK 3 : WINDOWS CONTROLS
7	Programs using windows common controls
8	Programs using menus, built in dialog boxes, Image lis, tree and list views
9	Programs using toolbars, statusbars, progressbars, tab controls, graphics and file handling
	BLOCK 4 : ASP.NET
10	Writing ASP programs using HttpRequest and HttpResponse
11	Develop an application for ASP web controls, list controls, validation and rich controls, Data controls
12	Develop an application for HTML server controls, custom controls, logging and error handling
	BLOCK 5 : ADO.NET

13	Database applications using ADO.NET
14	Accessing a database using SQL commands, Data binding controls, DataList

Text Book:

1. Visual Basic .Net programming, Steve Holzner, Dreamtech press.
2. The complete reference for ASP.Net, Mathew macdonald, TMH

Books for Reference:

1. Visual Basic .Net programming Bible, Bill Evjen, JasonBeres, Wiley dreamtech press.

III SEMESTER

Course Code	Title of the Course
31	**CRYPTOGRAPHY AND NETWORK SECURITY**

Course Objective:

- To understand the computer security concepts
- To understand the Data Encryption Standard mechanism

Course Outcome:

- Able to know AES, RSA cryptography principles
- Able to know Digital Signatures, E-mail security

Unit No.	Contents
	BLOCK 1: COMPUTER SECURITY INTRODUCTION
1	Introduction: The OSI security architecture, security attacks,
2	security services, security mechanisms, A model for network security
3	Classical Encryption Techniques: symmetric cipher model, substitution techniques
	BLOCK 2 : BLOCK CIPHERS AND DES
4	Block cipher principle, the data encryption standard, The strength of DES,
5	Differential and Linear cryptanalysis, Block cipher design principles
6	Advanced Encryption Standard: Finite Field arithmetic , AES structure, AES transformation function, Implementation
	BLOCK 3 : PUBLIC KEY CRYPTOGRAPHY AND RSA
7	Principles of public-key cryptosystems, The RSA algorithms
8	Other public key cryptosystems: Diffie-Helman key Exchange, Elgamel cryptographic system
9	Elliptic curve cryptography, pseudorandom number generation based on asymmetric cipher
	BLOCK 4 : MESSAGE AUTHENTICATION CODES
10	Message authentication requirements, functions, message authentication codes
11	Security of MACs, MAC based Hash functions, MAC based ciphers
12	Digital Signatures: ElGamal Digital Signature scheme, schnorr digital

	signature schemes, digital signature standard
	BLOCK 5 : TRANSPORT LEVEL SECURITY
13	Web security considerations, Socket layer and transport layer and transport layer security
14	Electronic mail security: pretty good privacy, IP security overview, IP security policy, encapsulating security payload

Text Book:

1. William Stallings, "Cryptography and Network Security Principles and Practice", Pearson, 5th Edition.

Book for Reference:

1. William Stallings - "Data Communication" - Pearson

Course Code	Title of the Course
32	**CLOUD COMPUTING**

Course objective:

- To understand service oriented architecture and virtual storage applications

Course outcome:

- Able to know cloud computing Environments
- Able to know cloud virtualization technology

Unit No.	Contents
	BLOCK 1: CLOUD COMPUTING BASICS
1	Introduction: History, working with cloud computing, pros and cons of cloud computing, Benefits, ,
2	Developing cloud services, pros and cons of cloud service development, types of cloud service development
3	Discovering cloud services development services and tools
	BLOCK 2 : CLOUD COMPUTING FOR EVERYONE
4	Centralizing Email communications, collaborating to-do lists
5	Collaborating on household budgets, contact lists, communications across community
6	Collaborating on schedules, collaborating on group project and events, cloud computing for corporation
	BLOCK 3 : CLOUD SERVICES
7	Exploring on-line calendar applications, Exploring online scheduling applications, Exploring online planning and task management
8	Collaborations with event management, contact management, project management, word processing and databases
9	Storing and sharing files and other online content
	BLOCK 4 : CLOUD COMPUTING ENVIRONMENT
10	Classification of cloud Implementation, Amazon web services, IaaS, VMware vCloud, Google AppEngine,PaaS, Windows Azure Platform, SaaS/PaaS, Microsoft Live

11	Comparison of cloud computing platforms
	BLOCK 5 : CLOUD VIRTUALIZATION TECHNOLOGY
12	Introduction : Virtualization Defined, Benefits, server virtualization
13	Hypervisor management software, Logical Partitioning, VIO Server, virtual Infrastructure requirements
14	Cloud virtualization: Introduction, storage virtualization, Storage area networks, cloud server virtualization, virtualized Data Centre

Text Book:

1. Michael Miller, Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online, Que Publishing, August 2008.

Book for Reference:

1. Haley Beard, Cloud Computing Best Practices for Managing and Measuring Processes for On-demand Computing, Applications and Data Centers in the Cloud with SLAs, Emereo Pty Limited, July 2008.

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Course Code	Title of the Course
33	**WEB TECHNOLOGY**

Course Objective:

- To understand the wide range of web technologies both client side and server side to provide exposure to the students in developing Rich Internet Applications.

Course Outcome:

- Able to know client side and server side programming

Unit No.	Contents
	BLOCK 1: HTML,XHTML AND STYLE SHEETS
1	**Introduction:** HTML, XML and WWW, Basic HTML, document Body, text, hyperlinks
2	Lists, using color and images, tables, multimedia objects
3	Style sheets: using styles, examples, formatting blocks of information
	BLOCK 2 : CLIENT SIDE PROGRAMMING
4	Introduction: Dynamic HTML, Java script, variables, string manipulations, mathematical functions, operators, arrays, functions
5	Regular expressions, cookies and Events
6	Dynamic HTML with Java script: Data validation, messages and confirmations, writing to a different frame, Rollover buttons, moving images
	BLOCK 3 : HOST OBJECTS
7	Browsers and DOM, DOM history and levels, Intrinsic event handling,
8	Representing web Data: XML, Documents and vocabularies, versions and declarations, namespaces
9	Java script and XML: Ajax, DOM based XML processing, SAX,XSL,XSLT,XPATH
	BLOCK 4 : SERVER SIDE PROGRAMMING
10	Java Servlets, history of web applications, The power of Servlets,

	HTTP servlet basics, the servlet API, page generations
11	The servlet Lifecycle: The servlet alternative, servlet reloading, Init and Destroy, single thread model, background processing, load on startup, client side caching, server side caching
12	Retrieving information: the servlet, the server, the client
	BLOCK 5 : JSP TECHNOLOGY
13	Introduction: Need, HTTP and servlet basics, HTTP request/response model, Servlets, anatomy of a JSP page, JSP application design with MVC
14	Setting up JSP Environment: Installing the JSDK, Installing Tomcat server, testing tomcat, creating, installing, running a JSP page

Text Books:

1. Web Programming: Building Internet applications, Chris Bates, Wiley India
2. Web technologies – A computer science perspective, Jeffrey C Jackson, Pearson Education,2006
3. Java server pages, Hans Bergsten, o'reilly,2010

Reference Books:

1. Robert W.Sebesta "Programming the world wide web" Pearson Education
2. Java servlet Programming, Joson Hunter, o'Reilly,2010
3. Bates, Developing web applications, wiley,2006.

Course Code	Title of the Course
34	**LAB – WEB TECHNOLOGY**

Course Objective:

- To understand the wide range of web technologies both client side and server side to provide exposure to the students in developing Rich Internet Applications.

Course Outcome:

- Able to develop client side and server side programming

Unit No.	Contents
	BLOCK 1: HTML,XHTML,STYLESHEETS
1	Programs using Basic HTML, text and hyperlinks
2	Programs using multimedia objects, XHTML
3	Programs using Style sheets
	BLOCK 2 : CLIENT SIDE PROGRAMS
4	Programs using JavaScript, Dynamic HTML, operators, arrays, cookies
5	Programs using Java script data validation, messages and confirmations
6	Programs using Rollover Buttons, writing to a Different frame, moving images
	BLOCK 3 : BROWSERS AND DOM
7	Programs using Intrinsic event handling, document tree
8	Representing web data, XML
9	Programs using Java script and XML, XSL, XSLT
	BLOCK 4 : SERVER SIDE PROGRAMS

10	Programs with simple Java Servlets
11	Programs with Init and Destroy, single thread model, client side caching and server side caching
12	Programs using Retrieving information from The Servlet-The Server-The Client
	BLOCK 5 : JSP PROGRAMS
13	Programs with simple JSP applications
14	Developing a program to access a database from a JSP page

Text Books:

1. Web Programming: Building Internet applications, Chris Bates, Wiley India
2. Web technologies – A computer science perspective, Jeffrey C Jackson, Pearson Education,2006
3. Java server pages, Hans Bergsten, o'reilly,2010

Reference Books:

1. Robert W.Sebesta "Programming the world wide web" Pearson Education
2. Java servlet Programming, Joson Hunter, o'Reilly,2010
3. Bates, Developing web applications, wiley,2006.

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IV SEMESTER

Course Code	Title of the Course
41	**DATA MINING AND WAREHOUSING**

Course objective:

- To learn design of data store of warehousing, Retrieving and mining information of warehouse

Course outcome:

- Able to know data mining techniques and trends
- Able to know associative rules, clustering techniques and web mining.

Unit No.	Contents
	BLOCK 1: DATAWAREHOUSING
1	Introduction: definition, architecture, warehouse schema, warehouse server, OLAP operations
2	Data warehouse technology: Hardware and operating system, warehousing software, Extraction tools, Transformation tools
3	Case studies: data warehousing in Government, tourism, Industry, Genomics data
	BLOCK 2 : DATA MINING
4	Introduction: definition, techniques, current trends in data mining
5	Different forms of knowledge: Data selection, cleaning, Integration, Transformation, Reduction and Enrichment
6	Data : types of data, data quality, data preprocessing, measures of similarity and dissimilarity, Exploration, summary statistics, visualization
	BLOCK 3 : ASSOCIATION RULES
7	Introduction: methods to discover association rule, apriori algorithm partition

	algorithm, pincher search algorithm
8	Dynamic Item set algorithm, FP Tree growth algorithm
9	Classification: Decision tree classification, bayesian classification, classification by Back propagation
	BLOCK 4 : CLUSTERING TECHNIQUES
10	Introduction: clustering paradigms, partitioning algorithms, K means & K mediod algorithms, CLARA, CLARANS, Hierarchial clustering, DBSCAN,BIRCH, clustering algorithms, STIRR, ROCK, CACTUS
11	Machine Learning: supervised learning, unsupervised learning, machine learning and data mining
12	Neural networks: Introduction, use of NN, working of NN Genetic algorithm
	BLOCK 5 : WEB MINING
13	Introduction : web content mining, web structure mining, web usage mining, text mining, Text clustering, Temporal, spatial, visual data mining, knowledge mining
14	Tools and techniques: using weka, Rapidminer and matlab

Text Book:

1. Arun K Pujari, Data mining Techniques, University press,2008
2. CSR Prabhu, Data warehousing-concepts, techniques and applications, Prentice Hall of India.

Book for Reference:

1. Jaiwan han, Michelinne Kambar, Data Mining: Concepts and Techniques, Harcourt India/Morgan Kauffman Publishers ,2008
2. Alex Berson, Stephen J Smith, Data warehousing, data mining&OLAP, TMH, 2004.

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Course Code	Title of the Course
42	**MOBILE APPLICATION DEVELOPMENT**

Course objective:

- To provide an overall knowledge about mobile devices, communication methodologies and its application development

Course outcome:

- Able to know mobile ecosystem, mobile information architecture
- Able to know J2ME architecture and development and case studies

Unit No.	Contents
	BLOCK 1: MOBILE ECOSYSTEM
1	Introduction: The mobile ecosystem, operators, networks
2	Devices : platforms, operating systems
3	Applications : application frameworks, applications, services
	BLOCK 2 : MOBILE DEVICE PROFILES
4	Categories : SMS, mobile websites, mobile web widgets
5	Native applications: Games, utility apps, location based services(LBS)
6	Apps : Informative apps, Enterprise apps
	BLOCK 3 : MOBILE INFORMATION ARCHITECTURE

7	Introduction : sitemaps, click streams, wireframes, prototyping, architecture
8	Mobile design : Interpreting design, Elements of mobile design
9	Mobile design tools : Designing for different device/ screens
	BLOCK 4 : J2ME
10	Introduction : J2ME architecture and development environment, small computing device requirements, Run-time environment, MIDlet programming
11	Languages : J2ME, J2ME SDK, J2ME wireless toolkit
	BLOCK 5 : CASE STUDY
12	Introduction : Google Android introduction, Android development Environment
13	Development framework, SDK, Eclipse, Emulator, Android AVD
14	Project framework: Apple IOS, RIM Blackberry, Samsung Bada, Nokia Symbian, Microsoft windows phone

Text Book:

1. Brian Fling, Mobile Design and Development, OReilly media,2009.
2. James Keogh, J2ME, The Complete Reference, TataMcHill,2003

Book for Reference:

1. Pei zheng and Lionel Ni, Smart phone and Next Generation Mobile Computing, Elsevier, 2006
2. Mark L.Murphy, Beginning Android", Apress 2009.

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Course Code	Title of the Course
43	**ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS**

Course objective:

- To familiarize the basic concepts in artificial intelligence
- To understand the basic concepts in expert systems

Course outcome:

- Apply forward and backward reasoning algorithms, searching algorithms and optimization algorithms to create problem solving agents.
- Represent knowledge using propositional logic, inference rules and Bayesian networks.
- Design simple expert systems.
- Design algorithms for robotics and machine vision.

Unit No.	Contents
	BLOCK 1: PROBLEMS ANDSEARCH
1	Introduction: Concept of AI, approaches –Application areas Problem formulation- -Forward & Backward reasoning-Graphs & Trees
2	Measuring Problem solving agents: problem solving performance
3	Search Strategies-local search algorithms and optimization problems, Genetic Algorithms, terminology.
	BLOCK 2 : KNOWLEDGE REPRESENTATION
4	Relational knowledge & Procedural knowledge Propositional Logic – Syntax & semantics –Inference rules –Inference methods

5	Knowledge engineering process –Handling uncertain knowledge
6	Bayesian networks –Learning –Pattern recognition.
	BLOCK 3 : KNOWLEDGE BASED SYSTEMS
7	Expert systems–Components, Characteristic features of expert systems
8	Rule based system architecture-Using domain knowledge
9	Expert system shell -Explaining the reasoning and knowledge acquisition-Applications
	BLOCK 4 : AI IN ROBOTICS
10	State space search -Block word & robot example -Path selection -Monkey & Banana problem AND –OR graph -Means end analysis in a robotic problem -Robot problem solving as a production system -Triangle table-Robot learning
11	Robot task planning -Phases in task planning -Symbolic spatial relationships -Obstacle avoidance -Graph planning.
	BLOCK 5 : MACHINE VISION
12	Introduction –Functions in a vision system –Imaging devices –Lighting–A-D conversion–Quantization–Encoding image storage–Image data reduction
13	Segmentation techniques –Feature extraction –object recognition
14	Training the vision system –Robotic applications of machine vision

Text books

1 .Stuart Russel, Peter Norvig,“Artificial Intelligence: A Modern Approach-2 /e”,2003,Pearson Education.
2. Elaine Rich, Kevin Knight,“Artificial Intelligence”2/e,1991, TMH.

References

1. Dan W.Patterson, “Introduction to Artificial Intelligence & Expert Systems”, Seventh Indian Reprint 1999, EEE, PHI.

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Course Code	Title of the Course
44	**PROJECT**

1. The students will be allowed to work an any project based on the concepts studied in core courses
2. The following list of parameters taken into account for the evaluation of project work.
   Total marks: 100 (Internal : 25 marks, External:75 marks)

Minutes of the Meeting of the Board of Studies in Computer Science for the Master of Computer Applications (M.C.A), M.Sc(Information Technology), M.Sc. (Computer Science), Post Graduate Diploma in Computer Applications (P.G.D.C.A), Bachelor of Computer Applications (B.C.A), B.Sc (Information Technology), B.Sc. (Computer Science) Programmes to be offered through Open Distance Learning (ODL) Mode held at The Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 04.09.2017, (11.00 A.M).

Members Present

1.	Dr. V. Palanisamy	-	Chairman
2.	Dr. E.Ramaraj	-	Member
3.	Dr. K.Kuppusamy	-	Member
4.	Dr. T.Meyyappan	-	Member
5.	Dr. S.S.Dhenakaran	-	Member
6.	Dr. K.Mahesh	-	Special Invitee
7.	Dr. A. Padmapriya	-	Special Invitee
8.	Dr. P. Prabhu	-	Member
9.	Mr.S.Balasubramanian	-	Member

After the deliberation and discussion the board resolved the following:

1. The Board considered the curriculum design and detailed syllabi of Computer Science programmes, prepared as per the norms and the Board scrutinized and necessary modifications are specified.

2. The Board resolved to approve curriculum design, detailed syllabi and other regulations for the Master of Computer Applications (M.C.A), M.Sc(Information Technology), M.Sc. (Computer Science), Post Graduate Diploma in Computer Applications (P.G.D.C.A), Bachelor of Computer Applications (B.C.A), B.Sc (Information Technology), B.Sc. (Computer Science) programmes to be offered from 2018-2019 academic year onwards by the Directorate of Distance Education of Alagappa University, Karaikudi.

Dr. V. Palanisamy	Dr. E.Ramaraj	Dr. K.Kuppusamy
Dr. T.Meyyappan	Dr. S.S.Dhenakaran	Dr. K.Mahesh
Dr. A. Padmapriya	Dr. P. Prabhu	Mr.S.Balasubramanian